The Negative-Viscosity Effect in Ferrofluids

A Complete First-Principles Reconstruction
of Every Equation

COMPLETE EDITION

Volumes I–VII · Chapters 1–69

A self-contained graduate textbook built on the papers

M. I. Shliomis and K. I. Morozov,
Negative viscosity of ferrofluid under alternating magnetic field,
Physics of Fluids **6**, 2855–2861 (1994)

J.-C. Bacri, R. Perzynski, M. I. Shliomis and G. I. Burde,
"Negative-Viscosity" Effect in a Magnetic Fluid,
Physical Review Letters **75**, 2128–2131 (1995)

All 43 equations of the first paper and all 10 of the second,
derived from first principles with no omitted step.

Preface: How This Book Works

What this volume is

This is the first instalment of a single, continuously growing textbook whose final goal is stated once and never modified:

> *Every equation appearing in the two papers listed on the title page will be derived completely, from first principles, with no omitted algebraic step, no appeal to "standard results", and no unexplained symbol.*

Because the physics of the two papers rests on continuum mechanics, statistical mechanics, magnetostatics and rotational Brownian motion — none of which may be assumed known — the book must first build those tools. That is what Volume I does. It assumes only elementary algebra, trigonometry, and the ability to differentiate and integrate a function of one variable. Everything else, including vector calculus, index notation, the Navier–Stokes equations and Stokes' law for a rotating sphere, is constructed here from scratch.

Volume I ends with the first equation of the Shliomis–Morozov paper, Eq. (SM-1), derived in the eight-part format that will be used for every paper equation from here on.

A note on continuity

This document is Volume I: it is the beginning of the derivation project, and no earlier material precedes it. Everything derived here is therefore new, and every later volume will continue from exactly the point at which this one stops, preserving all numbering, notation and conventions established below. The command **Continue** will always produce the next volume, never a restart and never a repetition.

The two numbering systems

To keep cross-references unambiguous forever, two distinct systems are used.

1. **Paper equations** carry the tag of the source paper:
   - **(SM-n)** is equation n of Shliomis & Morozov (1994), $n = 1, \dots, 43$;
   - **(B-n)** is equation n of Bacri, Perzynski, Shliomis & Burde (1995), $n = 1, \dots, 10$.

   These tags never change, in any volume.

2. **Derived and intermediate equations** are numbered **(chapter.section.number)**, e.g. (6.3.11). Every intermediate step of every derivation receives such a number so that it can be referred to later.

The eight-part format for paper equations

Every equation taken from the papers is presented in exactly this order:

1. **Context** — why the equation is introduced at this point.
2. **Symbols** — every symbol: meaning, mathematical type, units, whether constant or variable.
3. **Assumptions** — every assumption, stated explicitly.
4. **Mathematical Background** — any mathematics not yet available is derived first.
5. **Step-by-Step Derivation** — every algebraic manipulation, without compression.
6. **Physical Interpretation** — what the equation means, and what each term does.
7. **Connection to the Next Equation** — how it leads into what follows.
8. **Consistency Checks** — limits, dimensions, numerical values.

Units

Both papers use the Gaussian (CGS) system, as was universal in magnetism in 1994. We use it throughout, without exception, so that every number in the papers can be reproduced directly:

Quantity	CGS unit	Meaning
length	cm	centimetre
mass	g	gram
time	s	second
force	dyn $= \mathrm{g\,cm\,s^{-2}}$	10^{-5} N
energy	erg = dyn cm	10^{-7} J
dynamic viscosity	P (poise) $= \mathrm{g\,cm^{-1}s^{-1}}$	1 P = 0.1 Pa s; 1 cP $= 10^{-2}$ P
magnetic field H	Oe (oersted)	
magnetisation M, induction B	G (gauss)	1 G $\leftrightarrow$ 1 Oe in vacuum
magnetic moment m	$\mathrm{erg\,G^{-1}}$	
Boltzmann constant k	1.38×10^{-16} $\mathrm{erg\,K^{-1}}$	

A crucial convenience of Gaussian units: the energy of a magnetic moment m in a field H is $-mH$ with m in $\mathrm{erg\,G^{-1}}$ and H in Oe, so that the dimensionless ratio mH/kT contains no extra factors such as μ_0.

Typographic conventions

Vectors are bold upright-italic: $\boldsymbol{v}, \boldsymbol{H}, \boldsymbol{M}, \boldsymbol{\Omega}$. Their magnitudes are the same letter in italic: $v = |\boldsymbol{v}|$, $H = |\boldsymbol{H}|$. Unit vectors carry a hat, $\hat{\boldsymbol{h}} = \boldsymbol{H}/H$, except that we follow the papers in writing $\boldsymbol{h}$ (unbolded in the original) for the unit vector along the field; we will always write it $\hat{\boldsymbol{h}}$ to avoid ambiguity. Tensors are written in components, σ_{ij}. The symbol $\equiv$ means "is defined as".

Note to the complete edition

This single volume contains the seven instalments in which the work was originally written, unaltered in content and unbroken in numbering. Chapters run continuously from 1 to 69; equations carry the same tags they have always carried — (SM-n) and (B-n) for the papers' own equations, and chapter.section.number for every derived and intermediate step — so that every cross-reference made in any volume remains exactly valid here. The half-title pages that separate the volumes have been kept: each states briefly where the previous instalment ended, and they serve as signposts through a long argument. The table of contents that follows covers the whole work.

Master Notation Table

Every symbol below is defined once, in the section indicated, and keeps that meaning in every volume of this book. Symbols not yet introduced in Volume I are listed here in advance so that the table can serve as a permanent reference.

Symbol	Meaning	Type	Units (CGS)
t	time	scalar, variable	s
$\boldsymbol{r}$, x_i	position vector / its components	vector	cm
$\boldsymbol{v}$	fluid velocity field	vector field	$\mathrm{cm\,s^{-1}}$
p	pressure	scalar field	$\mathrm{dyn\,cm^{-2}}$
ρ	mass density of the fluid	scalar	$\mathrm{g\,cm^{-3}}$
ρ_s	mass density of the solid particle material	scalar, constant	$\mathrm{g\,cm^{-3}}$
η	dynamic (shear) viscosity of the carrier liquid	scalar	P
$\nu = \eta/\rho$	kinematic viscosity	scalar	$\mathrm{cm^2\,s^{-1}}$
$\Delta\eta$	rotational (vortical) viscosity: the field-induced addition to η	scalar	P
$\boldsymbol{\Omega}$	local angular velocity of the fluid, $\boldsymbol{\Omega} = \frac{1}{2}\nabla \times \boldsymbol{v}$	vector field	$\mathrm{s^{-1}}$
$\boldsymbol{\Theta}$, $\boldsymbol{\omega}_p$	macroscopic angular velocity of the particles ($\boldsymbol{\Theta}$ in SM, $\boldsymbol{\omega}_p$ in B)	vector field	$\mathrm{s^{-1}}$
d	particle diameter	scalar, constant	cm
$a = d/2$	particle radius	scalar, constant	cm
$V = \pi d^3/6$	volume of one particle	scalar, constant	$\mathrm{cm^3}$
n	number of particles per unit volume	scalar	$\mathrm{cm^{-3}}$
$\varphi = nV$	volume fraction of solid particles	scalar, dimensionless	—
I	total moment of inertia of the particles per unit volume	scalar	$\mathrm{g\,cm^{-1}}$
τ_s	inertial relaxation time of particle rotation	scalar	s
τ_B	Brownian rotational relaxation time	scalar	s
τ_h	hydrodynamic time scale l^2/ν	scalar	s
$\boldsymbol{H}$	magnetic field	vector field	Oe
H_0	amplitude of the alternating field	scalar, constant	Oe
ω	angular frequency of the field, $\omega = 2\pi f$	scalar, constant	$\mathrm{s^{-1}}$
f	frequency of the field	scalar	Hz
$\boldsymbol{M}$	magnetisation (magnetic moment per unit volume)	vector field	G
$\boldsymbol{M}_0$	instantaneous-equilibrium magnetisation	vector field	G

Symbol	Meaning	Type	Units (CGS)
M_s	spontaneous magnetisation of the particle material	scalar, constant	G
$m = M_s V$	magnetic moment of one particle	scalar, constant	$\mathrm{erg\,G^{-1}}$
$\boldsymbol{e}$	unit vector along a particle's magnetic moment	vector, variable	—
$\hat{\boldsymbol{h}}$	unit vector along $\boldsymbol{H}$	vector	—
$\xi = mH/kT$	Langevin parameter (field energy / thermal energy)	scalar, dimensionless	—
$\xi_0 = mH_0/kT$	Langevin parameter built on the field amplitude	scalar	—
ξ_e	effective-field Langevin parameter (Bacri et al.)	scalar	—
$L(\xi)$	Langevin function $\coth\xi - \xi^{-1}$	function	—
χ	initial (linear) magnetic susceptibility	scalar	—
kT	thermal energy	scalar	erg
T	absolute temperature	scalar	K
β	angle between $\boldsymbol{H}$ and $\boldsymbol{\Omega}$	scalar	rad
α	phase lag of $\boldsymbol{M}$ behind $\boldsymbol{H}$	scalar	rad
σ_{ij}	stress tensor	2nd-rank tensor	$\mathrm{dyn\,cm^{-2}}$
e_{ij}	rate-of-strain tensor	2nd-rank tensor	$\mathrm{s^{-1}}$
δ_{ij}	Kronecker delta	2nd-rank tensor	—
ε_{ijk}	Levi-Civita symbol	3rd-rank tensor	—
ζ_r	rotational friction coefficient of one particle	scalar	erg s
D_r	rotational diffusion coefficient	scalar	$\mathrm{s^{-1}}$
$W(\boldsymbol{e}, t)$	orientational distribution function	scalar field on the sphere	—
g, F, G, $\gamma(t)$	auxiliary functions of Secs. II–III of SM	functions	—

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Part I

Mathematical Foundations

Chapter 1

The Language of Vectors and Indices

1.1 Why we need a language before we need physics

Both papers are written in the language of three-dimensional vector calculus: their very first sentence contains $\boldsymbol{\Omega} = \nabla \times \boldsymbol{v}/2$, and their central object is a *cross product*, $\boldsymbol{M} \times \boldsymbol{H}$. None of this can be used until it has been built. This chapter builds it. Nothing in it is specific to ferrofluids; everything in it will be used constantly for the rest of the book.

1.2 Scalars, vectors and tensors

Definition 1.2.1 (Scalar)**.** A *scalar* is a quantity specified by a single number together with a unit, whose value does not change when the coordinate axes are rotated. Temperature T, pressure p and density ρ are scalars.

Definition 1.2.2 (Vector)**.** A *vector* is a quantity specified in three-dimensional space by three numbers (A_1, A_2, A_3) — its *components* along three mutually perpendicular axes — with the property that under a rotation of the axes the three numbers transform exactly as the components of a displacement do. Velocity $\boldsymbol{v}$, magnetic field $\boldsymbol{H}$ and angular velocity $\boldsymbol{\Omega}$ are vectors.

The transformation requirement is not decoration: it is what guarantees that an equation between vectors expresses a fact about nature and not about our choice of axes. Writing $\boldsymbol{A} = \boldsymbol{B}$ is the simultaneous statement of the three equations $A_1 = B_1$, $A_2 = B_2$, $A_3 = B_3$ in *every* coordinate system at once.

Definition 1.2.3 (Second-rank tensor)**.** A *second-rank tensor* is an object with two indices, T_{ij} $(i, j = 1, 2, 3)$, i.e. nine numbers, transforming under rotations as a product of two vector components does. The stress tensor σ_{ij} of Chapter 5 is the example we shall need.

Physically, a second-rank tensor is what one needs whenever a vector cause produces a vector effect that is *not* parallel to it. A force per unit area acting on a surface depends on the orientation of that surface: two directions are involved, hence two indices.

1.3 Index notation and the summation convention

Writing $\boldsymbol{A} \cdot \boldsymbol{B} = A_1B_1 + A_2B_2 + A_3B_3$ becomes intolerable in long calculations. We therefore adopt *index notation*: a vector $\boldsymbol{A}$ is represented by its typical component A_i, where the free index i silently runs over $1, 2, 3$.

Definition 1.3.1 (Einstein summation convention)**.** Whenever an index appears *exactly twice* in a single product, a sum over $1, 2, 3$ on that index is understood, and the summation sign is omitted:

$$A_i B_i \equiv \sum_{i=1}^{3} A_i B_i = A_1 B_1 + A_2 B_2 + A_3 B_3. \tag{1.3.1}$$

A twice-repeated index is called a *dummy* (or bound) index; a once-appearing index is a *free* index.

Two rules follow immediately and will be used without further comment:

1. A dummy index may be renamed at will: $A_i B_i = A_k B_k$, because both are the same sum of three terms.
2. Every term of a valid equation must carry the *same* free indices. An equation $A_i = B_{ij} C_j$ is meaningful (free index i on both sides, j summed); an equation $A_i = B_j$ is not.

Thus the scalar product is

$$\boldsymbol{A} \cdot \boldsymbol{B} = A_i B_i, \tag{1.3.2}$$

a scalar (no free index), and the squared length of a vector is $A^2 = A_i A_i$.

1.4 The Kronecker delta

Definition 1.4.1 (Kronecker delta)**.**

$$\delta_{ij} \equiv \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases} \tag{1.4.1}$$

Written as a matrix, δ_{ij} is the 3×3 identity. Its single most used property is the *substitution property*: for any indexed quantity A_j,

$$\delta_{ij} A_j = \delta_{i1} A_1 + \delta_{i2} A_2 + \delta_{i3} A_3 = A_i, \tag{1.4.2}$$

because exactly one of the three deltas equals 1 (the one with $j = i$) and the other two vanish. In words: *contracting an index with a Kronecker delta renames that index.* Two further consequences, obtained the same way:

$$\delta_{ii} = \delta_{11} + \delta_{22} + \delta_{33} = 3, \tag{1.4.3}$$

$$\delta_{ij} \delta_{jk} = \delta_{ik}. \tag{1.4.4}$$

The first is the trace of the identity in three dimensions; the second follows by applying (1.4.2) with $A_j \to \delta_{jk}$.

1.5 The Levi-Civita symbol and the cross product

Definition 1.5.1 (Levi-Civita symbol)**.**

$$\varepsilon_{ijk} \equiv \begin{cases} +1, & (ijk) \text{ an even (cyclic) permutation of } (123):\ 123,\ 231,\ 312, \\ -1, & (ijk) \text{ an odd permutation of } (123):\ 132,\ 321,\ 213, \\ 0, & \text{any two indices equal.} \end{cases} \tag{1.5.1}$$

Its defining property is *total antisymmetry*: interchanging any two indices changes the sign,

$$\varepsilon_{ijk} = -\varepsilon_{jik} = -\varepsilon_{ikj} = -\varepsilon_{kji}, \qquad \varepsilon_{ijk} = \varepsilon_{jki} = \varepsilon_{kij}. \tag{1.5.2}$$

The second line says a *cyclic* shift (two interchanges) leaves the sign unchanged. The vanishing of ε_{ijk} when two indices coincide is forced by antisymmetry: if $\varepsilon_{iik} = -\varepsilon_{iik}$, then $\varepsilon_{iik} = 0$.

Definition 1.5.2 (Cross product)**.**

$$(\boldsymbol{A} \times \boldsymbol{B})_i \equiv \varepsilon_{ijk} A_j B_k. \tag{1.5.3}$$

Let us check that this reproduces the familiar formula. Take $i = 1$. The sum over j, k contains only terms with $\varepsilon_{1jk} \neq 0$, i.e. $(j, k) = (2, 3)$ giving $\varepsilon_{123} = +1$ and $(j, k) = (3, 2)$ giving $\varepsilon_{132} = -1$:

$$(\boldsymbol{A} \times \boldsymbol{B})_1 = \varepsilon_{123} A_2 B_3 + \varepsilon_{132} A_3 B_2 = A_2 B_3 - A_3 B_2, \tag{1.5.4}$$

which is indeed the first component of the elementary cross product. The other two components follow by cyclic permutation, $1 \to 2 \to 3 \to 1$.

Two properties follow at once from antisymmetry (1.5.2):

$$(\boldsymbol{A} \times \boldsymbol{B})_i = \varepsilon_{ijk} A_j B_k = -\varepsilon_{ikj} A_j B_k = -\varepsilon_{ijk} A_k B_j = -(\boldsymbol{B} \times \boldsymbol{A})_i, \tag{1.5.5}$$

$$\boldsymbol{A} \times \boldsymbol{A} = \boldsymbol{0}, \tag{1.5.6}$$

where in (1.5.5) we first used the antisymmetry of ε in its last two indices, then *renamed the dummy indices* $j \leftrightarrow k$ (legal, because both are summed).

Also, contracting a cross product with one of its own factors gives zero:

$$\boldsymbol{A} \cdot (\boldsymbol{A} \times \boldsymbol{B}) = A_i \varepsilon_{ijk} A_j B_k = 0, \tag{1.5.7}$$

because ε_{ijk} is antisymmetric in $i \leftrightarrow j$ while $A_i A_j$ is symmetric, and the contraction of a symmetric with an antisymmetric pair of indices always vanishes. (Proof of that statement: if $S_{ij} = S_{ji}$ and $A_{ij} = -A_{ji}$, then $S_{ij} A_{ij} = S_{ji} A_{ij} = -S_{ji} A_{ji} = -S_{ij} A_{ij}$ after renaming, so $S_{ij} A_{ij} = 0$.) Equation (1.5.7) is the statement that $\boldsymbol{A} \times \boldsymbol{B}$ is perpendicular to $\boldsymbol{A}$, and by the same argument to $\boldsymbol{B}$: the cross product points along the normal to the plane of the two vectors. This is why torques and angular velocities — quantities that characterise *rotation in a plane* — are represented by vectors along the rotation axis.

1.6 The ε–δ identity

The single most useful algebraic identity in this book is

$$\boxed{\varepsilon_{ijk} \varepsilon_{klm} = \delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}} \tag{1.6.1}$$

(the repeated index k is summed).

Proof. Both sides are numbers determined by the four free indices i, j, l, m, so we compare them case by case.

Case 1: $i = j$. The left side contains $\varepsilon_{ijk} = \varepsilon_{iik} = 0$, so it vanishes. The right side becomes $\delta_{il}\delta_{im} - \delta_{im}\delta_{il} = 0$. Both sides vanish. The same argument applies if $l = m$.

Case 2: $i \neq j$ and $l \neq m$. For fixed $i \neq j$, the sum over k has exactly one nonzero term, namely the k that differs from both i and j; similarly ε_{klm} is nonzero only for the k differing from both l and m. Hence the left side is nonzero only if $\{i, j\} = \{l, m\}$ as sets.

Case 2a: $i = l$, $j = m$ (with $i \neq j$). Then the left side is $\varepsilon_{ijk}\varepsilon_{kij} = \varepsilon_{ijk}\varepsilon_{ijk}$ (cyclic shift), which is $(\pm 1)^2 = 1$ for the single surviving k. The right side is $\delta_{ii}^{(\text{no sum})}\delta_{jj}^{(\text{no sum})} - \delta_{ij}\delta_{ji} = 1 \cdot 1 - 0 = 1$. Equal.

Case 2b: $i = m$, $j = l$ (with $i \neq j$). The left side is $\varepsilon_{ijk}\varepsilon_{kji} = -\varepsilon_{ijk}\varepsilon_{kij} = -1$ by Case 2a and one interchange. The right side is $\delta_{ij}\delta_{ji} - \delta_{ii}\delta_{jj} = 0 - 1 = -1$. Equal.

Case 2c: $\{i, j\} \neq \{l, m\}$. The left side vanishes, as argued. On the right, $\delta_{il}\delta_{jm}$ and $\delta_{im}\delta_{jl}$ both vanish, since each requires $\{i, j\} = \{l, m\}$. Equal.

All cases agree, so (1.6.1) holds identically. ■

1.7 Triple products

Result 1.7.1 (Vector triple product)**.**

$$\boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C}) = \boldsymbol{B}\,(\boldsymbol{A} \cdot \boldsymbol{C}) - \boldsymbol{C}\,(\boldsymbol{A} \cdot \boldsymbol{B}). \tag{1.7.1}$$

Derivation. Write the ith component using (1.5.3) twice. The inner product $(\boldsymbol{B} \times \boldsymbol{C})_k = \varepsilon_{klm}B_lC_m$, hence

$$\begin{aligned}
[\boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C})]_i &= \varepsilon_{ijk}A_j(\boldsymbol{B} \times \boldsymbol{C})_k && && (1.7.2)\\
&= \varepsilon_{ijk}A_j\,\varepsilon_{klm}B_lC_m && && (1.7.3)\\
&= \varepsilon_{ijk}\varepsilon_{klm}\,A_jB_lC_m && && (1.7.4)\\
&= (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl})A_jB_lC_m && \text{by (1.6.1)} && (1.7.5)\\
&= \delta_{il}\delta_{jm}A_jB_lC_m - \delta_{im}\delta_{jl}A_jB_lC_m && && (1.7.6)\\
&= A_mB_iC_m - A_lB_lC_i && \text{by the substitution property (1.4.2)} && (1.7.7)\\
&= B_i(A_mC_m) - C_i(A_lB_l) && && (1.7.8)\\
&= B_i(\boldsymbol{A} \cdot \boldsymbol{C}) - C_i(\boldsymbol{A} \cdot \boldsymbol{B}). && && (1.7.9)
\end{aligned}$$

Since this holds for every i, (1.7.1) follows. ■

Equation (1.7.1) is used in this book more than any other identity; in particular the combination $(\boldsymbol{e} \times \boldsymbol{H}) \times \boldsymbol{e}$, which converts a magnetic torque into a magnetic "force" on the orientation of a particle, is evaluated with it in Volume II.

Result 1.7.2 (Scalar triple product)**.**

$$\boldsymbol{A} \cdot (\boldsymbol{B} \times \boldsymbol{C}) = \varepsilon_{ijk}A_iB_jC_k = \boldsymbol{B} \cdot (\boldsymbol{C} \times \boldsymbol{A}) = \boldsymbol{C} \cdot (\boldsymbol{A} \times \boldsymbol{B}). \tag{1.7.10}$$

Derivation. $\boldsymbol{A} \cdot (\boldsymbol{B} \times \boldsymbol{C}) = A_i\varepsilon_{ijk}B_jC_k = \varepsilon_{ijk}A_iB_jC_k$. Now use the cyclic invariance $\varepsilon_{ijk} = \varepsilon_{jki}$ from (1.5.2): $\varepsilon_{ijk}A_iB_jC_k = \varepsilon_{jki}A_iB_jC_k = B_j\varepsilon_{jki}C_kA_i = \boldsymbol{B} \cdot (\boldsymbol{C} \times \boldsymbol{A})$, and once more for the third form. ■

This will be needed when a torque is projected onto the vorticity axis: the combination $\boldsymbol{\Omega} \cdot (\boldsymbol{e} \times \hat{\boldsymbol{h}})$ appears in the rotational diffusion problem of Volume II, and (1.7.10) lets us move the factors around freely.

Chapter 2

Vector Calculus

2.1 Fields

A *field* is a quantity defined at every point of space and at every instant. A *scalar field* $p(\boldsymbol{r}, t)$ assigns a number to each point (e.g. pressure); a *vector field* $\boldsymbol{v}(\boldsymbol{r}, t)$ assigns a vector (e.g. velocity). Continuum physics is entirely a theory of fields: instead of tracking 10^{20} molecules we track a handful of smooth functions.

Partial derivatives are abbreviated

$$\partial_i \equiv \frac{\partial}{\partial x_i}, \tag{2.1.1}$$

so that $\partial_i x_j = \delta_{ij}$: the coordinates are independent variables, and $\partial x_1/\partial x_2 = 0$ while $\partial x_1/\partial x_1 = 1$. This tiny identity is used repeatedly in Chapter 6.

2.2 Gradient

Definition 2.2.1 (Gradient)**.** For a scalar field f, $(\nabla f)_i \equiv \partial_i f$.

Why this is the right object. Consider moving by a small displacement $\mathrm{d}\boldsymbol{r}$ with components $\mathrm{d}x_i$. The change in f is, by the chain rule for functions of several variables,

$$\mathrm{d}f = \frac{\partial f}{\partial x_1}\mathrm{d}x_1 + \frac{\partial f}{\partial x_2}\mathrm{d}x_2 + \frac{\partial f}{\partial x_3}\mathrm{d}x_3 = \partial_i f \,\mathrm{d}x_i = \nabla f \cdot \mathrm{d}\boldsymbol{r}. \tag{2.2.1}$$

Thus ∇f is the unique vector whose scalar product with any displacement gives the resulting change of f. Writing $\nabla f \cdot \mathrm{d}\boldsymbol{r} = |\nabla f|\,|\mathrm{d}\boldsymbol{r}|\cos\theta$, the change is largest when $\theta = 0$: *the gradient points in the direction of steepest increase, and its magnitude is the rate of increase in that direction.* Its units are those of f divided by cm.

2.3 Divergence and the divergence theorem

Definition 2.3.1 (Divergence)**.** For a vector field $\boldsymbol{A}$, $\nabla \cdot \boldsymbol{A} \equiv \partial_i A_i$, a scalar field.

Why this is the right object. Take a small rectangular box with one corner at $\boldsymbol{r}$ and edges $\delta x_1, \delta x_2, \delta x_3$ parallel to the axes. Compute the net outward flux of $\boldsymbol{A}$ through its surface — "how much of $\boldsymbol{A}$ leaves the box per unit time". Through the face at $x_1 + \delta x_1$ (outward normal

$+\hat{\boldsymbol{x}}_1$, area $\delta x_2 \delta x_3$) the flux is $A_1(x_1+\delta x_1, x_2, x_3)\,\delta x_2\delta x_3$; through the opposite face (outward normal $-\hat{\boldsymbol{x}}_1$) it is $-A_1(x_1,x_2,x_3)\,\delta x_2\delta x_3$. The sum is

$$\big[A_1(x_1+\delta x_1)-A_1(x_1)\big]\delta x_2\delta x_3 = \left[\frac{\partial A_1}{\partial x_1}\delta x_1 + O(\delta x_1^2)\right]\delta x_2\delta x_3 \tag{2.3.1}$$

$$\approx \frac{\partial A_1}{\partial x_1}\,\delta x_1\delta x_2\delta x_3. \tag{2.3.2}$$

The Taylor expansion in (2.3.1) is legitimate because A_1 is assumed differentiable and δx_1 is arbitrarily small; the neglected term is smaller than the retained one by a factor δx_1, which we let tend to zero. Adding the two remaining pairs of faces and dividing by the volume $\delta V = \delta x_1\delta x_2\delta x_3$,

$$\frac{\text{net outward flux}}{\delta V} = \partial_1 A_1 + \partial_2 A_2 + \partial_3 A_3 = \nabla\cdot\boldsymbol{A}. \tag{2.3.3}$$

So the divergence is the *outflow per unit volume*: a source (>0), a sink (<0), or neither ($=0$).

Result 2.3.1 (Divergence (Gauss) theorem)**.** *For a volume $\mathcal{V}$ bounded by a closed surface $\mathcal{S}$ with outward unit normal $\boldsymbol{n}$,*

$$\int_{\mathcal{V}} (\nabla\cdot\boldsymbol{A})\,\mathrm{d}V = \oint_{\mathcal{S}} \boldsymbol{A}\cdot\boldsymbol{n}\,\mathrm{d}S. \tag{2.3.4}$$

Derivation. Chop $\mathcal{V}$ into many tiny boxes. By (2.3.3), for each box the outward flux equals $(\nabla\cdot\boldsymbol{A})\delta V$. Sum over all boxes. Every internal face is shared by two neighbouring boxes and carries opposite outward normals for them, so the two contributions cancel exactly. Only the faces lying on the outer boundary $\mathcal{S}$ survive. Hence the sum of $(\nabla\cdot\boldsymbol{A})\delta V$ over the interior equals the flux through $\mathcal{S}$; in the limit of infinitely fine subdivision the sums become the integrals in (2.3.4). ■

This theorem is what converts a statement about a *surface* (fluxes, forces on a boundary) into a statement about a *volume* (local differential equations). It is the engine of Chapter 5.

2.4 Curl

Definition 2.4.1 (Curl)**.** $(\nabla\times\boldsymbol{A})_i \equiv \varepsilon_{ijk}\partial_j A_k$.

Why this is the right object. Consider a small rectangle in the x_1x_2 plane with corners (x_1,x_2), $(x_1+\delta x_1, x_2)$, $(x_1+\delta x_1, x_2+\delta x_2)$, $(x_1, x_2+\delta x_2)$, and compute the *circulation* $\oint \boldsymbol{A}\cdot\mathrm{d}\boldsymbol{l}$ counterclockwise:

$$\oint \boldsymbol{A}\cdot\mathrm{d}\boldsymbol{l} = \underbrace{A_1(x_1,x_2)\,\delta x_1}_{\text{bottom}} + \underbrace{A_2(x_1+\delta x_1,x_2)\,\delta x_2}_{\text{right}} - \underbrace{A_1(x_1,x_2+\delta x_2)\,\delta x_1}_{\text{top}} - \underbrace{A_2(x_1,x_2)\,\delta x_2}_{\text{left}} \tag{2.4.1}$$

$$= \big[A_2(x_1+\delta x_1,x_2)-A_2(x_1,x_2)\big]\delta x_2 - \big[A_1(x_1,x_2+\delta x_2)-A_1(x_1,x_2)\big]\delta x_1 \tag{2.4.2}$$

$$= \left(\frac{\partial A_2}{\partial x_1}\delta x_1\right)\delta x_2 - \left(\frac{\partial A_1}{\partial x_2}\delta x_2\right)\delta x_1 + O(\delta^3) \tag{2.4.3}$$

$$= (\partial_1 A_2 - \partial_2 A_1)\,\delta x_1\delta x_2. \tag{2.4.4}$$

The signs of the top and left segments are negative because the path traverses them in the direction opposite to the coordinate axis. Dividing by the enclosed area $\delta x_1\delta x_2$,

$$\frac{\text{circulation}}{\text{area}} = \partial_1 A_2 - \partial_2 A_1 = (\nabla\times\boldsymbol{A})_3. \tag{2.4.5}$$

Thus the component of the curl along a given axis is the circulation per unit area in the plane perpendicular to that axis: *the curl measures local rotation.* That is precisely why $\nabla \times \boldsymbol{v}$ will turn out to be twice the local angular velocity of the fluid (Chapter 3), which is the content of the very first formula of the Shliomis–Morozov paper.

The same box-and-cancellation argument as before converts (2.4.5) into *Stokes' theorem*,

$$\int_{\mathcal{S}} (\nabla \times \boldsymbol{A}) \cdot \boldsymbol{n} \, \mathrm{d}S = \oint_{\partial \mathcal{S}} \boldsymbol{A} \cdot \mathrm{d}\boldsymbol{l}, \tag{2.4.6}$$

where now the interior edges shared by adjacent small loops are traversed twice in opposite directions and cancel.

2.5 Laplacian

Definition 2.5.1 (Laplacian)**.** $\nabla^2 f \equiv \nabla \cdot (\nabla f) = \partial_i \partial_i f$, and for a vector field, componentwise, $(\nabla^2 \boldsymbol{A})_i = \partial_j \partial_j A_i$.

Its meaning: $\nabla^2 f$ at a point is proportional to the amount by which the average of f over a small sphere around the point exceeds f at the point itself. To see this in one line for one dimension, expand $f(x \pm \delta) = f \pm f'\delta + \frac{1}{2} f''\delta^2 + \dots$ and add:

$$\tfrac{1}{2}\big[f(x+\delta) + f(x-\delta)\big] - f(x) = \tfrac{1}{2} f''(x)\delta^2. \tag{2.5.1}$$

A positive Laplacian means the neighbourhood is "higher" than the point, so diffusion-like processes ($\partial_t f = D\nabla^2 f$) push f up there. In the Navier–Stokes equation the term $\eta \nabla^2 \boldsymbol{v}$ plays exactly this role for momentum: viscosity diffuses momentum from fast to slow layers.

2.6 Two identities we shall need

Result 2.6.1.

$$\nabla \times (\nabla \times \boldsymbol{A}) = \nabla(\nabla \cdot \boldsymbol{A}) - \nabla^2 \boldsymbol{A}. \tag{2.6.1}$$

Derivation.

$$\begin{aligned}
[\nabla \times (\nabla \times \boldsymbol{A})]_i &= \varepsilon_{ijk} \partial_j (\nabla \times \boldsymbol{A})_k = \varepsilon_{ijk} \partial_j \big(\varepsilon_{klm} \partial_l A_m\big) && (2.6.2)\\
&= \varepsilon_{ijk}\varepsilon_{klm} \, \partial_j \partial_l A_m && (2.6.3)\\
&= (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}) \partial_j \partial_l A_m && (2.6.4)\\
&= \partial_j \partial_i A_j - \partial_j \partial_j A_i && (2.6.5)\\
&= \partial_i (\partial_j A_j) - \partial_j \partial_j A_i && (2.6.6)\\
&= [\nabla(\nabla \cdot \boldsymbol{A})]_i - (\nabla^2 \boldsymbol{A})_i. && (2.6.7)
\end{aligned}$$

In the fifth line the order of the partial derivatives was exchanged, $\partial_j \partial_i = \partial_i \partial_j$, which is permitted for fields with continuous second derivatives (Clairaut's theorem). ■

Result 2.6.2.

$$\nabla \cdot (\nabla \times \boldsymbol{A}) = 0. \tag{2.6.8}$$

Derivation. $\partial_i(\varepsilon_{ijk}\partial_j A_k) = \varepsilon_{ijk}\partial_i\partial_j A_k$. The factor $\partial_i\partial_j$ is symmetric under $i \leftrightarrow j$ while ε_{ijk} is antisymmetric; by the symmetric–antisymmetric lemma proved after (1.5.7), the contraction vanishes. ■

Identity (2.6.8) guarantees that any velocity field written as a curl is automatically incompressible — a fact we exploit in Chapter 6 to write down the flow around a rotating sphere without solving a differential equation.

Chapter 3

Kinematics: How a Fluid Element Moves, Deforms and Spins

3.1 Why this chapter exists

The opening sentence of Shliomis & Morozov reads: "Moving with a velocity $\boldsymbol{v}$, each element of the liquid volume rotates with a local angular velocity $\boldsymbol{\Omega} = \nabla \times \boldsymbol{v}/2$." That statement is not a definition plucked from the air; it is a theorem about the local structure of an arbitrary velocity field. This chapter proves it. Without it, nothing in either paper — which is *entirely* about the difference between the spin of the particles and the spin of the fluid — can even be formulated.

3.2 The material derivative

Let $f(\boldsymbol{r}, t)$ be any property carried by the fluid. A fluid element that is at $\boldsymbol{r}$ at time t is at $\boldsymbol{r} + \boldsymbol{v}\,\mathrm{d}t$ at time $t + \mathrm{d}t$. The change of f *following the element* is therefore

$$\mathrm{d}f = f(\boldsymbol{r} + \boldsymbol{v}\,\mathrm{d}t,\ t + \mathrm{d}t) - f(\boldsymbol{r}, t) \tag{3.2.1}$$

$$= \frac{\partial f}{\partial t}\mathrm{d}t + \frac{\partial f}{\partial x_i}\, v_i\,\mathrm{d}t + O(\mathrm{d}t^2), \tag{3.2.2}$$

using the multivariable Taylor expansion (legitimate for differentiable f and infinitesimal $\mathrm{d}t$). Dividing by $\mathrm{d}t$ and letting $\mathrm{d}t \to 0$,

$$\boxed{\frac{\mathrm{D}f}{\mathrm{D}t} \equiv \frac{\partial f}{\partial t} + (\boldsymbol{v}\cdot\nabla)f} \tag{3.2.3}$$

This is the *material* (substantial, convective) derivative. The first term is the change at a fixed point in space; the second is the change caused by the element being swept into a region where f has a different value. In both papers the symbol $\mathrm{d}/\mathrm{d}t$ in the equations of motion always means (3.2.3); Shliomis & Morozov state this explicitly below their Eq. (SM-26), writing $\mathrm{d}/\mathrm{d}t = \partial/\partial t + (\boldsymbol{v}_0\cdot\nabla)$.

3.3 Local decomposition of a velocity field

Fix a point $\boldsymbol{r}$ and look at the velocity a small distance $\boldsymbol{\delta}$ away. Taylor expansion (each component separately) gives

$$v_i(\boldsymbol{r}+\boldsymbol{\delta}) = v_i(\boldsymbol{r}) + \big(\partial_j v_i\big)\delta_j + O(\delta^2). \tag{3.3.1}$$

The three terms will turn out to describe, in order: translation, deformation, and rotation. To separate the last two, split the velocity-gradient tensor $\partial_j v_i$ into symmetric and antisymmetric parts. This is always possible and always unique:

$$\partial_j v_i = \underbrace{\tfrac{1}{2}\big(\partial_j v_i + \partial_i v_j\big)}_{e_{ij}} + \underbrace{\tfrac{1}{2}\big(\partial_j v_i - \partial_i v_j\big)}_{a_{ij}}. \tag{3.3.2}$$

That the identity holds is checked by adding the two brackets: the terms $\frac{1}{2}\partial_i v_j$ cancel and $2\times\frac{1}{2}\partial_j v_i$ remains. Uniqueness follows because $e_{ij}=e_{ji}$ and $a_{ij}=-a_{ji}$, and a tensor that is both symmetric and antisymmetric is zero.

Definition 3.3.1 (Rate-of-strain tensor)**.** $e_{ij}\equiv\frac{1}{2}(\partial_j v_i+\partial_i v_j)$, units s^{-1}. It measures the rate at which the element is stretched and sheared.

3.4 The antisymmetric part is a rigid rotation

Result 3.4.1. *The antisymmetric part of the velocity gradient acts on $\boldsymbol{\delta}$ exactly as a rigid-body rotation with angular velocity*

$$\boxed{\boldsymbol{\Omega} = \tfrac{1}{2}\nabla\times\boldsymbol{v}} \tag{3.4.1}$$

that is, $a_{ij}\delta_j = (\boldsymbol{\Omega}\times\boldsymbol{\delta})_i$.

Step 1: what rigid rotation looks like. A rigid body rotating with angular velocity $\boldsymbol{\Omega}$ about a point has, at displacement $\boldsymbol{\delta}$ from that point, the velocity $\boldsymbol{\Omega}\times\boldsymbol{\delta}$. (Reason: the point moves on a circle of radius $\delta\sin\theta$ about the axis, where θ is the angle between $\boldsymbol{\Omega}$ and $\boldsymbol{\delta}$; its speed is $\Omega\,\delta\sin\theta = |\boldsymbol{\Omega}\times\boldsymbol{\delta}|$, and its direction is perpendicular to both the axis and $\boldsymbol{\delta}$, which is the direction of $\boldsymbol{\Omega}\times\boldsymbol{\delta}$ by (1.5.7).)

Step 2: evaluate $\boldsymbol{\Omega}\times\boldsymbol{\delta}$ with $\boldsymbol{\Omega}$ given by (3.4.1)**.** In components, using (1.5.3) and the definition of the curl,

$$\begin{aligned}(\boldsymbol{\Omega}\times\boldsymbol{\delta})_i &= \varepsilon_{ijk}\Omega_j\delta_k && (3.4.2)\\ &= \varepsilon_{ijk}\left(\tfrac{1}{2}\varepsilon_{jpq}\partial_p v_q\right)\delta_k && (3.4.3)\\ &= \tfrac{1}{2}\,\varepsilon_{ijk}\varepsilon_{jpq}\,\partial_p v_q\,\delta_k. && (3.4.4)\end{aligned}$$

To apply the ε–δ identity (1.6.1) we need the summed index (j) to sit in the *last* slot of the first symbol and the *first* slot of the second. Use cyclic invariance (1.5.2) on the first factor: $\varepsilon_{ijk}=\varepsilon_{jki}$. Then

$$\begin{aligned}\tfrac{1}{2}\,\varepsilon_{jki}\varepsilon_{jpq}\,\partial_p v_q\,\delta_k &= \tfrac{1}{2}\,\varepsilon_{kij}\varepsilon_{jpq}\,\partial_p v_q\,\delta_k && \text{(cyclic again: } \varepsilon_{jki}=\varepsilon_{kij}) && (3.4.5)\\ &= \tfrac{1}{2}\big(\delta_{kp}\delta_{iq}-\delta_{kq}\delta_{ip}\big)\partial_p v_q\,\delta_k && \text{by (1.6.1) with } (i,j,k,l,m)\to(k,i,j,p,q) && (3.4.6)\end{aligned}$$

$$= \tfrac{1}{2}\Big(\delta_{kp}\delta_{iq}\partial_p v_q \delta_k - \delta_{kq}\delta_{ip}\partial_p v_q \delta_k\Big) \tag{3.4.7}$$

$$= \tfrac{1}{2}\Big(\partial_k v_i\, \delta_k - \partial_i v_k\, \delta_k\Big) \qquad \text{substitution property (1.4.2)} \tag{3.4.8}$$

$$= \tfrac{1}{2}\big(\partial_k v_i - \partial_i v_k\big)\delta_k \tag{3.4.9}$$

$$= a_{ik}\delta_k, \tag{3.4.10}$$

which is exactly the antisymmetric part of (3.3.2) contracted with $\boldsymbol{\delta}$ (the dummy index is called k instead of j, which is allowed). ■

Step 3: the complete local picture. Substituting (3.3.2) into (3.3.1),

$$v_i(\boldsymbol{r}+\boldsymbol{\delta}) = \underbrace{v_i(\boldsymbol{r})}_{\text{translation}} + \underbrace{e_{ij}\delta_j}_{\text{deformation}} + \underbrace{(\boldsymbol{\Omega}\times\boldsymbol{\delta})_i}_{\text{rigid rotation}} . \tag{3.4.11}$$

Every flow, however complicated, looks locally like a translation plus a stretching plus a rigid spin. A small solid sphere suspended in the fluid cannot be stretched (it is rigid), and it is carried along by the translation; what it *can* do is spin. In the absence of any other torque it simply spins with the fluid, at the rate $\boldsymbol{\Omega}$. This is the physical content of the statement $\boldsymbol{\Theta} = \boldsymbol{\Omega}$ that opens the Shliomis–Morozov paper, and the whole of ferrofluid magnetoviscosity consists of making $\boldsymbol{\Theta} \neq \boldsymbol{\Omega}$.

Remark 3.4.1. The factor $\frac{1}{2}$ in (3.4.1) is not a convention: it is forced. The curl measures circulation per unit area, and for a rigid rotation $\boldsymbol{v} = \boldsymbol{\Omega}\times\boldsymbol{r}$ one finds (Exercise, done here for completeness)

$$(\nabla\times\boldsymbol{v})_i = \varepsilon_{ijk}\partial_j\big(\varepsilon_{kpq}\Omega_p x_q\big) = \varepsilon_{ijk}\varepsilon_{kpq}\Omega_p\,\partial_j x_q \tag{3.4.12}$$

$$= \varepsilon_{ijk}\varepsilon_{kpq}\Omega_p\delta_{jq} = (\delta_{ip}\delta_{jq} - \delta_{iq}\delta_{jp})\Omega_p\delta_{jq} \tag{3.4.13}$$

$$= \delta_{ip}\Omega_p\delta_{jj} - \delta_{jp}\Omega_p\delta_{ij} = 3\Omega_i - \Omega_i = 2\Omega_i. \tag{3.4.14}$$

So the curl of a rigid rotation is *twice* the angular velocity, and (3.4.1) inverts this.

3.5 Worked example: vorticity in the Poiseuille flow of the experiment

Bacri *et al.* measure viscosity in a horizontal capillary of diameter $d_{\text{tube}} = 1$ mm through which the ferrofluid is driven by a pressure difference. The flow is a cylindrical Poiseuille flow: with z along the tube axis and r the distance from it,

$$v_z(r) = \frac{\Delta p}{4\eta\ell}\big(R^2 - r^2\big), \qquad R = d_{\text{tube}}/2, \tag{3.5.1}$$

ℓ being the tube length and Δp the pressure drop. (This profile is *derived* in Volume II once the Navier–Stokes equation is available; here we only need its shape.) The velocity has a single component depending on a single coordinate, so the curl has one surviving component, along the azimuthal direction $\hat{\boldsymbol{\phi}}$:

$$\nabla\times\boldsymbol{v} = -\frac{\mathrm{d}v_z}{\mathrm{d}r}\,\hat{\boldsymbol{\phi}}, \qquad \boldsymbol{\Omega} = -\frac{1}{2}\frac{\mathrm{d}v_z}{\mathrm{d}r}\,\hat{\boldsymbol{\phi}} = \frac{\Delta p\, r}{4\eta\ell}\,\hat{\boldsymbol{\phi}}. \tag{3.5.2}$$

Three facts, all of which matter later:

1. $\boldsymbol{\Omega}$ is *azimuthal*: it is perpendicular to the tube axis. The solenoid in the experiment produces $\boldsymbol{H}$ *along* the tube axis. Hence $\boldsymbol{H} \perp \boldsymbol{\Omega}$, i.e. the angle between them is $\beta = \pi/2$ — exactly the case assumed in Eq. (SM-2) and the reason Shliomis & Morozov write "or in a cylindrical Poiseuille flow when the capillary tube is aligned along the axis of the magnetic coil".

2. $\boldsymbol{\Omega}$ vanishes on the axis and is maximal at the wall. The measured viscosity is therefore an average over a distribution of local vorticities; this is why a *Newtonian* response (flow rate $\propto$ pressure drop, their Fig. 1) is a nontrivial experimental check.

3. Numerically, Bacri *et al.* state that $\Omega < 60\ \mathrm{s}^{-1}$ throughout their experiments. With $\tau_B \simeq 1.6$ ms this gives $\Omega\tau_B < 60 \times 1.6 \times 10^{-3} \approx 0.1 \ll 1$, which is the small parameter that makes the entire perturbation theory of both papers legitimate.

Part II

Physical Foundations

Chapter 4

Mechanics of Rotating Bodies

4.1 Newton's laws, stated and used

Assumption 4.1.1 (Newton's laws)**.** (i) In an inertial frame a body with no net force on it moves with constant velocity. (ii) $\boldsymbol{F} = \mathrm{d}\boldsymbol{P}/\mathrm{d}t$ with $\boldsymbol{P} = m_p\boldsymbol{u}$ the linear momentum. (iii) Forces between two bodies are equal and opposite.

These are empirical postulates; they cannot be derived from anything more primitive, and we take them as the starting point of all mechanics in this book. Everything else in this chapter *is* derived.

4.2 Angular momentum and torque

Definition 4.2.1 (Angular momentum and torque)**.** For a point mass m_p at position $\boldsymbol{r}$ (measured from a chosen origin) moving with velocity $\boldsymbol{u}$,

$$\boldsymbol{L} \equiv \boldsymbol{r} \times m_p\boldsymbol{u} = \boldsymbol{r} \times \boldsymbol{P}, \qquad \boldsymbol{T} \equiv \boldsymbol{r} \times \boldsymbol{F}. \tag{4.2.1}$$

$\boldsymbol{L}$ has units $\mathrm{g\,cm^2 s^{-1}} = \mathrm{erg\,s}$; $\boldsymbol{T}$ has units $\mathrm{dyn\,cm} = \mathrm{erg}$.

Result 4.2.1.

$$\frac{\mathrm{d}\boldsymbol{L}}{\mathrm{d}t} = \boldsymbol{T}. \tag{4.2.2}$$

Derivation. Differentiate the definition with the product rule (valid componentwise, since $(\boldsymbol{A}\times\boldsymbol{B})_i = \varepsilon_{ijk}A_jB_k$ is bilinear):

$$\frac{\mathrm{d}\boldsymbol{L}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\big(\boldsymbol{r}\times\boldsymbol{P}\big) = \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t}\times\boldsymbol{P} + \boldsymbol{r}\times\frac{\mathrm{d}\boldsymbol{P}}{\mathrm{d}t} \tag{4.2.3}$$

$$= \boldsymbol{u}\times m_p\boldsymbol{u} + \boldsymbol{r}\times\boldsymbol{F} \qquad \text{(Newton II for the second term)} \tag{4.2.4}$$

$$= \boldsymbol{0} + \boldsymbol{T}, \tag{4.2.5}$$

because $\boldsymbol{u}\times\boldsymbol{u} = \boldsymbol{0}$. ■

Equation (4.2.2) is the rotational analogue of $\boldsymbol{F} = \mathrm{d}\boldsymbol{P}/\mathrm{d}t$ and is the equation obeyed by every particle in this book. For a *rigid body* rotating with angular velocity $\boldsymbol{\theta}$ about a symmetry axis it becomes $I\,\mathrm{d}\boldsymbol{\theta}/\mathrm{d}t = \boldsymbol{T}$, where I is the moment of inertia, defined next.

4.3 Moment of inertia, and its value for a sphere

For a rigid body rotating with angular velocity $\boldsymbol{\theta}$, each mass element $\mathrm{d}m$ at position $\boldsymbol{r}$ from the rotation axis moves with $\boldsymbol{u} = \boldsymbol{\theta} \times \boldsymbol{r}$. Its angular momentum is $\mathrm{d}\boldsymbol{L} = \boldsymbol{r} \times (\boldsymbol{\theta} \times \boldsymbol{r})\mathrm{d}m$. Using the triple-product identity (1.7.1) with $\boldsymbol{A} = \boldsymbol{r}$, $\boldsymbol{B} = \boldsymbol{\theta}$, $\boldsymbol{C} = \boldsymbol{r}$,

$$\boldsymbol{r} \times (\boldsymbol{\theta} \times \boldsymbol{r}) = \boldsymbol{\theta}\,(\boldsymbol{r} \cdot \boldsymbol{r}) - \boldsymbol{r}\,(\boldsymbol{r} \cdot \boldsymbol{\theta}) = \boldsymbol{\theta} r^2 - \boldsymbol{r}(\boldsymbol{r} \cdot \boldsymbol{\theta}). \tag{4.3.1}$$

For a spherically symmetric body the second term averages, over the body, to $\tfrac{1}{3}\boldsymbol{\theta} r^2$: indeed $\int x_i x_j \,\mathrm{d}m = \tfrac{1}{3}\delta_{ij}\int r^2 \mathrm{d}m$ by symmetry (the off-diagonal integrals vanish because the integrand is odd, and the three diagonal ones are equal and sum to $\int r^2 \mathrm{d}m$). Hence

$$\boldsymbol{L} = \int \Big(\boldsymbol{\theta} r^2 - \tfrac{1}{3}\boldsymbol{\theta} r^2\Big)\mathrm{d}m = \tfrac{2}{3}\boldsymbol{\theta}\int r^2 \,\mathrm{d}m \equiv I_p \boldsymbol{\theta}, \qquad I_p = \tfrac{2}{3}\int r^2 \mathrm{d}m. \tag{4.3.2}$$

Evaluation for a uniform sphere of radius a and density ρ_s. Write $\mathrm{d}m = \rho_s \,\mathrm{d}V = \rho_s\, 4\pi r^2 \,\mathrm{d}r$ for a spherical shell of radius r and thickness $\mathrm{d}r$ (its volume is surface area $\times$ thickness). Then

$$\int r^2 \,\mathrm{d}m = \int_0^a r^2\, \rho_s 4\pi r^2 \,\mathrm{d}r = 4\pi\rho_s \int_0^a r^4 \,\mathrm{d}r = 4\pi\rho_s \frac{a^5}{5}, \tag{4.3.3}$$

$$I_p = \frac{2}{3}\cdot\frac{4\pi\rho_s a^5}{5} = \frac{8\pi\rho_s a^5}{15}. \tag{4.3.4}$$

It is convenient to express this through the particle mass $m_p = \rho_s \cdot \tfrac{4}{3}\pi a^3$:

$$I_p = \frac{8\pi\rho_s a^5}{15} = \frac{2}{5}\left(\frac{4}{3}\pi a^3 \rho_s\right) a^2 = \frac{2}{5} m_p a^2, \tag{4.3.5}$$

where the middle step is verified by expanding: $\tfrac{2}{5}\cdot\tfrac{4}{3}\pi a^3\rho_s a^2 = \tfrac{8}{15}\pi\rho_s a^5$. ✓

Finally, in terms of the *diameter* $d = 2a$, which is the variable used in Eq. (SM-1):

$$I_p = \frac{8\pi\rho_s}{15}\left(\frac{d}{2}\right)^5 = \frac{8\pi\rho_s}{15}\cdot\frac{d^5}{32} = \frac{\pi\rho_s d^5}{60}. \tag{4.3.6}$$

Keep (4.3.6): it is one of the two ingredients of Eq. (SM-1).

4.4 Magnetic moment, its torque and its energy

Definition 4.4.1 (Magnetic moment)**.** A small current loop, or a uniformly magnetised grain, is characterised by a vector $\boldsymbol{m}$ — its *magnetic moment* — of magnitude equal to the product of the magnetisation and the volume, directed along the magnetisation. For a grain of spontaneous magnetisation M_s and volume V,

$$m = M_s V \tag{4.4.1}$$

(this is the relation used explicitly after Eq. (SM-29)). Units: $\mathrm{G\,cm^3} = \mathrm{erg\,G^{-1}}$.

Result 4.4.1 (Torque and energy of a dipole in a field)**.**

$$\boldsymbol{T} = \boldsymbol{m} \times \boldsymbol{H}, \qquad U = -\boldsymbol{m}\cdot\boldsymbol{H} = -mH\cos\theta. \tag{4.4.2}$$

Derivation of the torque. Model the moment as a planar rectangular current loop of sides b and c carrying current $\mathcal{I}$, so that $m = \mathcal{I}bc/c_{\text{light}}$ in Gaussian units; only the final structure matters here. Place it in a uniform field $\boldsymbol{H}$ with its normal $\hat{\boldsymbol{n}}$ at angle θ to $\boldsymbol{H}$. Each element of the loop feels the Ampère force $\mathrm{d}\boldsymbol{F} \propto \mathrm{d}\boldsymbol{l} \times \boldsymbol{H}$. For the two sides parallel to the axis about which the loop can tilt, the forces are equal and opposite (so the net force vanishes for a uniform field) but they do not act along the same line: they are separated by the distance $b\sin\theta$ projected perpendicular to $\boldsymbol{H}$. Their moment is therefore

$$T = (\text{force}) \times (\text{lever arm}) = \big(\mathcal{I}cH/c_{\text{light}}\big) \times \big(b\sin\theta\big) = mH\sin\theta, \tag{4.4.3}$$

directed so as to rotate $\boldsymbol{m}$ towards $\boldsymbol{H}$. Both the magnitude $mH\sin\theta$ and that direction are precisely what the cross product $\boldsymbol{m} \times \boldsymbol{H}$ encodes. ■

Derivation of the energy. Work must be done against the torque to increase θ:

$$U(\theta) - U(\tfrac{\pi}{2}) = \int_{\pi/2}^{\theta} T\,\mathrm{d}\theta' = \int_{\pi/2}^{\theta} (-mH\sin\theta')\,\mathrm{d}\theta' = mH\cos\theta'\Big|_{\pi/2}^{\theta} = mH\cos\theta, \tag{4.4.4}$$

Careful with the sign: the torque *restores* θ towards zero, so the torque component along increasing θ is $-mH\sin\theta$. Choosing the zero of energy at $\theta = \pi/2$ (moment perpendicular to field) we would obtain $U = +mH\cos\theta$, which has its minimum at $\theta = \pi$ — wrong. The error is that the work done *by* the field is $-\Delta U$; performing the bookkeeping correctly,

$$\mathrm{d}U = -T_{\text{field}}\,\mathrm{d}\theta = +mH\sin\theta\,\mathrm{d}\theta \implies U = -mH\cos\theta + \text{const}, \tag{4.4.5}$$

and with the constant chosen as zero, $U = -\boldsymbol{m}\cdot\boldsymbol{H}$, minimal when $\boldsymbol{m} \parallel \boldsymbol{H}$, as it must be. ■

The combination that will govern the statistical mechanics is the ratio of this energy scale to the thermal energy,

$$\xi \equiv \frac{mH}{kT}, \tag{4.4.6}$$

the *Langevin parameter*: dimensionless, positive, proportional to the field. It is the single most important variable of both papers.

4.5 Magnetisation of a suspension

If there are n particles per unit volume, each with moment m along a unit vector $\boldsymbol{e}$ that fluctuates by thermal agitation, the magnetisation — the magnetic moment per unit volume — is

$$\boldsymbol{M} = n\,m\,\langle \boldsymbol{e} \rangle, \tag{4.5.1}$$

where $\langle\cdot\rangle$ denotes an average over the particles contained in a physically small element of volume: small compared with the scale over which the flow varies, but containing enough particles for the average to be smooth. This "physically small element" idea, used verbatim by both papers, is the bridge between the single particle and the continuum. Note the immediate consequence

$$M \le nm = nM_sV = \varphi M_s, \tag{4.5.2}$$

the saturation magnetisation of the suspension, with $\varphi = nV$ the volume fraction of solid.

Chapter 5

The Equations of Fluid Motion

5.1 Conservation of mass: the continuity equation

Consider a fixed region $\mathcal{V}$ in space, bounded by $\mathcal{S}$. The mass inside is $\int_{\mathcal{V}} \rho \, \mathrm{d}V$; mass leaves only by being carried across the boundary, at the rate $\oint_{\mathcal{S}} \rho \boldsymbol{v} \cdot \boldsymbol{n} \, \mathrm{d}S$. Mass is neither created nor destroyed, so

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_{\mathcal{V}} \rho \, \mathrm{d}V = - \oint_{\mathcal{S}} \rho \boldsymbol{v} \cdot \boldsymbol{n} \, \mathrm{d}S. \tag{5.1.1}$$

Because $\mathcal{V}$ is fixed, the time derivative may be taken inside as a partial derivative; and the right-hand side may be converted by the divergence theorem (2.3.4) with $\boldsymbol{A} = \rho \boldsymbol{v}$:

$$\int_{\mathcal{V}} \frac{\partial \rho}{\partial t} \, \mathrm{d}V = - \int_{\mathcal{V}} \nabla \cdot (\rho \boldsymbol{v}) \, \mathrm{d}V \implies \int_{\mathcal{V}} \left[\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{v}) \right] \mathrm{d}V = 0. \tag{5.1.2}$$

Since $\mathcal{V}$ was arbitrary — the equality must hold for every region, including arbitrarily small ones — the integrand must vanish pointwise:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{v}) = 0. \tag{5.1.3}$$

(The step "integral zero for all volumes $\Rightarrow$ integrand zero" is valid for continuous integrands: if the integrand were positive at a point it would be positive in a small ball around it, and the integral over that ball would be positive.)

Assumption 5.1.1 (Incompressibility)**.** The carrier liquid is water, whose density changes by less than 10^{-4} under the pressures of the experiment. We therefore set $\rho = \text{const}$, whereupon (5.1.3) reduces to

$$\nabla \cdot \boldsymbol{v} = 0. \tag{5.1.4}$$

This is the first of the three relations quoted as Eq. (SM-9).

5.2 Conservation of momentum: the Cauchy equation

Apply Newton's second law to the fluid contained, at a given instant, in a volume $\mathcal{V}$. Its momentum is $\int \rho \boldsymbol{v} \, \mathrm{d}V$, and the forces on it are of two kinds:

- *body forces* $\boldsymbol{f}$ per unit volume, acting on the interior (gravity; in a ferrofluid, the magnetic force $(\boldsymbol{M} \cdot \nabla)\boldsymbol{H}$);

- *surface forces* exerted by the surrounding fluid across $\mathcal{S}$.

The surface force per unit area (the *traction*) on a surface element with normal $\boldsymbol{n}$ is a vector $\boldsymbol{t}(\boldsymbol{n})$ that depends on the orientation $\boldsymbol{n}$. Cauchy's insight is that this dependence must be linear:

$$t_i = \sigma_{ij} n_j, \tag{5.2.1}$$

which *defines* the stress tensor σ_{ij} (units $\mathrm{dyn\,cm^{-2}}$). The linearity follows from applying Newton's law to a shrinking tetrahedron: the surface terms scale as $(\text{length})^2$ while mass$\times$acceleration scales as $(\text{length})^3$, so in the limit the surface forces must balance among themselves, which forces the linear relation (5.2.1).

Newton's second law for $\mathcal{V}$ then reads

$$\int_{\mathcal{V}} \rho \,\frac{\mathrm{D}v_i}{\mathrm{D}t}\,\mathrm{d}V = \int_{\mathcal{V}} f_i \,\mathrm{d}V + \oint_{\mathcal{S}} \sigma_{ij} n_j \,\mathrm{d}S. \tag{5.2.2}$$

The material derivative appears on the left because we are following the fluid (Sec. 3.2). Convert the surface integral by the divergence theorem, applied to the vector field $A_j = \sigma_{ij}$ for each fixed i:

$$\oint_{\mathcal{S}} \sigma_{ij} n_j \,\mathrm{d}S = \int_{\mathcal{V}} \partial_j \sigma_{ij} \,\mathrm{d}V. \tag{5.2.3}$$

Since $\mathcal{V}$ is arbitrary, the integrands match:

$$\boxed{\rho \,\frac{\mathrm{D}v_i}{\mathrm{D}t} = f_i + \partial_j \sigma_{ij}} \tag{5.2.4}$$

This is the Cauchy equation of motion — Newton's second law for a continuum. Everything specific to a particular material now hides in σ_{ij}.

5.3 The Newtonian constitutive law

Assumption 5.3.1 (Newtonian fluid)**.** The stress depends linearly on the *rate* of deformation, isotropically, and not at all on the deformation itself (a fluid has no memory of its shape):

$$\sigma_{ij} = -p\,\delta_{ij} + 2\eta\, e_{ij} \qquad \text{with} \qquad e_{ij} = \tfrac{1}{2}(\partial_j v_i + \partial_i v_j), \tag{5.3.1}$$

for an incompressible fluid. Here p is the pressure and η the dynamic viscosity.

Three remarks that will matter enormously later:

1. The term $-p\delta_{ij}$ survives when the fluid is at rest: a fluid at rest supports only a normal, isotropic stress. The minus sign is because a positive pressure *pushes* on a surface, i.e. acts opposite to its outward normal.

2. Only the symmetric part e_{ij} appears. The antisymmetric part a_{ij} is a rigid rotation (Sec. 3.4), which cannot produce stress in an ordinary fluid, since rigidly rotating a fluid does not deform it.

3. **This last statement fails in a ferrofluid.** If a magnetic torque makes the particles spin at a rate $\boldsymbol{\Theta} \neq \boldsymbol{\Omega}$ different from the fluid, angular momentum is exchanged between the internal (spin) and external (orbital) degrees of freedom, and the stress tensor acquires an *antisymmetric* part. That extra piece is exactly the term $\frac{1}{2}\nabla \times (\boldsymbol{M} \times \boldsymbol{H})$ which appears in Eqs. (SM-12) and (B-3), and it is the origin of the entire negative-viscosity effect. We flag it here and derive it in Volume II.

5.4 The Navier–Stokes equation

Insert (5.3.1) into (5.2.4):

$$\partial_j \sigma_{ij} = \partial_j\big(-p\delta_{ij}\big) + \partial_j\big(2\eta e_{ij}\big) \tag{5.4.1}$$
$$= -\partial_i p + \eta\,\partial_j\big(\partial_j v_i + \partial_i v_j\big) \qquad (\eta \text{ constant};\ \delta_{ij}\partial_j = \partial_i) \tag{5.4.2}$$
$$= -\partial_i p + \eta\,\partial_j\partial_j v_i + \eta\,\partial_i\big(\partial_j v_j\big) \qquad (\text{exchanging } \partial_j\partial_i = \partial_i\partial_j) \tag{5.4.3}$$
$$= -\partial_i p + \eta\,\nabla^2 v_i, \tag{5.4.4}$$

the last term having vanished by incompressibility (5.1.4). Hence

$$\boxed{\rho\left[\frac{\partial \boldsymbol{v}}{\partial t} + (\boldsymbol{v}\cdot\nabla)\boldsymbol{v}\right] = -\nabla p + \eta\nabla^2\boldsymbol{v} + \boldsymbol{f}} \tag{5.4.5}$$

— the incompressible Navier–Stokes equation. Comparing with the fluid-motion equations actually used in the papers, Eq. (SM-6) and Eq. (B-3),

$$\rho\frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = -\nabla p + (\boldsymbol{M}\cdot\nabla)\boldsymbol{H} + \eta\Delta\boldsymbol{v} + \frac{I}{2\tau_s}\nabla\times(\boldsymbol{\Theta}-\boldsymbol{\Omega}), \tag{SM-6}$$

we can now read the structure completely: $\rho\,\mathrm{d}\boldsymbol{v}/\mathrm{d}t$ is the inertia term of (5.4.5) with the material derivative (3.2.3); $-\nabla p$ and $\eta\Delta\boldsymbol{v}$ (their Δ is our ∇^2) are the ordinary Newtonian terms; $(\boldsymbol{M}\cdot\nabla)\boldsymbol{H}$ is the body force $\boldsymbol{f}$ exerted by a *non-uniform* magnetic field on the magnetised medium; and the final term is the new, ferrofluid-specific antisymmetric stress announced in remark 3 above. Every symbol of (SM-6) has now been accounted for except the last term, whose derivation is the business of Volume II.

5.5 The Reynolds number and the Stokes limit

Compare the inertial term $\rho(\boldsymbol{v}\cdot\nabla)\boldsymbol{v}$, of order $\rho U^2/\ell$, with the viscous term $\eta\nabla^2\boldsymbol{v}$, of order $\eta U/\ell^2$, where U and ℓ are the characteristic velocity and length:

$$\mathrm{Re} \equiv \frac{\rho U^2/\ell}{\eta U/\ell^2} = \frac{\rho U\ell}{\eta}. \tag{5.5.1}$$

For a colloidal particle, $\ell \sim d = 10^{-6}$ cm, and the relevant velocity is that of its surface when it rotates at the fluid vorticity, $U \sim \Omega d/2 \sim 60\times 5\times 10^{-7} \approx 3\times 10^{-5}$ cm s^{-1}. With $\rho \approx 1$ g cm^{-3} and $\eta = 10^{-2}$ P,

$$\mathrm{Re} \sim \frac{1\times 3\times 10^{-5}\times 10^{-6}}{10^{-2}} = 3\times 10^{-9} \lll 1. \tag{5.5.2}$$

Inertia is utterly negligible at the scale of a particle. The Navier–Stokes equation then reduces to the *Stokes equations*

$$0 = -\nabla p + \eta\nabla^2\boldsymbol{v}, \qquad \nabla\cdot\boldsymbol{v} = 0, \tag{5.5.3}$$

which are linear — a decisive simplification, and the basis of the next chapter.

Chapter 6

Stokes Flow Around a Rotating Sphere

6.1 The problem and why we must solve it

A magnetic grain forced to rotate at angular velocity $\boldsymbol{\theta}$ inside a liquid at rest drags the liquid with it and feels a retarding torque. The relation between the two,

$$\boldsymbol{T}_{\text{visc}} = -\zeta_r \boldsymbol{\theta}, \tag{6.1.1}$$

defines the *rotational friction coefficient* ζ_r, and *every* quantitative statement in both papers depends on its value: it fixes τ_s in Eq. (SM-1), τ_B in Eq. (SM-5), the constant $6\eta\varphi$ in Eqs. (SM-19), (B-1) and (SM-37), and the numerical prefactor $\frac{3}{2}\eta\varphi$ in Eq. (SM-2). So we derive it in full.

6.2 Setting up

Let a rigid sphere of radius a, centred at the origin, rotate with constant angular velocity $\boldsymbol{\theta}$ in an unbounded liquid that is at rest at infinity. We must solve the Stokes equations (5.5.3) subject to

$$\boldsymbol{v} = \boldsymbol{\theta} \times \boldsymbol{r} \qquad \text{at } r = a \qquad \text{(no-slip on the sphere)}, \tag{6.2.1}$$

$$\boldsymbol{v} \to \boldsymbol{0} \qquad \text{as } r \to \infty \qquad \text{(fluid at rest far away)}. \tag{6.2.2}$$

The no-slip condition (6.2.1) is the experimental fact that a viscous liquid does not slide over a solid surface: the fluid in contact with the sphere moves with the sphere's surface, whose velocity is $\boldsymbol{\theta} \times \boldsymbol{r}$ by Sec. 3.4, Step 1.

6.3 The solution

Result 6.3.1.

$$\boldsymbol{v}(\boldsymbol{r}) = \frac{a^3}{r^3}\,\boldsymbol{\theta} \times \boldsymbol{r}, \qquad p = const \tag{6.3.1}$$

solves the problem.

Step 1: it satisfies the boundary conditions. At $r = a$ the prefactor $a^3/r^3 = 1$, so $\boldsymbol{v} = \boldsymbol{\theta} \times \boldsymbol{r}$: (6.2.1) holds. As $r \to \infty$, $|\boldsymbol{v}| \sim \theta a^3/r^2 \to 0$: (6.2.2) holds.

Step 2: it is incompressible. In components, $v_i = a^3\varepsilon_{ijk}\theta_j x_k r^{-3}$, hence

$$\partial_i v_i = a^3\varepsilon_{ijk}\theta_j\,\partial_i\big(x_k r^{-3}\big) = a^3\varepsilon_{ijk}\theta_j\left[\delta_{ik}r^{-3} + x_k\,\partial_i r^{-3}\right]. \tag{6.3.2}$$

We shall need $\partial_i r^{-3}$ repeatedly. Since $r = (x_l x_l)^{1/2}$,

$$\partial_i r = \frac{1}{2}(x_l x_l)^{-1/2}\cdot 2x_i = \frac{x_i}{r}, \qquad \Longrightarrow \qquad \partial_i r^{-3} = -3r^{-4}\,\partial_i r = -3\frac{x_i}{r^5}. \tag{6.3.3}$$

Substituting into (6.3.2),

$$\partial_i v_i = a^3\theta_j\left[\varepsilon_{ijk}\delta_{ik}r^{-3} - 3\varepsilon_{ijk}\frac{x_k x_i}{r^5}\right] = 0, \tag{6.3.4}$$

because $\varepsilon_{ijk}\delta_{ik} = \varepsilon_{iji} = 0$ (two equal indices) and $\varepsilon_{ijk}x_k x_i = 0$ (antisymmetric contracted with symmetric).

Step 3: it satisfies the Stokes equation with constant pressure. Observe first that

$$\frac{\boldsymbol{\theta}\times\boldsymbol{r}}{r^3} = -\boldsymbol{\theta}\times\nabla\left(\frac{1}{r}\right) = \nabla\times\left(\frac{\boldsymbol{\theta}}{r}\right), \tag{6.3.5}$$

which is verified as follows. From (6.3.3) with the exponent -1, $\partial_i r^{-1} = -x_i/r^3$, i.e. $\nabla(1/r) = -\boldsymbol{r}/r^3$; therefore $-\boldsymbol{\theta}\times\nabla(1/r) = \boldsymbol{\theta}\times\boldsymbol{r}/r^3$. ✓ For the second equality, with $\boldsymbol{\theta}$ a constant vector,

$$\left[\nabla\times\frac{\boldsymbol{\theta}}{r}\right]_i = \varepsilon_{ijk}\partial_j\big(\theta_k r^{-1}\big) = \varepsilon_{ijk}\theta_k\left(-\frac{x_j}{r^3}\right) = \frac{1}{r^3}\varepsilon_{ikj}\theta_k x_j = \frac{(\boldsymbol{\theta}\times\boldsymbol{r})_i}{r^3}, \tag{6.3.6}$$

where one interchange of the last two indices of ε supplied the compensating minus sign. ✓

Now take the Laplacian of (6.3.1) written in the form $\boldsymbol{v} = a^3\nabla\times(\boldsymbol{\theta}/r)$. Since ∇^2 commutes with $\nabla\times$ (both are built from partial derivatives, which commute),

$$\nabla^2\boldsymbol{v} = a^3\,\nabla\times\left(\boldsymbol{\theta}\,\nabla^2\frac{1}{r}\right) = \mathbf{0}, \tag{6.3.7}$$

because $1/r$ is a harmonic function away from the origin:

$$\nabla^2\frac{1}{r} = \partial_i\partial_i r^{-1} = \partial_i\Big(-\frac{x_i}{r^3}\Big) = -\left[\frac{\delta_{ii}}{r^3} + x_i\left(-3\frac{x_i}{r^5}\right)\right] = -\left[\frac{3}{r^3} - \frac{3r^2}{r^5}\right] = 0 \quad (r>0). \tag{6.3.8}$$

So $\eta\nabla^2\boldsymbol{v} = \mathbf{0}$, and (5.5.3) is satisfied by $\nabla p = \mathbf{0}$, i.e. constant pressure, which we set to zero (a constant pressure exerts no net force or torque on a closed surface). ■

By the uniqueness theorem for Stokes flow (two solutions with the same boundary data would differ by a flow with zero boundary velocity and hence zero dissipation, therefore zero), (6.3.1) is *the* solution.

6.4 The traction on the sphere

We need $t_i = \sigma_{ij}n_j$ at $r = a$, with $\boldsymbol{n} = \boldsymbol{r}/a$ the outward unit normal and $\sigma_{ij} = -p\delta_{ij} + \eta(\partial_j v_i + \partial_i v_j)$ (from (5.3.1), writing $2e_{ij}$ out). With $p = 0$,

$$t_i = \eta\big(\partial_j v_i\big)n_j + \eta\big(\partial_i v_j\big)n_j. \tag{6.4.1}$$

We evaluate the two pieces separately. First,

$$\partial_j v_i = a^3 \varepsilon_{ikl} \theta_k \, \partial_j \big(x_l r^{-3} \big) = a^3 \varepsilon_{ikl} \theta_k \left[\frac{\delta_{lj}}{r^3} - 3 \frac{x_l x_j}{r^5} \right]. \tag{6.4.2}$$

Contract with $n_j = x_j/a$ and set $r = a$ (so $x_j = a n_j$, $x_l x_j n_j = x_l a$):

$$\begin{aligned}
\big(\partial_j v_i\big) n_j &= a^3 \varepsilon_{ikl} \theta_k \left[\frac{n_l}{a^3} - 3 \frac{(a n_l)(a n_j) n_j}{a^5} \right] && (6.4.3)\\
&= a^3 \varepsilon_{ikl} \theta_k \left[\frac{n_l}{a^3} - 3 \frac{a^2 n_l}{a^5} \right] && \text{using } n_j n_j = 1 \quad (6.4.4)\\
&= a^3 \varepsilon_{ikl} \theta_k \, \frac{n_l - 3 n_l}{a^3} && (6.4.5)\\
&= -2 \, \varepsilon_{ikl} \theta_k n_l = -2 (\boldsymbol{\theta} \times \boldsymbol{n})_i. && (6.4.6)
\end{aligned}$$

Second,

$$\partial_i v_j = a^3 \varepsilon_{jkl} \theta_k \left[\frac{\delta_{li}}{r^3} - 3 \frac{x_l x_i}{r^5} \right], \tag{6.4.7}$$

so, again at $r = a$ with $x_i = a n_i$,

$$\begin{aligned}
\big(\partial_i v_j\big) n_j &= a^3 \varepsilon_{jkl} \theta_k n_j \left[\frac{\delta_{li}}{a^3} - 3 \frac{(a n_l)(a n_i)}{a^5} \right] && (6.4.8)\\
&= \varepsilon_{jkl} \theta_k n_j \delta_{li} - 3 \, \varepsilon_{jkl} \theta_k n_j n_l n_i && (6.4.9)\\
&= \varepsilon_{jki} \theta_k n_j - 0 && \text{since } \varepsilon_{jkl} n_j n_l = 0 \quad (6.4.10)\\
&= -\varepsilon_{ikj} \theta_k n_j && \text{one interchange } j \leftrightarrow i \text{ in } \varepsilon_{jki} = -\varepsilon_{ikj} \quad (6.4.11)\\
&= -(\boldsymbol{\theta} \times \boldsymbol{n})_i. && (6.4.12)
\end{aligned}$$

Adding (6.4.6) and (6.4.12) in (6.4.1),

$$\boldsymbol{t} = \eta \big[-2(\boldsymbol{\theta} \times \boldsymbol{n}) - (\boldsymbol{\theta} \times \boldsymbol{n}) \big] = -3\eta \, (\boldsymbol{\theta} \times \boldsymbol{n}). \tag{6.4.13}$$

The traction is everywhere *opposite* to the local surface velocity $a(\boldsymbol{\theta} \times \boldsymbol{n})$: the liquid resists, as it must.

6.5 The torque

The torque about the centre is $\boldsymbol{T} = \oint \boldsymbol{r} \times \boldsymbol{t} \, \mathrm{d}S$ with $\boldsymbol{r} = a\boldsymbol{n}$ on the surface:

$$\begin{aligned}
T_i &= \oint a \, \varepsilon_{ijk} n_j t_k \, \mathrm{d}S = -3\eta a \oint \varepsilon_{ijk} n_j (\boldsymbol{\theta} \times \boldsymbol{n})_k \, \mathrm{d}S && (6.5.1)\\
&= -3\eta a \oint \big[\boldsymbol{n} \times (\boldsymbol{\theta} \times \boldsymbol{n}) \big]_i \, \mathrm{d}S && (6.5.2)\\
&= -3\eta a \oint \Big[\theta_i (\boldsymbol{n} \cdot \boldsymbol{n}) - n_i (\boldsymbol{n} \cdot \boldsymbol{\theta}) \Big] \mathrm{d}S && \text{by (1.7.1)} \quad (6.5.3)\\
&= -3\eta a \left[\theta_i \oint \mathrm{d}S - \theta_j \oint n_i n_j \, \mathrm{d}S \right] && \text{using } \boldsymbol{n} \cdot \boldsymbol{n} = 1. \quad (6.5.4)
\end{aligned}$$

The first integral is the sphere's area, $4\pi a^2$. For the second we use the symmetry result

$$\oint n_i n_j \, \mathrm{d}S = \frac{4\pi a^2}{3} \delta_{ij}, \tag{6.5.5}$$

which is proved exactly as in Sec. 4.3: by symmetry the integral must be proportional to δ_{ij} (there is no preferred direction on a sphere), say $C\delta_{ij}$; taking the trace, $\oint n_i n_i \,\mathrm{d}S = \oint 1 \,\mathrm{d}S = 4\pi a^2$ on the left and $3C$ on the right, so $C = 4\pi a^2/3$. Substituting into (6.5.4),

$$T_i = -3\eta a \left[4\pi a^2 \theta_i - \frac{4\pi a^2}{3}\theta_j \delta_{ij}\right] = -3\eta a \left[4\pi a^2 \theta_i - \frac{4\pi a^2}{3}\theta_i\right] \tag{6.5.6}$$

$$= -3\eta a \cdot 4\pi a^2 \theta_i \left(1 - \frac{1}{3}\right) = -3\eta a \cdot 4\pi a^2 \theta_i \cdot \frac{2}{3} = -8\pi\eta a^3 \theta_i. \tag{6.5.7}$$

Therefore

$$\boxed{\boldsymbol{T}_{\text{visc}} = -8\pi\eta a^3\, \boldsymbol{\theta} \;\equiv -\zeta_r \boldsymbol{\theta}, \qquad \zeta_r = 8\pi\eta a^3} \tag{6.5.8}$$

6.6 Two equivalent forms of ζ_r used by the papers

Express ζ_r through the particle *volume* $V = \frac{4}{3}\pi a^3$:

$$\zeta_r = 8\pi\eta a^3 = 6\eta \cdot \frac{4}{3}\pi a^3 = 6\eta V. \tag{6.6.1}$$

(Check the arithmetic: $6 \times \frac{4}{3}\pi = 8\pi$. ✓) And in terms of the diameter $d = 2a$,

$$\zeta_r = 8\pi\eta \frac{d^3}{8} = \pi\eta d^3. \tag{6.6.2}$$

Both forms appear in the papers, usually multiplied by the number density n so as to refer to unit volume of suspension:

$$n\zeta_r = 6\eta\, nV = 6\eta\varphi. \tag{6.6.3}$$

This is the origin of the constant $6\eta\varphi$ in the torque-balance equation (B-1) of Bacri *et al.*,

$$6\eta\varphi\big(\boldsymbol{\omega}_p - \boldsymbol{\Omega}\big) = \boldsymbol{M} \times \boldsymbol{H}, \tag{B-1}$$

and of the identity $I/\tau_s = 6\eta\varphi$ quoted between Eqs. (SM-18) and (SM-19). We shall verify that identity explicitly in the next chapter.

Remark 6.6.1 (Why $8\pi\eta a^3$ and not $6\pi\eta a$). The familiar Stokes drag $\boldsymbol{F} = -6\pi\eta a \boldsymbol{u}$ refers to *translation* and has units of (viscosity)×(length); rotational friction relates a torque to an angular velocity and therefore has units (viscosity)×(length)^3. The two coefficients are different numbers because the flow patterns are different — the rotating sphere's flow (6.3.1) decays as $1/r^2$, far faster than the $1/r$ decay around a translating sphere.

Part III

The Papers, Equation by Equation

Chapter 7

Equation (SM-1): The Inertial Relaxation Time of Particle Rotation

The first equation of Shliomis & Morozov is

$$\tau_s = \frac{\rho_s d^2}{60\,\eta}. \tag{SM-1}$$

We now derive it in the eight-part format.

7.1 Context

The paper's argument begins by recalling why a stationary magnetic field increases the viscosity of a ferrofluid. The chain of reasoning is:

1. Each element of liquid rotates locally with $\boldsymbol{\Omega} = \frac{1}{2}\nabla \times \boldsymbol{v}$ (our Chapter 3).

2. *Without* a field, the suspended particles rotate with the same angular velocity, $\boldsymbol{\Theta} = \boldsymbol{\Omega}$.

3. That equality is not a postulate: it is the outcome of a very fast relaxation. If a particle's spin $\boldsymbol{\Theta}$ momentarily differs from $\boldsymbol{\Omega}$, the viscous torque drives the difference to zero. Equation (SM-1) is the time constant of that decay.

4. The point of computing τ_s is to show that it is *fantastically* small (10^{-11} s), so that particle *inertia* may be dropped everywhere in the theory. This is what later allows Eq. (SM-8) to be replaced by the algebraic relation (SM-11), and it is the reason the moment of inertia I disappears from the final results.

So (SM-1) is a *licence to neglect something*, and its derivation must be quantitative enough to justify that licence.

7.2 Symbols

Symbol	Meaning	Type	Units
τ_s	relaxation time for the particle's angular velocity to equalise with the fluid's	scalar, positive, constant	s
ρ_s	mass density of the solid magnetic material of a particle (subscript s = solid)	scalar constant	$\mathrm{g\,cm^{-3}}$
d	particle diameter	scalar constant	cm
η	dynamic viscosity of the *carrier liquid*	scalar constant	$\mathrm{P = g\,cm^{-1}s^{-1}}$
$\boldsymbol{\Theta}$	macroscopic angular velocity of the particles, averaged over a physically small volume element	vector field	$\mathrm{s^{-1}}$
$\boldsymbol{\Omega}$	local angular velocity of the fluid, $\frac{1}{2}\nabla \times \boldsymbol{v}$	vector field	$\mathrm{s^{-1}}$
I_p	moment of inertia of *one* particle	scalar constant	$\mathrm{g\,cm^2}$
I	sum of the particles' moments of inertia *per unit volume*, $I = nI_p$	scalar constant	$\mathrm{g\,cm^{-1}}$
ζ_r	rotational friction coefficient of one particle, $8\pi\eta a^3$	scalar constant	erg s

A dimensional check of the target formula, performed before any derivation:

$$\left[\frac{\rho_s d^2}{\eta}\right] = \frac{\mathrm{g\,cm^{-3}}\cdot\mathrm{cm^2}}{\mathrm{g\,cm^{-1}s^{-1}}} = \frac{\mathrm{g\,cm^{-1}}}{\mathrm{g\,cm^{-1}s^{-1}}} = \mathrm{s}.\checkmark \tag{7.2.1}$$

So (SM-1) is at least dimensionally a time.

7.3 Assumptions

1. The particle is a *rigid sphere* of uniform density ρ_s and diameter d. (Real grains are faceted crystallites of ≈ 10 nm; the sphere is an idealisation accurate to the accuracy of the numerical prefactor.)

2. The particle is small enough that the flow around it is a Stokes flow, $\mathrm{Re} \lll 1$ (verified numerically in Sec. 5.5).

3. The suspension is dilute enough that each particle feels only the carrier liquid, not its neighbours. (Strictly, $\varphi = 0.2$ in the experiment is not dilute; the papers acknowledge this by using the *measured* zero-field viscosity as the reference η.)

4. The only torque acting is the viscous one. Brownian and magnetic torques are excluded from this particular calculation because we are asking a different question: *if* the spin differs from the fluid's, how fast does the difference decay under viscous friction alone?

5. The fluid vorticity $\boldsymbol{\Omega}$ is constant over the time τ_s. Since we will find $\tau_s \sim 10^{-11}$ s and hydrodynamic times are $\gtrsim 10^{-3}$ s, this is overwhelmingly satisfied.

6. Particle inertia is retained here — it is the whole point — but nowhere else in the theory.

7.4 Mathematical Background: the linear relaxation equation

We shall meet the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}t} = -\frac{y}{\tau}, \tag{7.4.1}$$

where $\tau > 0$ is a constant. Solve it by separation of variables — legitimate wherever $y \neq 0$:

$$\frac{\mathrm{d}y}{y} = -\frac{\mathrm{d}t}{\tau}, \tag{7.4.2}$$

$$\int_{y_0}^{y} \frac{\mathrm{d}y'}{y'} = -\int_0^t \frac{\mathrm{d}t'}{\tau}, \tag{7.4.3}$$

$$\ln y - \ln y_0 = -\frac{t}{\tau}, \tag{7.4.4}$$

$$\ln \frac{y}{y_0} = -\frac{t}{\tau}, \tag{7.4.5}$$

$$y(t) = y_0\, e^{-t/\tau}. \tag{7.4.6}$$

The solution decays by the factor $e^{-1} \approx 0.368$ in the time τ; after 5τ less than 1% remains. Hence τ deserves the name *relaxation time*, and "inertia is negligible" means precisely "τ is short compared with every other time in the problem". Equation (7.4.6) applies componentwise to a vector equation $\mathrm{d}\boldsymbol{y}/\mathrm{d}t = -\boldsymbol{y}/\tau$, since the three components decouple.

7.5 Step-by-Step Derivation

Step 1: the equation of motion for the particle's spin. Apply (4.2.2) to one particle, whose angular momentum is $I_p\boldsymbol{\Theta}$ by (4.3.2):

$$I_p \frac{\mathrm{d}\boldsymbol{\Theta}}{\mathrm{d}t} = \boldsymbol{T}_{\text{total}}. \tag{7.5.1}$$

Step 2: identify the torque. By assumption 4, the only torque is viscous. But (6.5.8) was derived for a sphere rotating in a liquid *at rest at infinity*. Here the surrounding liquid is itself rotating locally at $\boldsymbol{\Omega}$. Because the Stokes equations are linear, we may superpose: work in a frame rotating with the fluid, in which the far liquid is at rest and the sphere spins at $\boldsymbol{\Theta} - \boldsymbol{\Omega}$. (The frame rotates so slowly — $\Omega \lesssim 60\ \mathrm{s}^{-1}$ — that fictitious forces are irrelevant at the particle scale.) Hence

$$\boldsymbol{T}_{\text{visc}} = -\zeta_r\big(\boldsymbol{\Theta} - \boldsymbol{\Omega}\big). \tag{7.5.2}$$

The sign says: if the particle spins faster than the fluid, the torque slows it down; if slower, the torque speeds it up.

Step 3: assemble the equation of motion.

$$I_p \frac{\mathrm{d}\boldsymbol{\Theta}}{\mathrm{d}t} = -\zeta_r\big(\boldsymbol{\Theta} - \boldsymbol{\Omega}\big). \tag{7.5.3}$$

This is precisely the structure of Eq. (SM-8) of the paper,

$$I \frac{\mathrm{d}\boldsymbol{\Theta}}{\mathrm{d}t} = \boldsymbol{M} \times \boldsymbol{H} - \frac{I}{\tau_s}\big(\boldsymbol{\Theta} - \boldsymbol{\Omega}\big), \tag{SM-8}$$

with the magnetic torque switched off and with everything referred to one particle rather than to unit volume. Comparing the two shows already that the paper's τ_s must equal I/ζ in the corresponding units; we now prove it.

Step 4: change variables to the slip. Define the *slip* (the quantity whose decay we are studying)

$$\boldsymbol{y} \equiv \boldsymbol{\Theta} - \boldsymbol{\Omega}. \tag{7.5.4}$$

Because $\boldsymbol{\Omega}$ is constant over the times of interest (assumption 5), $\mathrm{d}\boldsymbol{y}/\mathrm{d}t = \mathrm{d}\boldsymbol{\Theta}/\mathrm{d}t$, and (7.5.3) becomes

$$I_p \frac{\mathrm{d}\boldsymbol{y}}{\mathrm{d}t} = -\zeta_r \boldsymbol{y} \qquad \Longleftrightarrow \qquad \frac{\mathrm{d}\boldsymbol{y}}{\mathrm{d}t} = -\frac{\boldsymbol{y}}{I_p/\zeta_r}. \tag{7.5.5}$$

Step 5: read off the relaxation time. Comparing (7.5.5) with the standard form (7.4.1),

$$\boldsymbol{y}(t) = \boldsymbol{y}(0)\, e^{-t/\tau_s}, \qquad \boxed{\tau_s = \frac{I_p}{\zeta_r}} \tag{7.5.6}$$

Step 6: insert the two ingredients. From (4.3.6), $I_p = \pi\rho_s d^5/60$; from (6.6.2), $\zeta_r = \pi\eta d^3$. Therefore

$$\tau_s = \frac{\pi\rho_s d^5/60}{\pi\eta d^3} \tag{7.5.7}$$

$$= \frac{\pi\rho_s d^5}{60} \cdot \frac{1}{\pi\eta d^3} \qquad \text{(division = multiplication by the reciprocal)} \tag{7.5.8}$$

$$= \frac{\rho_s d^5}{60\,\eta\, d^3} \qquad (\pi \text{ cancels}) \tag{7.5.9}$$

$$= \frac{\rho_s d^2}{60\,\eta}. \qquad (d^5/d^3 = d^2) \tag{7.5.10}$$

That is

$$\tau_s = \frac{\rho_s d^2}{60\eta}, \tag{SM-1}$$

which is Eq. (SM-1) exactly, prefactor included. ■

Step 7: the per-unit-volume version, and the identity $I/\tau_s = 6\eta\varphi$. The paper writes I for the total moment of inertia *per unit volume*, $I = nI_p$. Dividing numerator and denominator of (7.5.6) by the same factor n leaves the ratio unchanged, so

$$\tau_s = \frac{I_p}{\zeta_r} = \frac{nI_p}{n\zeta_r} = \frac{I}{n\zeta_r}, \tag{7.5.11}$$

whence, using (6.6.3),

$$\boxed{\frac{I}{\tau_s} = n\zeta_r = 6\eta\varphi} \tag{7.5.12}$$

This is exactly the identity $I/\tau_s = 6\eta\varphi$ that Shliomis & Morozov invoke without comment between their Eqs. (SM-18) and (SM-19), and it is what converts the last term of (SM-8) into the friction term of (B-1). It also shows why the combination I/τ_s, rather than I or τ_s separately, is what survives in the final theory: it contains no inertia at all, only viscosity and concentration.

7.6 Consistency Checks

Numerical value. With the paper's numbers $d = 10^{-6}$ cm (= 10 nm), $\rho_s = 6\ \mathrm{g\,cm^{-3}}$, $\eta = 10^{-2}\ \mathrm{g\,cm^{-1}s^{-1}}$:

$$\tau_s = \frac{6 \times (10^{-6})^2}{60 \times 10^{-2}} = \frac{6 \times 10^{-12}}{0.6} = 10^{-11}\ \mathrm{s}. \tag{7.6.1}$$

This reproduces the paper's quoted $\tau_s = 10^{-11}$ s exactly. ✓

Scaling. $\tau_s \propto d^2$: bigger particles carry more inertia relative to friction (inertia $\propto d^5$, friction $\propto d^3$). Doubling the diameter quadruples τ_s — still utterly negligible for any colloid.

Comparison with the other times of the problem.

Time	Value in the experiments	Ratio to τ_s
τ_s (inertia of particle spin)	10^{-11} s	1
τ_B (Brownian reorientation)	1.6×10^{-3} s (Bacri *et al.*)	$\sim 10^8$
$1/\omega$ (field period/2π, at 1 kHz)	1.6×10^{-4} s	$\sim 10^7$
$1/\Omega$ (hydrodynamic)	$\gtrsim 1.7 \times 10^{-2}$ s	$\sim 10^9$

Particle inertia is between seven and nine orders of magnitude faster than everything else. Whatever the particles are doing, they are always at instantaneous mechanical equilibrium.

7.7 Physical Interpretation

Equation (SM-1) says: *the memory of a magnetic nanoparticle's own spin lasts* 10^{-11} *seconds.* Three readings of the same fact:

- **As a competition of two effects.** $\tau_s = I_p/\zeta_r$ is (inertia)/(friction). Inertia tries to keep the particle spinning as it was; friction insists that it spin with the fluid. Friction wins by eight orders of magnitude.

- **As a licence.** Because τ_s is so short, the left-hand side of the spin equation (SM-8), $I\,\mathrm{d}\boldsymbol{\Theta}/\mathrm{d}t \sim I\Theta/\tau_{\mathrm{obs}}$, is smaller than the friction term $I\Theta/\tau_s$ by the factor $\tau_s/\tau_{\mathrm{obs}} \sim 10^{-8}$. This is precisely the estimate the paper makes when it writes "$I\,\mathrm{d}\boldsymbol{\Theta}/\mathrm{d}t \ll I\boldsymbol{\Theta}/\tau_s$, so Eq. (8) has the form (11)". Dropping the inertial term converts a differential equation for $\boldsymbol{\Theta}$ into an *algebraic* one, and this is the single most important simplification of the whole theory.

- **As a statement about what is *not* the cause of the effect.** One might imagine that the negative-viscosity effect involves storing energy in the rotational inertia of the grains, like a flywheel. Equation (SM-1) rules that out absolutely: a 10^{-11} s flywheel stores nothing. The energy that ultimately spins up the flow is stored in the *orientational* degrees of freedom of the magnetic moments and released through the magnetic torque, on the much longer time scale τ_B.

What if the term were removed? If instead we *kept* the inertial term as significant, Eq. (SM-8) would remain a differential equation, the particle spin would lag the torque by a phase, and there would be an additional resonance at $\omega \sim 1/\tau_s \sim 10^{11}$ s^{-1} — i.e. at ~ 16 GHz. Nothing in the 1 kHz experiments of Bacri *et al.* can see that; the physics discussed in these papers lives entirely in the regime $\omega\tau_s \lll 1$.

7.8 Connection to the Next Equation

Equation (SM-1) has established the *reference state*: in the absence of a magnetic field the particles spin with the fluid, $\boldsymbol{\Theta} = \boldsymbol{\Omega}$, essentially instantaneously, and therefore contribute no extra dissipation. The paper now asks what happens when a stationary field is switched on. The field grabs the magnetic moments and holds them; a particle whose moment is held cannot follow the fluid's rotation freely; the slip $\boldsymbol{\Omega} - \boldsymbol{\Theta}$ becomes nonzero; and maintaining a nonzero slip costs energy, which appears as an *extra* viscosity. Quantifying that is the content of the next equation,

$$\Delta\eta = \frac{3}{2}\,\eta\varphi\,\frac{\xi - \tanh\xi}{\xi + \tanh\xi}\,\sin^2\beta, \tag{SM-2}$$

Shliomis's 1972 formula for the rotational viscosity in a stationary field.

Deriving (SM-2) requires three ingredients that do not yet exist in this book and which will be constructed, from first principles, at the start of Volume II:

1. **The Boltzmann distribution and the Langevin function.** We must compute $\langle \boldsymbol{e} \rangle$ for dipoles in a field at temperature T, obtaining $L(\xi) = \coth\xi - 1/\xi$ and hence the equilibrium magnetisation $M_0 = nmL(\xi)$ used in Eq. (SM-10).

2. **Rotational Brownian motion.** We must derive the rotational diffusion (Fokker–Planck/Smoluchowski) equation for the orientation $\boldsymbol{e}$, obtain the Einstein relation $D_r = kT/\zeta_r$, and from it the Brownian time $\tau_B = 3\eta V/kT$ of Eq. (SM-5) — which, note, is already within reach: combining $D_r = kT/\zeta_r$ with $\zeta_r = 6\eta V$ from (6.6.1) gives $\tau_B = 1/(2D_r) = 3\eta V/kT$ immediately.

3. **The link between microscopic torque and macroscopic viscosity.** We must show that the averaged magnetic torque density $\overline{\boldsymbol{M} \times \boldsymbol{H}}$ enters the momentum equation as an antisymmetric stress and hence as an addition $\Delta\eta$ to the viscosity — the step that turns Eq. (SM-26) into (SM-27), and which in the papers appears in the compact forms $\overline{\boldsymbol{M} \times \boldsymbol{H}} = -6\eta\varphi(\boldsymbol{\Omega} - \boldsymbol{\Theta})$ and $\Delta\eta = \frac{3}{2}\eta\varphi(\boldsymbol{\Omega} - \boldsymbol{\Theta})/\boldsymbol{\Omega}$.

Volume II will carry out these three constructions in order and then derive (SM-2), (SM-3), (SM-4) and (SM-5) in full.

Chapter 8

Status and Road Map

8.1 What has been established so far

Result	Statement	Where derived
Index calculus	$\varepsilon_{ijk}\varepsilon_{klm} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}$ and consequences	Secs. 1.6–1.7
Divergence theorem	$\int \nabla \cdot \boldsymbol{A}\,\mathrm{d}V = \oint \boldsymbol{A} \cdot \boldsymbol{n}\,\mathrm{d}S$	Sec. 2.3
Local flow structure	translation + strain + rigid spin, with $\boldsymbol{\Omega} = \frac{1}{2}\nabla \times \boldsymbol{v}$	Sec. 3.4
Vorticity in the experiment	$\boldsymbol{\Omega} \perp \boldsymbol{H}$ in a capillary along the coil axis; $\Omega\tau_B \approx 0.1$	Sec. 3.5
Sphere inertia	$I_p = \pi\rho_s d^5/60 = \frac{2}{5}m_p a^2$	Sec. 4.3
Dipole mechanics	$\boldsymbol{T} = \boldsymbol{m} \times \boldsymbol{H}$, $U = -\boldsymbol{m} \cdot \boldsymbol{H}$, $\xi = mH/kT$	Sec. 4.4
Continuity	$\nabla \cdot \boldsymbol{v} = 0$ [first relation of (SM-9)]	Sec. 5.1
Cauchy / Navier–Stokes	$\rho\,\mathrm{D}\boldsymbol{v}/\mathrm{D}t = -\nabla p + \eta\nabla^2\boldsymbol{v} + \boldsymbol{f}$	Secs. 5.2–5.4
Stokes regime	$\mathrm{Re} \sim 3 \times 10^{-9}$ at the particle scale	Sec. 5.5
Rotational friction	$\zeta_r = 8\pi\eta a^3 = 6\eta V = \pi\eta d^3$; $n\zeta_r = 6\eta\varphi$	Ch. 6
Eq. (SM-1)	$\tau_s = \rho_s d^2/60\eta = 10^{-11}$ s; $I/\tau_s = 6\eta\varphi$	Ch. 7

8.2 The remaining equations, in the order they will be derived

Volume II and its successors will proceed strictly in the papers' own order. For reference, the full list of what remains:

- **Shliomis & Morozov:** (SM-2) rotational viscosity in a stationary field; (SM-3), (SM-4) its weak- and strong-field limits; (SM-5) $\tau_B = 3\eta V/kT$; (SM-6)–(SM-9) the full ferrohydrodynamic system; (SM-10) the Langevin magnetisation; (SM-11)–(SM-13) elimination of the particle spin; (SM-14)–(SM-15) the alternating field and linear susceptibility; (SM-16)–(SM-24) the circular-polarisation decomposition; (SM-25)–(SM-29) the averaged torque and the central result $\Delta\eta = \frac{1}{4}\eta\varphi\xi^2(1-\omega^2\tau_B^2)/(1+\omega^2\tau_B^2)^2$; (SM-30)–(SM-34) the second, direct variant of the solution and the formula $\Delta\eta = \frac{3}{2}\eta\varphi(\Omega-\Theta)/\Omega$; (SM-35)–(SM-43) arbitrary field amplitude, the functions F, G, $\gamma(t)$, and the numerical results.
- **Bacri *et al.*:** (B-1), (B-2) torque balance and $\Delta\eta$; (B-3), (B-4) the hydrodynamic and

magnetodynamic pair; (B-5)–(B-8) the effective field method of Martsenyuk, Raikher and Shliomis; (B-9) the averaged torque and the function g; (B-10) the grouping of the magnetic term with the viscous one.

Every one of these will be derived here with the same standard of completeness: nothing asserted, nothing skipped, every symbol defined, every approximation justified.

End of Volume I. Volume II begins with the Boltzmann distribution for a dipole in a field and proceeds to Eq. (SM-2).

THE NEGATIVE-VISCOSITY EFFECT IN FERROFLUIDS

A Complete First-Principles Reconstruction of Every Equation

VOLUME II

Statistical Mechanics of the Orientations,
Rotational Brownian Motion,
and Equations (SM-2)–(SM-5)

Where Volume I ended. Volume I built the mathematical apparatus (index notation, vector calculus, the local decomposition $\boldsymbol{\Omega} = \frac{1}{2}\nabla \times \boldsymbol{v}$), the continuum equations (continuity, Cauchy, Navier–Stokes), and the Stokes-flow result $\zeta_r = 8\pi\eta a^3 = 6\eta V = \pi\eta d^3$ for the rotational friction of a sphere. It closed with the eight-part derivation of

$$\tau_s = \frac{\rho_s d^2}{60\eta} = 10^{-11}\ \text{s}, \qquad \text{and the identity} \qquad \frac{I}{\tau_s} = n\zeta_r = 6\eta\varphi. \tag{SM-1}$$

Volume II now supplies the three missing ingredients announced in Sec. 7.9 — the Boltzmann/Langevin statistics of the moments, rotational Brownian motion, and the link between magnetic torque and viscosity — and then derives Eqs. (SM-2), (SM-3), (SM-4) and (SM-5) in the papers' own order. All numbering, notation and conventions of Volume I continue unchanged; chapter numbers resume at 9, page numbers at 34.

Part IV

Statistical Mechanics of the Orientations

Chapter 9

The Boltzmann Distribution

9.1 Why a distribution is needed at all

A magnetic grain of diameter $d = 10$ nm carries a magnetic moment $m = M_s V$ which, in a field H, has the orientational energy $-mH\cos\theta$ of Eq. (4.4.2). If nothing else acted, every moment would simply align with the field and the suspension would be saturated at any field strength whatsoever. That is not what happens: the magnetisation of a ferrofluid rises gradually with H over a range of hundreds of oersteds. The reason is that the grain is immersed in a liquid whose molecules bombard it $\sim 10^{12}$ times per second from random directions, delivering random torques that continually knock the moment away from the field direction. The competition is between two energies:

$$\underbrace{mH}_{\text{magnetic}} \qquad \text{versus} \qquad \underbrace{kT}_{\text{thermal}}, \tag{9.1.1}$$

whose ratio is exactly the Langevin parameter $\xi = mH/kT$ defined in Eq. (4.4.6). We therefore need the statistical law that says how often a given orientation occurs. That law is the Boltzmann distribution, which we now derive.

9.2 The two postulates of statistical mechanics

Assumption 9.2.1 (Equal a priori probabilities)**.** An isolated system in equilibrium is equally likely to be found in each of the microstates accessible to it at its given energy.

Assumption 9.2.2 (Boltzmann's entropy)**.** The entropy of an isolated system with $\mathcal{N}(E)$ accessible microstates at energy E is

$$S(E) = k\ln\mathcal{N}(E). \tag{9.2.1}$$

These two statements cannot be derived from mechanics; they are the postulates on which statistical mechanics rests, and we adopt them. Everything below *is* derived.

We also need the thermodynamic definition of absolute temperature. Consider two systems, with energies E_1 and E_2 and $E_1 + E_2 = E_{\text{tot}}$ fixed, allowed to exchange energy. The number of joint microstates is $\mathcal{N}_1(E_1)\mathcal{N}_2(E_{\text{tot}} - E_1)$, and by Assumption 9.2.1 the observed value of E_1 is overwhelmingly the one that maximises this product, i.e. that maximises $S_1(E_1)+S_2(E_{\text{tot}}-E_1)$. Setting the derivative to zero,

$$\frac{\partial S_1}{\partial E_1} - \frac{\partial S_2}{\partial E_2} = 0 \qquad \Longrightarrow \qquad \frac{\partial S_1}{\partial E_1} = \frac{\partial S_2}{\partial E_2}. \tag{9.2.2}$$

So two systems in thermal equilibrium share the common value of $\partial S/\partial E$. This is precisely the property a thermometer measures, and we *define*

$$\frac{1}{T} \equiv \frac{\partial S}{\partial E}. \tag{9.2.3}$$

9.3 Derivation of the Boltzmann factor

Let a small system $\mathcal{A}$ (here: one magnetic grain, whose only relevant variable is the direction $\boldsymbol{e}$ of its moment) be in contact with a large reservoir $\mathcal{R}$ (the carrier liquid) at temperature T. The compound system is isolated, with total energy E_{tot}.

Step 1. Suppose $\mathcal{A}$ is in one definite microstate i of energy E_i. Then the reservoir has energy $E_{\mathrm{tot}} - E_i$ and $\mathcal{N}_{\mathcal{R}}(E_{\mathrm{tot}} - E_i)$ microstates available. By Assumption 9.2.1 applied to the *compound* system, the probability of finding $\mathcal{A}$ in state i is proportional to the number of ways the rest of the world can accommodate it:

$$p_i \propto \mathcal{N}_{\mathcal{R}}(E_{\mathrm{tot}} - E_i). \tag{9.3.1}$$

Step 2. Take the logarithm and use (9.2.1):

$$\ln p_i = \mathrm{const} + \frac{1}{k} S_{\mathcal{R}}(E_{\mathrm{tot}} - E_i). \tag{9.3.2}$$

Step 3. Expand in E_i. This is legitimate because the reservoir is enormous: $E_i \ll E_{\mathrm{tot}}$, so E_i is a tiny displacement of the reservoir's energy and a first-order Taylor expansion is accurate:

$$S_{\mathcal{R}}(E_{\mathrm{tot}} - E_i) = S_{\mathcal{R}}(E_{\mathrm{tot}}) - E_i \left.\frac{\partial S_{\mathcal{R}}}{\partial E}\right|_{E_{\mathrm{tot}}} + O(E_i^2) \tag{9.3.3}$$

$$= S_{\mathcal{R}}(E_{\mathrm{tot}}) - \frac{E_i}{T} + O(E_i^2), \tag{9.3.4}$$

where the definition (9.2.3) was used. The neglected term involves $\partial^2 S/\partial E^2 = -(1/T^2)\partial T/\partial E$, which vanishes for a reservoir so large that absorbing E_i does not change its temperature. That is the defining property of a reservoir.

Step 4. Therefore

$$\ln p_i = \mathrm{const}' - \frac{E_i}{kT} \qquad \Longrightarrow \qquad \boxed{p_i = \frac{1}{Z}\, e^{-E_i/kT}} \tag{9.3.5}$$

with Z (the partition function) fixed by $\sum_i p_i = 1$. ■

Note what (9.3.5) says physically: states of *lower* energy are exponentially more probable, and the scale on which "lower" is measured is kT. Energy differences much smaller than kT are essentially invisible; differences much larger than kT are decisive.

9.4 Orientations: the solid-angle measure

The magnetic moment of a rigid dipole is specified by a unit vector $\boldsymbol{e}$, i.e. by a point on the unit sphere. In spherical polar coordinates with the polar axis along the applied field $\hat{\boldsymbol{h}}$,

$$\boldsymbol{e} = (\sin\theta\cos\phi,\ \sin\theta\sin\phi,\ \cos\theta), \qquad 0 \le \theta \le \pi, \quad 0 \le \phi < 2\pi, \tag{9.4.1}$$

where θ is the angle between $\boldsymbol{e}$ and $\hat{\boldsymbol{h}}$, so that $\boldsymbol{e}\cdot\hat{\boldsymbol{h}} = \cos\theta$.

The set of orientations is continuous, so instead of probabilities p_i we need a probability *density* $W(\boldsymbol{e})$ such that $W\,\mathrm{d}\Omega_e$ is the probability of finding the moment within the element of solid angle $\mathrm{d}\Omega_e$. We must therefore know $\mathrm{d}\Omega_e$.

Derivation of $\mathrm{d}\Omega_e = \sin\theta\,\mathrm{d}\theta\,\mathrm{d}\phi$. On a sphere of radius 1, increasing θ by $\mathrm{d}\theta$ moves a point a distance $\mathrm{d}\theta$ along a meridian (arc length = radius × angle = $1\cdot\mathrm{d}\theta$). Increasing ϕ by $\mathrm{d}\phi$ moves it along a circle of latitude whose radius is not 1 but $\sin\theta$ (the perpendicular distance from the polar axis), so the arc length is $\sin\theta\,\mathrm{d}\phi$. The two displacements are perpendicular, so the patch is a rectangle of area

$$\mathrm{d}\Omega_e = \mathrm{d}\theta\cdot\sin\theta\,\mathrm{d}\phi = \sin\theta\,\mathrm{d}\theta\,\mathrm{d}\phi. \tag{9.4.2}$$

Check: $\int \mathrm{d}\Omega_e = \int_0^{2\pi}\mathrm{d}\phi\int_0^{\pi}\sin\theta\,\mathrm{d}\theta = 2\pi\cdot\big[-\cos\theta\big]_0^{\pi} = 2\pi\cdot 2 = 4\pi$, the total solid angle. ✓

It is almost always convenient to substitute

$$u \equiv \cos\theta, \qquad \mathrm{d}u = -\sin\theta\,\mathrm{d}\theta, \tag{9.4.3}$$

which converts $\int_0^{\pi}(\cdots)\sin\theta\,\mathrm{d}\theta$ into $\int_{-1}^{1}(\cdots)\,\mathrm{d}u$ — the sign from $\mathrm{d}u$ being cancelled by the reversal of the limits ($\theta=0\Rightarrow u=1$, $\theta=\pi\Rightarrow u=-1$).

9.5 The equilibrium orientational distribution

Insert the dipole energy $U = -\boldsymbol{m}\cdot\boldsymbol{H} = -mH\cos\theta$ of Eq. (4.4.2) into the Boltzmann law (9.3.5):

$$W_0(\boldsymbol{e}) = C\exp\!\left(\frac{mH\cos\theta}{kT}\right) = C\,e^{\xi\cos\theta} = C\,e^{\xi\,\boldsymbol{e}\cdot\hat{\boldsymbol{h}}}, \tag{9.5.1}$$

with $\xi = mH/kT$ as defined in (4.4.6). The subscript 0 marks the *equilibrium* distribution, i.e. the one holding when the fluid is at rest; all of Volume II's later work consists of computing the small correction to (9.5.1) produced by flow.

Normalisation. Require $\int W_0\,\mathrm{d}\Omega_e = 1$:

$$1 = C\int_0^{2\pi}\mathrm{d}\phi\int_0^{\pi} e^{\xi\cos\theta}\sin\theta\,\mathrm{d}\theta \tag{9.5.2}$$

$$= 2\pi C\int_{-1}^{1} e^{\xi u}\,\mathrm{d}u \qquad \text{by (9.4.3)} \tag{9.5.3}$$

$$= 2\pi C\left[\frac{e^{\xi u}}{\xi}\right]_{-1}^{1} \tag{9.5.4}$$

$$= 2\pi C\,\frac{e^{\xi}-e^{-\xi}}{\xi} \tag{9.5.5}$$

$$= \frac{4\pi C\sinh\xi}{\xi}, \qquad \text{using } \sinh\xi = \tfrac{1}{2}(e^{\xi}-e^{-\xi}) \tag{9.5.6}$$

so that

$$C = \frac{\xi}{4\pi\sinh\xi}, \qquad W_0(\theta) = \frac{\xi}{4\pi\sinh\xi}\,e^{\xi\cos\theta}. \tag{9.5.7}$$

Check the field-free limit. As $\xi\to 0$, $\sinh\xi\to\xi$ and $e^{\xi\cos\theta}\to 1$, so $W_0\to 1/4\pi$: all directions equally likely, as they must be when there is no field. ✓

The integral appearing above will recur so often that we name it:

$$Z(\xi) \equiv \int e^{\xi\, \boldsymbol{e}\cdot\hat{\boldsymbol{h}}}\, \mathrm{d}\Omega_e = \frac{4\pi \sinh \xi}{\xi}. \tag{9.5.8}$$

Chapter 10

The Langevin Function and the Equilibrium Magnetisation

10.1 Context

The magnetisation of the suspension was defined in Eq. (4.5.1) as $\boldsymbol{M} = nm\langle \boldsymbol{e} \rangle$. With the distribution (9.5.7) in hand we can now compute $\langle \boldsymbol{e} \rangle$ exactly. The result is the function called $L(\xi)$ in both papers, and it enters Eq. (SM-10), Eq. (B-4) and, through them, essentially every later formula.

10.2 Computation of $\langle \boldsymbol{e} \rangle$

Step 1: only the component along the field survives. By symmetry the distribution (9.5.7) depends on θ but not on ϕ. The components $e_x = \sin\theta\cos\phi$ and $e_y = \sin\theta\sin\phi$ therefore average to zero, since $\int_0^{2\pi} \cos\phi \, \mathrm{d}\phi = \int_0^{2\pi} \sin\phi \, \mathrm{d}\phi = 0$. Hence $\langle \boldsymbol{e} \rangle = \langle \cos\theta \rangle \, \hat{\boldsymbol{h}}$, and only one integral is needed.

Step 2: set up the integral.

$$\langle \cos\theta \rangle = \int \cos\theta \, W_0 \, \mathrm{d}\Omega_e = \frac{\displaystyle\int_0^{2\pi} \mathrm{d}\phi \int_0^{\pi} \cos\theta \, e^{\xi\cos\theta} \sin\theta \, \mathrm{d}\theta}{\displaystyle\int_0^{2\pi} \mathrm{d}\phi \int_0^{\pi} e^{\xi\cos\theta} \sin\theta \, \mathrm{d}\theta} \tag{10.2.1}$$

$$= \frac{\displaystyle\int_{-1}^{1} u \, e^{\xi u} \, \mathrm{d}u}{\displaystyle\int_{-1}^{1} e^{\xi u} \, \mathrm{d}u}. \tag{10.2.2}$$

The factors 2π and C cancel between numerator and denominator, which is why writing the average as a ratio of unnormalised integrals is safe.

Step 3: the denominator. Already computed:

$$\mathcal{D} \equiv \int_{-1}^{1} e^{\xi u} \, \mathrm{d}u = \frac{e^{\xi} - e^{-\xi}}{\xi} = \frac{2\sinh\xi}{\xi}. \tag{10.2.3}$$

Step 4: the numerator, by the differentiation trick. Notice that

$$u\,e^{\xi u} = \frac{\partial}{\partial \xi} e^{\xi u}, \tag{10.2.4}$$

because differentiating $e^{\xi u}$ with respect to the *parameter* ξ brings down a factor u. Since the limits of integration do not depend on ξ, differentiation and integration may be interchanged, and

$$\mathcal{N} \equiv \int_{-1}^{1} u\,e^{\xi u}\,\mathrm{d}u = \frac{\mathrm{d}}{\mathrm{d}\xi}\int_{-1}^{1} e^{\xi u}\,\mathrm{d}u = \frac{\mathrm{d}}{\mathrm{d}\xi}\left(\frac{2\sinh\xi}{\xi}\right) \tag{10.2.5}$$

$$= 2\left[\frac{\cosh\xi}{\xi} - \frac{\sinh\xi}{\xi^2}\right] \qquad \text{(quotient rule; } \sinh' = \cosh\text{)} \tag{10.2.6}$$

$$= \frac{2(\xi\cosh\xi - \sinh\xi)}{\xi^2}. \tag{10.2.7}$$

Step 4′: the same numerator by integration by parts (independent check). With $\mathrm{d}v = e^{\xi u}\mathrm{d}u$, $v = e^{\xi u}/\xi$, and the other factor u:

$$\int_{-1}^{1} u\,e^{\xi u}\,\mathrm{d}u = \left[\frac{u\,e^{\xi u}}{\xi}\right]_{-1}^{1} - \int_{-1}^{1}\frac{e^{\xi u}}{\xi}\,\mathrm{d}u \tag{10.2.8}$$

$$= \frac{e^{\xi} + e^{-\xi}}{\xi} - \frac{1}{\xi}\cdot\frac{e^{\xi} - e^{-\xi}}{\xi} \tag{10.2.9}$$

$$= \frac{2\cosh\xi}{\xi} - \frac{2\sinh\xi}{\xi^2} = \frac{2(\xi\cosh\xi - \sinh\xi)}{\xi^2}. \quad \checkmark \tag{10.2.10}$$

(In the second line the boundary term was evaluated as $\frac{1}{\xi}[1\cdot e^{\xi} - (-1)e^{-\xi}] = (e^{\xi}+e^{-\xi})/\xi$.)

Step 5: the ratio.

$$\langle\cos\theta\rangle = \frac{\mathcal{N}}{\mathcal{D}} = \frac{2(\xi\cosh\xi - \sinh\xi)/\xi^2}{2\sinh\xi/\xi} \tag{10.2.11}$$

$$= \frac{\xi\cosh\xi - \sinh\xi}{\xi^2}\cdot\frac{\xi}{\sinh\xi} \tag{10.2.12}$$

$$= \frac{\xi\cosh\xi - \sinh\xi}{\xi\,\sinh\xi} \tag{10.2.13}$$

$$= \frac{\cosh\xi}{\sinh\xi} - \frac{1}{\xi} \tag{10.2.14}$$

$$= \coth\xi - \frac{1}{\xi}. \tag{10.2.15}$$

Result 10.2.1 (Langevin function)**.**

$$\boxed{\langle\cos\theta\rangle = L(\xi) \equiv \coth\xi - \frac{1}{\xi}} \tag{10.2.16}$$

10.3 Properties of $L(\xi)$ that will be needed

(a) Weak fields: the series. We need $\coth\xi$ for small ξ. Use the power series of the hyperbolic functions (each obtained from the exponential series):

$$\sinh\xi = \xi + \frac{\xi^3}{6} + \frac{\xi^5}{120} + \cdots, \qquad \cosh\xi = 1 + \frac{\xi^2}{2} + \frac{\xi^4}{24} + \cdots \tag{10.3.1}$$

Then

$$\coth\xi = \frac{\cosh\xi}{\sinh\xi} = \frac{1+\dfrac{\xi^2}{2}+\dfrac{\xi^4}{24}}{\xi\left(1+\dfrac{\xi^2}{6}+\dfrac{\xi^4}{120}\right)} \tag{10.3.2}$$

$$= \frac{1}{\xi}\,(1+a)\,(1+b)^{-1}, \qquad a=\frac{\xi^2}{2}+\frac{\xi^4}{24}, \quad b=\frac{\xi^2}{6}+\frac{\xi^4}{120} \tag{10.3.3}$$

$$= \frac{1}{\xi}\,(1+a)\left(1-b+b^2-\cdots\right) \qquad \text{geometric series, } |b|<1 \tag{10.3.4}$$

$$= \frac{1}{\xi}\Big(1+(a-b)+(b^2-ab)+O(\xi^6)\Big). \tag{10.3.5}$$

Now evaluate the brackets term by term:

$$a-b = \left(\frac{\xi^2}{2}-\frac{\xi^2}{6}\right)+\left(\frac{\xi^4}{24}-\frac{\xi^4}{120}\right) = \frac{\xi^2}{3}+\frac{\xi^4}{30}, \tag{10.3.6}$$

$$b^2-ab = \frac{\xi^4}{36}-\frac{\xi^2}{2}\cdot\frac{\xi^2}{6}+O(\xi^6) = \frac{\xi^4}{36}-\frac{\xi^4}{12} = -\frac{\xi^4}{18}, \tag{10.3.7}$$

where $\frac{1}{2}-\frac{1}{6}=\frac{1}{3}$ and $\frac{1}{24}-\frac{1}{120}=\frac{5-1}{120}=\frac{1}{30}$, and $\frac{1}{36}-\frac{1}{12}=\frac{1-3}{36}=-\frac{1}{18}$. Adding the two ξ^4 pieces, $\frac{1}{30}-\frac{1}{18}=\frac{3-5}{90}=-\frac{1}{45}$, so

$$\coth\xi = \frac{1}{\xi}+\frac{\xi}{3}-\frac{\xi^3}{45}+O(\xi^5), \qquad \boxed{L(\xi)=\frac{\xi}{3}-\frac{\xi^3}{45}+O(\xi^5)} \tag{10.3.8}$$

The leading behaviour $L\approx\xi/3$ is the statement that in weak fields the alignment is proportional to the field — linear response — and the factor $1/3$ is the average of $\cos^2\theta$ over a sphere, i.e. the fact that only one of three equivalent directions is favoured.

(b) Strong fields. Write $\coth\xi = \dfrac{e^{\xi}+e^{-\xi}}{e^{\xi}-e^{-\xi}} = \dfrac{1+e^{-2\xi}}{1-e^{-2\xi}}$. For $\xi\gg 1$ the exponentials are negligible and $\coth\xi\to 1$, so

$$L(\xi)\to 1-\frac{1}{\xi} \qquad (\xi\gg 1). \tag{10.3.9}$$

Saturation: all moments aligned, $\langle\cos\theta\rangle\to 1$, approached slowly (as $1/\xi$) because thermal agitation keeps knocking moments off the axis.

(c) The identity that produces the form of Eq. (SM-2). Since $\coth\xi = 1/\tanh\xi$,

$$L(\xi) = \frac{1}{\tanh\xi}-\frac{1}{\xi} = \frac{\xi-\tanh\xi}{\xi\tanh\xi}, \qquad \text{equivalently} \qquad \xi L(\xi) = \frac{\xi-\tanh\xi}{\tanh\xi}. \tag{10.3.10}$$

This innocent rearrangement is precisely what will convert the result of Chapter 15 into the exact form printed as Eq. (SM-2); keep it in view.

10.4 The equilibrium magnetisation

Combining $\boldsymbol{M} = nm\langle\boldsymbol{e}\rangle$ [Eq. (4.5.1)] with (10.2.16):

$$\boxed{\boldsymbol{M}_0 = n\,m\,L(\xi)\,\hat{\boldsymbol{h}}, \qquad \hat{\boldsymbol{h}}=\boldsymbol{H}/H, \qquad \xi=\frac{mH}{kT}} \tag{10.4.1}$$

This is exactly the content of the paper's Eq. (SM-10),

$$\boldsymbol{M}_0 = nmL(\xi)\boldsymbol{H}/H, \qquad \xi = mH(t)/kT, \qquad L(\xi) = \coth\xi - \xi^{-1}, \tag{SM-10}$$

and of the first half of Bacri *et al.*'s Eq. (B-4), where the same object is written $\boldsymbol{M}_0(t) = \varphi M_s L(\xi)\hat{\boldsymbol{h}}$. The two forms agree because $nm = nM_sV = \varphi M_s$ by Eqs. (4.4.1) and the definition $\varphi = nV$.

Remark 10.4.1 (What "instantaneous equilibrium" means). When the field varies in time, $\boldsymbol{M}_0(t)$ is defined by inserting the *instantaneous* $H(t)$ into (10.4.1). It is the magnetisation the suspension *would* have if it could reorganise infinitely fast, i.e. if $\tau_B = 0$. The real magnetisation $\boldsymbol{M}(t)$ lags behind it, and the whole of the negative-viscosity effect lives in that lag. Both papers stress this point in the text following (SM-10) and (B-4).

Weak-field form. Using (10.3.8) to leading order,

$$\boldsymbol{M}_0 = nm\frac{\xi}{3}\hat{\boldsymbol{h}} = \frac{nm}{3}\cdot\frac{mH}{kT}\hat{\boldsymbol{h}} = \frac{nm^2}{3kT}\boldsymbol{H} \equiv \chi\boldsymbol{H}, \qquad \chi = \frac{nm^2}{3kT}, \tag{10.4.2}$$

which is the paper's Eq. (SM-15). We have thus already derived, as a by-product, an equation that will be formally presented in its own place later. That is allowed and unavoidable: (SM-15) is nothing but the small-ξ limit of (SM-10).

Part V

Rotational Brownian Motion

Chapter 11

The Rotational Diffusion (Smoluchowski) Equation

11.1 Context and strategy

The Boltzmann distribution of Chapter 9 describes *equilibrium*. But the central question of both papers is intrinsically out of equilibrium: what happens to the orientations when the field oscillates, or when the fluid shears? We need the *dynamical* equation whose stationary solution is (9.5.1). That equation is the rotational analogue of the diffusion equation, and it is derived here from three ingredients only:

1. conservation of probability;
2. the deterministic drift produced by known torques (viscous, magnetic) in the inertialess limit established by Eq. (SM-1);
3. Fick's law for the diffusive part of the flux, with an as-yet-unknown diffusion coefficient D_r, whose value is then *fixed* by requiring consistency with Chapter 9.

Step 3 is the Einstein argument, and it is the reason τ_B ends up containing the viscosity η.

11.2 Calculus on the unit sphere

Orientation space is the surface of the unit sphere. We need the gradient and divergence *restricted to that surface*; they differ from the ordinary three-dimensional operators because there is no "radial" direction to differentiate along.

Definition 11.2.1 (Surface gradient)**.** For a function $f(\boldsymbol{e})$ defined on the unit sphere,

$$\nabla_s f \equiv \hat{\boldsymbol{\theta}}\,\frac{\partial f}{\partial \theta} + \hat{\boldsymbol{\phi}}\,\frac{1}{\sin\theta}\frac{\partial f}{\partial \phi}, \tag{11.2.1}$$

where $\hat{\boldsymbol{\theta}}$ and $\hat{\boldsymbol{\phi}}$ are the unit vectors of increasing θ and ϕ.

The factor $1/\sin\theta$ is not a convention: it is forced by the geometry established in (9.4.2). Moving through the angle $\mathrm{d}\phi$ carries one a *distance* $\sin\theta\,\mathrm{d}\phi$, and a gradient is a change per unit *distance*, so the derivative with respect to ϕ must be divided by $\sin\theta$. The same reasoning applied to θ needs no factor, because the distance is $\mathrm{d}\theta$ itself.

Equivalently and more usefully, $\nabla_s f$ is the ordinary three-dimensional gradient with its radial part projected out:

$$\nabla_s f = (1 - \boldsymbol{e}\boldsymbol{e}) \cdot \frac{\partial f}{\partial \boldsymbol{e}}, \qquad \text{i.e.} \qquad (\nabla_s f)_i = \big(\delta_{ij} - e_i e_j\big) \frac{\partial f}{\partial e_j}. \tag{11.2.2}$$

The operator $\delta_{ij} - e_i e_j$ is the projector onto the plane tangent to the sphere at $\boldsymbol{e}$: acting on any vector $\boldsymbol{A}$ it gives $\boldsymbol{A} - \boldsymbol{e}(\boldsymbol{e} \cdot \boldsymbol{A})$, which is $\boldsymbol{A}$ with its component along $\boldsymbol{e}$ removed. (Check that it is a projector: $(\delta_{ij} - e_i e_j)(\delta_{jk} - e_j e_k) = \delta_{ik} - e_i e_k - e_i e_k + e_i e_k (e_j e_j) = \delta_{ik} - e_i e_k$, using $e_j e_j = 1$. ✓)

Two special cases will be used constantly:

$$\nabla_s\big(\boldsymbol{e} \cdot \boldsymbol{A}\big) = \boldsymbol{A} - \boldsymbol{e}\,(\boldsymbol{e} \cdot \boldsymbol{A}) \qquad \text{for a constant vector } \boldsymbol{A}, \tag{11.2.3}$$

$$\nabla_s e_i = \hat{\boldsymbol{x}}_i - \boldsymbol{e}\, e_i. \tag{11.2.4}$$

Both follow from (11.2.2) with $\partial(\boldsymbol{e} \cdot \boldsymbol{A})/\partial e_j = A_j$ and $\partial e_i / \partial e_j = \delta_{ij}$.

Definition 11.2.2 (Surface divergence)**.** For a tangential vector field $\boldsymbol{j}(\boldsymbol{e})$ on the sphere,

$$\nabla_s \cdot \boldsymbol{j} \equiv \frac{1}{\sin\theta} \frac{\partial}{\partial \theta}\big(\sin\theta\, j_\theta\big) + \frac{1}{\sin\theta} \frac{\partial j_\phi}{\partial \phi}. \tag{11.2.5}$$

Derivation of (11.2.5)**.** Repeat the box argument of Sec. 2.3, now for a patch on the unit sphere bounded by the coordinate curves θ, $\theta + \mathrm{d}\theta$, ϕ, $\phi + \mathrm{d}\phi$. Its area is $\sin\theta\, \mathrm{d}\theta\, \mathrm{d}\phi$ by (9.4.2). The outflow through the edge at $\theta + \mathrm{d}\theta$ is (flux density)$\times$(edge length) $= j_\theta(\theta + \mathrm{d}\theta) \sin(\theta + \mathrm{d}\theta) \mathrm{d}\phi$, and the inflow at θ is $j_\theta(\theta) \sin\theta\, \mathrm{d}\phi$; their difference is $\partial_\theta\big(j_\theta \sin\theta\big) \mathrm{d}\theta\, \mathrm{d}\phi$. Note that the edge length itself depends on θ, which is why $\sin\theta$ sits *inside* the derivative. For the ϕ pair, the edge length is $\mathrm{d}\theta$ (independent of ϕ), and the net outflow is $\partial_\phi j_\phi\, \mathrm{d}\phi\, \mathrm{d}\theta$. Adding and dividing by the area gives (11.2.5). ■

Definition 11.2.3 (Surface Laplacian)**.** $\nabla_s^2 f \equiv \nabla_s \cdot (\nabla_s f) = \dfrac{1}{\sin\theta} \dfrac{\partial}{\partial\theta}\Big(\sin\theta \dfrac{\partial f}{\partial\theta}\Big) + \dfrac{1}{\sin^2\theta} \dfrac{\partial^2 f}{\partial \phi^2}$.

Result 11.2.1 (Green's identity on a closed surface)**.** *For any two smooth functions f, g on the sphere,*

$$\int f\, \nabla_s^2 g\, \mathrm{d}\Omega_e = -\int \nabla_s f \cdot \nabla_s g\, \mathrm{d}\Omega_e = \int g\, \nabla_s^2 f\, \mathrm{d}\Omega_e. \tag{11.2.6}$$

Derivation. Apply the surface divergence theorem — proved exactly as in Sec. 2.3, by summing patches and cancelling shared edges — to the tangential field $f \nabla_s g$. A closed surface has *no boundary*, so there is no boundary term at all:

$$0 = \int \nabla_s \cdot \big(f \nabla_s g\big) \mathrm{d}\Omega_e = \int \Big(\nabla_s f \cdot \nabla_s g + f \nabla_s^2 g\Big) \mathrm{d}\Omega_e, \tag{11.2.7}$$

which is the first equality; the second follows by exchanging f and g in the symmetric middle expression. ■

The absence of boundary terms is a genuine simplification peculiar to orientation space, and we shall use it without comment when integrating by parts over $\mathrm{d}\Omega_e$.

11.3 Conservation of probability

Let $W(\boldsymbol{e}, t)$ be the probability density of orientations, normalised by $\int W \mathrm{d}\Omega_e = 1$ at all times. Orientations are neither created nor destroyed: a moment can only change direction by moving continuously across the sphere. The same bookkeeping that produced the continuity equation (5.1.3) in Sec. 5.1 — now applied to a patch of the sphere instead of a box in space — gives

$$\frac{\partial W}{\partial t} + \nabla_s \cdot \boldsymbol{j} = 0, \tag{11.3.1}$$

where $\boldsymbol{j}$ is the probability flux (units: probability per unit solid angle per second, flowing tangentially).

11.4 The flux: drift

A particle whose instantaneous angular velocity is $\boldsymbol{\omega}$ carries its moment according to

$$\dot{\boldsymbol{e}} = \boldsymbol{\omega} \times \boldsymbol{e}, \tag{11.4.1}$$

which is Sec. 3.4, Step 1 applied to the unit vector $\boldsymbol{e}$ rigidly attached to the grain. (Here the "rigid dipole" assumption enters: the moment is frozen into the crystal, so rotating the grain rotates the moment. Its validity is checked numerically in Sec. 17.6.)

By Eq. (SM-1) the particle has no inertia on any relevant time scale, so its angular velocity follows instantaneously from the balance of torques:

$$0 = \underbrace{\boldsymbol{T}_{\mathrm{mag}}}_{m\,\boldsymbol{e}\times\boldsymbol{H}} \; \underbrace{-\zeta_r\big(\boldsymbol{\omega} - \boldsymbol{\Omega}\big)}_{\text{viscous, Eq. (6.5.8)}} \qquad \Longrightarrow \qquad \boldsymbol{\omega} = \boldsymbol{\Omega} + \frac{m}{\zeta_r}\big(\boldsymbol{e} \times \boldsymbol{H}\big). \tag{11.4.2}$$

Substituting into (11.4.1),

$$\dot{\boldsymbol{e}} = \boldsymbol{\Omega} \times \boldsymbol{e} + \frac{m}{\zeta_r}\big(\boldsymbol{e} \times \boldsymbol{H}\big) \times \boldsymbol{e}. \tag{11.4.3}$$

The second term simplifies beautifully. By the triple product (1.7.1) with $\boldsymbol{A} = \boldsymbol{e}$, $\boldsymbol{B} = \boldsymbol{e}$, $\boldsymbol{C} = \boldsymbol{H}$ — more directly, using $(\boldsymbol{A} \times \boldsymbol{B}) \times \boldsymbol{C} = -\boldsymbol{C} \times (\boldsymbol{A} \times \boldsymbol{B})$ —

$$\begin{aligned}
\big(\boldsymbol{e} \times \boldsymbol{H}\big) \times \boldsymbol{e} &= -\boldsymbol{e} \times \big(\boldsymbol{e} \times \boldsymbol{H}\big) && && (11.4.4)\\
&= -\Big[\boldsymbol{e}\,(\boldsymbol{e} \cdot \boldsymbol{H}) - \boldsymbol{H}\,(\boldsymbol{e} \cdot \boldsymbol{e})\Big] && \text{by (1.7.1)} && (11.4.5)\\
&= \boldsymbol{H} - \boldsymbol{e}\,(\boldsymbol{e} \cdot \boldsymbol{H}) && \text{using } \boldsymbol{e}\cdot\boldsymbol{e} = 1 && (11.4.6)\\
&= \nabla_s\big(\boldsymbol{e} \cdot \boldsymbol{H}\big) && \text{by (11.2.3).} && (11.4.7)
\end{aligned}$$

Since the orientational energy is $U(\boldsymbol{e}) = -m\,\boldsymbol{e} \cdot \boldsymbol{H}$, we have $m\nabla_s(\boldsymbol{e} \cdot \boldsymbol{H}) = -\nabla_s U$, and (11.4.3) becomes

$$\dot{\boldsymbol{e}} = \boldsymbol{\Omega} \times \boldsymbol{e} - \frac{1}{\zeta_r}\nabla_s U. \tag{11.4.8}$$

This is a completely transparent statement: the orientation is carried around by the flow, and in addition slides "downhill" on the energy landscape $U(\boldsymbol{e})$ at a rate proportional to the slope and inversely proportional to the friction. The corresponding probability flux is $W\dot{\boldsymbol{e}}$.

11.5 The flux: diffusion, and the Einstein relation

The random molecular kicks are not contained in (11.4.8). Their effect is to spread the distribution out, exactly as molecular kicks spread a dye in water. We model them by Fick's law: a flux down the gradient of concentration,

$$\boldsymbol{j}_{\text{diff}} = -D_r \nabla_s W, \tag{11.5.1}$$

where D_r is the *rotational diffusion coefficient* (units s^{-1}, since angles are dimensionless). The total flux is

$$\boldsymbol{j} = W\Big(\boldsymbol{\Omega} \times \boldsymbol{e}\Big) - \frac{W}{\zeta_r}\nabla_s U - D_r \nabla_s W. \tag{11.5.2}$$

Result 11.5.1 (Einstein relation)**.**

$$\boxed{D_r = \frac{kT}{\zeta_r}} \tag{11.5.3}$$

Derivation. The value of D_r is not free: drift and diffusion are two aspects of the same molecular bombardment, and they must be so related that the *equilibrium* distribution of Chapter 9 is reproduced. In equilibrium the fluid is at rest ($\boldsymbol{\Omega} = 0$) and nothing flows anywhere, so $\boldsymbol{j} = 0$ pointwise. Setting $\boldsymbol{\Omega} = 0$ and $W = W_0 = Ce^{-U/kT}$ in (11.5.2):

$$0 = -\frac{W_0}{\zeta_r}\nabla_s U - D_r \nabla_s W_0. \tag{11.5.4}$$

Compute $\nabla_s W_0$ by the chain rule:

$$\nabla_s W_0 = C\,\nabla_s e^{-U/kT} = C\,e^{-U/kT}\left(-\frac{\nabla_s U}{kT}\right) = -\frac{W_0}{kT}\nabla_s U. \tag{11.5.5}$$

Substituting (11.5.5) into (11.5.4),

$$0 = -\frac{W_0}{\zeta_r}\nabla_s U + \frac{D_r W_0}{kT}\nabla_s U = W_0\left(\frac{D_r}{kT} - \frac{1}{\zeta_r}\right)\nabla_s U. \tag{11.5.6}$$

Since this must hold for every orientation, and neither W_0 nor $\nabla_s U$ vanishes identically, the bracket must vanish: $D_r/kT = 1/\zeta_r$, i.e. (11.5.3). ■

Inserting the Stokes value $\zeta_r = 6\eta V$ of Eq. (6.6.1),

$$D_r = \frac{kT}{6\eta V} = \frac{kT}{8\pi\eta a^3}. \tag{11.5.7}$$

Thermal energy makes orientations wander; viscosity and size resist. Note that D_r contains η in the denominator: *the same* friction coefficient governs both the systematic response to a magnetic torque and the random wandering. That is the content of the fluctuation–dissipation idea, and it is why a purely hydrodynamic quantity (η) will appear in the final magnetic relaxation time.

11.6 The Smoluchowski equation

Substituting (11.5.2) into (11.3.1) and using (11.5.3) to eliminate ζ_r in favour of D_r:

$$\boxed{\frac{\partial W}{\partial t} = -\nabla_s \cdot \Big[W\big(\boldsymbol{\Omega} \times \boldsymbol{e}\big)\Big] + D_r\,\nabla_s \cdot \left[\nabla_s W + \frac{W}{kT}\nabla_s U\right]} \tag{11.6.1}$$

This is the "Fokker–Planck equation for rotary diffusion of colloidal particles" referred to (but not written out) by Bacri *et al.* just before their Eq. (B-5). Every subsequent result about magnetisation dynamics — the phenomenological equation (SM-7), the effective-field equation (B-6), the exact weak-field results — is a consequence of (11.6.1).

Consistency check. With $\boldsymbol{\Omega} = 0$, the stationary solution is obtained by setting the bracket to zero, which by (11.5.5) is solved by $W = Ce^{-U/kT}$: the Boltzmann distribution (9.5.1) is recovered. ✓

Chapter 12

Free Relaxation of the Magnetisation and the Value of τ_B

12.1 The question

Switch off the field and the flow. An initially aligned assembly of moments randomises. How fast? The answer defines the Brownian rotational relaxation time and is the content of Eq. (SM-5); we derive its *value* here, and present the equation formally in Chapter 17, in the papers' own order.

12.2 The equation of motion for $\langle \boldsymbol{e} \rangle$

With $\boldsymbol{\Omega} = 0$ and $U = 0$ the Smoluchowski equation (11.6.1) reduces to pure diffusion on the sphere,

$$\frac{\partial W}{\partial t} = D_r \nabla_s^2 W. \tag{12.2.1}$$

Multiply by e_i and integrate over all orientations:

$$\frac{\mathrm{d}\langle e_i \rangle}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t} \int e_i W \, \mathrm{d}\Omega_e = \int e_i \frac{\partial W}{\partial t} \, \mathrm{d}\Omega_e \qquad (e_i \text{ is a fixed label, not a function of } t) \tag{12.2.2}$$

$$= D_r \int e_i \, \nabla_s^2 W \, \mathrm{d}\Omega_e \tag{12.2.3}$$

$$= D_r \int W \, \nabla_s^2 e_i \, \mathrm{d}\Omega_e \qquad \text{by Green's identity (11.2.6).} \tag{12.2.4}$$

Everything now depends on one small calculation.

Result 12.2.1.

$$\nabla_s^2 e_i = -2 e_i. \tag{12.2.5}$$

Derivation. By isotropy it suffices to prove it for the component along the polar axis, $e_z = \cos\theta$; the result then holds for any component because we may always choose the polar axis along the direction in question. With $f = \cos\theta$ (no ϕ dependence, so the second term of the surface Laplacian vanishes):

$$\frac{\partial f}{\partial \theta} = -\sin\theta, \tag{12.2.6}$$

$$\sin\theta\frac{\partial f}{\partial\theta} = -\sin^2\theta, \tag{12.2.7}$$

$$\frac{\partial}{\partial\theta}\Big(\sin\theta\frac{\partial f}{\partial\theta}\Big) = \frac{\partial}{\partial\theta}\big(-\sin^2\theta\big) = -2\sin\theta\cos\theta, \tag{12.2.8}$$

$$\nabla_s^2 f = \frac{1}{\sin\theta}\big(-2\sin\theta\cos\theta\big) = -2\cos\theta = -2f. \quad \blacksquare \tag{12.2.9}$$

(In the language of spherical harmonics, $\boldsymbol{e}$ is a set of three $\ell = 1$ harmonics and the eigenvalue of ∇_s^2 is $-\ell(\ell+1) = -2$. We have derived the special case we need without invoking that theory.)

12.3 The relaxation law and τ_B

Substituting (12.2.5) into (12.2.4),

$$\frac{\mathrm{d}\langle e_i\rangle}{\mathrm{d}t} = D_r\int W(-2e_i)\,\mathrm{d}\Omega_e = -2D_r\langle e_i\rangle, \tag{12.3.1}$$

that is, in vector form and in terms of the magnetisation $\boldsymbol{M} = nm\langle\boldsymbol{e}\rangle$,

$$\frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t} = -2D_r\boldsymbol{M} = -\frac{\boldsymbol{M}}{\tau_B}, \qquad \boxed{\tau_B \equiv \frac{1}{2D_r}} \tag{12.3.2}$$

By Sec. 7.4 the solution is $\boldsymbol{M}(t) = \boldsymbol{M}(0)e^{-t/\tau_B}$: the magnetisation decays exponentially with time constant τ_B. Inserting the Einstein relation (11.5.3) and $\zeta_r = 6\eta V$:

$$\tau_B = \frac{1}{2D_r} = \frac{\zeta_r}{2kT} = \frac{6\eta V}{2kT}, \tag{12.3.3}$$

that is

$$\boxed{\tau_B = \frac{3\eta V}{kT}} \tag{12.3.4}$$

which is Eq. (SM-5) and the same quantity quoted in Bacri *et al.* beneath their Eq. (B-4). We shall return to it formally in Chapter 17.

Chapter 13

The Magnetisation Relaxation Equation

13.1 Context

Solving the Smoluchowski equation (11.6.1) in full generality is hard. Both papers therefore work with a *macroscopic* equation for $\boldsymbol{M}$ alone — Eq. (SM-7) in Shliomis & Morozov, Eq. (B-4) in Bacri *et al.*:

$$\frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t} = \boldsymbol{\Theta} \times \boldsymbol{M} - \frac{1}{\tau_B}\big(\boldsymbol{M} - \boldsymbol{M}_0\big). \tag{SM-7}$$

This chapter derives it, states precisely in which regime it is exact, and identifies exactly where it becomes an approximation — the very point on which Bacri *et al.* later base their criticism and their use of the effective-field method.

13.2 Derivation from the Smoluchowski equation

Take moments of (11.6.1), i.e. multiply by e_i and integrate. Using probability conservation in the form $\partial_t W = -\nabla_s \cdot \boldsymbol{j}$ and integrating by parts (no boundary terms, Sec. 11.2):

$$\frac{\mathrm{d}\langle e_i\rangle}{\mathrm{d}t} = -\int e_i\, \nabla_s \cdot \boldsymbol{j}\, \mathrm{d}\Omega_e = \int \boldsymbol{j} \cdot \nabla_s e_i\, \mathrm{d}\Omega_e, \tag{13.2.1}$$

with $\nabla_s e_i = \hat{\boldsymbol{x}}_i - \boldsymbol{e}\, e_i$ from (11.2.4). We treat the three contributions to $\boldsymbol{j}$ in (11.5.2) one at a time.

(i) The flow term.

$$\int W(\boldsymbol{\Omega}\times\boldsymbol{e})\cdot\big(\hat{\boldsymbol{x}}_i - \boldsymbol{e}e_i\big)\mathrm{d}\Omega_e = \int W\Big[(\boldsymbol{\Omega}\times\boldsymbol{e})_i - e_i\,(\boldsymbol{\Omega}\times\boldsymbol{e})\cdot\boldsymbol{e}\Big]\mathrm{d}\Omega_e \tag{13.2.2}$$

$$= \int W(\boldsymbol{\Omega}\times\boldsymbol{e})_i\,\mathrm{d}\Omega_e \qquad \text{by (1.5.7)} \tag{13.2.3}$$

$$= \big(\boldsymbol{\Omega}\times\langle\boldsymbol{e}\rangle\big)_i. \tag{13.2.4}$$

The averaging passed through the cross product because $\boldsymbol{\Omega}$ is a constant on the scale of the orientational average.

(ii) The diffusion term.

$$\int (-D_r \nabla_s W)\cdot \nabla_s e_i \,\mathrm{d}\Omega_e = +D_r \int W \nabla_s^2 e_i \,\mathrm{d}\Omega_e \qquad \text{Green's identity (11.2.6)} \tag{13.2.5}$$

$$= -2D_r \langle e_i \rangle \qquad \text{by (12.2.5).} \tag{13.2.6}$$

(iii) The magnetic term. Using $-\nabla_s U/\zeta_r = (m/\zeta_r)\nabla_s(\boldsymbol{e}\cdot\boldsymbol{H})$ and (11.2.3):

$$\int W \frac{m}{\zeta_r}\Big[\boldsymbol{H} - \boldsymbol{e}(\boldsymbol{e}\cdot\boldsymbol{H})\Big]\cdot\big(\hat{\boldsymbol{x}}_i - \boldsymbol{e}e_i\big)\mathrm{d}\Omega_e = \frac{m}{\zeta_r}\int W\Big[H_i - e_i(\boldsymbol{e}\cdot\boldsymbol{H})\Big]\mathrm{d}\Omega_e \tag{13.2.7}$$

$$= \frac{m}{\zeta_r}\Big[H_i - \big\langle e_i\, e_j\big\rangle H_j\Big]. \tag{13.2.8}$$

(The projector acts twice and, being idempotent, gives the same bracket; the cross-terms $\boldsymbol{e}\cdot\hat{\boldsymbol{x}}_i\, e_i(\boldsymbol{e}\cdot\boldsymbol{H})$ recombine to reproduce (13.2.7), as follows from $\boldsymbol{e}\cdot\boldsymbol{e}=1$.)

Here is the crucial structural point: the equation for the first moment $\langle e_i\rangle$ contains the *second* moment $\langle e_i e_j\rangle$. The hierarchy does not close. To obtain a closed macroscopic equation one must say something about $\langle e_i e_j\rangle$, and every such statement is a *closure approximation*.

The weak-field closure. If $\xi \ll 1$ the orientations are nearly isotropic, and for an isotropic distribution

$$\langle e_i e_j\rangle = \tfrac{1}{3}\delta_{ij} \tag{13.2.9}$$

(proved exactly as in Sec. 4.3: the average must be proportional to δ_{ij} by symmetry, and taking the trace gives $\langle e_i e_i\rangle = 1 = 3\times\tfrac{1}{3}$). Then (13.2.8) becomes

$$\frac{m}{\zeta_r}\left[H_i - \frac{H_i}{3}\right] = \frac{2m}{3\zeta_r}H_i = 2D_r\cdot\frac{mH_i}{3kT} = 2D_r\,\langle e_i\rangle_{\mathrm{eq}}, \tag{13.2.10}$$

where the Einstein relation (11.5.3) was used in the middle step and, in the last, the equilibrium weak-field average $\langle\boldsymbol{e}\rangle_{\mathrm{eq}} = L(\xi)\hat{\boldsymbol{h}} \simeq (\xi/3)\hat{\boldsymbol{h}} = m\boldsymbol{H}/3kT$ from (10.3.8).

Assembling. Adding (13.2.4), (13.2.6) and (13.2.10) in (13.2.1), and using $2D_r = 1/\tau_B$:

$$\frac{\mathrm{d}\langle\boldsymbol{e}\rangle}{\mathrm{d}t} = \boldsymbol{\Omega}\times\langle\boldsymbol{e}\rangle - \frac{1}{\tau_B}\Big(\langle\boldsymbol{e}\rangle - \langle\boldsymbol{e}\rangle_{\mathrm{eq}}\Big). \tag{13.2.11}$$

Multiplying by nm and using $\boldsymbol{M} = nm\langle\boldsymbol{e}\rangle$, $\boldsymbol{M}_0 = nm\langle\boldsymbol{e}\rangle_{\mathrm{eq}}$:

$$\boxed{\frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t} = \boldsymbol{\Omega}\times\boldsymbol{M} - \frac{1}{\tau_B}\big(\boldsymbol{M} - \boldsymbol{M}_0\big)} \tag{13.2.12}$$

■

13.3 From $\boldsymbol{\Omega}$ to $\boldsymbol{\Theta}$: why the paper writes $\boldsymbol{\Theta}\times\boldsymbol{M}$

In (13.2.12) the rotation carrying the moments around is the *fluid* vorticity $\boldsymbol{\Omega}$, because in deriving it we split the particle's angular velocity (11.4.2) into a flow part and a magnetic part and then treated the magnetic part separately (it produced the relaxation term). An equivalent

bookkeeping keeps the *whole* mean particle angular velocity in the convective term. Since by definition the mean particle spin is

$$\boldsymbol{\Theta} = \boldsymbol{\Omega} + \frac{1}{6\eta\varphi}\big(\boldsymbol{M}\times\boldsymbol{H}\big) \tag{SM-11}$$

— which is (11.4.2) averaged over orientations, multiplied by n, and written with $n\zeta_r = 6\eta\varphi$ from Eq. (6.6.3) — the two forms differ by

$$\boldsymbol{\Theta}\times\boldsymbol{M} = \boldsymbol{\Omega}\times\boldsymbol{M} + \frac{1}{6\eta\varphi}\big(\boldsymbol{M}\times\boldsymbol{H}\big)\times\boldsymbol{M} = \boldsymbol{\Omega}\times\boldsymbol{M} - \frac{1}{6\eta\varphi}\,\boldsymbol{M}\times\big(\boldsymbol{M}\times\boldsymbol{H}\big). \tag{13.3.1}$$

Thus writing $\boldsymbol{\Theta}\times\boldsymbol{M}$, as the paper does in (SM-7), is the same as writing $\boldsymbol{\Omega}\times\boldsymbol{M}$ plus the extra nonlinear term displayed in the paper's Eq. (SM-13),

$$\frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t} = \boldsymbol{\Omega}\times\boldsymbol{M} - \frac{1}{\tau_B}\big(\boldsymbol{M}-\boldsymbol{M}_0\big) - \frac{\tau_s}{I}\boldsymbol{M}\times\big(\boldsymbol{M}\times\boldsymbol{H}\big), \tag{SM-13}$$

since $\tau_s/I = 1/6\eta\varphi$ by Eq. (7.5.12) of Volume I. Both papers use both forms interchangeably; Bacri *et al.*'s Eq. (B-4) is exactly (SM-13) with $1/6\eta\varphi$ written out. We have now shown they are one equation.

13.4 Exactly where the approximation lies

Remark 13.4.1 (Accuracy of (SM-7)). The derivation above is *exact* to first order in ξ: the only step that required an assumption was the closure (13.2.9). For finite ξ the true second moment is not $\frac{1}{3}\delta_{ij}$ but is elongated along the field, and the relaxation term is no longer a single exponential with the single time τ_B. Consequently:

- In a *stationary* field of arbitrary strength, (SM-7) nevertheless reproduces the correct answer for the rotational viscosity remarkably well, which is why Bacri *et al.* write that "the phenomenological magnetization equation (4) is working well for arbitrary intensity of a stationary magnetic field".
- In an *alternating* field of finite amplitude and frequency, it does not. This is the failure that leads Bacri *et al.* to replace it with the effective-field equation (B-5)–(B-6) of Martsenyuk, Raikher and Shliomis, derived from the Fokker–Planck equation (11.6.1). That derivation is the business of a later volume; the point to register now is that the discrepancy between the theoretical Fig. 4 of Shliomis & Morozov and the experimental Fig. 2 of Bacri *et al.* traces back to this one closure step.

Part VI

Rotational Viscosity in a Stationary Field

Chapter 14

From Magnetic Torque to Viscosity

14.1 Context

We can now compute the magnetisation in any given flow. What we actually want is a *viscosity*. The bridge is the subject of this chapter, and it is the third of the three ingredients promised in Sec. 7.9. The logic has three links:

1. A magnetic torque acting on the suspended particles is transmitted to the liquid as a *body couple*.
2. A body couple is equivalent, in the momentum equation, to an antisymmetric stress, i.e. to the extra force density $\frac{1}{2}\nabla \times (\boldsymbol{M} \times \boldsymbol{H})$ that appears in (SM-12) and (B-3).
3. When that extra force is proportional to $\nabla^2 \boldsymbol{v}$, it simply renormalises the viscosity — and the coefficient is $\Delta\eta$.

14.2 Link 1: torque balance on the particles

The particles are inertialess (Eq. SM-1). Per unit volume of suspension, the magnetic torque density is $\boldsymbol{M} \times \boldsymbol{H}$ — this follows by summing $\boldsymbol{T} = \boldsymbol{m} \times \boldsymbol{H}$ of Eq. (4.4.2) over the n particles in unit volume and using $\boldsymbol{M} = nm\langle \boldsymbol{e} \rangle$. The viscous torque density is $-n\zeta_r(\boldsymbol{\Theta} - \boldsymbol{\Omega}) = -6\eta\varphi(\boldsymbol{\Theta} - \boldsymbol{\Omega})$ by Eq. (6.6.3). Their sum vanishes:

$$6\eta\varphi\big(\boldsymbol{\omega}_p - \boldsymbol{\Omega}\big) = \boldsymbol{M} \times \boldsymbol{H}, \tag{B-1}$$

which is Eq. (B-1) of Bacri *et al.* exactly ($\boldsymbol{\omega}_p$ is their notation for $\boldsymbol{\Theta}$). Equivalently

$$\overline{\boldsymbol{M} \times \boldsymbol{H}} = -6\eta\varphi\big(\boldsymbol{\Omega} - \boldsymbol{\Theta}\big), \tag{14.2.1}$$

the relation quoted by Shliomis & Morozov below their Eq. (SM-39). The overbar will matter only later, when the field oscillates and the torque must be averaged over the fast time; for a stationary field it may be ignored.

14.3 Link 2: a body couple is an antisymmetric stress

Result 14.3.1. *A body-couple density* $\boldsymbol{\Lambda}$ *(torque per unit volume, delivered as couples rather than as forces) is equivalent, in the linear-momentum equation, to the extra force density*

$$\boldsymbol{f}_{extra} = \tfrac{1}{2}\nabla \times \boldsymbol{\Lambda}. \tag{14.3.1}$$

Derivation. Write the stress as symmetric plus antisymmetric parts, $\sigma_{ij} = \sigma^s_{ij} + \sigma^a_{ij}$. Consider the angular-momentum balance of a small volume. The torque exerted on it by surface stresses is $\oint \varepsilon_{ijk} x_j \sigma_{kl} n_l \,\mathrm{d}S$; converting by the divergence theorem (2.3.4),

$$\oint \varepsilon_{ijk} x_j \sigma_{kl} n_l \,\mathrm{d}S = \int \partial_l\big(\varepsilon_{ijk} x_j \sigma_{kl}\big)\mathrm{d}V \tag{14.3.2}$$

$$= \int \varepsilon_{ijk}\Big(\delta_{jl}\sigma_{kl} + x_j \partial_l \sigma_{kl}\Big)\mathrm{d}V \tag{14.3.3}$$

$$= \int \varepsilon_{ijk}\sigma_{kj}\,\mathrm{d}V + \int \varepsilon_{ijk} x_j \big(\partial_l \sigma_{kl}\big)\mathrm{d}V. \tag{14.3.4}$$

The *second* integral is just $\int \boldsymbol{r} \times (\text{force density})\,\mathrm{d}V$, i.e. the torque already accounted for by the linear-momentum equation. The *first* is a genuine internal torque density, $\varepsilon_{ijk}\sigma_{kj}$, and it is nonzero only if σ has an antisymmetric part. Since the intrinsic spin angular momentum of the particles is utterly negligible (it is of order $I\Theta$, and I is the quantity shown in Volume I to be irrelevant), angular momentum balance requires this internal torque density to cancel the imposed couple:

$$\varepsilon_{ijk}\sigma^a_{kj} + \Lambda_i = 0. \tag{14.3.5}$$

Solve for σ^a with the ansatz $\sigma^a_{kj} = c\,\varepsilon_{kjl}\Lambda_l$ (the most general antisymmetric tensor built from $\boldsymbol{\Lambda}$). Then

$$\varepsilon_{ijk}\sigma^a_{kj} = c\,\varepsilon_{ijk}\varepsilon_{kjl}\Lambda_l = c\,\varepsilon_{kij}\varepsilon_{kjl}\Lambda_l \qquad \text{(cyclic: } \varepsilon_{ijk} = \varepsilon_{kij}\text{)} \tag{14.3.6}$$

$$= c\big(\delta_{ij}\delta_{jl} - \delta_{il}\delta_{jj}\big)\Lambda_l \qquad \text{by (1.6.1)} \tag{14.3.7}$$

$$= c\big(\delta_{il} - 3\delta_{il}\big)\Lambda_l = -2c\,\Lambda_i. \tag{14.3.8}$$

Comparing with (14.3.5), $-2c = -1$, so $c = \tfrac{1}{2}$ and

$$\sigma^a_{ij} = \tfrac{1}{2}\varepsilon_{ijl}\Lambda_l. \tag{14.3.9}$$

The force density it produces is its divergence:

$$\partial_j \sigma^a_{ij} = \tfrac{1}{2}\varepsilon_{ijl}\partial_j\Lambda_l = \tfrac{1}{2}\big(\nabla \times \boldsymbol{\Lambda}\big)_i. \quad \blacksquare \tag{14.3.10}$$

With $\boldsymbol{\Lambda} = \boldsymbol{M} \times \boldsymbol{H}$ this is exactly the last term of the paper's fluid-motion equation,

$$\rho\frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = -\nabla p + (\boldsymbol{M}\cdot\nabla)\boldsymbol{H} + \eta\Delta\boldsymbol{v} + \tfrac{1}{2}\nabla\times\big(\boldsymbol{M}\times\boldsymbol{H}\big), \tag{SM-12}$$

which is also Eq. (B-3). The term promised in remark 3 of Sec. 5.3 of Volume I is now derived. Note also that (SM-12) and (SM-6) are the same equation: substituting $\boldsymbol{\Theta}-\boldsymbol{\Omega} = (\tau_s/I)(\boldsymbol{M}\times\boldsymbol{H})$ from (SM-11) into the last term of (SM-6), $\frac{I}{2\tau_s}\nabla\times(\boldsymbol{\Theta}-\boldsymbol{\Omega})$, gives $\frac{1}{2}\nabla\times(\boldsymbol{M}\times\boldsymbol{H})$ identically.

14.4 Link 3: the extra force renormalises the viscosity

Assumption 14.4.1 (Proportional slip)**.** The slip between fluid and particle spins is parallel to the vorticity and proportional to it,

$$\boldsymbol{\Omega} - \boldsymbol{\Theta} = g\,\boldsymbol{\Omega}, \tag{14.4.1}$$

with g a dimensionless scalar, constant throughout the flow. This holds whenever the field is uniform and the orientational statistics are the same everywhere — true in all situations considered in these papers.

Substituting (14.2.1) and (14.4.1) into the last term of (SM-12):

$$\tfrac{1}{2}\nabla\times\big(\boldsymbol{M}\times\boldsymbol{H}\big)=\tfrac{1}{2}\nabla\times\Big[-6\eta\varphi\, g\,\boldsymbol{\Omega}\Big] \tag{14.4.2}$$

$$=-3\eta\varphi g\,\nabla\times\boldsymbol{\Omega} \tag{14.4.3}$$

$$=-3\eta\varphi g\,\nabla\times\big(\tfrac{1}{2}\nabla\times\boldsymbol{v}\big) \qquad \text{by (3.4.1)} \tag{14.4.4}$$

$$=-\tfrac{3}{2}\eta\varphi g\,\nabla\times\big(\nabla\times\boldsymbol{v}\big) \tag{14.4.5}$$

$$=-\tfrac{3}{2}\eta\varphi g\Big[\nabla(\nabla\cdot\boldsymbol{v})-\nabla^2\boldsymbol{v}\Big] \qquad \text{by (2.6.1)} \tag{14.4.6}$$

$$=+\tfrac{3}{2}\eta\varphi g\,\nabla^2\boldsymbol{v}, \qquad \text{incompressibility (5.1.4)} \tag{14.4.7}$$

so that the two viscous-looking terms of (SM-12) combine:

$$\eta\nabla^2\boldsymbol{v}+\tfrac{3}{2}\eta\varphi g\,\nabla^2\boldsymbol{v}=\Big(\eta+\tfrac{3}{2}\eta\varphi g\Big)\nabla^2\boldsymbol{v}. \tag{14.4.8}$$

The fluid therefore behaves exactly like an ordinary Newtonian fluid whose viscosity is $\eta+\Delta\eta$ with

$$\boxed{\Delta\eta=\frac{3}{2}\,\eta\varphi\, g=\frac{3}{2}\,\eta\varphi\,\frac{\Omega-\Theta}{\Omega}} \tag{14.4.9}$$

This is simultaneously Eq. (SM-34) of Shliomis & Morozov,

$$\Delta\eta=\frac{3}{2}\eta\varphi\frac{\Omega-\Theta}{\Omega}, \tag{SM-34}$$

and Eq. (B-2) of Bacri *et al.*,

$$\Delta\eta=\frac{3}{2}\eta\varphi\frac{\Omega-\omega_p}{\Omega}. \tag{B-2}$$

The identical grouping appears as Bacri *et al.*'s Eq. (B-10),

$$\eta\Delta\boldsymbol{v}-3\eta\varphi g\,\operatorname{curl}\boldsymbol{\Omega}=-2\eta\Big(1+\tfrac{3}{2}\varphi g\Big)\operatorname{curl}\boldsymbol{\Omega}, \tag{B-10}$$

which is (14.4.7) and (14.4.8) written with $\nabla^2\boldsymbol{v}=-\nabla\times(\nabla\times\boldsymbol{v})=-2\operatorname{curl}\boldsymbol{\Omega}$.

14.5 Physical interpretation of the master formula

Equation (14.4.9) is the conceptual heart of both papers, and it deserves to be read slowly.

- **If the particles spin with the fluid** ($\Theta=\Omega$, $g=0$) there is no extra viscosity. Nothing is being resisted; the suspension flows as if the grains were tiny pieces of the liquid.
- **If the particles are held still** by a strong field ($\Theta=0$, $g=1$) then $\Delta\eta=\frac{3}{2}\eta\varphi$: the maximum possible magnetoviscous increase. The liquid must flow *past* frozen obstacles, which dissipates extra energy.
- **If the particles spin faster than the fluid** ($\Theta>\Omega$, $g<0$) then $\Delta\eta<0$. The grains are now *driving* the flow rather than resisting it. This is the negative-viscosity effect, and (14.4.9) shows that it is not a paradox at all: it is what the same formula says when the sign of the slip is reversed. The energy comes from the alternating field, not from nowhere.

Equation (SM-2), to which we now turn, is nothing but (14.4.9) evaluated for a stationary field.

Chapter 15

Equation (SM-2): Rotational Viscosity in a Stationary Field

The equation to be derived is

$$\Delta\eta = \frac{3}{2}\,\eta\varphi\,\frac{\xi - \tanh\xi}{\xi + \tanh\xi}\,\sin^2\beta. \tag{SM-2}$$

15.1 Context

Shliomis & Morozov open by recalling the known behaviour of a ferrofluid in a *stationary* field: the viscosity increases. Equation (SM-2), derived by Shliomis in 1972 [their Ref. 2], is the quantitative statement of that fact. It is the reference against which the entire alternating-field analysis is set: the paper's central claim is that when the field oscillates fast enough, the same formalism produces a $\Delta\eta$ of the *opposite* sign. Bacri *et al.* quote the same physics in their Eq. (B-2) and use the measured saturation value $\eta_r(\infty, 0) \simeq 2.5$ as the calibration of their experiment.

15.2 Symbols

Symbol	Meaning	Type	Units
$\Delta\eta$	rotational (vortical) viscosity: the field-induced addition to the viscosity	scalar	P
η	viscosity of the suspension in zero field (the reference)	scalar	P
$\varphi = nV$	volume fraction of solid particles	dimensionless	—
$\xi = mH/kT$	Langevin parameter	dimensionless	—
$\tanh\xi$	hyperbolic tangent, $(e^{\xi} - e^{-\xi})/(e^{\xi} + e^{-\xi})$	function	—
β	angle between $\boldsymbol{H}$ and $\boldsymbol{\Omega}$	scalar	rad
$\boldsymbol{\Omega}, \boldsymbol{\Theta}$	angular velocities of fluid and particles	vectors	s^{-1}
$\boldsymbol{M}_0 = nmL(\xi)\hat{\boldsymbol{h}}$	equilibrium magnetisation	vector	G
$\boldsymbol{m}_1$	first-order (in $\Omega\tau_B$) correction to $\boldsymbol{M}$	vector	G
$\zeta \equiv \tau_B H M_0/6\eta\varphi$	dimensionless combination appearing in the solution	scalar	—

15.3 Assumptions

1. The field is *stationary*: $\boldsymbol{H}$ constant in time and uniform in space.
2. Rigid dipoles: the moment is frozen in the crystal (checked in Sec. 17.6).
3. The magnetisation obeys the relaxation equation (13.2.12), with its closure limitations as discussed in Sec. 13.4.
4. The flow is slow in the sense $\Omega\tau_B \ll 1$, so that the deviation of $\boldsymbol{M}$ from $\boldsymbol{M}_0$ may be treated to *first order* only. Bacri *et al.* verify this explicitly: $\tau_B \simeq 1.6$ ms and $\Omega < 60\ \mathrm{s}^{-1}$, hence $\Omega\tau_B \lesssim 0.1$.
5. Steady state: $\mathrm{d}\boldsymbol{M}/\mathrm{d}t = 0$. The field does not change and the flow is stationary, so after a few τ_B the magnetisation settles.
6. Interparticle magnetic and hydrodynamic interactions are neglected; the effect is computed per particle and multiplied by n. (Shliomis & Morozov note that $\Delta\eta$ and the Einstein correction $\frac{5}{2}\eta\varphi$ are of the same order, i.e. that we are in the dilute-suspension regime.)

15.4 Mathematical Background

Everything required has been built: the master formula (14.4.9), the relaxation equation (13.2.12), the definition (11.4.2) of $\boldsymbol{\Theta}$, the Langevin function (10.2.16) and the identity (10.3.10). One new piece of geometry is needed.

Choice of axes. Let the vorticity define the z axis and let the field lie in the xz plane at angle β from it:

$$\boldsymbol{\Omega} = \Omega\,\hat{\boldsymbol{z}}, \qquad \hat{\boldsymbol{h}} = \big(\sin\beta,\, 0,\, \cos\beta\big). \tag{15.4.1}$$

This is fully general: any two vectors span a plane, which we call xz. Two elementary cross products will be needed:

$$\hat{\boldsymbol{z}} \times \hat{\boldsymbol{h}} = \hat{\boldsymbol{z}} \times \big(\sin\beta\,\hat{\boldsymbol{x}} + \cos\beta\,\hat{\boldsymbol{z}}\big) = \sin\beta\,(\hat{\boldsymbol{z}} \times \hat{\boldsymbol{x}}) = \sin\beta\,\hat{\boldsymbol{y}}, \tag{15.4.2}$$

$$\hat{\boldsymbol{y}} \times \hat{\boldsymbol{h}} = \sin\beta\,(\hat{\boldsymbol{y}} \times \hat{\boldsymbol{x}}) + \cos\beta\,(\hat{\boldsymbol{y}} \times \hat{\boldsymbol{z}}) = -\sin\beta\,\hat{\boldsymbol{z}} + \cos\beta\,\hat{\boldsymbol{x}}, \tag{15.4.3}$$

using $\hat{\boldsymbol{z}} \times \hat{\boldsymbol{x}} = \hat{\boldsymbol{y}}$, $\hat{\boldsymbol{y}} \times \hat{\boldsymbol{x}} = -\hat{\boldsymbol{z}}$, $\hat{\boldsymbol{y}} \times \hat{\boldsymbol{z}} = \hat{\boldsymbol{x}}$.

15.5 Step-by-Step Derivation

Step 1: write the equation to be solved. Take (13.2.12) in the $\boldsymbol{\Theta}$ form (SM-7) and set $\mathrm{d}\boldsymbol{M}/\mathrm{d}t = 0$:

$$0 = \boldsymbol{\Theta} \times \boldsymbol{M} - \frac{1}{\tau_B}\big(\boldsymbol{M} - \boldsymbol{M}_0\big), \qquad \boldsymbol{\Theta} = \boldsymbol{\Omega} + \frac{1}{6\eta\varphi}\big(\boldsymbol{M} \times \boldsymbol{H}\big). \tag{15.5.1}$$

Step 2: split $\boldsymbol{M}$ into equilibrium plus a small correction.

$$\boldsymbol{M} = M_0\hat{\boldsymbol{h}} + \boldsymbol{m}_1, \qquad M_0 = nmL(\xi), \qquad |\boldsymbol{m}_1| \sim \Omega\tau_B M_0 \ll M_0. \tag{15.5.2}$$

The ordering $|\boldsymbol{m}_1|/M_0 \sim \Omega\tau_B$ follows because the only thing driving $\boldsymbol{M}$ away from $\boldsymbol{M}_0$ is the flow, whose strength relative to the restoring relaxation is exactly $\Omega\tau_B$.

Step 3: evaluate the magnetic torque to first order.

$$\boldsymbol{M} \times \boldsymbol{H} = \big(M_0\hat{\boldsymbol{h}} + \boldsymbol{m}_1\big) \times H\hat{\boldsymbol{h}} \tag{15.5.3}$$

$$= M_0 H\,\big(\hat{\boldsymbol{h}} \times \hat{\boldsymbol{h}}\big) + H\,\big(\boldsymbol{m}_1 \times \hat{\boldsymbol{h}}\big) \tag{15.5.4}$$

$$= H\,\boldsymbol{m}_1 \times \hat{\boldsymbol{h}}. \tag{15.5.5}$$

The equilibrium part contributes *nothing*: a magnetisation parallel to the field feels no torque. This single fact is why the whole effect is proportional to the small quantity $\boldsymbol{m}_1$ — and, later, why an oscillating field can reverse its sign.

Step 4: expand $\boldsymbol{\Theta} \times \boldsymbol{M}$ and drop second-order terms. Insert (15.5.5) into $\boldsymbol{\Theta}$ and then form $\boldsymbol{\Theta} \times \boldsymbol{M}$:

$$\boldsymbol{\Theta} \times \boldsymbol{M} = \left[\Omega\hat{\boldsymbol{z}} + \frac{H}{6\eta\varphi}\big(\boldsymbol{m}_1 \times \hat{\boldsymbol{h}}\big)\right] \times \Big[M_0\hat{\boldsymbol{h}} + \boldsymbol{m}_1\Big] \tag{15.5.6}$$

$$= \Omega M_0\big(\hat{\boldsymbol{z}} \times \hat{\boldsymbol{h}}\big) + \Omega\big(\hat{\boldsymbol{z}} \times \boldsymbol{m}_1\big) + \frac{HM_0}{6\eta\varphi}\big(\boldsymbol{m}_1 \times \hat{\boldsymbol{h}}\big) \times \hat{\boldsymbol{h}} + O(m_1^2). \tag{15.5.7}$$

Now compare the orders. The retained terms are $O(\Omega M_0)$; the term $\Omega(\hat{\boldsymbol{z}} \times \boldsymbol{m}_1)$ is $O(\Omega m_1) = O(\Omega^2\tau_B M_0)$, i.e. smaller by the factor $\Omega\tau_B \ll 1$ (Assumption 4). It is therefore dropped, along with $O(m_1^2)$. This is the *only* approximation made in the algebra, and it is precisely the linearisation in the vorticity.

Step 5: simplify the third term. Using the triple product (1.7.1):

$$\big(\boldsymbol{m}_1 \times \hat{\boldsymbol{h}}\big) \times \hat{\boldsymbol{h}} = -\hat{\boldsymbol{h}} \times \big(\boldsymbol{m}_1 \times \hat{\boldsymbol{h}}\big) \tag{15.5.8}$$

$$= -\Big[\boldsymbol{m}_1\big(\hat{\boldsymbol{h}} \cdot \hat{\boldsymbol{h}}\big) - \hat{\boldsymbol{h}}\big(\hat{\boldsymbol{h}} \cdot \boldsymbol{m}_1\big)\Big] \tag{15.5.9}$$

$$= -\boldsymbol{m}_1 + \hat{\boldsymbol{h}}\big(\hat{\boldsymbol{h}} \cdot \boldsymbol{m}_1\big) \tag{15.5.10}$$

$$= -\boldsymbol{m}_1^{\perp}, \tag{15.5.11}$$

where $\boldsymbol{m}_1^{\perp}$ denotes the part of $\boldsymbol{m}_1$ perpendicular to the field. The magnetic torque acts only on the perpendicular part — again the statement that a moment along the field feels nothing.

Step 6: assemble the linearised equation. Substituting (15.5.7), (15.5.11) and (15.4.2) into (15.5.1), and noting that $\boldsymbol{M} - \boldsymbol{M}_0 = \boldsymbol{m}_1$:

$$0 = \Omega M_0 \sin\beta\,\hat{\boldsymbol{y}} - \frac{HM_0}{6\eta\varphi}\boldsymbol{m}_1^{\perp} - \frac{\boldsymbol{m}_1}{\tau_B}. \tag{15.5.12}$$

Step 7: separate parallel and perpendicular components. Decompose $\boldsymbol{m}_1 = m_1^{\parallel}\hat{\boldsymbol{h}} + \boldsymbol{m}_1^{\perp}$. The source term $\Omega M_0 \sin\beta\,\hat{\boldsymbol{y}}$ is purely perpendicular to $\hat{\boldsymbol{h}}$, because $\hat{\boldsymbol{h}}$ has no y component by our choice of axes (15.4.1): $\hat{\boldsymbol{y}} \cdot \hat{\boldsymbol{h}} = 0$. Projecting (15.5.12) onto $\hat{\boldsymbol{h}}$:

$$0 = 0 - 0 - \frac{m_1^{\parallel}}{\tau_B} \qquad \Longrightarrow \qquad m_1^{\parallel} = 0. \tag{15.5.13}$$

(The field cannot change the *magnitude* of the magnetisation to this order; it can only tilt it. Physically obvious, and here proved.)

Step 8: solve for the perpendicular part. What remains of (15.5.12) is a scalar equation along $\hat{\boldsymbol{y}}$. Write $\boldsymbol{m}_1 = m_1\hat{\boldsymbol{y}}$:

$$0 = \Omega M_0 \sin\beta - \frac{HM_0}{6\eta\varphi} m_1 - \frac{m_1}{\tau_B} \tag{15.5.14}$$

$$\Omega M_0 \sin\beta = m_1 \left[\frac{1}{\tau_B} + \frac{HM_0}{6\eta\varphi}\right] \tag{15.5.15}$$

$$= \frac{m_1}{\tau_B}\left[1 + \frac{\tau_B H M_0}{6\eta\varphi}\right]. \tag{15.5.16}$$

Define the dimensionless combination

$$\zeta \equiv \frac{\tau_B H M_0}{6\eta\varphi}, \tag{15.5.17}$$

so that

$$\boxed{m_1 = \frac{\tau_B \Omega M_0 \sin\beta}{1+\zeta}} \tag{15.5.18}$$

Step 9: evaluate ζ. This is the step where all the physics of Chapters 9–12 enters at once. Insert $\tau_B = 3\eta V/kT$ from (12.3.4), $M_0 = nmL(\xi)$ from (10.4.1), and $\varphi = nV$:

$$\zeta = \frac{1}{6\eta\varphi}\cdot\frac{3\eta V}{kT}\cdot H \cdot nmL(\xi) \tag{15.5.19}$$

$$= \frac{3\eta V\, n\, m\, L(\xi) H}{6\eta\,(nV)\,kT} \tag{15.5.20}$$

$$= \frac{3\,m\,L(\xi)\,H}{6\,kT} \qquad (\eta,\ n,\ V \text{ all cancel}) \tag{15.5.21}$$

$$= \frac{1}{2}\cdot\frac{mH}{kT}\,L(\xi) \tag{15.5.22}$$

$$= \frac{\xi L(\xi)}{2}. \tag{15.5.23}$$

The cancellation of η, n and V is remarkable and important: ζ depends on the field *only* through ξ, not on the particle size, concentration, or viscosity separately.

Step 10: compute the slip $\boldsymbol{\Omega} - \boldsymbol{\Theta}$. From (15.5.1),

$$\boldsymbol{\Omega} - \boldsymbol{\Theta} = -\frac{1}{6\eta\varphi}\big(\boldsymbol{M}\times\boldsymbol{H}\big) = -\frac{H}{6\eta\varphi}\,\boldsymbol{m}_1\times\hat{\boldsymbol{h}} = -\frac{Hm_1}{6\eta\varphi}\big(\hat{\boldsymbol{y}}\times\hat{\boldsymbol{h}}\big), \tag{15.5.24}$$

and with (15.4.3),

$$\boldsymbol{\Omega} - \boldsymbol{\Theta} = \frac{Hm_1}{6\eta\varphi}\Big[\sin\beta\,\hat{\boldsymbol{z}} - \cos\beta\,\hat{\boldsymbol{x}}\Big]. \tag{15.5.25}$$

Step 11: project onto the vorticity. The master formula (14.4.9) requires the component of the slip along $\boldsymbol{\Omega}$, i.e. along $\hat{\boldsymbol{z}}$; the $\hat{\boldsymbol{x}}$ component represents a particle spin *transverse* to the vorticity, which produces no stress opposing the shear and therefore does not renormalise the viscosity (it is a symmetric-stress effect, absent from (14.4.7) because there $\nabla\times$ acts on the component along $\boldsymbol{\Omega}$). Hence

$$g = \frac{\big(\boldsymbol{\Omega} - \boldsymbol{\Theta}\big)\cdot\hat{\boldsymbol{z}}}{\Omega} = \frac{Hm_1\sin\beta}{6\eta\varphi\,\Omega}. \tag{15.5.26}$$

Step 12: insert m_1. Using (15.5.18),

$$g = \frac{H\sin\beta}{6\eta\varphi\,\Omega}\cdot\frac{\tau_B\Omega M_0\sin\beta}{1+\zeta} \tag{15.5.27}$$

$$= \frac{\tau_B H M_0}{6\eta\varphi}\cdot\frac{\sin^2\beta}{1+\zeta} \qquad (\Omega \text{ cancels}) \tag{15.5.28}$$

$$= \frac{\zeta}{1+\zeta}\sin^2\beta \qquad \text{by the definition (15.5.17).} \tag{15.5.29}$$

The cancellation of Ω is the mathematical expression of the fact that $\Delta\eta$ is a genuine *viscosity*: a material constant, independent of the shear rate. It is also why the fluid remains *Newtonian*, as Bacri *et al.* verify experimentally in their Fig. 1.

Step 13: express $\zeta/(1+\zeta)$ through ξ. With $\zeta = \xi L/2$ from (15.5.23),

$$\frac{\zeta}{1+\zeta} = \frac{\xi L/2}{1+\xi L/2} \tag{15.5.30}$$

$$= \frac{\xi L/2}{\dfrac{2+\xi L}{2}} \qquad \text{(common denominator in the denominator)} \tag{15.5.31}$$

$$= \frac{\xi L(\xi)}{2+\xi L(\xi)}. \tag{15.5.32}$$

Step 14: convert to the tanh **form of the paper.** Use the identity (10.3.10), $\xi L(\xi) = (\xi - \tanh\xi)/\tanh\xi$. Write $T \equiv \tanh\xi$ for brevity:

$$\frac{\xi L}{2+\xi L} = \frac{\dfrac{\xi - T}{T}}{2 + \dfrac{\xi - T}{T}} \tag{15.5.33}$$

$$= \frac{\dfrac{\xi - T}{T}}{\dfrac{2T + \xi - T}{T}} \qquad \text{(common denominator } T\text{)} \tag{15.5.34}$$

$$= \frac{\xi - T}{2T + \xi - T} \qquad (T \text{ cancels}) \tag{15.5.35}$$

$$= \frac{\xi - T}{\xi + T} = \frac{\xi - \tanh\xi}{\xi + \tanh\xi}. \tag{15.5.36}$$

Step 15: assemble. Substituting (15.5.36) into (15.5.29) and then into the master formula (14.4.9):

$$g = \frac{\xi - \tanh\xi}{\xi + \tanh\xi}\sin^2\beta, \qquad \Delta\eta = \frac{3}{2}\eta\varphi\, g, \tag{15.5.37}$$

that is

$$\boxed{\Delta\eta = \frac{3}{2}\,\eta\varphi\,\frac{\xi - \tanh\xi}{\xi + \tanh\xi}\,\sin^2\beta} \tag{SM-2}$$

which is Eq. (SM-2) exactly, with every factor accounted for. ■

15.6 Consistency Checks

Sign. For all $\xi > 0$ one has $\tanh\xi < \xi$ (since $\tanh\xi = \xi - \xi^3/3 + \cdots$), so the numerator is positive and $\Delta\eta > 0$: a stationary field always *increases* the viscosity. Bacri *et al.* give the thermodynamic reason in one sentence: a constant field can be produced by a permanent magnet, which does no work, so it cannot pump energy into the fluid and cannot reduce the dissipation.

Field parallel to the vorticity. $\beta = 0 \Rightarrow \Delta\eta = 0$. If the moments are aligned along the rotation axis, rotating about that axis does not move them, so the field exerts no torque and there is no extra friction. Both papers state exactly this beneath their Eq. (SM-2).

Zero field. $\xi \to 0$: numerator $\to 0$, so $\Delta\eta \to 0$. ✓

Saturation. $\xi \to \infty$: $\tanh\xi \to 1$, so the fraction $\to (\xi-1)/(\xi+1) \to 1$ and $\Delta\eta \to \frac{3}{2}\eta\varphi\sin^2\beta$. ✓

Order of magnitude against experiment. For Bacri *et al.*, $\varphi = 0.20$ and $\beta = \pi/2$, so the saturation prediction is $\Delta\eta/\eta = \frac{3}{2} \times 0.20 = 0.30$. The measured saturation is $\eta_r(\infty, 0) \simeq 2.5$ — eight times larger. This is *not* a failure of the derivation but a well-understood consequence of Assumption 6: at $\varphi = 0.2$ the suspension is concentrated, and interparticle interactions amplify the magnetoviscous effect far beyond the dilute prediction. The papers handle this by treating $\eta_r(\infty, 0)$ as an experimental normalisation — which is exactly what Bacri *et al.* do when they plot $g = \eta_r(\xi, \omega\tau_B)/\eta_r(\infty, 0)$ in their Figs. 3–5. The *shape* of the curve, which is what the theory predicts reliably, is what is compared.

15.7 Physical Interpretation

Read the derivation backwards and the mechanism is transparent.

The vorticity tries to rotate every grain at Ω. Rotating the grain drags its frozen moment away from the field direction, creating the transverse magnetisation m_1 of (15.5.18). A transverse magnetisation means a magnetic torque $\boldsymbol{M} \times \boldsymbol{H}$, which acts *against* the rotation. So the grain settles at a compromise spin $\Theta < \Omega$: it turns more slowly than the liquid around it. The liquid must then shear past it, and shearing past an obstacle costs energy. That extra dissipation, expressed as an effective viscosity, is $\Delta\eta$.

The structure of the answer, $\zeta/(1+\zeta)$, is the signature of a competition between two rates: $\zeta = \tau_B/\tau_{\mathrm{mag}}$ compares the Brownian randomisation time with the time $6\eta\varphi/(HM_0)$ in which the magnetic torque would restore alignment. When $\zeta \ll 1$ (weak field) Brownian motion wins, the moments barely notice the field, and $\Delta\eta \to 0$. When $\zeta \gg 1$ (strong field) the field wins completely, the moments are locked in space, the grains cannot rotate at all ($\Theta \to 0$), and $\Delta\eta$ saturates at $\frac{3}{2}\eta\varphi$.

What if a term were removed? Delete the magnetic feedback in $\boldsymbol{\Theta}$ — i.e. set $\zeta = 0$ in the denominator of (15.5.18) while keeping it in the numerator of (15.5.29) — and one gets $\Delta\eta \propto \xi L$ growing without bound in strong fields, which is unphysical. The denominator is the statement that the grains cannot be slowed by more than 100%: Θ cannot go below zero in a stationary field.

15.8 Connection to the Next Equation

Equation (SM-2) contains two limits of practical importance, which the paper displays immediately as (SM-3) and (SM-4): the weak-field quadratic growth and the strong-field saturation. We derive them next.

Chapter 16

Equations (SM-3) and (SM-4): The Two Limits

16.1 Equation (SM-3): weak fields

$$\Delta\eta = \tfrac{1}{4}\eta\varphi\xi^2, \qquad \xi \ll 1. \tag{SM-3}$$

Context. This is the regime of small fields, and it is the form in which the alternating-field theory of Sec. II of Shliomis & Morozov is worked out analytically. Every result of that section — (SM-28), (SM-29), (SM-40) — reduces to (SM-3) at zero frequency, and the authors use that fact repeatedly as a check.

Symbols. All defined; $\beta = \pi/2$ is understood, as the authors state just before Eq. (SM-3) ("below we will assume that $\beta = \pi/2$").

Assumptions. Those of Chapter 15, plus $\xi \ll 1$, i.e. $mH \ll kT$. For the Co-ferrite grains of Bacri *et al.*, $\xi = 5.25 \times 10^{-3} H(\mathrm{Oe})$, so $\xi \ll 1$ means $H \ll 190$ Oe.

Mathematical background. The series (10.3.8) for $L(\xi)$, and the elementary expansion $\tanh\xi = \xi - \xi^3/3 + 2\xi^5/15 - \cdots$, obtained from tanh = sinh / cosh by the same division performed in Sec. 10.3.

Step-by-step derivation. Work from the L-form (15.5.32), which is algebraically simpler than the tanh form:

$$\Delta\eta = \frac{3}{2}\eta\varphi\,\frac{\xi L(\xi)}{2 + \xi L(\xi)}, \qquad \sin^2\beta = 1. \tag{16.1.1}$$

For small ξ, $L(\xi) = \xi/3 + O(\xi^3)$, hence

$$\xi L(\xi) = \frac{\xi^2}{3} + O(\xi^4), \tag{16.1.2}$$

which is small compared with 2. Therefore

$$\frac{\xi L}{2 + \xi L} = \frac{\xi^2/3}{2 + \xi^2/3} \tag{16.1.3}$$

$$= \frac{\xi^2}{3} \cdot \frac{1}{2} \cdot \frac{1}{1+\xi^2/6} \qquad \text{(factor 2 out of the denominator)} \tag{16.1.4}$$

$$= \frac{\xi^2}{6}\Big(1 - \frac{\xi^2}{6} + \cdots\Big) \qquad \text{geometric expansion} \tag{16.1.5}$$

$$= \frac{\xi^2}{6} + O(\xi^4). \tag{16.1.6}$$

Hence

$$\Delta\eta = \frac{3}{2}\eta\varphi \cdot \frac{\xi^2}{6} = \frac{3\eta\varphi\xi^2}{12} = \frac{1}{4}\eta\varphi\xi^2. \tag{SM-3}$$

■

Cross-check from the tanh **form.** Using $\tanh\xi \approx \xi - \xi^3/3$:

$$\frac{\xi - \tanh\xi}{\xi + \tanh\xi} \approx \frac{\xi^3/3}{2\xi - \xi^3/3} \approx \frac{\xi^3/3}{2\xi} = \frac{\xi^2}{6}, \tag{16.1.7}$$

the same result. ✓ Note how the leading terms cancel in the numerator — this is why the effect is $O(\xi^2)$ and not $O(\xi)$.

Physical interpretation. $\Delta\eta \propto H^2$. Two factors of the field are needed: one to create the transverse magnetisation $m_1 \propto \chi H$, and one to convert it into a torque $\propto m_1 H$. A single power of H cannot appear, because reversing the field ($H \to -H$) cannot change a dissipation — $\Delta\eta$ must be an even function of $\boldsymbol{H}$. Equation (SM-3) is that symmetry statement made quantitative.

Connection. The opposite limit follows immediately.

16.2 Equation (SM-4): strong fields

$$\Delta\eta = \tfrac{3}{2}\eta\varphi, \qquad \xi \gg 1. \tag{SM-4}$$

Context. The saturation value. It is the reference against which all the paper's later results are normalised: in Sec. III the quantity plotted is $\Delta\eta(\xi, \omega\tau_B)/\Delta\eta(\infty, 0)$ with $\Delta\eta(\infty, 0) = \frac{3}{2}\eta\varphi$, and Bacri *et al.* do the same with their measured $\eta_r(\infty, 0) = 2.5$.

Assumptions. Those of Chapter 15, plus $\xi \gg 1$ ($H \gg 190$ Oe for the experimental grains) and $\beta = \pi/2$.

Step-by-step derivation. From (10.3.9), $L(\xi) \to 1 - 1/\xi$, so $\xi L(\xi) \to \xi - 1$, which is large. Then

$$\frac{\xi L}{2 + \xi L} = \frac{\xi - 1}{2 + \xi - 1} = \frac{\xi - 1}{\xi + 1} \tag{16.2.1}$$

$$= \frac{1 - 1/\xi}{1 + 1/\xi} \qquad \text{(divide top and bottom by } \xi) \tag{16.2.2}$$

$$\longrightarrow 1 \qquad \text{as } \xi \to \infty. \tag{16.2.3}$$

Equivalently, from the tanh form: $\tanh\xi \to 1$, so the fraction is $(\xi-1)/(\xi+1) \to 1$ — identically the same expression, as it must be. Therefore

$$\Delta\eta \to \frac{3}{2}\eta\varphi. \tag{SM-4}$$

■

Physical interpretation. In a saturating field the moments — and, being rigid dipoles, the grains themselves — are locked in space: $\Theta = 0$. The master formula (14.4.9) with $g = 1$ then gives $\frac{3}{2}\eta\varphi$ directly, without any calculation. Equation (SM-4) is thus the statement that a strongly magnetised ferrofluid behaves like a suspension of *non-rotating* spheres.

A revealing comparison. Shliomis & Morozov point out that $\frac{3}{2}\eta\varphi$ is of the same order as the Einstein correction $\frac{5}{2}\eta\varphi$ for the ordinary (translational) viscosity of a suspension of spheres. The two are physically distinct — one comes from the particles refusing to *rotate*, the other from their refusing to *deform* — but both are first order in φ, confirming that we are consistently working to leading order in a dilute suspension, and that no larger effect has been overlooked.

Connection. Both limits are controlled by the ratio mH/kT, so the next thing the paper must specify is the *time* scale on which the moments respond, since an alternating field will be compared against it. That is Eq. (SM-5).

Chapter 17

Equation (SM-5): The Brownian Relaxation Time

$$\tau_B = 3\eta V/kT. \tag{SM-5}$$

17.1 Context

Everything so far concerned a stationary field. The paper now turns to the alternating field $\boldsymbol{H} = (H_0 \cos \omega t, 0, 0)$, and the decisive question becomes: can the magnetisation follow the field? The answer depends on comparing the field period with the time the moments need to reorganise. Equation (SM-5) is that time. The dimensionless product $\omega\tau_B$ that it defines is, together with ξ, one of the two variables in terms of which every remaining result of both papers is expressed — the axes of Figs. 3, 4 and 5 of both papers are precisely ξ and $\omega\tau_B$.

17.2 Symbols

Symbol	Meaning	Type	Units
τ_B	Brownian time of rotational diffusion: the time for the magnetisation to relax by mechanical rotation of the grains	scalar, constant	s
η	viscosity of the carrier liquid	scalar	P
$V = \pi d^3/6$	volume of one particle	scalar	cm^3
kT	thermal energy	scalar	erg
$D_r = kT/6\eta V$	rotational diffusion coefficient	scalar	s^{-1}
τ_N	Néel time (internal reorientation of the moment)	scalar	s
K	magnetic anisotropy constant of the particle material	scalar	$\mathrm{erg\,cm}^{-3}$
d_s	crossover diameter at which $\tau_N \sim \tau_B$	scalar	cm

17.3 Assumptions

1. Rigid dipole: the moment is frozen in the crystal lattice, so reorienting the moment requires physically rotating the grain (justified in Sec. 17.6).
2. The grain is a sphere obeying Stokes rotational friction, $\zeta_r = 6\eta V$.
3. The liquid is a Newtonian continuum on the scale of the grain — an idealisation, since $d = 10$ nm is only ~ 30 water molecules across, but one that experiment vindicates.
4. Dilute suspension: each grain diffuses independently.

17.4 Derivation

Fully carried out in Chapter 12. Collecting the three ingredients:

$$\text{Einstein relation (11.5.3):} \qquad D_r = \frac{kT}{\zeta_r}, \tag{17.4.1}$$

$$\text{Stokes friction (6.6.1):} \qquad \zeta_r = 6\eta V, \tag{17.4.2}$$

$$\text{Relaxation of the first moment (12.3.2):} \qquad \tau_B = \frac{1}{2D_r}, \tag{17.4.3}$$

whence

$$\tau_B = \frac{1}{2}\cdot\frac{\zeta_r}{kT} = \frac{6\eta V}{2kT} = \frac{3\eta V}{kT}. \tag{SM-5}$$

Dimensional check: $[\eta V/kT] = (\mathrm{g\,cm^{-1}s^{-1}}\cdot\mathrm{cm^3})/\mathrm{erg} = (\mathrm{g\,cm^2 s^{-1}})/(\mathrm{g\,cm^2 s^{-2}}) = \mathrm{s}$. ✓

17.5 Numerical Values

An isolated 10-nm grain in water. $d = 10^{-6}$ cm gives $V = \pi d^3/6 = \pi\times10^{-18}/6 = 5.24\times10^{-19}$ cm^3; with $\eta = 10^{-2}$ P and $kT = 1.38\times10^{-16}\times300 = 4.14\times10^{-14}$ erg,

$$\tau_B = \frac{3\times10^{-2}\times5.24\times10^{-19}}{4.14\times10^{-14}} = \frac{1.57\times10^{-20}}{4.14\times10^{-14}} = 3.8\times10^{-7}\ \mathrm{s}. \tag{17.5.1}$$

Shliomis & Morozov state that for such particles τ_B "is usually not higher than 10^{-5} s", which our value comfortably satisfies. ✓

The experimental fluid of Bacri *et al.* They *measure* $\tau_B \simeq 1.6$ ms by transient birefringence — four thousand times larger. Both factors in ηV are responsible:

- The relevant η is not that of pure water but of the concentrated suspension in which each grain finds itself; the measured zero-field value is 77 cP = 0.77 P, i.e. 77 times larger.
- $\tau_B \propto V \propto d^3$, so the *largest* grains of a polydisperse sample dominate the measured relaxation, and each grain carries a surfactant/hydration layer that increases its hydrodynamic diameter above the magnetic core diameter.

Solving $\tau_B = 3\eta V/kT$ for the effective diameter with $\eta = 0.77$ P and $\tau_B = 1.6\times10^{-3}$ s:

$$V = \frac{\tau_B kT}{3\eta} = \frac{1.6\times10^{-3}\times4.14\times10^{-14}}{2.31} = 2.9\times10^{-17}\ \mathrm{cm^3} \implies d = \left(\frac{6V}{\pi}\right)^{1/3} = 3.8\times10^{-6}\ \mathrm{cm}, \tag{17.5.2}$$

i.e. an effective hydrodynamic diameter of about 38 nm for a magnetic core of 10 nm. That is a large but entirely plausible figure for a 20% ferrofluid, in which small clusters and the surfactant shell both contribute. The important point for the theory is that τ_B enters only through the measurable combination $\omega\tau_B$, and Bacri *et al.* accordingly use their *measured* value, converting frequency to the dimensionless variable by

$$\omega\tau_B = 2\pi f \tau_B = 10^{-2} f \ \text{(Hz)}, \tag{17.5.3}$$

which is the calibration printed in their text. Check: $2\pi \times 1.6 \times 10^{-3} = 1.005 \times 10^{-2}$. ✓

Where the interesting physics lies. $\omega\tau_B = 1$ corresponds, by (17.5.3), to $f = 100$ Hz. The experiments run from 0 to 1480 Hz, i.e. from $\omega\tau_B = 0$ to $\omega\tau_B \approx 15$: exactly the range in which the sign change predicted in the following chapters occurs. This is not a coincidence — the ferrofluid was chosen so that the effect would be accessible at audio frequencies.

17.6 Physical Interpretation, and the Néel Alternative

τ_B is the time in which thermal collisions rotate a grain through an angle of order one radian. It grows with η (a stiffer liquid resists), grows with V (a bigger grain has more surface to drag), and falls with T (more vigorous kicks).

There is, however, a second way for the magnetisation to relax, which does *not* require the grain to move: the moment can rotate *inside* the crystal, jumping over the magnetic anisotropy barrier. This is the Néel mechanism, with time

$$\tau_N = \tau_0 \exp\!\left(\frac{KV}{kT}\right), \qquad \tau_0 \sim 10^{-9}\ \text{s}, \tag{17.6.1}$$

where K is the anisotropy energy per unit volume, so that KV is the height of the barrier. If $\tau_N \ll \tau_B$ the moment relaxes without the grain turning, no mechanical torque is transmitted to the liquid, and the entire magnetoviscous effect disappears. Shliomis & Morozov make this point explicitly: only particles with "frozen-in" moments contribute, and the crossover diameter d_s at which the two rates are comparable falls near the sizes commonly used in ferrofluids.

Numerical check for the experimental fluid. Bacri *et al.* chose Co-ferrite precisely to be safe on this point, quoting $K = 2 \times 10^6$ erg cm^{-3}. With $V = 5.24 \times 10^{-19}$ cm^3 for a 10 nm grain:

$$KV = 2 \times 10^6 \times 5.24 \times 10^{-19} = 1.05 \times 10^{-12}\ \text{erg}, \tag{17.6.2}$$

$$\frac{KV}{kT} = \frac{1.05 \times 10^{-12}}{4.14 \times 10^{-14}} = 25.3, \tag{17.6.3}$$

$$\tau_N = 10^{-9} \times e^{25.3} \approx 10^{-9} \times 9.8 \times 10^{10} \approx 98\ \text{s}. \tag{17.6.4}$$

The Néel time is about 100 seconds — eight orders of magnitude longer than τ_B, and eleven orders longer than the field period. The moment is, for all experimental purposes, welded to the crystal. This is exactly the claim Bacri *et al.* make when they write that the large anisotropy constant "induces a rigid dipole behaviour of the particles". Assumption 1 of this chapter, and Assumption 2 of Chapter 15, are thereby justified quantitatively.

17.7 Connection to the Next Equation

With τ_B defined, the two dimensionless variables of the entire problem are in place:

$$\xi = \frac{mH}{kT} \quad \text{(how strong the field is)}, \qquad \omega\tau_B \quad \text{(how fast it alternates)}. \tag{17.7.1}$$

The paper now writes down the complete system of ferrohydrodynamic equations that must be solved in the $(\xi, \omega\tau_B)$ plane — Eqs. (SM-6) to (SM-10), which are the fluid-motion equation, the magnetisation equation, the equation of rotational motion of the particles, the Maxwell constraints, and the Langevin formula. Every one of these five has in fact already been derived in Volumes I and II; Volume III will assemble them formally in the paper's own order, then proceed to the elimination (SM-11)–(SM-13), the alternating field (SM-14)–(SM-15), and the circular-polarisation solution beginning at (SM-16).

Chapter 18

Status and Road Map

18.1 What has been established in Volume II

Result	Statement	Where derived
Boltzmann law	$p \propto e^{-E/kT}$, from the two postulates	Sec. 9.3
Orientational equilibrium	$W_0 = \dfrac{\xi}{4\pi \sinh \xi} e^{\xi \cos\theta}$	Sec. 9.5
Langevin function	$\langle \cos\theta \rangle = L(\xi) = \coth \xi - 1/\xi$	Sec. 10.2
Series and identity	$L \simeq \xi/3 - \xi^3/45$; $L \to 1 - 1/\xi$; $\xi L = \dfrac{\xi - \tanh \xi}{\tanh \xi}$	Sec. 10.3
Eq. (SM-10)	$\boldsymbol{M}_0 = nmL(\xi)\hat{\boldsymbol{h}}$ (formally restated in Vol. III)	Sec. 10.4
Eq. (SM-15)	$\chi = nm^2/3kT$ (formally restated in Vol. III)	Sec. 10.4
Surface calculus	∇_s, $\nabla_{s'}$, ∇_s^2, Green's identity	Sec. 11.2
Smoluchowski equation	$\partial_t W = -\nabla_s \cdot [W(\boldsymbol{\Omega} \times \boldsymbol{e})] + D_r \nabla_s \cdot [\nabla_s W + W \nabla_s U/kT]$	Sec. 11.6
Einstein relation	$D_r = kT/\zeta_r = kT/6\eta V$	Sec. 11.5
Eigenvalue	$\nabla_s^2 \boldsymbol{e} = -2\boldsymbol{e}$	Sec. 12.2
Eq. (SM-7)/(B-4)	$\mathrm{d}\boldsymbol{M}/\mathrm{d}t = \boldsymbol{\Theta} \times \boldsymbol{M} - (\boldsymbol{M} - \boldsymbol{M}_0)/\tau_B$, exact for $\xi \ll 1$	Ch. 13
Eq. (SM-11)	$\boldsymbol{\Theta} = \boldsymbol{\Omega} + (\boldsymbol{M} \times \boldsymbol{H})/6\eta\varphi$	Sec. 13.3
Eq. (SM-13)	equivalent nonlinear form of (SM-7)	Sec. 13.3
Couple → stress	body couple $\boldsymbol{\Lambda} \Leftrightarrow$ force $\frac{1}{2}\nabla \times \boldsymbol{\Lambda}$	Sec. 14.3
Eq. (SM-12)/(B-3)	the ferrohydrodynamic momentum equation	Sec. 14.3
Eq. (B-1)	$6\eta\varphi(\boldsymbol{\omega}_p - \boldsymbol{\Omega}) = \boldsymbol{M} \times \boldsymbol{H}$	Sec. 14.2
Eq. (SM-34)/(B-2)/(B-10)	$\Delta\eta = \frac{3}{2}\eta\varphi(\Omega - \Theta)/\Omega$	Sec. 14.4
Eq. (SM-2)	$\Delta\eta = \frac{3}{2}\eta\varphi \dfrac{\xi - \tanh \xi}{\xi + \tanh \xi} \sin^2 \beta$	Ch. 15
Eq. (SM-3)	$\Delta\eta = \frac{1}{4}\eta\varphi\xi^2$, $\xi \ll 1$	Sec. 16.1
Eq. (SM-4)	$\Delta\eta = \frac{3}{2}\eta\varphi$, $\xi \gg 1$	Sec. 16.2
Eq. (SM-5)	$\tau_B = 3\eta V/kT$; $\omega\tau_B = 10^{-2} f(\mathrm{Hz})$ experimentally	Ch. 17

Note that several equations of the papers (SM-7, SM-10, SM-11, SM-12, SM-13, SM-15, SM-34, B-1, B-2, B-3, B-4, B-10) have been *derived* in Volume II because they were needed as tools. They will be *presented* in the eight-part format, in the papers' own order, when we reach them — with cross-references back to the derivations given here, and with no repetition of the algebra.

18.2 Next

Volume III will cover, in order:

- Eq. (SM-6): the fluid-motion equation with the spin-coupling term, in its original form;
- Eqs. (SM-7), (SM-8): the magnetisation equation and the equation of rotational motion of the particles;
- Eq. (SM-9): the Maxwell constraints $\nabla \cdot \boldsymbol{v} = 0$, $\nabla \times \boldsymbol{H} = 0$, $\nabla \cdot (\boldsymbol{H} + 4\pi \boldsymbol{M}) = 0$, each derived from the Maxwell equations for a nonconducting medium;
- Eq. (SM-10): the Langevin magnetisation, formally;
- Eqs. (SM-11)–(SM-13): elimination of the particle spin, including the explicit estimate $I\,\mathrm{d}\boldsymbol{\Theta}/\mathrm{d}t \ll I\boldsymbol{\Theta}/\tau_s$;
- Eqs. (SM-14)–(SM-15): the alternating field and the linear magnetisation law.

After that, Volume IV takes up the circular-polarisation decomposition (SM-16)–(SM-24), which is where the sign reversal of $\Delta\eta$ finally emerges.

End of Volume II. Volume III begins with Eq. (SM-6).

THE NEGATIVE-VISCOSITY EFFECT IN FERROFLUIDS

A Complete First-Principles Reconstruction of Every Equation

VOLUME III

The Ferrohydrodynamic System Assembled,
Elimination of the Particle Spin,
and the Alternating Field: Equations (SM-6)–(SM-15)

Where Volume II ended. Volume II built the statistical mechanics of the orientations (Boltzmann law, Langevin function $L(\xi) = \coth\xi - 1/\xi$, equilibrium magnetisation $\boldsymbol{M}_0 = nmL(\xi)\hat{\boldsymbol{h}}$), rotational Brownian motion (the Smoluchowski equation, the Einstein relation $D_r = kT/\zeta_r$, and $\tau_B = 3\eta V/kT$), the magnetisation relaxation equation, and the bridge from magnetic torque to viscosity, $\Delta\eta = \frac{3}{2}\eta\varphi(\Omega - \Theta)/\Omega$. With those it derived Eqs. (SM-2), (SM-3), (SM-4) and (SM-5).

Volume III now presents, in the papers' own order and in the eight-part format, the complete system of ferrohydrodynamic equations (SM-6)–(SM-10), the elimination of the particle spin (SM-11)–(SM-13), and the specification of the alternating field (SM-14)–(SM-15). Several of these were derived in Volume II as tools; where that is so, the derivation is *cited*, not repeated, and the chapter concentrates on the context, the assumptions, the physical reading and the numerical checks. Two genuinely new derivations appear here: the Kelvin force $(\boldsymbol{M}\cdot\nabla)\boldsymbol{H}$ in Chapter 19, and the magnetostatic constraints (SM-9) in Chapter 22. Chapter numbers resume at 19, page numbers at 78.

Part VII

The Ferrohydrodynamic System, Assembled

Chapter 19

Equation (SM-6): The Fluid-Motion Equation

The equation is

$$\rho \frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = -\nabla p + (\boldsymbol{M} \cdot \nabla)\boldsymbol{H} + \eta \Delta \boldsymbol{v} + \frac{I}{2\tau_s} \nabla \times \big(\boldsymbol{\Theta} - \boldsymbol{\Omega}\big). \tag{SM-6}$$

19.1 Context

Shliomis & Morozov open Sec. II by stating that "the system of ferrohydrodynamic equations accounting for the connection between the magnetic and rotational mechanical degrees of freedom consists of the fluid motion equation, the magnetization equation, and the equation of rotational motion of the particles". Equation (SM-6) is the first of these three. It is the ordinary Navier–Stokes equation of Sec. 5.4 with two additional terms: one describing the force a non-uniform field exerts on a magnetised medium, and one describing the momentum transferred to the liquid when the particles spin at a rate different from the local vorticity. The second is the term that carries the entire magnetoviscous effect; the first plays no role in the papers' calculations (the applied field is uniform) but must be present for the equation to be correct in general, and we derive it.

19.2 Symbols

Symbol	Meaning	Type	Units
ρ	mass density of the ferrofluid as a whole (carrier plus grains)	scalar	$\mathrm{g\,cm^{-3}}$
$\boldsymbol{v}$	velocity field of the ferrofluid, treated as a single continuum	vector field	$\mathrm{cm\,s^{-1}}$
$\mathrm{d}/\mathrm{d}t$	material derivative $\partial_t + (\boldsymbol{v}\cdot\nabla)$, Eq. (3.2.3)	operator	$\mathrm{s^{-1}}$
p	pressure	scalar field	$\mathrm{dyn\,cm^{-2}}$
$\boldsymbol{M}$	magnetisation	vector field	G
$\boldsymbol{H}$	magnetic field acting inside the fluid	vector field	Oe
η	viscosity of the suspension in zero field	scalar	P
Δ	the authors' notation for the Laplacian ∇^2	operator	$\mathrm{cm^{-2}}$
I	moment of inertia of the particles per unit volume, $I = nI_p$	scalar	$\mathrm{g\,cm^{-1}}$
τ_s	inertial spin relaxation time, Eq. (SM-1)	scalar	s
$\boldsymbol{\Theta}$	mean angular velocity of the particles	vector field	$\mathrm{s^{-1}}$
$\boldsymbol{\Omega}$	local angular velocity of the fluid, $\frac{1}{2}\nabla\times\boldsymbol{v}$	vector field	$\mathrm{s^{-1}}$

Remark 19.2.1 (One velocity field for two phases). Writing a single $\boldsymbol{v}$ for a suspension of solid grains in a liquid is an approximation whose justification is the enormous separation of scales: the grains are 10 nm, the flow varies over 1 mm, a ratio of 10^5. On any scale the flow can resolve, grain and liquid move together. What the grains *do* retain as an independent degree of freedom is their *spin* $\boldsymbol{\Theta}$, because a torque can make them rotate differently from the fluid without making them translate differently. This is the entire physical content of ferrohydrodynamics, and it is why $\boldsymbol{\Theta}$ appears in (SM-6) while no second velocity does.

19.3 Assumptions

1. Incompressible ($\nabla\cdot\boldsymbol{v}=0$) and Newtonian carrier, so that the ordinary viscous term is $\eta\nabla^2\boldsymbol{v}$ (Sec. 5.4).

2. The suspension is treated as a single continuum with one velocity field, as in the remark above.

3. The magnetic grains are rigid dipoles, so a magnetic torque on the moment is a mechanical torque on the grain (Sec. 17.6).

4. Gravity and buoyancy are omitted: they are balanced in the horizontal capillary of the experiment and drop out of the equation for the deviation from hydrostatic equilibrium.

5. The intrinsic spin angular momentum of the grains is negligible compared with their orbital angular momentum, as established by the smallness of τ_s (Eq. SM-1).

19.4 Mathematical Background: the Kelvin Force

Result 19.4.1 (Gradient of a scalar product, constant first factor)**.** *For a* constant *vector* $\boldsymbol{A}$ *and a vector field* $\boldsymbol{B}(\boldsymbol{r})$*,*

$$\nabla\big(\boldsymbol{A}\cdot\boldsymbol{B}\big) = (\boldsymbol{A}\cdot\nabla)\boldsymbol{B} + \boldsymbol{A}\times(\nabla\times\boldsymbol{B}). \tag{19.4.1}$$

Derivation. Work in components. The left-hand side is

$$\Big[\nabla(\boldsymbol{A}\cdot\boldsymbol{B})\Big]_i = \partial_i\big(A_jB_j\big) = A_j\,\partial_iB_j, \tag{19.4.2}$$

the derivative passing over A_j because $\boldsymbol{A}$ is constant. For the right-hand side, the first term is immediately

$$\big[(\boldsymbol{A}\cdot\nabla)\boldsymbol{B}\big]_i = A_j\partial_jB_i. \tag{19.4.3}$$

The second term is evaluated with the ε–δ identity (1.6.1):

$$\begin{aligned}\big[\boldsymbol{A}\times(\nabla\times\boldsymbol{B})\big]_i &= \varepsilon_{ijk}A_j\big(\nabla\times\boldsymbol{B}\big)_k &(19.4.4)\\ &= \varepsilon_{ijk}A_j\,\varepsilon_{klm}\partial_lB_m &(19.4.5)\\ &= \varepsilon_{ijk}\varepsilon_{klm}\,A_j\partial_lB_m &(19.4.6)\\ &= \big(\delta_{il}\delta_{jm}-\delta_{im}\delta_{jl}\big)A_j\partial_lB_m &(19.4.7)\\ &= A_j\partial_iB_j - A_j\partial_jB_i. &(19.4.8)\end{aligned}$$

Adding (19.4.3) and (19.4.8), the terms $A_j\partial_jB_i$ cancel and what remains is $A_j\partial_iB_j$, which is (19.4.2). ■

Result 19.4.2 (Kelvin force density)**.**

$$\boldsymbol{f}_{mag} = (\boldsymbol{M}\cdot\nabla)\boldsymbol{H}. \tag{19.4.9}$$

Derivation, in four steps. *Step 1.* A single rigid dipole of moment $\boldsymbol{m}$ at position $\boldsymbol{r}$ has the potential energy $U(\boldsymbol{r}) = -\boldsymbol{m}\cdot\boldsymbol{H}(\boldsymbol{r})$ of Eq. (4.4.2). Its *orientation* is fixed at the instant considered, so the only $\boldsymbol{r}$ dependence is through the field.

Step 2. The force on a body is minus the gradient of its potential energy (this is the definition of a conservative force; it follows from requiring that the work done, $\boldsymbol{F}\cdot\mathrm{d}\boldsymbol{r}$, equal $-\mathrm{d}U$ for every displacement):

$$\boldsymbol{F} = -\nabla U = \nabla\big(\boldsymbol{m}\cdot\boldsymbol{H}\big). \tag{19.4.10}$$

Step 3. Apply (19.4.1) with $\boldsymbol{A}=\boldsymbol{m}$ (constant) and $\boldsymbol{B}=\boldsymbol{H}$:

$$\boldsymbol{F} = (\boldsymbol{m}\cdot\nabla)\boldsymbol{H} + \boldsymbol{m}\times\big(\nabla\times\boldsymbol{H}\big). \tag{19.4.11}$$

The second term vanishes because $\nabla\times\boldsymbol{H}=0$ in the non-conducting, quasi-static situation — this is the second relation of Eq. (SM-9), derived independently in Chapter 22. Hence $\boldsymbol{F}=(\boldsymbol{m}\cdot\nabla)\boldsymbol{H}$.

Step 4. Sum over the n grains in unit volume. The field and its derivatives are the same for all of them (they vary over millimetres, the grains are nanometres apart), so the sum acts only on the moments:

$$\boldsymbol{f}_{\mathrm{mag}} = n\big\langle(\boldsymbol{m}\cdot\nabla)\big\rangle\boldsymbol{H} = \Big(\big(n\langle\boldsymbol{m}\rangle\big)\cdot\nabla\Big)\boldsymbol{H} = (\boldsymbol{M}\cdot\nabla)\boldsymbol{H}, \tag{19.4.12}$$

using $\boldsymbol{M}=n\langle\boldsymbol{m}\rangle$ from Eq. (4.5.1). ■

Dimensional check. $[\boldsymbol{M}][\boldsymbol{H}]/\mathrm{cm} = \mathrm{G{\cdot}Oe/cm}$. In Gaussian units G·Oe is an energy density, $\mathrm{erg\,cm^{-3}}$, so the quotient is $\mathrm{erg\,cm^{-4}} = \mathrm{dyn\,cm^{-3}}$: a force per unit volume. ✓

19.5 Step-by-Step Derivation of (SM-6)

Step 1: start from Cauchy's equation. From Eq. (5.2.4),

$$\rho \frac{\mathrm{d}v_i}{\mathrm{d}t} = f_i + \partial_j \sigma_{ij}, \tag{19.5.1}$$

where $\boldsymbol{f}$ collects all body forces and σ_{ij} is the stress.

Step 2: identify the body force. The only body force is magnetic, so $\boldsymbol{f} = (\boldsymbol{M}\cdot\nabla)\boldsymbol{H}$ by (19.4.9).

Step 3: split the stress. As established in Sec. 14.3, the stress of a ferrofluid has a symmetric Newtonian part and an antisymmetric part produced by the body couple exerted on the liquid by the spinning grains:

$$\sigma_{ij} = \underbrace{-p\delta_{ij} + 2\eta e_{ij}}_{\text{Newtonian}} + \underbrace{\tfrac{1}{2}\varepsilon_{ijl}\Lambda_l}_{\text{antisymmetric}}, \qquad \boldsymbol{\Lambda} = \text{body-couple density.} \tag{19.5.2}$$

Taking the divergence, the Newtonian part gives $-\nabla p + \eta\nabla^2\boldsymbol{v}$ exactly as in Sec. 5.4, and the antisymmetric part gives $\frac{1}{2}\nabla\times\boldsymbol{\Lambda}$ by Eq. (14.3.1).

Step 4: identify the couple density. The couple delivered to the liquid is the viscous torque the grains exert on it, which is minus the torque the liquid exerts on the grains. From Eq. (6.5.8) summed over unit volume, the liquid exerts $-n\zeta_r(\boldsymbol{\Theta}-\boldsymbol{\Omega})$ on the grains, so the grains exert

$$\boldsymbol{\Lambda} = +n\zeta_r\big(\boldsymbol{\Theta}-\boldsymbol{\Omega}\big) = \frac{I}{\tau_s}\big(\boldsymbol{\Theta}-\boldsymbol{\Omega}\big) \tag{19.5.3}$$

on the liquid, where the identity $n\zeta_r = I/\tau_s = 6\eta\varphi$ of Eq. (7.5.12) was used. Note the direction: if the particles spin *faster* than the fluid ($\boldsymbol{\Theta} > \boldsymbol{\Omega}$) they hand angular momentum *to* the fluid. That single sign is the germ of the negative-viscosity effect.

Step 5: assemble. Substituting Steps 2–4 into (19.5.1):

$$\rho\frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = -\nabla p + (\boldsymbol{M}\cdot\nabla)\boldsymbol{H} + \eta\nabla^2\boldsymbol{v} + \frac{I}{2\tau_s}\nabla\times\big(\boldsymbol{\Theta}-\boldsymbol{\Omega}\big), \tag{SM-6}$$

which is Eq. (SM-6) with every term derived. ■

19.6 Physical Interpretation

Read term by term, from left to right:

- $\rho\,\mathrm{d}\boldsymbol{v}/\mathrm{d}t$: the inertia of a fluid parcel. Under the conditions of these experiments it is small (Sec. 5.5) but it is kept because (SM-6) is meant to be a general equation.

- $-\nabla p$: the ordinary pressure gradient. In the experiment it is what drives the flow, being supplied by the difference h in ferrofluid level between the ends of the capillary.
- $(\boldsymbol{M}\cdot\nabla)\boldsymbol{H}$: the Kelvin force. A magnetised medium is pulled towards regions of stronger field. It is what makes ferrofluids climb magnets and form the famous spikes; it plays *no* role in these two papers because the solenoid field is uniform over the capillary, so $\nabla\boldsymbol{H}\approx 0$.
- $\eta\nabla^2\boldsymbol{v}$: viscous diffusion of momentum.
- $\frac{I}{2\tau_s}\nabla\times(\boldsymbol{\Theta}-\boldsymbol{\Omega})$: the new term. It is nonzero *only* when the grains spin at a rate different from the fluid, which happens *only* when a magnetic torque acts on them. It is a genuine internal source of linear momentum for the fluid, fed by the field.

What if the last term were removed? The equation would reduce to ordinary Navier–Stokes, the viscosity would be the field-free η, and neither the McTague effect nor the negative-viscosity effect would exist. Every result of both papers lives in that one term.

19.7 Connection to the Next Equation

Equation (SM-6) contains three unknown fields: $\boldsymbol{v}$, $\boldsymbol{M}$ and $\boldsymbol{\Theta}$ (the pressure being fixed by incompressibility). It cannot be solved alone. The magnetisation obeys its own dynamics, which is the next equation, (SM-7).

Chapter 20

Equation (SM-7): The Magnetisation Equation

$$\frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t} = \boldsymbol{\Theta} \times \boldsymbol{M} - \frac{1}{\tau_B}\big(\boldsymbol{M} - \boldsymbol{M}_0\big). \tag{SM-7}$$

20.1 Context

The second member of the ferrohydrodynamic triple. It says how the magnetisation of the suspension changes: it is carried around by the rotating particles, and it relaxes towards its instantaneous equilibrium value with the Brownian time τ_B. It is the equation that introduces *memory* into the problem, and without memory — that is, if τ_B were zero — there would be no negative-viscosity effect at all, because $\boldsymbol{M}$ would always be parallel to $\boldsymbol{H}$ and the torque $\boldsymbol{M} \times \boldsymbol{H}$ would vanish identically.

20.2 Symbols

All have appeared: $\boldsymbol{M}$ (magnetisation, G), $\boldsymbol{M}_0 = nmL(\xi)\hat{\boldsymbol{h}}$ (instantaneous-equilibrium magnetisation, Eq. (10.4.1), G), $\boldsymbol{\Theta}$ (mean particle angular velocity, s^{-1}), $\tau_B = 3\eta V/kT$ (Brownian relaxation time, Eq. (12.3.4), s). The derivative $\mathrm{d}/\mathrm{d}t$ is again the material derivative (3.2.3), since $\boldsymbol{M}$ is a property carried by the moving fluid.

20.3 Assumptions

1. Rigid dipoles (no Néel relaxation): checked numerically in Sec. 17.6.

2. Dilute in the magnetic sense: no dipole–dipole interaction between grains, so that each moment feels only the applied field.

3. Monodisperse grains: a single m and a single τ_B. Real samples are polydisperse, which smears the relaxation over a range of times; this is one of the reasons the measured curves are smoother than the theoretical ones.

4. **The closure assumption.** As shown in Sec. 13.2, the exact first-moment equation contains $\langle e_i e_j \rangle$, and (SM-7) follows only if that second moment is replaced by its isotropic

value $\frac{1}{3}\delta_{ij}$. The equation is therefore *exact* for $\xi \ll 1$ and a controlled approximation otherwise.

20.4 Derivation

Carried out in full in Sec. 13.2, by multiplying the Smoluchowski equation (11.6.1) by $\boldsymbol{e}$ and integrating over orientations. The three contributions were

$$\text{flow:} \quad \boldsymbol{\Omega} \times \langle \boldsymbol{e} \rangle \qquad \text{Eq. (13.2.4)}, \tag{20.4.1}$$

$$\text{diffusion:} \quad -2D_r \langle \boldsymbol{e} \rangle \qquad \text{Eq. (13.2.6)}, \tag{20.4.2}$$

$$\text{magnetic:} \quad +2D_r \langle \boldsymbol{e} \rangle_{\text{eq}} \qquad \text{Eq. (13.2.10)}, \tag{20.4.3}$$

whose sum, multiplied by nm and written with $2D_r = 1/\tau_B$, is Eq. (13.2.12):

$$\frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t} = \boldsymbol{\Omega} \times \boldsymbol{M} - \frac{1}{\tau_B}\big(\boldsymbol{M} - \boldsymbol{M}_0\big). \tag{20.4.4}$$

Section 13.3 then showed that replacing $\boldsymbol{\Omega}$ by the *total* mean particle spin $\boldsymbol{\Theta}$ merely re-books the magnetic torque term, so that (20.4.4) and (SM-7) are the same statement. Nothing is repeated here.

20.5 Physical Interpretation

The convective term $\boldsymbol{\Theta} \times \boldsymbol{M}$. If the grains rotate rigidly with angular velocity $\boldsymbol{\Theta}$, the moments frozen in them rotate too, and a vector rotating with angular velocity $\boldsymbol{\Theta}$ obeys $\mathrm{d}\boldsymbol{M}/\mathrm{d}t = \boldsymbol{\Theta} \times \boldsymbol{M}$ (Sec. 3.4, Step 1). This term conserves $|\boldsymbol{M}|$: indeed $\boldsymbol{M} \cdot (\boldsymbol{\Theta} \times \boldsymbol{M}) = 0$ by (1.5.7), so it can only turn the magnetisation, never change its length.

The relaxation term. Everything that *does* change $|\boldsymbol{M}|$ is in $-(\boldsymbol{M} - \boldsymbol{M}_0)/\tau_B$: thermal agitation degrades any alignment not sustained by the field, and the field rebuilds it, the balance being $\boldsymbol{M}_0$. The single time constant τ_B is the closure's simplification of what is in reality a spectrum.

Why memory matters. Set $\tau_B \to 0$: the relaxation term becomes infinitely stiff, forcing $\boldsymbol{M} = \boldsymbol{M}_0 \parallel \boldsymbol{H}$ at every instant. Then $\boldsymbol{M} \times \boldsymbol{H} = 0$, the coupling term in (SM-6) vanishes, $\boldsymbol{\Theta} = \boldsymbol{\Omega}$, and the ferrofluid is an ordinary liquid. *The entire subject exists because $\tau_B \neq 0$.*

20.6 Connection to the Next Equation

Equation (SM-7) contains $\boldsymbol{\Theta}$, which is still undetermined. The third member of the system supplies it: Newton's law for the rotation of the particles, (SM-8).

Chapter 21

Equation (SM-8): Rotational Motion of the Particles

$$I\,\frac{\mathrm{d}\boldsymbol{\Theta}}{\mathrm{d}t} = \boldsymbol{M}\times\boldsymbol{H} - \frac{I}{\tau_s}\left(\boldsymbol{\Theta}-\boldsymbol{\Omega}\right). \tag{SM-8}$$

21.1 Context

The third equation of the system. It is Newton's second law for rotation [Eq. (4.2.2)] applied to the grains contained in unit volume, with the two torques that act on them: magnetic and viscous. Volume I derived its field-free version as Eq. (7.5.3); here the magnetic torque is restored and the equation is written per unit volume, exactly as the paper does.

21.2 Symbols

I is the sum of the particles' moments of inertia over unit volume, $I = nI_p = n\pi\rho_s d^5/60$ (units $\mathrm{g\,cm^{-1}}$); all other symbols are as before. Both sides have the units of a torque density, $\mathrm{erg\,cm^{-3}}$: $[I][\Theta]/[t] = \mathrm{g\,cm^{-1}s^{-2}} = \mathrm{dyn\,cm^{-2}} = \mathrm{erg\,cm^{-3}}$, and $[M][H] = \mathrm{G\,Oe} = \mathrm{erg\,cm^{-3}}$. ✓

21.3 Assumptions

1. Spherical grains, so that the moment of inertia is a scalar and $\boldsymbol{L} = I\boldsymbol{\Theta}$ (Sec. 4.3).

2. Stokes friction for the viscous torque (Chapter 6).

3. Rigid dipoles, so the magnetic torque on the moment is transmitted to the grain.

4. **The random Brownian torques average to zero.** This deserves emphasis. Each individual grain is battered by fluctuating molecular torques far larger than the systematic ones. But $\boldsymbol{\Theta}$ is defined as an average over the many grains in a physically small volume element (Sec. 4.5), and the random torques, having zero mean and being uncorrelated between grains, cancel in that average. Their effect does not disappear from the theory — it reappears, in the only place it can, as the relaxation term of (SM-7). *The systematic equation (SM-8) and the statistical equation (SM-7) are the two faces of the same molecular bombardment.*

21.4 Step-by-Step Derivation

Step 1. For a single grain, Eq. (4.2.2) with $\boldsymbol{L} = I_p\boldsymbol{\theta}$:

$$I_p\frac{\mathrm{d}\boldsymbol{\theta}}{\mathrm{d}t} = \boldsymbol{T}_{\mathrm{mag}} + \boldsymbol{T}_{\mathrm{visc}} = \boldsymbol{m}\times\boldsymbol{H} - \zeta_r\big(\boldsymbol{\theta}-\boldsymbol{\Omega}\big), \tag{21.4.1}$$

the magnetic torque from Eq. (4.4.2) and the viscous one from Eq. (6.5.8), the latter written in the frame co-rotating with the fluid as explained in Sec. 7.5, Step 2.

Step 2. Sum (21.4.1) over the n grains in unit volume and divide by nothing (we *want* the per-unit-volume form). Averaging termwise:

$$nI_p\frac{\mathrm{d}\langle\boldsymbol{\theta}\rangle}{\mathrm{d}t} = n\langle\boldsymbol{m}\rangle\times\boldsymbol{H} - n\zeta_r\big(\langle\boldsymbol{\theta}\rangle-\boldsymbol{\Omega}\big). \tag{21.4.2}$$

The field $\boldsymbol{H}$ came out of the average because it is common to all grains, and $\boldsymbol{\Omega}$ likewise.

Step 3. Introduce the macroscopic symbols $I \equiv nI_p$, $\boldsymbol{\Theta} \equiv \langle\boldsymbol{\theta}\rangle$, $\boldsymbol{M} \equiv n\langle\boldsymbol{m}\rangle$:

$$I\frac{\mathrm{d}\boldsymbol{\Theta}}{\mathrm{d}t} = \boldsymbol{M}\times\boldsymbol{H} - n\zeta_r\big(\boldsymbol{\Theta}-\boldsymbol{\Omega}\big). \tag{21.4.3}$$

Step 4. Replace $n\zeta_r$ by I/τ_s using the identity (7.5.12):

$$I\frac{\mathrm{d}\boldsymbol{\Theta}}{\mathrm{d}t} = \boldsymbol{M}\times\boldsymbol{H} - \frac{I}{\tau_s}\big(\boldsymbol{\Theta}-\boldsymbol{\Omega}\big), \tag{SM-8}$$

which is Eq. (SM-8). ■

21.5 Physical Interpretation

The equation is a tug of war. The magnetic torque $\boldsymbol{M}\times\boldsymbol{H}$ tries to twist the grains so as to align their moments with the field; the viscous torque tries to make them turn with the liquid. The inertial term on the left says how fast the compromise is reached — and by Eq. (SM-1) the answer is 10^{-11} s, i.e. instantly. Chapter 24 exploits exactly that.

21.6 Connection to the Next Equation

Equations (SM-6)–(SM-8) involve the fields $\boldsymbol{v}$, p, $\boldsymbol{M}$, $\boldsymbol{\Theta}$ and $\boldsymbol{H}$. Two things are still missing: the constraints that $\boldsymbol{v}$ and $\boldsymbol{H}$ must satisfy (incompressibility and magnetostatics), and the explicit formula for $\boldsymbol{M}_0$. These are Eqs. (SM-9) and (SM-10).

Chapter 22

Equation (SM-9): The Constraints

$$\nabla \cdot \boldsymbol{v} = 0, \qquad \nabla \times \boldsymbol{H} = 0, \qquad \nabla \cdot \big(\boldsymbol{H} + 4\pi \boldsymbol{M}\big) = 0. \tag{SM-9}$$

22.1 Context

The paper introduces these three relations with the remark that they indicate "that the fluid is considered to be incompressible and nonconducting". They are not dynamical equations; they are constraints that the fields must satisfy at every instant. The first was derived in Volume I; the other two are magnetostatics and are derived here.

22.2 Symbols

$\boldsymbol{B} = \boldsymbol{H} + 4\pi\boldsymbol{M}$ is the magnetic induction (G) in Gaussian units; $\boldsymbol{E}$ the electric field; $\boldsymbol{D}$ the electric displacement; $\boldsymbol{J}$ the free current density; $c = 3 \times 10^{10}\ \mathrm{cm\,s^{-1}}$ the speed of light. All other symbols are as before.

22.3 Assumptions

Assumption 22.3.1 (Maxwell's equations)**.** In Gaussian units, for a medium at rest,

$$\nabla \cdot \boldsymbol{D} = 4\pi\rho_{\mathrm{free}}, \qquad \nabla \times \boldsymbol{H} = \frac{4\pi}{c}\boldsymbol{J} + \frac{1}{c}\frac{\partial \boldsymbol{D}}{\partial t}, \tag{22.3.1}$$

$$\nabla \cdot \boldsymbol{B} = 0, \qquad \nabla \times \boldsymbol{E} = -\frac{1}{c}\frac{\partial \boldsymbol{B}}{\partial t}, \tag{22.3.2}$$

with $\boldsymbol{B} = \boldsymbol{H} + 4\pi\boldsymbol{M}$.

These are empirical laws — Gauss's law, the Ampère–Maxwell law, the absence of magnetic monopoles, and Faraday's law — distilled from two centuries of experiment. They cannot be derived from anything more primitive, and we adopt them in the same spirit as Newton's laws in Sec. 4.1. Everything below is derived.

Assumption 22.3.2 (Non-conducting medium)**.** The carrier is water with dissolved counterions; on the time scales of the experiment no significant free current flows in response to the alternating field. Hence $\boldsymbol{J} = 0$.

Assumption 22.3.3 (Quasi-static regime)**.** The displacement current is negligible.

Justification of Assumption 22.3.3, quantitatively. Compare the two terms on the right of the Ampère–Maxwell law. With a field varying at angular frequency ω and a system of linear size L, the displacement-current term is of order $\omega D/c$ and the curl on the left of order H/L. Their ratio is

$$\frac{\omega D L}{c H} \sim \frac{\omega L}{c} = \frac{2\pi L}{\lambda}, \qquad \lambda = \frac{2\pi c}{\omega} = \frac{c}{f}. \tag{22.3.3}$$

For the highest frequency used by Bacri *et al.*, $f = 1480$ Hz,

$$\lambda = \frac{3 \times 10^{10}}{1.48 \times 10^{3}} = 2.0 \times 10^{7} \text{ cm} = 200 \text{ km}, \tag{22.3.4}$$

while the apparatus is $L \sim 10$ cm. The ratio is $2\pi \times 10/2 \times 10^{7} \approx 3 \times 10^{-6}$. The field fills the solenoid as if it were static, adjusting itself instantaneously to the coil current. ✓

22.4 Step-by-Step Derivation

First relation. Conservation of mass with constant density gives $\nabla \cdot \boldsymbol{v} = 0$; this is Eq. (5.1.4), derived in Sec. 5.1. ✓

Second relation. Insert Assumptions 22.3.2 and 22.3.3 into the Ampère–Maxwell law (22.3.1):

$$\nabla \times \boldsymbol{H} = \frac{4\pi}{c} \underbrace{\boldsymbol{J}}_{=0} + \frac{1}{c} \underbrace{\frac{\partial \boldsymbol{D}}{\partial t}}_{\text{negligible}} = 0. \tag{22.4.1}$$

■

A useful consequence, used in Step 3 of Sec. 19.4 and again in Chapter 27: a curl-free field is a gradient,

$$\nabla \times \boldsymbol{H} = 0 \quad \Longrightarrow \quad \boldsymbol{H} = -\nabla \psi, \tag{22.4.2}$$

for some scalar magnetic potential ψ. (Reason: by Stokes' theorem (2.4.6) the circulation of $\boldsymbol{H}$ around every closed loop vanishes, so the line integral between two points is path-independent and defines a potential.)

Third relation. The absence of magnetic monopoles, $\nabla \cdot \boldsymbol{B} = 0$ from (22.3.2), together with the constitutive definition $\boldsymbol{B} = \boldsymbol{H} + 4\pi \boldsymbol{M}$, gives directly

$$\nabla \cdot \big(\boldsymbol{H} + 4\pi \boldsymbol{M} \big) = 0, \qquad \text{equivalently} \qquad \nabla \cdot \boldsymbol{H} = -4\pi \nabla \cdot \boldsymbol{M}. \tag{22.4.3}$$

■

22.5 Physical Interpretation: why $\boldsymbol{H}$ inside is not the applied field

Equation (22.4.3) says that the magnetisation acts as a source of $\boldsymbol{H}$: wherever $\boldsymbol{M}$ has a divergence — in practice, at the surface of the sample, where $\boldsymbol{M}$ drops abruptly to zero — there are effective "magnetic charges" which create an additional field inside the sample. That field opposes $\boldsymbol{M}$ and is called the demagnetising field. It is the reason a magnetised body does not experience the field one applied to it.

Why this does not spoil the experiment. Bacri *et al.* place a capillary of diameter 1 mm and length several centimetres *along* the axis of the solenoid, so that $\boldsymbol{H}$, $\boldsymbol{M}$ and the tube axis are all parallel. Inside such a long cylinder with $\boldsymbol{M}$ uniform and axial,

- $\nabla \cdot \boldsymbol{M} = 0$ in the interior (a uniform vector has zero divergence), so there are no volume charges;
- the only surface charges lie on the two end faces, at a distance $\gtrsim 10^2$ diameters from the measuring region, so their field there is negligible (it falls off as the inverse square of the distance).

Hence $\boldsymbol{H}$ inside the capillary equals the applied solenoid field, and the demagnetising correction — which for a long cylinder magnetised along its axis is formally zero — may be ignored. This is why the papers can treat H as the externally imposed quantity without further comment. Had the field been applied *across* the capillary, the demagnetising factor would have been 2π per unit M and the internal field would have differed appreciably from the applied one.

A second reason $\boldsymbol{H}$ may be treated as given. Suppose we tried to keep the coupling: $\boldsymbol{M}$ depends on $\boldsymbol{H}$ through the Langevin law, and $\boldsymbol{H}$ depends on $\boldsymbol{M}$ through (22.4.3). The problem would be self-consistent and nonlinear. The geometry above cuts that loop, which is a piece of experimental design, not of mathematics.

22.6 Connection to the Next Equation

The system is now closed except for one thing: the explicit formula for the instantaneous equilibrium magnetisation $\boldsymbol{M}_0$ appearing in (SM-7). That is (SM-10).

Chapter 23

Equation (SM-10): The Langevin Magnetisation

$$\boldsymbol{M}_0 = nmL(\xi)\frac{\boldsymbol{H}}{H}, \qquad \xi = \frac{mH(t)}{kT}, \qquad L(\xi) = \coth\xi - \xi^{-1}. \tag{SM-10}$$

23.1 Context

The last member of the system (SM-6)–(SM-10). It supplies the equilibrium towards which (SM-7) relaxes, and it is the only place where the temperature enters explicitly. Bacri *et al.* write the same formula beneath their Eq. (B-4) as $\boldsymbol{M}_0(t) = \varphi M_s L(\xi)\hat{\boldsymbol{h}}$; the two agree because $nm = nM_sV = \varphi M_s$.

23.2 Symbols

Symbol	Meaning	Type	Units
$\boldsymbol{M}_0$	instantaneous-equilibrium magnetisation	vector field	G
n	number of grains per unit volume	scalar	cm^{-3}
$m = M_sV$	magnetic moment of one grain	scalar	$\mathrm{erg\,G}^{-1}$
$L(\xi)$	Langevin function	function	—
$\xi = mH(t)/kT$	Langevin parameter, evaluated at the *instantaneous* field	scalar	—
$\hat{\boldsymbol{h}} = \boldsymbol{H}/H$	unit vector along the field	vector	—

23.3 Assumptions

1. Thermal equilibrium of the orientations *at the instantaneous field* — i.e. $\boldsymbol{M}_0$ is defined as the magnetisation the sample would have if τ_B were zero. It is a mathematical construct, not the actual magnetisation.

2. No interparticle magnetic interaction.

3. Monodisperse, rigid dipoles.

23.4 Derivation

Carried out in full in Chapter 10: the Boltzmann distribution (9.5.7) gives $\langle \cos\theta \rangle = L(\xi)$ by Eq. (10.2.16), and multiplication by nm gives (10.4.1), which *is* (SM-10). Nothing is repeated.

23.5 Physical Interpretation and a Warning About the Word "Equilibrium"

Shliomis & Morozov insert a paragraph of clarification at exactly this point, and it is worth restating because it is the most common source of confusion in the subject:

> $\boldsymbol{M}_0[\boldsymbol{H}(t)]$ is *not* the magnetisation. It is the value the magnetisation would take if the field were changing slowly enough for the orientations to keep up. The real $\boldsymbol{M}(t)$ differs from it precisely because $\tau_B \neq 0$ in (SM-7).

The difference $\boldsymbol{M} - \boldsymbol{M}_0$ is what drives everything: it is proportional to the transverse magnetisation, hence to the torque $\boldsymbol{M} \times \boldsymbol{H}$, hence to the slip $\boldsymbol{\Omega} - \boldsymbol{\Theta}$, hence to $\Delta\eta$. Chapter 15 computed that chain for a stationary field; Volumes IV and V will compute it for an alternating one.

Numerical calibration of ξ for the experimental fluid. With $M_s = 400$ G and $d = 10$ nm ($V = 5.24 \times 10^{-19}$ cm^3),

$$m = M_s V = 400 \times 5.24 \times 10^{-19} = 2.10 \times 10^{-16} \ \text{erg}\,\text{G}^{-1}, \tag{23.5.1}$$

$$\xi = \frac{mH}{kT} = \frac{2.10 \times 10^{-16}}{4.14 \times 10^{-14}} H = 5.1 \times 10^{-3}\, H \ \text{(Oe)}. \tag{23.5.2}$$

Bacri *et al.* quote $\xi_0 = M_s V H/kT = 5.25 \times 10^{-3} H(\text{Oe})$. Working backwards from their number, $V = 5.43 \times 10^{-19}$ cm^3 and $d = 10.1$ nm — confirming both their stated grain size and our arithmetic. ✓ Note the practical consequence: their maximum field of 2000 Oe corresponds to $\xi \approx 10.5$, well into the nonlinear region of $L(\xi)$, which is why the weak-field theory of Sec. II of Shliomis & Morozov cannot describe the experiment quantitatively and Sec. III is needed.

23.6 Connection to the Next Equation

The system (SM-6)–(SM-10) is now complete and closed: five relations for $\boldsymbol{v}, p, \boldsymbol{M}, \boldsymbol{\Theta}, \boldsymbol{H}$. The paper's next move is to simplify it by eliminating $\boldsymbol{\Theta}$, which is possible because of the smallness of τ_s. That is (SM-11).

Part VIII

Elimination of the Spin, and the Alternating Field

Chapter 24

Equation (SM-11): The Inertialess Spin

$$\boldsymbol{\Theta} = \boldsymbol{\Omega} + \big(\tau_s / I\big)\big(\boldsymbol{M} \times \boldsymbol{H}\big). \tag{SM-11}$$

24.1 Context

This is where the licence bought in Volume I is spent. Equation (SM-1) established that $\tau_s \sim 10^{-11}$ s; the paper now argues that the inertial term of (SM-8) is therefore negligible, converting a differential equation for $\boldsymbol{\Theta}$ into an algebraic one. From this point on, $\boldsymbol{\Theta}$ is no longer an independent unknown: it is whatever the torques say it is.

24.2 Symbols

All previously defined. Note $\tau_s / I = 1/n\zeta_r = 1/6\eta\varphi$ by Eq. (7.5.12), so (SM-11) may equivalently be written

$$\boldsymbol{\Theta} = \boldsymbol{\Omega} + \frac{1}{6\eta\varphi}\big(\boldsymbol{M} \times \boldsymbol{H}\big), \tag{24.2.1}$$

which is the form used in Volume II [Eq. (11.4.2) averaged, and Sec. 13.3] and the form in which Bacri *et al.* write it as their Eq. (B-1).

24.3 Assumptions

The single assumption is that the inertial term of (SM-8) is negligible:

$$\left| I \frac{\mathrm{d}\boldsymbol{\Theta}}{\mathrm{d}t} \right| \ll \left| \frac{I}{\tau_s}\big(\boldsymbol{\Theta} - \boldsymbol{\Omega}\big) \right|. \tag{24.3.1}$$

Justification, quantitatively. Estimate each side. The left-hand side is of order $I\,\Theta_{\mathrm{typ}}/t_{\mathrm{typ}}$, where t_{typ} is the shortest time on which $\boldsymbol{\Theta}$ actually varies. In these experiments that is the field period, $t_{\mathrm{typ}} = 1/\omega$. The right-hand side is of order $I\,\Theta_{\mathrm{typ}}/\tau_s$. The ratio is therefore

$$\frac{|I\,\mathrm{d}\boldsymbol{\Theta}/\mathrm{d}t|}{|I(\boldsymbol{\Theta} - \boldsymbol{\Omega})/\tau_s|} \sim \omega\tau_s. \tag{24.3.2}$$

At the highest experimental frequency, $f = 1480$ Hz,

$$\omega\tau_s = 2\pi \times 1480 \times 10^{-11} = 9.3 \times 10^{-8}. \tag{24.3.3}$$

The inertial term is a hundred-millionth of the friction term. This is precisely the paper's statement "$I\,\mathrm{d}\boldsymbol{\Theta}/\mathrm{d}t \ll I\boldsymbol{\Theta}/\tau_s$, so Eq. (8) has the form (11)". ✓

24.4 Step-by-Step Derivation

Step 1. Write (SM-8) again:

$$I\frac{\mathrm{d}\boldsymbol{\Theta}}{\mathrm{d}t} = \boldsymbol{M} \times \boldsymbol{H} - \frac{I}{\tau_s}\big(\boldsymbol{\Theta} - \boldsymbol{\Omega}\big). \tag{24.4.1}$$

Step 2. Set the left-hand side to zero by (24.3.1):

$$0 = \boldsymbol{M} \times \boldsymbol{H} - \frac{I}{\tau_s}\big(\boldsymbol{\Theta} - \boldsymbol{\Omega}\big). \tag{24.4.2}$$

Step 3. Solve for $\boldsymbol{\Theta}$. Add $\frac{I}{\tau_s}(\boldsymbol{\Theta} - \boldsymbol{\Omega})$ to both sides:

$$\frac{I}{\tau_s}\big(\boldsymbol{\Theta} - \boldsymbol{\Omega}\big) = \boldsymbol{M} \times \boldsymbol{H}. \tag{24.4.3}$$

Multiply both sides by τ_s/I:

$$\boldsymbol{\Theta} - \boldsymbol{\Omega} = \frac{\tau_s}{I}\big(\boldsymbol{M} \times \boldsymbol{H}\big), \tag{24.4.4}$$

and add $\boldsymbol{\Omega}$:

$$\boldsymbol{\Theta} = \boldsymbol{\Omega} + \frac{\tau_s}{I}\big(\boldsymbol{M} \times \boldsymbol{H}\big). \tag{SM-11}$$

■

Equation (24.4.3), written with $I/\tau_s = 6\eta\varphi$, is exactly Eq. (B-1) of Bacri *et al.*,

$$6\eta\varphi\big(\boldsymbol{\omega}_p - \boldsymbol{\Omega}\big) = \boldsymbol{M} \times \boldsymbol{H}, \tag{B-1}$$

already used in Sec. 14.2.

24.5 Physical Interpretation

The grains have no memory of their own rotation. At every instant the magnetic torque and the viscous torque balance exactly, and the spin is whatever makes that true. Three readings:

- **No magnetic torque** $\Rightarrow \boldsymbol{\Theta} = \boldsymbol{\Omega}$: the grains turn with the liquid, and nothing happens.
- **The slip is proportional to the torque**, with the proportionality constant $1/6\eta\varphi$: more viscous liquid or more particles means less slip for a given torque.
- **The sign of the slip is the sign of the effect.** A stationary field always produces a torque opposing the rotation ($\Theta < \Omega$, $\Delta\eta > 0$), because $\boldsymbol{M}$ lags *behind* $\boldsymbol{H}$ in the direction of rotation. An alternating field can produce the opposite, and that is the subject of the volumes to come.

24.6 Connection to the Next Equation

With $\boldsymbol{\Theta}$ eliminated, both (SM-6) and (SM-7) can be rewritten without it. Doing so gives (SM-12) and (SM-13), the reduced pair the papers actually solve.

Chapter 25

Equation (SM-12): The Reduced Momentum Equation

$$\rho\,\frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = -\nabla p + (\boldsymbol{M}\cdot\nabla)\boldsymbol{H} + \eta\Delta\boldsymbol{v} + \tfrac{1}{2}\nabla\times\big(\boldsymbol{M}\times\boldsymbol{H}\big). \tag{SM-12}$$

25.1 Context

Equation (SM-6) with $\boldsymbol{\Theta}$ eliminated by (SM-11). It is also, term for term, Eq. (B-3) of Bacri *et al.* The virtue of this form is that the mechanical unknown $\boldsymbol{\Theta}$ has disappeared entirely: the fluid equation now couples only to the magnetisation.

25.2 Symbols and Assumptions

As in Chapters 19 and 24; nothing new.

25.3 Step-by-Step Derivation

Step 1. Take the last term of (SM-6) and substitute (SM-11) in the form (24.4.3), $\boldsymbol{\Theta}-\boldsymbol{\Omega} = (\tau_s/I)(\boldsymbol{M}\times\boldsymbol{H})$:

$$\frac{I}{2\tau_s}\nabla\times\big(\boldsymbol{\Theta}-\boldsymbol{\Omega}\big) = \frac{I}{2\tau_s}\nabla\times\Big[\frac{\tau_s}{I}\big(\boldsymbol{M}\times\boldsymbol{H}\big)\Big] \tag{25.3.1}$$

$$= \frac{I}{2\tau_s}\cdot\frac{\tau_s}{I}\,\nabla\times\big(\boldsymbol{M}\times\boldsymbol{H}\big) \quad (\tau_s/I \text{ is a constant, so it passes through } \nabla\times) \tag{25.3.2}$$

$$= \tfrac{1}{2}\nabla\times\big(\boldsymbol{M}\times\boldsymbol{H}\big). \tag{25.3.3}$$

Step 2. Substitute (25.3.3) into (SM-6):

$$\rho\frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = -\nabla p + (\boldsymbol{M}\cdot\nabla)\boldsymbol{H} + \eta\Delta\boldsymbol{v} + \tfrac{1}{2}\nabla\times\big(\boldsymbol{M}\times\boldsymbol{H}\big). \tag{SM-12}$$

■

Note that we arrived at the same expression by an entirely independent route in Sec. 14.3, where $\frac{1}{2}\nabla\times(\boldsymbol{M}\times\boldsymbol{H})$ was obtained directly from the requirement of angular-momentum

balance for a body couple. That the two derivations agree is a strong internal check on the whole construction: the couple transmitted by friction and the couple imposed by the field are the same couple.

25.4 Physical Interpretation

The magnetic field enters the momentum balance in exactly two ways:

- through $(\boldsymbol{M} \cdot \nabla)\boldsymbol{H}$, if the field is *non-uniform* — a force;
- through $\frac{1}{2}\nabla \times (\boldsymbol{M} \times \boldsymbol{H})$, if the magnetisation is *not parallel* to the field — a couple.

In these papers the first is zero by construction (uniform solenoid field) and the second is everything. It vanishes when $\boldsymbol{M} \parallel \boldsymbol{H}$, i.e. in perfect equilibrium; it is nonzero only because the magnetisation lags.

25.5 Connection to the Next Equation

The same substitution must now be made in the magnetisation equation, giving (SM-13).

Chapter 26

Equation (SM-13): The Reduced Magnetisation Equation

$$\frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t} = \boldsymbol{\Omega} \times \boldsymbol{M} - \frac{1}{\tau_B}\big(\boldsymbol{M} - \boldsymbol{M}_0\big) - \frac{\tau_s}{I}\boldsymbol{M} \times \big(\boldsymbol{M} \times \boldsymbol{H}\big). \tag{SM-13}$$

26.1 Context

Equation (SM-7) with $\boldsymbol{\Theta}$ eliminated. Together with (SM-12) it forms the pair the authors describe as "the equivalent system (10)–(13)", which "allows the calculation of the effective viscosity of the ferrofluid in the stationary and alternating magnetic field". Bacri *et al.* print exactly this equation as their Eq. (B-4), writing $1/6\eta\varphi$ in place of τ_s/I.

26.2 Symbols and Assumptions

As before. The new feature is that the equation is now *nonlinear* in $\boldsymbol{M}$: the last term is cubic in the magnetisation–field product. This nonlinearity is not an artefact — it is the feedback by which the magnetisation changes the particle spin, which in turn changes the magnetisation.

26.3 Step-by-Step Derivation

Step 1. Start from (SM-7) and substitute (SM-11):

$$\frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t} = \boldsymbol{\Theta} \times \boldsymbol{M} - \frac{1}{\tau_B}\big(\boldsymbol{M} - \boldsymbol{M}_0\big), \qquad \boldsymbol{\Theta} = \boldsymbol{\Omega} + \frac{\tau_s}{I}\big(\boldsymbol{M} \times \boldsymbol{H}\big). \tag{26.3.1}$$

Step 2. Expand the convective term, using the distributivity of the cross product over addition:

$$\boldsymbol{\Theta} \times \boldsymbol{M} = \Big[\boldsymbol{\Omega} + \frac{\tau_s}{I}\big(\boldsymbol{M} \times \boldsymbol{H}\big)\Big] \times \boldsymbol{M} \tag{26.3.2}$$

$$= \boldsymbol{\Omega} \times \boldsymbol{M} + \frac{\tau_s}{I}\Big[\big(\boldsymbol{M} \times \boldsymbol{H}\big) \times \boldsymbol{M}\Big]. \tag{26.3.3}$$

Step 3. Reverse the order of the last cross product, using the antisymmetry (1.5.5), $\boldsymbol{A}\times\boldsymbol{B} = -\boldsymbol{B}\times\boldsymbol{A}$:

$$\big(\boldsymbol{M} \times \boldsymbol{H}\big) \times \boldsymbol{M} = -\boldsymbol{M} \times \big(\boldsymbol{M} \times \boldsymbol{H}\big). \tag{26.3.4}$$

Step 4. Substitute (26.3.4) into (26.3.3) and then into Step 1:

$$\frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t} = \boldsymbol{\Omega} \times \boldsymbol{M} - \frac{\tau_s}{I}\boldsymbol{M} \times \big(\boldsymbol{M} \times \boldsymbol{H}\big) - \frac{1}{\tau_B}\big(\boldsymbol{M} - \boldsymbol{M}_0\big), \tag{SM-13}$$

which is (SM-13) with the terms in the paper's order. ■

26.4 Physical Interpretation of the New Term

Use the triple product (1.7.1) to expand it:

$$\boldsymbol{M} \times \big(\boldsymbol{M} \times \boldsymbol{H}\big) = \boldsymbol{M}\big(\boldsymbol{M} \cdot \boldsymbol{H}\big) - \boldsymbol{H}\big(\boldsymbol{M} \cdot \boldsymbol{M}\big) \tag{26.4.1}$$

$$= -M^2\left[\boldsymbol{H} - \hat{\boldsymbol{M}}\big(\hat{\boldsymbol{M}} \cdot \boldsymbol{H}\big)\right], \qquad \hat{\boldsymbol{M}} = \boldsymbol{M}/M, \tag{26.4.2}$$

i.e. it is proportional to *minus* the component of $\boldsymbol{H}$ perpendicular to $\boldsymbol{M}$. With the overall minus sign in (SM-13), the term therefore pushes $\boldsymbol{M}$ *towards* $\boldsymbol{H}$. It is a second alignment mechanism, distinct from the Brownian relaxation term: here the alignment is achieved by physically rotating the grains, whereas the relaxation term achieves it by thermal reorientation of the population. The characteristic rate of the mechanical mechanism is

$$\frac{\tau_s}{I}MH = \frac{MH}{6\eta\varphi}, \tag{26.4.3}$$

and the ratio of this rate to the Brownian rate $1/\tau_B$ is exactly the dimensionless combination $\zeta = \tau_B MH/6\eta\varphi = \xi L(\xi)/2$ met in Eq. (15.5.17). The competition between these two alignment mechanisms is what produced the factor $\zeta/(1+\zeta)$ in Eq. (SM-2).

26.5 Connection to the Next Equation

The system (SM-10)–(SM-13) is now the complete, closed problem. What remains is to specify *which* field is applied. That is (SM-14).

Chapter 27

Equation (SM-14): The Alternating Field

$$H_x = H_0 \cos \omega t, \qquad H_y = H_z = 0. \tag{SM-14}$$

27.1 Context

Everything until now has been general. Equation (SM-14) specialises the problem to the case the paper is about: a spatially uniform, *linearly polarised* field oscillating harmonically along a fixed direction. The authors introduce it with the sentence "Thus the problem is to express the viscosity $\Delta\eta$ as a function of amplitude and frequency of the field". From here on, the two control parameters are H_0 (through $\xi_0 = mH_0/kT$) and ω (through $\omega\tau_B$).

27.2 Symbols

Symbol	Meaning	Type	Units
H_0	amplitude of the field, i.e. its maximum value	scalar, constant	Oe
ω	angular frequency, $\omega = 2\pi f$	scalar, constant	s^{-1}
f	ordinary frequency	scalar	Hz
t	time	scalar	s
$\xi_0 = mH_0/kT$	Langevin parameter formed with the *amplitude*	dimensionless	—
$\omega\tau_B$	dimensionless frequency	dimensionless	—

27.3 Assumptions

1. **Spatial uniformity.** H_0 does not depend on position, so the Kelvin force $(\boldsymbol{M}\cdot\nabla)\boldsymbol{H}$ vanishes. Realised experimentally by placing the short capillary in the middle of a long solenoid.

2. **Linear polarisation.** The field oscillates along a fixed axis, rather than rotating. This is the essential feature: a linearly polarised field *picks out no sense of rotation*, and the whole mechanism of the effect depends on that fact, as the next volume shows.

3. **Harmonic time dependence.** A pure cosine. Section III of the paper relaxes this and treats a square wave instead, for computational convenience (Eqs. SM-36 onwards), and the two give quantitatively similar answers — a robustness check worth noting.

4. **The field is perpendicular to the vorticity**, $\beta = \pi/2$. The authors state this just before Eq. (SM-3): "below we will assume that $\beta = \pi/2$". It is realised in the experiment because the flow is a cylindrical Poiseuille flow whose vorticity is azimuthal while the field is axial (Sec. 3.5), so $\boldsymbol{H} \perp \boldsymbol{\Omega}$ automatically. It is also the case that maximises the effect, since (SM-2) carries a factor $\sin^2\beta$.

27.4 Mathematical Background: Period Averages

The function $H_0 \cos\omega t$ is periodic with period

$$T_0 = \frac{2\pi}{\omega} = \frac{1}{f}, \tag{27.4.1}$$

its mean over a period is zero,

$$\overline{\cos\omega t} = \frac{\omega}{2\pi}\int_0^{2\pi/\omega} \cos\omega t\,\mathrm{d}t = \frac{\omega}{2\pi}\left[\frac{\sin\omega t}{\omega}\right]_0^{2\pi/\omega} = \frac{\sin 2\pi - \sin 0}{2\pi} = 0, \tag{27.4.2}$$

and its mean square is one half,

$$\overline{\cos^2\omega t} = \frac{\omega}{2\pi}\int_0^{2\pi/\omega} \cos^2\omega t\,\mathrm{d}t \tag{27.4.3}$$

$$= \frac{\omega}{2\pi}\int_0^{2\pi/\omega} \frac{1+\cos 2\omega t}{2}\,\mathrm{d}t \qquad \text{using } \cos^2 x = \tfrac{1}{2}(1+\cos 2x) \tag{27.4.4}$$

$$= \frac{\omega}{2\pi}\left[\frac{t}{2} + \frac{\sin 2\omega t}{4\omega}\right]_0^{2\pi/\omega} = \frac{\omega}{2\pi}\cdot\frac{\pi}{\omega} = \frac{1}{2}. \tag{27.4.5}$$

Similarly

$$\overline{\sin\omega t\cos\omega t} = \overline{\tfrac{1}{2}\sin 2\omega t} = 0. \tag{27.4.6}$$

These three averages are used repeatedly from Eq. (SM-25) onwards, when the fast oscillation is averaged out of the torque; they are recorded here so that the algebra of Volume IV can proceed without interruption.

27.5 Physical Interpretation, and Why Linear Polarisation Is the Crux

Consider what a linearly polarised field does to a grain in a fluid *at rest*. During the first half-cycle the field points along $+x$ and the moment swings towards it; during the second half-cycle it points along $-x$ and the moment swings back. The grain therefore performs a rotational *oscillation*, not a rotation. Crucially, for every grain that happens to swing clockwise there is another, symmetrically placed, that swings counterclockwise: the field distinguishes the axis x, but not the sense of rotation about z. Averaged over the many grains in a volume element, the mean angular velocity vanishes,

$$\boldsymbol{\Theta} = 0 \qquad \text{in a quiescent fluid.} \tag{27.5.1}$$

Both papers stress this in their abstracts: "one half of the particles rotate clockwise and the other half counterclockwise. Hence the macroscopic angular velocity of the particles equals zero."

Now switch on a shear. The vorticity $\boldsymbol{\Omega}$ breaks the symmetry between the two senses — it is a pseudo-vector that distinguishes clockwise from counterclockwise — and the degeneracy is lifted. The mean particle spin becomes nonzero, and *it need not have the same sign as it would in a stationary field.* At high frequency it turns out to exceed Ω, so that $\boldsymbol{\Omega} - \boldsymbol{\Theta} < 0$ and, by the master formula (14.4.9), $\Delta\eta < 0$.

That, in one paragraph, is the entire physical mechanism of the papers. The next two volumes turn it into algebra.

Remark 27.5.1 (Circular polarisation is different). If instead $\boldsymbol{H} = (H_0 \cos\omega t,\ H_0 \sin\omega t,\ 0)$, the field itself would single out a sense of rotation, and the grains would be driven around even in a fluid at rest — a magnetic stirrer. That situation, mentioned in the opening sentence of Bacri *et al.*, produces a nonzero $\boldsymbol{\Theta}$ without any flow at all. The linearly polarised case is subtler and more interesting: nothing happens *until* there is a shear, and then the shear is amplified.

27.6 Connection to the Next Equation

To solve the system analytically the authors first restrict to weak fields, where the Langevin function may be linearised. That restriction is (SM-15).

Chapter 28

Equation (SM-15): The Linear Magnetisation Law

$$\boldsymbol{M}_0(t) = \chi \boldsymbol{H}(t), \qquad \chi = \frac{nm^2}{3kT}. \tag{SM-15}$$

28.1 Context

Section II of Shliomis & Morozov bears the title "Rotational viscosity under the low-amplitude alternating field". Equation (SM-15) is what "low-amplitude" means: the field is weak enough that the Langevin function may be replaced by its first term. The authors say so explicitly: this assumption "permits to evade the difficulties which are connected with the nonlinearity of the Langevin function (10)". With it, the whole of Sec. II becomes analytically soluble, and its result is the celebrated Eq. (SM-28).

28.2 Symbols

χ is the initial (linear) magnetic susceptibility: dimensionless in Gaussian units, since M and H are both measured in G/Oe. It is a material constant at fixed temperature, proportional to the concentration and to the *square* of the moment.

28.3 Assumptions

Assumption 28.3.1 (Weak field)**.** $mH_0 \ll kT$, i.e. $\xi_0 \ll 1$.

What this means numerically. With the calibration of Sec. 23.5, $\xi_0 = 5.25 \times 10^{-3} H_0(\mathrm{Oe})$, so $\xi_0 \ll 1$ requires

$$H_0 \ll \frac{1}{5.25 \times 10^{-3}} \approx 190 \text{ Oe}. \tag{28.3.1}$$

The experiments of Bacri *et al.* run to 2000 Oe, i.e. $\xi_0 \approx 10$. *Section II of the theory therefore does not apply quantitatively to the experiment.* This is not a criticism of either paper — it is exactly why Sec. III of Shliomis & Morozov exists, and why Bacri *et al.* compare their data with the finite-amplitude results of Figs. 4 and 5 rather than with Eq. (SM-28). The value of Sec. II is that it is exactly soluble and reveals the mechanism.

28.4 Derivation

Already obtained as Eq. (10.4.2). Repeating only the logic, not the algebra: the small-ξ series (10.3.8) gives $L(\xi) \simeq \xi/3$, so

$$\boldsymbol{M}_0 = nmL(\xi)\hat{\boldsymbol{h}} \simeq nm\frac{\xi}{3}\hat{\boldsymbol{h}} = \frac{nm}{3}\cdot\frac{mH}{kT}\hat{\boldsymbol{h}} = \frac{nm^2}{3kT}\boldsymbol{H}, \tag{28.4.1}$$

where $H\hat{\boldsymbol{h}} = \boldsymbol{H}$ was used in the last step. Hence $\chi = nm^2/3kT$. ■

Accuracy. The next term of the series (10.3.8) is $-\xi^3/45$, so the relative error of the linearisation is

$$\frac{\xi^3/45}{\xi/3} = \frac{\xi^2}{15}. \tag{28.4.2}$$

At $\xi = 0.5$ this is 1.7%; at $\xi = 1$ it is 6.7%; at $\xi = 3$ it is 60% and the expansion has failed. The linear theory is thus quantitatively reliable up to about $\xi \simeq 1$, i.e. $H_0 \simeq 200$ Oe for this fluid.

28.5 Physical Interpretation

Why $\chi \propto m^2$. Two factors of the moment appear for the same reason two factors of the field appeared in (SM-3): one because a large moment couples strongly to the field and is therefore well aligned, and one because a large moment contributes strongly to the magnetisation once aligned. The consequence is dramatic in practice — since $m = M_s V \propto d^3$, we have $\chi \propto nd^6$, so a small population of large grains dominates the susceptibility of a polydisperse ferrofluid. It is the same d^6 sensitivity that made the measured τ_B of Sec. 17.5 so much larger than the nominal one.

Why $\chi \propto 1/T$. This is Curie's law, and it says that the alignment achieved by a given field is inversely proportional to the strength of the thermal disorder opposing it.

The estimate for the experimental fluid. With $n = \varphi/V = 0.20/5.24\times10^{-19} = 3.8\times10^{17}$ cm^{-3} and $m = 2.10\times10^{-16}$ erg G^{-1}:

$$\chi = \frac{nm^2}{3kT} = \frac{3.8\times10^{17}\times\left(2.10\times10^{-16}\right)^2}{3\times4.14\times10^{-14}} = \frac{1.68\times10^{-14}}{1.24\times10^{-13}} = 0.14. \tag{28.5.1}$$

A susceptibility of order 0.1 confirms, incidentally, the claim of Sec. 22.6 that demagnetising corrections are modest even before the favourable geometry is taken into account: the internal field differs from the applied one by a factor $1/(1+4\pi\chi N)$ with N the demagnetising factor, and $N = 0$ for a long cylinder magnetised axially.

28.6 Connection to the Next Equation

Everything is now in place: a closed system (SM-10)–(SM-13), a specified field (SM-14), and a linear constitutive law (SM-15). The paper proceeds to solve it, and does so twice — first by decomposing the linearly polarised field into two counter-rotating circular ones, which makes

the mechanism visible, and then by a shorter but more formal direct route. The first solution begins at Eq. (SM-16),

$$\boldsymbol{H}_{\pm} = \big(H_0 \cos\omega t,\ \pm H_0 \sin\omega t,\ 0\big), \qquad \boldsymbol{H} = \tfrac{1}{2}\big(\boldsymbol{H}_{+} + \boldsymbol{H}_{-}\big),$$

and that is where Volume IV begins.

Chapter 29

Status and Road Map

29.1 What has been established in Volume III

Equation	Statement	Where derived
(19.4.1)	$\nabla(\boldsymbol{A}\cdot\boldsymbol{B}) = (\boldsymbol{A}\cdot\nabla)\boldsymbol{B} + \boldsymbol{A}\times(\nabla\times\boldsymbol{B})$, $\boldsymbol{A}$ constant	Sec. 19.4
(19.4.6)	Kelvin force $\boldsymbol{f}_{\rm mag} = (\boldsymbol{M}\cdot\nabla)\boldsymbol{H}$	Sec. 19.4
(SM-6)	fluid motion with spin coupling $\frac{I}{2\tau_s}\nabla\times(\boldsymbol{\Theta}-\boldsymbol{\Omega})$	Ch. 19
(SM-7)	$\mathrm{d}\boldsymbol{M}/\mathrm{d}t = \boldsymbol{\Theta}\times\boldsymbol{M} - (\boldsymbol{M}-\boldsymbol{M}_0)/\tau_B$	Ch. 20 (derived Sec. 13.2)
(SM-8)	$I\,\mathrm{d}\boldsymbol{\Theta}/\mathrm{d}t = \boldsymbol{M}\times\boldsymbol{H} - (I/\tau_s)(\boldsymbol{\Theta}-\boldsymbol{\Omega})$	Ch. 21
(SM-9)	$\nabla\cdot\boldsymbol{v} = 0$; $\nabla\times\boldsymbol{H} = 0$; $\nabla\cdot(\boldsymbol{H}+4\pi\boldsymbol{M}) = 0$	Ch. 22
(SM-10)	$\boldsymbol{M}_0 = nmL(\xi)\boldsymbol{H}/H$; $\xi = 5.1\times10^{-3}H(\mathrm{Oe})$ for the experimental fluid	Ch. 23 (derived Ch. 10)
(SM-11)/(B-1)	$\boldsymbol{\Theta} = \boldsymbol{\Omega} + (\tau_s/I)(\boldsymbol{M}\times\boldsymbol{H})$; justified by $\omega\tau_s \sim 10^{-7}$	Ch. 24
(SM-12)/(B-3)	momentum equation with $\frac{1}{2}\nabla\times(\boldsymbol{M}\times\boldsymbol{H})$	Ch. 25
(SM-13)/(B-4)	nonlinear magnetisation equation	Ch. 26
(SM-14)	$\boldsymbol{H} = (H_0\cos\omega t, 0, 0)$; period averages of cos, $\cos^2$, sin cos	Ch. 27
(SM-15)	$\boldsymbol{M}_0 = \chi\boldsymbol{H}$, $\chi = nm^2/3kT = 0.14$; valid for $H_0 \ll 190$ Oe	Ch. 28

29.2 Cumulative progress

Of the 43 equations of Shliomis & Morozov, numbers 1–15 are now complete; of the 10 equations of Bacri *et al.*, numbers 1, 2, 3, 4 and 10 are complete. What remains:

- **Volume IV** — Eqs. (SM-16)–(SM-24): the decomposition of the linearly polarised field into two counter-rotating circular components; the stationary solution for each; the phase lag α; the appearance of the transverse magnetisation M_y when a vortex breaks the degeneracy; and the relation $\tan\alpha = \omega\tau_B$.

- **Volume V** — Eqs. (SM-25)–(SM-34): the averaged magnetic torque, the separation of

fast and slow time scales, and the central result

$$\Delta\eta = \frac{1}{4}\eta\varphi\xi^2\,\frac{1-\omega^2\tau_B^2}{\left(1+\omega^2\tau_B^2\right)^2},$$

whose sign change at $\omega\tau_B = 1$ *is* the negative-viscosity effect; together with the second variant of the solution and the formula $\boldsymbol{\Theta} = \boldsymbol{\Omega}\big[1 - \frac{\xi^2}{6}\frac{1-\xi^2\tau_B^2}{(1+\omega^2\tau_B^2)^2}\big]$.

- **Volume VI** — Eqs. (SM-35)–(SM-43): arbitrary field amplitude; the functions $F(t)$, $G(t)$, $\gamma(t)$; the rectangular-modulation solution; and the numerical results of Figs. 3–5.
- **Volume VII** — Eqs. (B-5)–(B-9): the effective-field method of Martsenyuk, Raikher and Shliomis, derived from the Smoluchowski equation (11.6.1); the function $g(\xi_0, \omega\tau_B)$; and the comparison with the experimental isolines.

End of Volume III. Volume IV begins with Eq. (SM-16).

THE NEGATIVE-VISCOSITY EFFECT IN FERROFLUIDS

A Complete First-Principles Reconstruction of Every Equation

VOLUME IV

The Circular-Polarisation Decomposition
and the Birth of the Transverse Magnetisation:
Equations (SM-16)–(SM-24)

Where Volume III ended. Volume III assembled the complete ferrohydrodynamic system (SM-6)–(SM-10), eliminated the particle spin to obtain the reduced pair (SM-12)–(SM-13), specified the applied field $\boldsymbol{H} = (H_0 \cos \omega t, 0, 0)$ in (SM-14), and adopted the linear magnetisation law $\boldsymbol{M}_0 = \chi \boldsymbol{H}$, $\chi = nm^2/3kT$, in (SM-15). It ended by quoting the first line of the solution to come.

Volume IV carries out that solution. The strategy of Shliomis & Morozov's "first variant" is to write the linearly polarised field as the sum of two counter-rotating circular fields, solve for each separately — a problem in which nothing depends on time in the co-rotating frame — and then superpose. The payoff is Eq. (SM-23): a transverse magnetisation *proportional to the vorticity*, which is the object that will produce the torque, the slip and finally the negative viscosity. Chapter numbers resume at 30, page numbers at 113.

Part IX

The Circular-Polarisation Solution

Chapter 30

Equation (SM-16): Decomposition into Two Rotating Fields

$$\boldsymbol{H}_{\pm} = \big(H_0 \cos\omega t,\ \pm H_0 \sin\omega t,\ 0\big), \qquad \boldsymbol{H} = \tfrac{1}{2}\big(\boldsymbol{H}_{+} + \boldsymbol{H}_{-}\big). \tag{SM-16}$$

30.1 Context

The authors introduce their Sec. II A with the words: "In the first the linear-polarized field (14) is presented as a sum of two fields, which are circularly polarized opposite each other. This approach is rather visual and in our opinion it makes clear the physical nature of the problem." That is exactly right, and it is why we follow it before the shorter formal route of Sec. II B (which will be Volume V, Chapter 44).

The reason the decomposition helps is this. In the linearly polarised field of (SM-14) *everything* depends on time: the field grows, dies, reverses. In a circularly polarised field nothing does — the field merely rotates, and in a frame turning with it the problem is completely static. A static problem is one we can solve exactly. Superposing the two solutions then reconstructs the oscillating case.

30.2 Symbols

Symbol	Meaning	Type	Units
$\boldsymbol{H}_{+}$	left-hand (counterclockwise) circularly polarised field, rotating from $+x$ towards $+y$	vector field	Oe
$\boldsymbol{H}_{-}$	right-hand (clockwise) circularly polarised field	vector field	Oe
H_0	amplitude of each rotating field, equal to the amplitude of the linear field	scalar	Oe
ω	angular frequency of rotation	scalar	s^{-1}

Note that $|\boldsymbol{H}_{\pm}| = H_0$ at *all* times:

$$|\boldsymbol{H}_{\pm}|^2 = H_0^2 \cos^2\omega t + H_0^2 \sin^2\omega t = H_0^2, \tag{30.2.1}$$

by the Pythagorean identity. A circularly polarised field has constant magnitude and rotating direction; a linearly polarised one has fixed direction and oscillating magnitude. They are

opposite in every respect, which is why the decomposition is illuminating rather than merely algebraic.

30.3 Assumptions

None beyond those of (SM-14): spatial uniformity and harmonic time dependence. The decomposition itself is an identity, not an approximation.

Remark 30.3.1 (But the *use* of the decomposition does assume linearity). Splitting the field is exact. Splitting the *response* — writing $\boldsymbol{M} = \frac{1}{2}(\boldsymbol{M}_+ + \boldsymbol{M}_-)$ — requires that the magnetisation depend linearly on the field, i.e. Assumption 32.3.1 of Chapter 32. The authors return to this point explicitly beneath their Eq. (SM-24), and we discuss it in Sec. 37.6. Keep it in mind: it is the single subtlety of the whole chapter.

30.4 Verification of the Identity

Step 1: add the two fields componentwise.

$$\boldsymbol{H}_+ + \boldsymbol{H}_- = \big(H_0\cos\omega t,\ +H_0\sin\omega t,\ 0\big) + \big(H_0\cos\omega t,\ -H_0\sin\omega t,\ 0\big) \tag{30.4.1}$$

$$= \big(H_0\cos\omega t + H_0\cos\omega t,\ \ H_0\sin\omega t - H_0\sin\omega t,\ \ 0\big) \tag{30.4.2}$$

$$= \big(2H_0\cos\omega t,\ 0,\ 0\big). \tag{30.4.3}$$

Step 2: halve.

$$\tfrac{1}{2}\big(\boldsymbol{H}_+ + \boldsymbol{H}_-\big) = \big(H_0\cos\omega t,\ 0,\ 0\big) = \boldsymbol{H}, \tag{30.4.4}$$

which is exactly the field (SM-14). ■

30.5 Physical Interpretation

Picture two vectors of equal length H_0, both starting along $+x$ at $t = 0$, one turning counterclockwise and one clockwise at the same rate ω. At every instant their y components are equal and opposite and cancel, while their x components are equal and add. The sum therefore stays on the x axis and its length oscillates as $2H_0\cos\omega t$. This is the standard fact that a linear oscillation is the superposition of two opposite circular motions — the same statement that in optics decomposes plane-polarised light into left- and right-circular components.

Why this is the physically right decomposition here. The suspended grains care about *rotation*. A circularly polarised field drags the moments around continuously and therefore drives a steady, unidirectional particle spin. The + field drives spin along $+z$; the − field drives spin along $-z$. In the linearly polarised sum the two drives are exactly equal and opposite — which is precisely the statement, made qualitatively in Sec. 27.5, that half the grains swing one way and half the other, so that $\boldsymbol{\Theta} = 0$ in a quiescent fluid. What the decomposition buys us is a way to see *quantitatively* how a shear unbalances the two.

30.6 Connection to the Next Equation

We must now solve the ferrohydrodynamic system for one circular field. The form of that solution is (SM-17).

Chapter 31

Equation (SM-17): The Steadily Rotating Solution

$$\boldsymbol{M}_{\pm} = \big[M\cos(\omega t-\alpha),\ \pm M\sin(\omega t-\alpha),\ 0\big], \qquad \boldsymbol{\Theta}_{\pm} = \big(0,0,\pm\Theta\big). \tag{SM-17}$$

31.1 Context

This is the *ansatz*: the form of the solution, to be justified now and verified by substitution in Chapter 32. The authors say: "For each of these fields taken separately, Eqs. (6)–(8) admit a solution, in which the fluid is quiescent ($\boldsymbol{\Omega} = 0$) and the magnetization rotates with an angular frequency of the field, lagging behind it at some angle α. The ferroparticles rotate with a stationary angular velocity, which is oriented perpendicularly to the plane of the field rotation."

31.2 Symbols

Symbol	Meaning	Type	Units
M	magnitude of the rotating magnetisation, constant in time	scalar	G
α	phase lag of the magnetisation behind the field	scalar, $0 \le \alpha < \pi/2$	rad
Θ	magnitude of the steady particle spin	scalar	s^{-1}
$\omega t - \alpha$	instantaneous angle of $\boldsymbol{M}$ in the xy plane	scalar	rad

31.3 Assumptions

1. **The fluid is at rest**, $\boldsymbol{\Omega} = 0$. This chapter deals with the unperturbed problem; the vortex is switched on in Chapter 34.

2. **Steady state in the rotating frame.** All transients have decayed; this requires only that we wait a few τ_B after switching on, which is $\sim$ms.

3. **The field amplitude is weak**, so that $\boldsymbol{M}_0 = \chi \boldsymbol{H}$ (SM-15) holds and the response is linear.

31.4 Why the Solution Must Have This Form

The argument is a symmetry argument and is worth spelling out, because it is what makes the ansatz inevitable rather than lucky.

Step 1: the problem has a symmetry. Consider the + field. Rotate the entire problem about the z axis by an angle δ and simultaneously shift time by δ/ω. The field is unchanged:

$$\boldsymbol{H}_+\big(t+\delta/\omega\big) = H_0\big(\cos(\omega t+\delta),\, \sin(\omega t+\delta),\, 0\big) = \mathsf{R}_z(\delta)\,\boldsymbol{H}_+(t), \tag{31.4.1}$$

where $\mathsf{R}_z(\delta)$ is the rotation matrix. All the other ingredients of the problem — the isotropic carrier liquid, the spherical grains, the temperature — are unchanged by a rotation. So the problem is invariant under the combined operation.

Step 2: therefore the solution is too. A unique steady solution of an invariant problem must be invariant. Hence $\boldsymbol{M}(t+\delta/\omega) = \mathsf{R}_z(\delta)\boldsymbol{M}(t)$ for every δ, which says precisely that $\boldsymbol{M}$ rotates rigidly in the xy plane at angular velocity ω with a fixed magnitude. Writing its angle as $\omega t - \alpha$ gives the first line of (SM-17). Because $\boldsymbol{H}$ began on the x axis, α is the amount by which $\boldsymbol{M}$ trails it.

Step 3: the direction of $\boldsymbol{\Theta}$. Likewise $\boldsymbol{\Theta}$ must be invariant under R_z, and the only vectors invariant under all rotations about z are those along z. Hence $\boldsymbol{\Theta} = (0, 0, \Theta)$, with Θ a constant. This is the paper's "oriented perpendicularly to the plane of the field rotation".

Step 4: the sign for the two fields. The − field is the mirror image of the + field in the xz plane ($y \to -y$). The magnetisation, being a sum of moments carried by the grains, transforms the same way, giving the $\pm$ in front of $M\sin(\omega t-\alpha)$. The spin $\boldsymbol{\Theta}$ is a pseudo-vector along z and reverses under this mirror, giving $\pm\Theta$. The magnitudes M, α, Θ are the *same* for both fields — a fact we shall use in Chapter 33 and lose in Chapter 34, where the vortex destroys the mirror symmetry.

31.5 Physical Interpretation of the Lag

Why must $\boldsymbol{M}$ lag rather than lead? Because relaxation is dissipative. The magnetisation is always trying to reach $\boldsymbol{M}_0$, which points along the present field; but reaching it takes a time τ_B, during which the field has moved on by $\omega\tau_B$ radians. The magnetisation is therefore permanently behind by an angle of that order, and we shall find in Chapter 32 that the relation is exactly $\tan\alpha = \omega\tau_B$ when the back-reaction is neglected.

This lag is the whole story: a magnetisation parallel to the field exerts no torque, by (1.5.7). Only because $\alpha \neq 0$ is there a torque $\boldsymbol{M}\times\boldsymbol{H}$ at all — and that torque is what spins the particles.

31.6 Connection to the Next Equation

Substituting the ansatz into the equations of motion turns three vector equations into three scalar relations among M, α and Θ. Those are (SM-18).

Chapter 32

Equation (SM-18): The Three Relations

$$M = \chi H_0 \cos\alpha, \qquad \Theta = \big(\tau_s/I\big) M H_0 \sin\alpha, \qquad \tan\alpha = \big(\omega - \Theta\big)\tau_B. \tag{SM-18}$$

32.1 Context

The paper says simply: "Substituting (15)–(17) into Eqs. (7) and (8), we obtain (18)." We now perform that substitution completely.

32.2 Symbols

All defined. $\chi = nm^2/3kT$ is the susceptibility (SM-15); $\tau_s/I = 1/6\eta\varphi$ by Eq. (7.5.12).

32.3 Assumptions

Assumption 32.3.1 (Weak field)**.** $\xi_0 = mH_0/kT \ll 1$, so that $\boldsymbol{M}_0 = \chi \boldsymbol{H}$ (SM-15). Without this the Langevin nonlinearity would forbid the superposition of Chapter 37.

Plus Assumptions 1–2 of Chapter 31 (quiescent fluid, steady state).

32.4 Mathematical Background: the co-rotating basis

Because everything rotates at the rate ω, it pays to introduce two unit vectors that rotate *with the magnetisation*. Write

$$\phi \equiv \omega t - \alpha \tag{32.4.1}$$

for the instantaneous angle of $\boldsymbol{M}$, and define

$$\hat{\boldsymbol{e}}_r \equiv \big(\cos\phi,\ \sin\phi,\ 0\big), \qquad \hat{\boldsymbol{e}}_t \equiv \big(-\sin\phi,\ \cos\phi,\ 0\big). \tag{32.4.2}$$

The subscripts stand for "radial" (along $\boldsymbol{M}$) and "tangential" (perpendicular to it, in the direction of increasing ϕ).

Property 1: they are orthonormal.

$$\hat{\boldsymbol{e}}_r \cdot \hat{\boldsymbol{e}}_r = \cos^2\phi + \sin^2\phi = 1, \tag{32.4.3}$$

$$\hat{\boldsymbol{e}}_t \cdot \hat{\boldsymbol{e}}_t = \sin^2\phi + \cos^2\phi = 1, \tag{32.4.4}$$

$$\hat{\boldsymbol{e}}_r \cdot \hat{\boldsymbol{e}}_t = -\cos\phi\sin\phi + \sin\phi\cos\phi = 0. \tag{32.4.5}$$

Property 2: their time derivatives. Since $\mathrm{d}\phi/\mathrm{d}t = \omega$,

$$\frac{\mathrm{d}\hat{\boldsymbol{e}}_r}{\mathrm{d}t} = \big(-\sin\phi,\ \cos\phi,\ 0\big)\,\omega = \omega\,\hat{\boldsymbol{e}}_t, \tag{32.4.6}$$

$$\frac{\mathrm{d}\hat{\boldsymbol{e}}_t}{\mathrm{d}t} = \big(-\cos\phi,\ -\sin\phi,\ 0\big)\,\omega = -\omega\,\hat{\boldsymbol{e}}_r. \tag{32.4.7}$$

Property 3: cross products with $\hat{\boldsymbol{z}}$.

$$\hat{\boldsymbol{z}} \times \hat{\boldsymbol{e}}_r = \hat{\boldsymbol{e}}_t, \qquad \hat{\boldsymbol{z}} \times \hat{\boldsymbol{e}}_t = -\hat{\boldsymbol{e}}_r, \qquad \hat{\boldsymbol{e}}_r \times \hat{\boldsymbol{e}}_t = \hat{\boldsymbol{z}}. \tag{32.4.8}$$

The first follows from $\hat{\boldsymbol{z}} \times \hat{\boldsymbol{x}} = \hat{\boldsymbol{y}}$ and $\hat{\boldsymbol{z}} \times \hat{\boldsymbol{y}} = -\hat{\boldsymbol{x}}$: $\hat{\boldsymbol{z}} \times (\cos\phi, \sin\phi, 0) = (-\sin\phi, \cos\phi, 0) = \hat{\boldsymbol{e}}_t$. The others are obtained the same way.

Property 4: the field in this basis. The + field is $\boldsymbol{H}_+ = H_0(\cos\omega t, \sin\omega t, 0)$, i.e. a vector of length H_0 at angle $\omega t = \phi + \alpha$. Using the addition formulas $\cos(\phi+\alpha) = \cos\phi\cos\alpha - \sin\phi\sin\alpha$ and $\sin(\phi+\alpha) = \sin\phi\cos\alpha + \cos\phi\sin\alpha$,

$$\boldsymbol{H}_+ = H_0\big(\cos(\phi+\alpha),\ \sin(\phi+\alpha),\ 0\big) \tag{32.4.9}$$

$$= H_0\Big[\cos\alpha\,\big(\cos\phi, \sin\phi, 0\big) + \sin\alpha\,\big(-\sin\phi, \cos\phi, 0\big)\Big] \tag{32.4.10}$$

$$= H_0\big(\cos\alpha\,\hat{\boldsymbol{e}}_r + \sin\alpha\,\hat{\boldsymbol{e}}_t\big). \tag{32.4.11}$$

The geometric content: the field is ahead of the magnetisation by the angle α, so it has a component $H_0\cos\alpha$ along $\boldsymbol{M}$ and a component $H_0\sin\alpha$ perpendicular to it, on the leading side.

32.5 Step-by-Step Derivation

We work with the + field throughout; the − case follows by the mirror symmetry of Sec. 31.4, Step 4.

Step 1: write the ansatz in the rotating basis. By (SM-17) and (32.4.2),

$$\boldsymbol{M} = M\,\hat{\boldsymbol{e}}_r, \qquad \boldsymbol{\Theta} = \Theta\,\hat{\boldsymbol{z}}. \tag{32.5.1}$$

Step 2: the left-hand side of (SM-7). M is constant, so only the basis vector varies:

$$\frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t} = M\frac{\mathrm{d}\hat{\boldsymbol{e}}_r}{\mathrm{d}t} = M\omega\,\hat{\boldsymbol{e}}_t, \tag{32.5.2}$$

using (32.4.6).

Step 3: the convective term. With (32.4.8),

$$\boldsymbol{\Theta} \times \boldsymbol{M} = \Theta M\big(\hat{\boldsymbol{z}} \times \hat{\boldsymbol{e}}_r\big) = \Theta M\, \hat{\boldsymbol{e}}_t. \tag{32.5.3}$$

Step 4: the relaxation term. By (SM-15), $\boldsymbol{M}_0 = \chi \boldsymbol{H}_+$, so with (32.4.11)

$$-\frac{1}{\tau_B}\big(\boldsymbol{M} - \boldsymbol{M}_0\big) = -\frac{1}{\tau_B}\Big[M\hat{\boldsymbol{e}}_r - \chi H_0 \cos\alpha\, \hat{\boldsymbol{e}}_r - \chi H_0 \sin\alpha\, \hat{\boldsymbol{e}}_t\Big]. \tag{32.5.4}$$

Step 5: assemble (SM-7). Setting (32.5.2) equal to the sum of (32.5.3) and (32.5.4):

$$M\omega\, \hat{\boldsymbol{e}}_t = \Theta M\, \hat{\boldsymbol{e}}_t - \frac{M - \chi H_0 \cos\alpha}{\tau_B}\, \hat{\boldsymbol{e}}_r + \frac{\chi H_0 \sin\alpha}{\tau_B}\, \hat{\boldsymbol{e}}_t. \tag{32.5.5}$$

Step 6: separate the components. Because $\hat{\boldsymbol{e}}_r$ and $\hat{\boldsymbol{e}}_t$ are independent (orthonormal, Property 1), the coefficients of each must match separately. This is the crucial move and it is legitimate exactly because the basis is orthonormal: dotting (32.5.5) with $\hat{\boldsymbol{e}}_r$ and with $\hat{\boldsymbol{e}}_t$ gives two independent scalar equations.

Radial ($\hat{\boldsymbol{e}}_r$) component:

$$0 = -\frac{M - \chi H_0 \cos\alpha}{\tau_B} \qquad \Longrightarrow \qquad \boxed{M = \chi H_0 \cos\alpha} \tag{32.5.6}$$

which is the first relation of (SM-18). ✓

Tangential ($\hat{\boldsymbol{e}}_t$) component:

$$M\omega = \Theta M + \frac{\chi H_0 \sin\alpha}{\tau_B}. \tag{32.5.7}$$

Step 7: extract the phase relation. Substitute $\chi H_0 = M/\cos\alpha$ from (32.5.6) into (32.5.7):

$$M\omega = \Theta M + \frac{M}{\tau_B}\frac{\sin\alpha}{\cos\alpha} \tag{32.5.8}$$

$$= \Theta M + \frac{M \tan\alpha}{\tau_B}. \tag{32.5.9}$$

Divide throughout by M (nonzero) and rearrange:

$$\omega - \Theta = \frac{\tan\alpha}{\tau_B}, \tag{32.5.10}$$

$$\boxed{\tan\alpha = \big(\omega - \Theta\big)\tau_B} \tag{32.5.11}$$

which is the third relation of (SM-18). ✓

Step 8: the particle spin, from (SM-11). With the fluid at rest, $\boldsymbol{\Omega} = 0$, Eq. (SM-11) gives $\boldsymbol{\Theta} = (\tau_s/I)(\boldsymbol{M} \times \boldsymbol{H})$. Compute the torque with (32.5.1), (32.4.11) and (32.4.8):

$$\boldsymbol{M} \times \boldsymbol{H}_+ = M\hat{\boldsymbol{e}}_r \times H_0\big(\cos\alpha\, \hat{\boldsymbol{e}}_r + \sin\alpha\, \hat{\boldsymbol{e}}_t\big) \tag{32.5.12}$$

$$= MH_0 \cos\alpha\, \big(\hat{\boldsymbol{e}}_r \times \hat{\boldsymbol{e}}_r\big) + MH_0 \sin\alpha\, \big(\hat{\boldsymbol{e}}_r \times \hat{\boldsymbol{e}}_t\big) \tag{32.5.13}$$

$$= \boldsymbol{0} + MH_0 \sin\alpha\, \hat{\boldsymbol{z}} \tag{32.5.14}$$

$$= MH_0 \sin\alpha\, \hat{\boldsymbol{z}}. \tag{32.5.15}$$

Hence

$$\boldsymbol{\Theta} = \frac{\tau_s}{I} MH_0 \sin\alpha\, \hat{\boldsymbol{z}} \qquad \Longrightarrow \qquad \boxed{\Theta = \frac{\tau_s}{I} MH_0 \sin\alpha} \tag{32.5.16}$$

the second relation of (SM-18). ✓ ■

32.6 Physical Interpretation

The three relations of (SM-18) form a small self-consistent loop, and it is worth tracing it round once.

- (32.5.6): the magnetisation is smaller than the equilibrium value χH_0 by the factor $\cos\alpha$. *Why:* only the component of the driving field along $\boldsymbol{M}$ can sustain it; the perpendicular component is spent turning it. The bigger the lag, the weaker the magnetisation.
- (32.5.15): the torque is $MH_0 \sin\alpha$, proportional to the *perpendicular* component of the field. It vanishes if $\alpha = 0$ (perfect following, no dissipation) and is maximal at $\alpha = \pi/2$.
- (32.5.16): that torque, divided by the friction, is the spin.
- (32.5.11): and the spin feeds back on the lag. The particle, by turning, carries the magnetisation partway around, so the magnetisation only has to relax through the *relative* rotation $\omega - \Theta$, not the full ω. Hence $\tan\alpha = (\omega - \Theta)\tau_B$, not $\omega\tau_B$.

Remark 32.6.1 (The physical meaning of $\omega - \Theta$). This difference is the frequency of the field *as seen by a grain*. If a grain could be dragged around at exactly the field frequency, $\Theta = \omega$, it would see a static field, there would be no lag, no torque — and hence, by (32.5.16), no spin, contradicting the assumption. The self-consistent compromise is $0 < \Theta < \omega$, and we shall see in Chapter 33 that Θ is in fact smaller than ω by the factor $\xi^2/3$: for weak fields the grains are barely dragged at all.

32.7 Connection to the Next Equation

The three relations of (SM-18) still contain the auxiliary unknown α. The paper now eliminates it, obtaining explicit expressions for M, Θ and $\tan\alpha$ in terms of $\omega\tau_B$ and the field strength. That is (SM-19).

Chapter 33

Equation (SM-19): Eliminating the Phase Angle

$$M = \frac{\chi H_0}{\sqrt{1+\omega^2\tau_B^2}}, \qquad \Theta = \omega\,\frac{\xi^2/3}{1+\omega^2\tau_B^2}, \qquad \tan\alpha = \omega\tau_B\left(1 - \frac{\xi^2/3}{1+\omega^2\tau_B^2}\right). \tag{SM-19}$$

33.1 Context

The paper writes: "Since $\xi = mH_0/kT\sqrt{2}$ is small (we defined here the Langevin parameter over the 'acting' magnetic field $H_0/\sqrt{2}$), using the expression $I/\tau_s = 6\eta\varphi$, one can eliminate the angle α from the solution (18). Up to the ξ^2 terms we have (19)." Two things must therefore be explained before we can derive anything: what the "acting field" means, and why the small parameter is ξ^2.

33.2 Symbols: the acting field and its Langevin parameter

Definition 33.2.1 (Acting field)**.** For a field oscillating as $H_0\cos\omega t$, the *acting* (root-mean-square) field is

$$H_{\text{act}} \equiv \sqrt{\overline{H^2}} = \sqrt{H_0^2\,\overline{\cos^2\omega t}} = \sqrt{H_0^2/2} = \frac{H_0}{\sqrt{2}}, \tag{33.2.1}$$

using the period average $\overline{\cos^2\omega t} = \frac{1}{2}$ derived as Eq. (27.4.5).

Throughout Sec. II of Shliomis & Morozov — and therefore throughout this volume and the next — the symbol ξ denotes the Langevin parameter formed with this acting field:

$$\boxed{\xi \equiv \frac{mH_0}{kT\sqrt{2}} \qquad\Longleftrightarrow\qquad \xi^2 = \frac{m^2H_0^2}{2(kT)^2}} \tag{33.2.2}$$

Remark 33.2.1 (A notational trap)*.* This is *not* the same ξ as in Sec. I of the paper, where $\xi = mH/kT$ was formed with the instantaneous field, nor the same as the $\xi_0 = mH_0/kT$ of Sec. III and of Bacri *et al.* The relation is $\xi_{\text{Sec. II}} = \xi_0/\sqrt{2}$. The authors chose it so that the results of Sec. II reduce, at $\omega = 0$, to the stationary formula (SM-3) with *no* stray factors — as we verify in Sec. 33.6. In this book we keep the paper's convention, flagging it wherever confusion is possible; the master notation table of Volume I records both.

33.3 Assumptions

1. Weak field: $\xi \ll 1$, and we work to order ξ^2 inclusive.

2. Everything from Chapter 32.

Why ξ^2 and not ξ? Because Θ, by (32.5.16), is proportional to MH_0, and M is itself proportional to χH_0. Two powers of the field are unavoidable — the same parity argument as for (SM-3) in Sec. 16.1. The first correction beyond leading order is therefore $O(\xi^2)$, and that is the order at which the paper works.

33.4 Step-by-Step Derivation

Step 1: solve for $\cos\alpha$ and $\sin\alpha$ in terms of $\tan\alpha$. For any angle with $\tan\alpha = x$ and $-\pi/2 < \alpha < \pi/2$,

$$\cos\alpha = \frac{1}{\sqrt{1+x^2}}, \qquad \sin\alpha = \frac{x}{\sqrt{1+x^2}}. \tag{33.4.1}$$

Derivation: from $\sin^2 + \cos^2 = 1$, divide by $\cos^2\alpha$ to get $\tan^2\alpha + 1 = 1/\cos^2\alpha$, hence $\cos\alpha = (1+x^2)^{-1/2}$ (positive root, since α is in the first quadrant here), and then $\sin\alpha = \cos\alpha\tan\alpha = x/\sqrt{1+x^2}$.

Step 2: zeroth approximation. By (32.5.11), $\tan\alpha = (\omega - \Theta)\tau_B$. We do not yet know Θ; but (32.5.16) shows it is of order ξ^2 relative to ω (Step 4 below confirms this). To leading order, therefore,

$$\tan\alpha = \omega\tau_B + O(\xi^2). \tag{33.4.2}$$

Step 3: the magnitude M. Insert (33.4.2) into (33.4.1) and then into (32.5.6):

$$M = \chi H_0 \cos\alpha = \frac{\chi H_0}{\sqrt{1+\omega^2\tau_B^2}}, \tag{33.4.3}$$

the first relation of (SM-19). ✓ (The $O(\xi^2)$ correction to α would give an $O(\xi^2)$ correction to M, i.e. an $O(\xi^3)$ correction to the torque, beyond the order kept.)

Step 4: the spin Θ. From (32.5.16) with $\tau_s/I = 1/6\eta\varphi$:

$$\Theta = \frac{MH_0\sin\alpha}{6\eta\varphi} \tag{33.4.4}$$

$$= \frac{H_0}{6\eta\varphi}\cdot\frac{\chi H_0}{\sqrt{1+\omega^2\tau_B^2}}\cdot\frac{\omega\tau_B}{\sqrt{1+\omega^2\tau_B^2}} \qquad \text{using (33.4.3) and (33.4.1)} \tag{33.4.5}$$

$$= \frac{\chi H_0^2\,\omega\tau_B}{6\eta\varphi\left(1+\omega^2\tau_B^2\right)}. \tag{33.4.6}$$

Step 5: evaluate the prefactor. This is the step where all the physics of Volume II enters. We must simplify

$$\frac{\chi H_0^2 \tau_B}{6\eta\varphi}. \tag{33.4.7}$$

Insert $\chi = nm^2/3kT$ from (SM-15), $\tau_B = 3\eta V/kT$ from (SM-5), and $\varphi = nV$:

$$\frac{\chi H_0^2 \tau_B}{6\eta\varphi} = \frac{1}{6\eta(nV)} \cdot \frac{nm^2}{3kT} \cdot H_0^2 \cdot \frac{3\eta V}{kT} \tag{33.4.8}$$

$$= \frac{n\, m^2 H_0^2\, 3\eta V}{6\eta\, nV\, 3kT\, kT} \qquad \text{(collecting all factors)} \tag{33.4.9}$$

$$= \frac{m^2 H_0^2}{6(kT)^2} \qquad (n,\ V,\ \eta \text{ and the 3's cancel}) \tag{33.4.10}$$

$$= \frac{1}{6} \cdot \frac{m^2 H_0^2}{(kT)^2} \tag{33.4.11}$$

$$= \frac{1}{6} \cdot 2\xi^2 \qquad \text{by (33.2.2)} \tag{33.4.12}$$

$$= \frac{\xi^2}{3}. \tag{33.4.13}$$

This is the calculation that justifies the paper's definition of ξ over the acting field: with that definition, and only with it, the prefactor comes out as the clean $\xi^2/3$ printed in (SM-19).

Step 6: assemble Θ. Substituting (33.4.13) into (33.4.6):

$$\boxed{\Theta = \omega\, \frac{\xi^2/3}{1 + \omega^2\tau_B^2}} \tag{33.4.14}$$

the second relation of (SM-19). ✓ Note that this confirms, a posteriori, the ordering assumed in Step 2: $\Theta/\omega = O(\xi^2)$.

Step 7: the corrected phase. Now return to the exact relation (32.5.11) and insert (33.4.14):

$$\tan\alpha = \big(\omega - \Theta\big)\tau_B \tag{33.4.15}$$

$$= \omega\tau_B \left(1 - \frac{\Theta}{\omega}\right) \tag{33.4.16}$$

$$= \boxed{\omega\tau_B \left(1 - \frac{\xi^2/3}{1 + \omega^2\tau_B^2}\right)} \tag{33.4.17}$$

the third relation of (SM-19). ✓ ■

33.5 Consistency Checks

Zero frequency. $\omega = 0$: then $\alpha = 0$, $M = \chi H_0$ and $\Theta = 0$. A static field produces no lag, full magnetisation, and no spin — correct, since a permanent magnet does no work (Sec. 15.6).

Infinite frequency. $\omega\tau_B \to \infty$: $\alpha \to \pi/2$, $M \to \chi H_0/\omega\tau_B \to 0$, and $\Theta \to \omega\xi^2/(3\omega^2\tau_B^2) \to 0$. The moments cannot follow at all; the magnetisation dies and with it the spin. This is the behaviour the authors say "was expected" in their introduction — and it is why the negative-viscosity effect, which lives at $\omega\tau_B \gtrsim 1$ but not ∞, is a surprise.

Maximum spin. Where is Θ largest? Differentiate $\omega/(1+\omega^2\tau_B^2)$ with respect to ω:

$$\frac{\mathrm{d}}{\mathrm{d}\omega}\frac{\omega}{1+\omega^2\tau_B^2} = \frac{\big(1+\omega^2\tau_B^2\big) - \omega\big(2\omega\tau_B^2\big)}{\big(1+\omega^2\tau_B^2\big)^2} = \frac{1-\omega^2\tau_B^2}{\big(1+\omega^2\tau_B^2\big)^2}, \tag{33.5.1}$$

which vanishes at $\omega\tau_B = 1$. The particles are dragged around fastest when the field frequency matches the Brownian rate. That the same combination $1-\omega^2\tau_B^2$ will appear in the numerator of the central result (SM-28) is not a coincidence; we shall meet it again in Volume V.

Dimensions. $[\chi H_0] = \mathrm{G}$ ✓; $[\Theta] = \mathrm{s}^{-1}$ since ξ and $\omega\tau_B$ are pure numbers and ω is s^{-1} ✓.

33.6 Physical Interpretation

Equation (33.4.3) is the classical Debye relaxation law: a system with a single relaxation time τ_B responds to a drive of frequency ω with an amplitude reduced by $(1+\omega^2\tau_B^2)^{-1/2}$ and a lag $\arctan\omega\tau_B$. It is the same formula that describes a dielectric, an RC circuit, or an overdamped oscillator. The ferrofluid is, magnetically, a Debye relaxator.

Equation (33.4.14) is more specific to our problem: it says the grains are dragged around at a rate that is only a fraction $\xi^2/3$ of the field frequency. In a weak field the moments are barely aligned, so the torque is weak, so the mechanical rotation is slow. Nevertheless it is *nonzero and steady*: a rotating field is a rotary motor for ferrofluid grains. That is the single most important qualitative fact of this chapter, because in the next two chapters the two counter-rotating motors are put into competition.

33.7 Connection to the Next Equation

Everything so far assumed a quiescent fluid. The two circular components are then exact mirror images, their spins cancel, and the linear field does nothing. Now we switch on the vorticity, which breaks the mirror symmetry. That is (SM-20).

Chapter 34

Equation (SM-20): The Vortex Breaks the Symmetry

$$\begin{aligned} M_+ &= \chi H_0 \cos\alpha_+, & M_- &= \chi H_0 \cos\alpha_-, \\ \Theta_+ &= (\tau_s/I) M_+ H_0 \sin\alpha_+ + \Omega, & \Theta_- &= (\tau_s/I) M_- H_0 \sin\alpha_- - \Omega, \\ \tan\alpha_+ &= (\omega - \Theta_+)\tau_B, & \tan\alpha_- &= (\omega - \Theta_-)\tau_B. \end{aligned} \tag{SM-20}$$

34.1 Context

The paper's transition sentence is: "The degeneracy of the direction of rotation takes place only in the case of motionless fluid. Any flow with vorticity $\nabla \times \boldsymbol{v} = 2\boldsymbol{\Omega}$ ... eliminates this degeneracy because it destroys the symmetry between the 'plus' and 'minus' solutions. Let us show this."

34.2 Symbols

The subscripts $\pm$ now attach to *everything*: $M_\pm$, $\alpha_\pm$, $\Theta_\pm$ are the magnitude, lag and spin in the $+$ and $-$ circular fields respectively, and they are no longer equal. $\boldsymbol{\Omega} = (0, 0, \Omega)$ is the imposed hydrodynamic vorticity.

34.3 Assumptions

1. As before: weak field, steady state, rigid dipoles.
2. **The vorticity is along** z, i.e. perpendicular to the plane in which the field rotates. This is $\beta = \pi/2$ again, and it is the experimental geometry (Sec. 3.5).
3. The ansatz (SM-17) survives, with different constants for the two components. This requires the vorticity to be steady and uniform on the scale of a grain, which it is.

34.4 Step-by-Step Derivation

Step 1: what changes. The magnetisation equation (SM-7) is unchanged in form; it involves the particle spin $\boldsymbol{\Theta}$, which is still along z. The only equation that changes is (SM-11), which

now has $\boldsymbol{\Omega} \neq 0$:

$$\boldsymbol{\Theta} = \boldsymbol{\Omega} + \frac{\tau_s}{I}\big(\boldsymbol{M} \times \boldsymbol{H}\big). \tag{34.4.1}$$

Step 2: the + field. The ansatz is $\boldsymbol{M}_+ = M_+ \hat{\boldsymbol{e}}_r^{(+)}$ with $\hat{\boldsymbol{e}}_r^{(+)}$ at angle $\omega t - \alpha_+$, and $\boldsymbol{\Theta}_+ = \Theta_+ \hat{\boldsymbol{z}}$. Repeating Steps 1–7 of Sec. 32.5 verbatim — nothing in them used the value of $\boldsymbol{\Theta}$, only its direction — gives the same two relations with subscripts attached:

$$M_+ = \chi H_0 \cos\alpha_+, \tag{34.4.2}$$

$$\tan\alpha_+ = \big(\omega - \Theta_+\big)\tau_B. \tag{34.4.3}$$

Step 8, however, now reads

$$\boldsymbol{\Theta}_+ = \boldsymbol{\Omega} + \frac{\tau_s}{I} M_+ H_0 \sin\alpha_+ \, \hat{\boldsymbol{z}}, \tag{34.4.4}$$

whose z component is

$$\Theta_+ = \frac{\tau_s}{I} M_+ H_0 \sin\alpha_+ + \Omega. \tag{34.4.5}$$

Equations (34.4.2)–(34.4.5) are the left column of (SM-20). ✓

Step 3: the − field, and a careful word about signs. For the − field the magnetisation rotates clockwise and the spin points along $-z$. Following the paper we write the ansatz with explicit signs, as in (SM-17):

$$\boldsymbol{M}_- = \big[M_- \cos(\omega t - \alpha_-),\ -M_- \sin(\omega t - \alpha_-),\ 0\big], \qquad \boldsymbol{\Theta}_- = \big(0, 0, -\Theta_-\big), \tag{34.4.6}$$

so that M_-, α_- and Θ_- are all *positive magnitudes*. The mirror operation $y \to -y$ maps the + problem onto the − problem *except for the vorticity*, which is a pseudo-vector along z and does *not* change sign under this mirror. Concretely, applying the mirror to the + results (34.4.2)–(34.4.5) and remembering that $\boldsymbol{\Omega}$ keeps its sign:

$$M_- = \chi H_0 \cos\alpha_-, \tag{34.4.7}$$

$$-\Theta_- = -\frac{\tau_s}{I} M_- H_0 \sin\alpha_- + \Omega \qquad \Longrightarrow \qquad \Theta_- = \frac{\tau_s}{I} M_- H_0 \sin\alpha_- - \Omega, \tag{34.4.8}$$

$$\tan\alpha_- = \big(\omega - \Theta_-\big)\tau_B. \tag{34.4.9}$$

These are the right column of (SM-20). ✓ ■

The asymmetry, isolated. Compare (34.4.5) and (34.4.8): the vorticity enters with a + sign in one and a − sign in the other. *That single sign difference is the entire origin of the effect.* Everything that follows in this volume and the next is bookkeeping applied to it.

34.5 Physical Interpretation

The authors put it in one sentence: "It can be seen that the rotation of the particles is faster when the field and the fluid are rotating in the same direction."

Consider a grain sitting in a shear flow that turns it counterclockwise at rate Ω. Now apply the + (counterclockwise) circular field. Field and flow cooperate: the grain turns at $\Theta_+ = \Omega +$ (magnetic drive). Apply instead the − field: they oppose, and the grain turns clockwise at Θ_- where the magnetic drive has been *reduced* by Ω.

Now recall that the linearly polarised field is the sum of the two. In a quiescent fluid the two drives were equal and cancelled. With a vortex they no longer cancel, and the surviving imbalance is what we are after. Whether that imbalance helps or hinders the flow — whether the grains end up turning slower or faster than the liquid — is not yet obvious from (SM-20); it takes until Volume V to settle. The answer will turn out to depend on frequency, and to change sign at $\omega\tau_B = 1$.

34.6 Connection to the Next Equation

Equations (SM-20) are still implicit: $\alpha_\pm$ appears on both sides. The paper now simplifies them by keeping only the leading terms, giving (SM-21).

Chapter 35

Equation (SM-21): The Principal Terms

$$\tan\alpha_+ = \big(\omega - \Omega\big)\tau_B, \qquad \tan\alpha_- = \big(\omega + \Omega\big)\tau_B. \tag{SM-21}$$

35.1 Context

"Retaining only the principal terms in the expression for $\tan\alpha_\pm$ in (20), we have (21)." We must say precisely which terms are principal and why the others may be dropped.

35.2 Assumptions

Assumption 35.2.1 (The magnetic drive is negligible compared with the vorticity in the phase relation)**.** In $\Theta_\pm = \pm\Omega +$ (magnetic drive), the magnetic drive is of order $\omega\xi^2/3$ by (33.4.14), and it is dropped from the expression for $\tan\alpha_\pm$ — but not the Ω.

Justification. Two separate points must be made.

(i) Why the magnetic term may be dropped. We are computing, ultimately, a quantity of order ξ^2 (the rotational viscosity) linear in Ω. Keeping the magnetic contribution to $\Theta_\pm$ inside $\tan\alpha_\pm$ would generate corrections of relative order ξ^2, i.e. terms of order $\xi^4\Omega$ in the final answer — beyond the accuracy of the weak-field theory.

(ii) Why the Ω may not *be dropped, even though $\Omega\tau_B \ll 1$.* Because it is the only thing that distinguishes the two components. Setting $\Omega = 0$ here would make $\alpha_+ = \alpha_-$, the two solutions would again be mirror images, and the entire effect would vanish identically. We are computing a quantity that is *first order* in Ω and we must therefore keep Ω to first order everywhere, however small it is. This is the standard structure of linear-response theory: the small parameter is retained to exactly the order of the response being computed.

35.3 Step-by-Step Derivation

Step 1. From (34.4.3) and (34.4.5):

$$\tan\alpha_+ = \big(\omega - \Theta_+\big)\tau_B \tag{35.3.1}$$

$$= \Big(\omega - \Omega - \underbrace{\tfrac{\tau_s}{I} M_+ H_0 \sin\alpha_+}_{O(\omega\xi^2)}\Big)\tau_B \tag{35.3.2}$$

$$= \big(\omega - \Omega\big)\tau_B + O(\xi^2). \tag{35.3.3}$$

Step 2. From (34.4.9) and (34.4.8):

$$\tan\alpha_- = \big(\omega - \Theta_-\big)\tau_B \tag{35.3.4}$$

$$= \Big(\omega + \Omega - \underbrace{\tfrac{\tau_s}{I} M_- H_0 \sin\alpha_-}_{O(\omega\xi^2)}\Big)\tau_B \tag{35.3.5}$$

$$= \big(\omega + \Omega\big)\tau_B + O(\xi^2). \tag{35.3.6}$$

Dropping the indicated corrections gives (SM-21). ■

35.4 Physical Interpretation

The result has a one-line reading which is worth more than the algebra: *a grain carried round by the flow sees the two circular fields at different frequencies.* In the frame of the rotating grain, the co-rotating field appears slowed to $\omega - \Omega$, and the counter-rotating one appears speeded to $\omega + \Omega$. This is a Doppler shift for rotations. Each component then behaves as a Debye relaxator at *its own* apparent frequency, and since the Debye response is a nonlinear function of frequency, the two responses no longer cancel.

That is the mechanism in its purest form, and everything remaining is the computation of how much they fail to cancel.

35.5 Connection to the Next Equation

With the two phase angles known, we can compute the transverse magnetisation generated by each component. That is (SM-22).

Chapter 36

Equation (SM-22): The Transverse Magnetisation of Each Component

$$M_y^{(+)} = \chi H_0 \cos\alpha_+ \sin(\omega t - \alpha_+) = \frac{\chi H_0}{1+(\omega-\Omega)^2\tau_B^2}\Big[\sin\omega t - (\omega-\Omega)\tau_B\cos\omega t\Big],$$
$$M_y^{(-)} = -\chi H_0 \cos\alpha_- \sin(\omega t - \alpha_-) = -\frac{\chi H_0}{1+(\omega+\Omega)^2\tau_B^2}\Big[\sin\omega t - (\omega+\Omega)\tau_B\cos\omega t\Big]. \quad \text{(SM-22)}$$

36.1 Context

"Owing to the difference between the angles α_+ and α_-, the superposition of the fields $\boldsymbol{H}_+$ and $\boldsymbol{H}_-$ generates the nonzero y component of magnetization, contrary to the case of quiescent fluid." The y component is the object we want: it is perpendicular to the applied linear field (which lies along x), and a magnetisation perpendicular to the field is exactly what produces a torque.

36.2 Symbols

$M_y^{(\pm)}$ is the y component of the magnetisation produced by the $\pm$ circular field *acting alone*, i.e. before the superposition of Chapter 37. Note that these are not halved yet: the factor $\frac{1}{2}$ enters in (SM-23).

36.3 Mathematical Background: a useful trigonometric reduction

Result 36.3.1. *For any angle α with $\tan\alpha = x$,*

$$\cos\alpha\,\sin(\omega t - \alpha) = \frac{\sin\omega t - x\cos\omega t}{1+x^2}. \tag{36.3.1}$$

Derivation. Expand the sine of a difference, $\sin(A-B) = \sin A\cos B - \cos A\sin B$, with $A = \omega t$, $B = \alpha$:

$$\cos\alpha\,\sin(\omega t - \alpha) = \cos\alpha\Big[\sin\omega t\cos\alpha - \cos\omega t\sin\alpha\Big] \tag{36.3.2}$$

$$= \cos^2\alpha\,\sin\omega t - \cos\alpha\sin\alpha\,\cos\omega t. \tag{36.3.3}$$

Now use (33.4.1):

$$\cos^2\alpha = \frac{1}{1+x^2}, \tag{36.3.4}$$

$$\cos\alpha\sin\alpha = \frac{1}{\sqrt{1+x^2}}\cdot\frac{x}{\sqrt{1+x^2}} = \frac{x}{1+x^2}. \tag{36.3.5}$$

Substituting into (36.3.3):

$$\cos\alpha\sin(\omega t-\alpha) = \frac{\sin\omega t}{1+x^2} - \frac{x\cos\omega t}{1+x^2} = \frac{\sin\omega t - x\cos\omega t}{1+x^2}. \quad \blacksquare \tag{36.3.6}$$

36.4 Step-by-Step Derivation

Step 1: read off M_y from the ansatz. By (SM-17), the y component of the magnetisation in the $\pm$ field is

$$M_y^{(\pm)} = \pm M_\pm \sin(\omega t - \alpha_\pm). \tag{36.4.1}$$

Step 2: insert the magnitudes. By (34.4.2) and (34.4.7), $M_\pm = \chi H_0\cos\alpha_\pm$, so

$$M_y^{(\pm)} = \pm\chi H_0\cos\alpha_\pm\sin(\omega t-\alpha_\pm), \tag{36.4.2}$$

which is the middle expression of each line of (SM-22). ✓

Step 3: apply the reduction. Use (36.3.1) with $x = x_\pm \equiv \tan\alpha_\pm$, and then (SM-21), which gives $x_+ = (\omega-\Omega)\tau_B$ and $x_- = (\omega+\Omega)\tau_B$:

$$M_y^{(+)} = \chi H_0\,\frac{\sin\omega t - (\omega-\Omega)\tau_B\cos\omega t}{1+(\omega-\Omega)^2\tau_B^2}, \tag{36.4.3}$$

$$M_y^{(-)} = -\chi H_0\,\frac{\sin\omega t - (\omega+\Omega)\tau_B\cos\omega t}{1+(\omega+\Omega)^2\tau_B^2}. \tag{36.4.4}$$

These are exactly the right-hand expressions of (SM-22). ■

36.5 Physical Interpretation

Each line of (36.4.3)–(36.4.4) contains two pieces:

- a term $\propto \sin\omega t$, which is *in quadrature* with the driving field $H_0\cos\omega t$; and
- a term $\propto \cos\omega t$, which is *in phase* with it.

In the language of linear response these are the dissipative and reactive parts. The essential structure to notice is that the two components differ *only* through the replacement $\omega \to \omega \mp \Omega$ — the rotational Doppler shift of Sec. 35.4. If Ω were zero the two lines would be identical apart from the overall minus sign, and their sum would be zero: no transverse magnetisation, no torque, nothing.

36.6 Connection to the Next Equation

Adding the two and expanding to first order in Ω gives the central result of this volume, (SM-23).

Chapter 37

Equations (SM-23) and (SM-24): The Transverse Magnetisation

$$M_y = \tfrac{1}{2}\big(M_y^{(+)} + M_y^{(-)}\big) = \Omega\tau_B\,\chi H_0 \cos^2\alpha\,\cos(\omega t - 2\alpha), \tag{SM-23}$$

$$\tan\alpha = \omega\tau_B. \tag{SM-24}$$

37.1 Context

This is the destination of the whole circular-polarisation construction. The paper prefaces it with the observation that "the Brownian rotational diffusion time for the particles with diameter 10^{-6} cm is usually not higher than 10^{-5} s. Therefore the inequality $\Omega\tau_B \ll 1$ is almost always valid", which is the licence for the linearisation performed below.

37.2 Symbols

M_y is the y component of the magnetisation in the *linearly polarised* field, i.e. the physically relevant quantity. α is now the phase lag evaluated at the *unshifted* frequency, defined by (SM-24). Note that this α is the α of (SM-19) with the ξ^2 correction discarded — a point the authors flag and which we discuss in Sec. 37.6.

37.3 Assumptions

Assumption 37.3.1 (Slow flow)**.** $\Omega\tau_B \ll 1$, so that M_y may be computed to first order in Ω.

Numerically, Bacri *et al.* have $\Omega < 60\ \mathrm{s}^{-1}$ and $\tau_B = 1.6$ ms, so $\Omega\tau_B < 0.1$ (Sec. 3.5). ✓

37.4 Mathematical Background: linearisation of a function of a shifted argument

We must add two expressions that differ only by $\omega \to \omega \mp \Omega$. Define

$$F(x) \equiv \frac{\sin\omega t - x\cos\omega t}{1 + x^2}, \tag{37.4.1}$$

so that, by (36.4.3)–(36.4.4), $M_y^{(+)} = \chi H_0 F(x_+)$ and $M_y^{(-)} = -\chi H_0 F(x_-)$ with $x_\pm = (\omega \mp \Omega)\tau_B$. Writing $x_0 \equiv \omega\tau_B$ and $\delta \equiv \Omega\tau_B$, we have $x_\pm = x_0 \mp \delta$ and therefore need

$$F(x_0 - \delta) - F(x_0 + \delta). \tag{37.4.2}$$

Taylor-expand each term about x_0 — legitimate because F is smooth and $\delta \ll 1$ by Assumption 37.3.1:

$$F(x_0 - \delta) = F(x_0) - \delta F'(x_0) + \tfrac{1}{2}\delta^2 F''(x_0) - \cdots, \tag{37.4.3}$$

$$F(x_0 + \delta) = F(x_0) + \delta F'(x_0) + \tfrac{1}{2}\delta^2 F''(x_0) + \cdots. \tag{37.4.4}$$

Subtracting, the even-order terms cancel identically and

$$F(x_0 - \delta) - F(x_0 + \delta) = -2\delta F'(x_0) + O(\delta^3). \tag{37.4.5}$$

Note the structure: the difference is *odd* in δ, hence odd in Ω — as it must be, since reversing the sense of the shear must reverse the sense of the transverse magnetisation.

The derivative. Differentiate (37.4.1) by the quotient rule, $(u/v)' = (u'v - uv')/v^2$, with $u = \sin\omega t - x\cos\omega t$ (so $u' = -\cos\omega t$, the time t being held fixed) and $v = 1 + x^2$ (so $v' = 2x$):

$$F'(x) = \frac{-\cos\omega t\big(1 + x^2\big) - \big(\sin\omega t - x\cos\omega t\big)2x}{\big(1 + x^2\big)^2} \tag{37.4.6}$$

$$= \frac{-\cos\omega t - x^2\cos\omega t - 2x\sin\omega t + 2x^2\cos\omega t}{\big(1 + x^2\big)^2} \tag{37.4.7}$$

$$= \frac{\big(x^2 - 1\big)\cos\omega t - 2x\sin\omega t}{\big(1 + x^2\big)^2}. \tag{37.4.8}$$

37.5 Step-by-Step Derivation of (SM-23)

Step 1: assemble the sum.

$$M_y = \tfrac{1}{2}\big(M_y^{(+)} + M_y^{(-)}\big) \tag{37.5.1}$$

$$= \tfrac{1}{2}\chi H_0\Big[F(x_+) - F(x_-)\Big] \tag{37.5.2}$$

$$= \tfrac{1}{2}\chi H_0\Big[F(x_0 - \delta) - F(x_0 + \delta)\Big] \tag{37.5.3}$$

$$= \tfrac{1}{2}\chi H_0\big(-2\delta\big)F'(x_0) \qquad \text{by (37.4.5)} \tag{37.5.4}$$

$$= -\chi H_0\,\Omega\tau_B\,F'(x_0). \tag{37.5.5}$$

Step 2: insert the derivative. With (37.4.8) at $x = x_0$:

$$M_y = -\chi H_0\Omega\tau_B\,\frac{\big(x_0^2 - 1\big)\cos\omega t - 2x_0\sin\omega t}{\big(1 + x_0^2\big)^2} \tag{37.5.6}$$

$$= \chi H_0\Omega\tau_B\,\frac{\big(1 - x_0^2\big)\cos\omega t + 2x_0\sin\omega t}{\big(1 + x_0^2\big)^2}, \tag{37.5.7}$$

the overall minus sign having been absorbed by reversing both terms in the numerator.

Step 3: recognise a double angle. Define α by $\tan\alpha = x_0 = \omega\tau_B$, which is Eq. (SM-24). The double-angle formulas in terms of the tangent are

$$\cos 2\alpha = \cos^2\alpha - \sin^2\alpha = \frac{1}{1+x_0^2} - \frac{x_0^2}{1+x_0^2} = \frac{1-x_0^2}{1+x_0^2}, \tag{37.5.8}$$

$$\sin 2\alpha = 2\sin\alpha\cos\alpha = 2\cdot\frac{x_0}{\sqrt{1+x_0^2}}\cdot\frac{1}{\sqrt{1+x_0^2}} = \frac{2x_0}{1+x_0^2}, \tag{37.5.9}$$

both obtained from (33.4.1). Therefore

$$1 - x_0^2 = \left(1+x_0^2\right)\cos 2\alpha, \qquad 2x_0 = \left(1+x_0^2\right)\sin 2\alpha. \tag{37.5.10}$$

Step 4: substitute. Insert (37.5.10) into (37.5.7):

$$M_y = \chi H_0 \Omega\tau_B \,\frac{\left(1+x_0^2\right)\left[\cos 2\alpha\cos\omega t + \sin 2\alpha\sin\omega t\right]}{\left(1+x_0^2\right)^2} \tag{37.5.11}$$

$$= \chi H_0 \Omega\tau_B \,\frac{\cos 2\alpha\cos\omega t + \sin 2\alpha\sin\omega t}{1+x_0^2}. \tag{37.5.12}$$

Step 5: collapse the bracket. The addition formula $\cos(A-B) = \cos A\cos B + \sin A\sin B$ with $A = \omega t$ and $B = 2\alpha$ turns the numerator into $\cos(\omega t - 2\alpha)$. And by (33.4.1), $1/(1+x_0^2) = \cos^2\alpha$. Hence

$$\boxed{M_y = \Omega\tau_B\,\chi H_0\,\cos^2\alpha\,\cos\left(\omega t - 2\alpha\right)} \tag{SM-23}$$

which is Eq. (SM-23) exactly. ■

37.6 Why (SM-24) Discards the ξ^2 Term

The paper is careful here, and so must we. In (SM-19) the phase was

$$\tan\alpha = \omega\tau_B\left(1 - \frac{\xi^2/3}{1+\omega^2\tau_B^2}\right), \tag{37.6.1}$$

yet (SM-24) uses simply $\tan\alpha = \omega\tau_B$. The authors explain: "which is obtained from (19) after discarding the ξ^2 term. The last procedure is necessary because in the presence of nonlinear terms the half of the sum $\boldsymbol{M}_+ + \boldsymbol{M}_-$ is not equal to the magnetization in the linearly polarized field. The principle of superposition is valid only within the framework of the linear response."

Unpacked, the argument is this. The step $\boldsymbol{M} = \frac{1}{2}(\boldsymbol{M}_+ + \boldsymbol{M}_-)$ used in Step 1 above is *only* valid if the magnetisation responds linearly to the field. The ξ^2 term in (SM-19) is precisely a signature of nonlinearity: it arises from the back-reaction of the particle spin, which is quadratic in the field. Retaining it while simultaneously using superposition would be inconsistent — it would keep a nonlinear correction in a framework that assumes linearity. The consistent procedure at this order is to use the strictly linear response, $\tan\alpha = \omega\tau_B$, throughout. Any ξ^2 refinements belong to the finite-amplitude theory of Sec. III of the paper, which abandons superposition altogether and which we shall reach in Volume VI.

37.7 Physical Interpretation

Equation (SM-23) is the pivot of the whole subject. Read it slowly.

(i) $M_y \propto \Omega$**.** No shear, no transverse magnetisation. In a quiescent fluid the linearly polarised field produces a magnetisation strictly along x, hence no torque, hence nothing. The vorticity is not a small perturbation to the effect — it is its cause.

(ii) $M_y \propto \tau_B$**.** No memory, no transverse magnetisation either. If the moments could follow the field instantaneously they would always be along x regardless of the flow.

(iii) $M_y \propto \chi H_0$**.** One power of the field, as expected for a linear response. The second power will arrive when this magnetisation is crossed with the field to make a torque, restoring the H_0^2 (i.e. ξ^2) scaling of the final answer.

(iv) The phase 2α **is the crux.** The transverse magnetisation oscillates as $\cos(\omega t - 2\alpha)$, i.e. lagging the field $H_0 \cos \omega t$ by *twice* the single-component lag. Why twice? Because M_y arises from the *difference* between two responses whose phases are shifted in opposite directions — effectively a derivative of the response with respect to frequency, which doubles the phase. This factor of two is the entire reason the effect can change sign, as follows.

The torque will be built from the product $M_y H_x \propto \cos(\omega t - 2\alpha) \cos \omega t$, whose time average is (by the addition formula and the period averages of Sec. 27.4)

$$\overline{\cos(\omega t - 2\alpha)\cos \omega t} = \overline{\cos 2\alpha \cos^2 \omega t + \sin 2\alpha \sin \omega t \cos \omega t} = \tfrac{1}{2} \cos 2\alpha. \tag{37.7.1}$$

And $\cos 2\alpha = (1 - \omega^2 \tau_B^2)/(1 + \omega^2 \tau_B^2)$ by (37.5.8) **changes sign at** $\omega \tau_B = 1$**.** A single lag α never exceeds $\pi/2$ and $\cos \alpha$ is always positive; but 2α passes through $\pi/2$ when $\alpha = \pi/4$, i.e. when $\omega \tau_B = 1$, and beyond that point the average torque reverses.

That is the negative-viscosity effect. Everything in Volume V is the careful execution of the calculation previewed in (37.7.1).

37.8 Connection to the Next Equation

Having a transverse magnetisation, the paper computes the torque it exerts on the fluid. That is (SM-25), the first equation of Volume V.

Chapter 38

Status and Road Map

38.1 What has been established in Volume IV

Equation	Statement	Where derived
(SM-16)	$\boldsymbol{H}_\pm = (H_0\cos\omega t, \pm H_0\sin\omega t, 0)$, $\boldsymbol{H} = \frac{1}{2}(\boldsymbol{H}_+ + \boldsymbol{H}_-)$	Ch. 30
(SM-17)	rotating ansatz with lag α and axial spin $\pm\Theta$, justified by symmetry	Ch. 31
(SM-18)	$M = \chi H_0\cos\alpha$; $\Theta = (\tau_s/I)MH_0\sin\alpha$; $\tan\alpha = (\omega-\Theta)\tau_B$	Ch. 32
(33.2.2)	the "acting field" convention $\xi = mH_0/kT\sqrt{2}$	Sec. 33.2
(SM-19)	$M = \chi H_0/\sqrt{1+\omega^2\tau_B^2}$; $\Theta = \omega(\xi^2/3)/(1+\omega^2\tau_B^2)$; corrected $\tan\alpha$	Ch. 33
(SM-20)	the same relations with $\pm\Omega$ added to the spins	Ch. 34
(SM-21)	$\tan\alpha_\pm = (\omega\mp\Omega)\tau_B$: the rotational Doppler shift	Ch. 35
(SM-22)	$M_y^{(\pm)}$ for each circular component	Ch. 36
(SM-23)	$M_y = \Omega\tau_B\chi H_0\cos^2\alpha\cos(\omega t - 2\alpha)$	Ch. 37
(SM-24)	$\tan\alpha = \omega\tau_B$, and why the ξ^2 term must be dropped	Sec. 37.6

38.2 Cumulative progress

Shliomis & Morozov: Eqs. (SM-1)–(SM-24) complete. Bacri *et al.*: Eqs. (B-1)–(B-4) and (B-10) complete. Remaining:

- **Volume V** — Eqs. (SM-25)–(SM-34): the instantaneous torque $\boldsymbol{M}\times\boldsymbol{H}$; the separation of fast and slow time scales and the averaged momentum equation (SM-26); the rotational viscosity (SM-27); the central result

$$\Delta\eta = \frac{1}{4}\eta\varphi\xi^2\,\frac{1-\omega^2\tau_B^2}{\left(1+\omega^2\tau_B^2\right)^2} \tag{SM-28}$$

  with its sign change at $\omega_0\tau_B = 1$, its minimum at $\omega_m\tau_B = \sqrt{3}$ and the value $-\Delta\eta(0)/8$ there; the high-frequency asymptote (SM-29); the second, direct variant of the solution

(SM-30)–(SM-32); and the formula for the mean particle spin (SM-33) leading back to (SM-34).

- **Volume VI** — Eqs. (SM-35)–(SM-43): arbitrary field amplitude.
- **Volume VII** — Eqs. (B-5)–(B-9): the effective-field method and the comparison with experiment.

End of Volume IV. Volume V begins with Eq. (SM-25).

THE NEGATIVE-VISCOSITY EFFECT IN FERROFLUIDS

A Complete First-Principles Reconstruction of Every Equation

VOLUME V

The Averaged Torque, the Two-Scale Expansion,
and the Central Result:
Equations (SM-25)–(SM-34)

Where Volume IV ended. Volume IV decomposed the linearly polarised field into two counter-rotating circular components, solved for each exactly, showed that a vortex Doppler-shifts their apparent frequencies to $\omega \mp \Omega$, and obtained from the difference of the two responses the transverse magnetisation

$$M_y = \Omega\tau_B\,\chi H_0 \cos^2\alpha\,\cos(\omega t - 2\alpha), \qquad \tan\alpha = \omega\tau_B. \qquad \text{(SM-23, SM-24)}$$

It closed by previewing the reason the effect can change sign: the lag of M_y is 2α, and $\cos 2\alpha$ reverses at $\omega\tau_B = 1$.

Volume V executes that calculation. It computes the instantaneous torque (SM-25), separates the fast and slow time scales to obtain the averaged momentum equation (SM-26), extracts the rotational viscosity (SM-27), and arrives at the central result (SM-28) — the equation for which both papers were written. It then gives the high-frequency asymptote (SM-29), re-derives the same result by the authors' shorter second route (SM-30)–(SM-32), and closes the logical circle by computing the mean particle spin (SM-33) and recovering the master formula (SM-34) of Volume II. Chapter numbers resume at 39, page numbers at 142.

Part X

The Alternating-Field Result

Chapter 39

Equation (SM-25): The Instantaneous Torque

$$\begin{aligned}\boldsymbol{M}\times\boldsymbol{H} &= -\Omega\tau_B\chi H_0^2\cos^2\alpha\,\cos\omega t\,\cos(\omega t-2\alpha)\\ &= -2\Omega\eta\varphi\xi^2\cos^2\alpha\big(\cos^2\omega t\,\cos 2\alpha+\sin\omega t\,\cos\omega t\,\sin 2\alpha\big).\end{aligned}\tag{SM-25}$$

39.1 Context

Volume IV produced a magnetisation with a component perpendicular to the field. A magnetisation not parallel to the field exerts a torque, by Eq. (4.4.2), and that torque is the mechanical agent of the whole effect: through (SM-12) it enters the momentum equation, and through (SM-11) it determines the particle spin. Equation (SM-25) computes it.

The result is written twice on purpose. The first line is the direct product; the second re-expresses it in the units natural to a viscosity ($\eta\varphi$) and separates the two time dependences, so that the averaging of the next chapter can be performed by inspection.

39.2 Symbols

Everything has appeared. Recall in particular

$$\xi=\frac{mH_0}{kT\sqrt{2}},\qquad \chi=\frac{nm^2}{3kT},\qquad \tau_B=\frac{3\eta V}{kT},\qquad \varphi=nV,\qquad \tan\alpha=\omega\tau_B.\tag{39.2.1}$$

39.3 Assumptions

Those of Volume IV: weak field ($\xi\ll 1$), slow flow ($\Omega\tau_B\ll 1$), rigid dipoles, steady state, $\boldsymbol{H}\perp\boldsymbol{\Omega}$. Nothing new is added.

39.4 Step-by-Step Derivation

Step 1: only the z component survives. The field lies along x, $\boldsymbol{H}=(H_0\cos\omega t,0,0)$, and the magnetisation lies in the xy plane, $\boldsymbol{M}=(M_x,M_y,0)$. By the component definition of

the cross product (1.5.3),

$$\big(\boldsymbol{M}\times\boldsymbol{H}\big)_x = M_yH_z - M_zH_y = 0 - 0 = 0, \tag{39.4.1}$$

$$\big(\boldsymbol{M}\times\boldsymbol{H}\big)_y = M_zH_x - M_xH_z = 0 - 0 = 0, \tag{39.4.2}$$

$$\big(\boldsymbol{M}\times\boldsymbol{H}\big)_z = M_xH_y - M_yH_x = 0 - M_yH_x = -M_yH_x. \tag{39.4.3}$$

So the torque is purely axial — along the same axis as the vorticity, which is exactly what is needed if it is to speed up or slow down the rotation.

Notice what has just happened: the longitudinal magnetisation M_x, which is by far the larger of the two components, has dropped out completely. A magnetisation parallel to the field does no mechanical work. Only the small transverse piece $M_y \propto \Omega$ matters.

Step 2: insert M_y and H_x. With (SM-23) and (SM-14),

$$\big(\boldsymbol{M}\times\boldsymbol{H}\big)_z = -\Big[\Omega\tau_B\chi H_0\cos^2\alpha\cos(\omega t-2\alpha)\Big]\Big[H_0\cos\omega t\Big] \tag{39.4.4}$$

$$= -\Omega\tau_B\chi H_0^2\cos^2\alpha\,\cos\omega t\,\cos(\omega t-2\alpha), \tag{39.4.5}$$

which is the first line of (SM-25). ✓

Step 3: expand the phase-shifted cosine. By the addition formula $\cos(A-B) = \cos A\cos B + \sin A\sin B$ with $A=\omega t$, $B=2\alpha$,

$$\cos\omega t\,\cos(\omega t-2\alpha) = \cos\omega t\Big[\cos\omega t\cos 2\alpha + \sin\omega t\sin 2\alpha\Big] = \cos^2\omega t\,\cos 2\alpha + \sin\omega t\cos\omega t\,\sin 2\alpha. \tag{39.4.6}$$

The two terms have quite different characters, as the next chapter will show: the first has a nonzero average over a period, the second does not.

Step 4: convert the prefactor. We must show

$$\tau_B\chi H_0^2 = 2\eta\varphi\xi^2. \tag{39.4.7}$$

Insert the definitions:

$$\tau_B\chi H_0^2 = \frac{3\eta V}{kT}\cdot\frac{nm^2}{3kT}\cdot H_0^2 \tag{39.4.8}$$

$$= \frac{3\,\eta\,V\,n\,m^2H_0^2}{3\,(kT)^2} \qquad \text{(collecting factors)} \tag{39.4.9}$$

$$= \eta\,(nV)\,\frac{m^2H_0^2}{(kT)^2} \qquad \text{(the 3's cancel)} \tag{39.4.10}$$

$$= \eta\varphi\cdot\frac{m^2H_0^2}{(kT)^2} \qquad (\varphi = nV) \tag{39.4.11}$$

$$= \eta\varphi\cdot 2\xi^2, \qquad \text{since } \xi^2 = \frac{m^2H_0^2}{2(kT)^2} \tag{39.4.12}$$

which is (39.4.7). ✓

Step 5: assemble. Substituting (39.4.6) and (39.4.7) into (39.4.5):

$$\boldsymbol{M}\times\boldsymbol{H} = -2\Omega\eta\varphi\xi^2\cos^2\alpha\Big(\cos^2\omega t\,\cos 2\alpha + \sin\omega t\cos\omega t\,\sin 2\alpha\Big)\hat{\boldsymbol{z}}, \tag{SM-25}$$

which is the second line of (SM-25). ■

39.5 Physical Interpretation

The sign. The torque is proportional to $-\Omega\cos 2\alpha$ (once the oscillating factor is averaged, as in Chapter 41). For $\omega\tau_B < 1$ we have $\cos 2\alpha > 0$ and the torque *opposes* the vorticity — the familiar magnetoviscous braking. For $\omega\tau_B > 1$, $\cos 2\alpha < 0$ and the torque acts *with* the vorticity. Nothing in the algebra forbids this; the field is doing work, and it is allowed to give as well as to take.

The two time scales. The authors point out immediately that (SM-25) "contains two time scales: the 'fast' one and the 'slow' one". The fast time appears explicitly, in the products of $\cos\omega t$ and $\sin\omega t$; the slow one is hidden inside Ω, which is set by the hydrodynamics and changes only on the scale of the flow. Sorting out the two is the business of Chapter 40.

Order of magnitude. The torque is of order $\eta\varphi\xi^2\Omega$. Compared with the viscous torque density $6\eta\varphi\Omega$ that would be needed to stop the particles rotating altogether, it is smaller by $\sim \xi^2/6$ — the same small factor that appeared in Θ/ω in Eq. (SM-19), and that will appear once more in (SM-33).

39.6 Connection to the Next Equation

To use (SM-25) in the momentum equation we must first establish that the fluid responds only to the *average* of this rapidly oscillating torque. That is the content of (SM-26).

Chapter 40

Equation (SM-26): Separation of the Time Scales

$$\rho \frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = -\nabla p + \overline{(\boldsymbol{M}\cdot\nabla)\boldsymbol{H}} + \eta\Delta\boldsymbol{v} + \tfrac{1}{2}\nabla\times\overline{\big(\boldsymbol{M}\times\boldsymbol{H}\big)}. \tag{SM-26}$$

40.1 Context

The magnetic torque (39.4.5) oscillates at 2ω (through $\cos^2\omega t$ and $\sin\omega t\cos\omega t$). The flow cannot possibly follow such rapid forcing: the liquid has inertia, and momentum diffuses only as fast as viscosity allows. Equation (SM-26) is the formal statement that what the flow feels is the *period average* of the torque, denoted by an overbar. Establishing it requires a two-scale (multiple-scale) expansion, which is a standard technique of applied mathematics but must, by our rules, be derived rather than invoked.

40.2 Symbols

Symbol	Meaning	Type	Units
$\boldsymbol{v}_0,\ p_0$	slowly varying parts of the velocity and pressure	fields	$\mathrm{cm\,s^{-1}}$, $\mathrm{dyn\,cm^{-2}}$
$\boldsymbol{v}_1,\ p_1$	small, fast-oscillating additions at frequency ω	fields	as above
$\overline{X}$	average of X over one field period $2\pi/\omega$	operator	—
$\tau_h \sim l^2/\nu$	characteristic hydrodynamic time	scalar	s
l	linear length scale of the flow (here the capillary radius)	scalar	cm
$\nu = \eta/\rho$	kinematic viscosity	scalar	$\mathrm{cm^2 s^{-1}}$

Definition 40.2.1 (Period average)**.**

$$\overline{X} \equiv \frac{\omega}{2\pi}\int_{t}^{t+2\pi/\omega} X(t')\,\mathrm{d}t'. \tag{40.2.1}$$

For a quantity that varies slowly, the average returns the quantity itself; for one that oscillates at ω or any multiple of it, the average is zero. Both facts are used below.

40.3 Assumptions

Assumption 40.3.1 (Scale separation)**.** The field period is much shorter than the hydrodynamic response time:

$$\frac{2\pi}{\omega} \ll \tau_h \sim \frac{l^2}{\nu} = \frac{\rho l^2}{\eta}. \tag{40.3.1}$$

Justification, with the experimental numbers. For Bacri *et al.*, the capillary radius is $l = 0.05$ cm, the zero-field viscosity is $\eta = 0.77$ P, and the density of a 20% suspension of ferrite in water is $\rho \approx 2\ \mathrm{g\,cm^{-3}}$. Hence

$$\tau_h = \frac{\rho l^2}{\eta} = \frac{2 \times (0.05)^2}{0.77} = \frac{5.0 \times 10^{-3}}{0.77} = 6.5 \times 10^{-3}\ \mathrm{s}. \tag{40.3.2}$$

Compare with the field period $2\pi/\omega = 1/f$:

f	$1/f$	$(1/f)/\tau_h$
52 Hz	19 ms	2.9
150 Hz	6.7 ms	1.0
645 Hz	1.6 ms	0.24
1480 Hz	0.68 ms	0.10

The separation is comfortable at the high frequencies — which is precisely where the negative-viscosity effect lives — and only marginal at the lowest ones, where the effect is positive and small anyway. It is honest to record this: the averaging procedure is a good approximation in the regime the papers care about, and a borderline one at 50 Hz. (Bacri *et al.* state the same criterion in the form $2\pi/\omega \ll \rho d^2/\eta$ before their Eq. (B-9), with d the tube diameter, which is four times more generous and comfortably satisfied throughout.)

Assumption 40.3.2 (No slow part in $\boldsymbol{M}$ and $\boldsymbol{H}$)**.** The magnetisation and field are *purely* fast: they oscillate at ω with no slowly varying component. This is evident from (SM-14) and (SM-23).

40.4 Step-by-Step Derivation

Step 1: split the hydrodynamic fields. Write

$$\boldsymbol{v} = \boldsymbol{v}_0 + \boldsymbol{v}_1, \qquad p = p_0 + p_1, \tag{40.4.1}$$

where $\boldsymbol{v}_0, p_0$ change only on the slow hydrodynamic time scale and $\boldsymbol{v}_1, p_1$ are the small responses at the field frequency. That such a split is possible, and that $\boldsymbol{v}_1 \ll \boldsymbol{v}_0$, follows from Assumption 40.3.1: a forcing of period T_0 acting on a system with response time $\tau_h \gg T_0$ produces a response smaller by the factor T_0/τ_h.

Step 2: substitute into the momentum equation. Insert (40.4.1) into (SM-12) and expand the material derivative $\mathrm{d}/\mathrm{d}t = \partial_t + (\boldsymbol{v}\cdot\nabla)$, keeping terms to first order in the small quantities $\boldsymbol{v}_1, p_1$:

$$\rho\left(\frac{\partial \boldsymbol{v}_0}{\partial t} + \frac{\partial \boldsymbol{v}_1}{\partial t} + (\boldsymbol{v}_1\cdot\nabla)\boldsymbol{v}_0\right) = -\nabla(p_0+p_1) + (\boldsymbol{M}\cdot\nabla)\boldsymbol{H} + \eta\Delta(\boldsymbol{v}_0+\boldsymbol{v}_1) + \tfrac{1}{2}\nabla\times(\boldsymbol{M}\times\boldsymbol{H}), \tag{40.4.2}$$

where now $\mathrm{d}/\mathrm{d}t = \partial/\partial t + (\boldsymbol{v}_0\cdot\nabla)$ acting on $\boldsymbol{v}_0$ has been written out and the term $(\boldsymbol{v}_1\cdot\nabla)\boldsymbol{v}_1$, being second order, dropped. This is the unnumbered equation printed by the authors immediately before their (SM-26).

Step 3: average over the fast time. Apply (40.2.1) to every term. During one field period the slow fields are effectively constant, so $\overline{\boldsymbol{v}_0} = \boldsymbol{v}_0$ and $\overline{p_0} = p_0$: they pass through the average untouched. The fast fields, on the other hand, are strictly periodic with the field period and have zero mean:

$$\overline{\boldsymbol{v}_1} = \boldsymbol{0}, \qquad \overline{p_1} = 0. \tag{40.4.3}$$

Why they have zero mean: $\boldsymbol{v}_1$ is driven linearly by the oscillating part of the torque, which by (39.4.6) contains only $\cos^2\omega t$ and $\sin\omega t\cos\omega t$; subtracting their averages leaves pure $\cos 2\omega t$ and $\sin 2\omega t$, both of zero mean by Eq. (27.4.2). A linear response to a zero-mean periodic drive is itself zero-mean.

Consequently *every term linear in* $\boldsymbol{v}_1$ *or* p_1 *averages to zero*:

$$\overline{\frac{\partial \boldsymbol{v}_1}{\partial t}} = \frac{\omega}{2\pi}\Big[\boldsymbol{v}_1\Big]_t^{t+2\pi/\omega} = \boldsymbol{0} \qquad \text{(the endpoints are equal, by periodicity)}, \tag{40.4.4}$$

$$\overline{(\boldsymbol{v}_1\cdot\nabla)\boldsymbol{v}_0} = (\overline{\boldsymbol{v}_1}\cdot\nabla)\boldsymbol{v}_0 = \boldsymbol{0} \qquad (\boldsymbol{v}_0 \text{ is constant over the average}), \tag{40.4.5}$$

$$\overline{\nabla p_1} = \nabla\overline{p_1} = 0, \qquad \overline{\eta\Delta\boldsymbol{v}_1} = \eta\Delta\overline{\boldsymbol{v}_1} = \boldsymbol{0}. \tag{40.4.6}$$

In the last two lines the spatial derivatives were interchanged with the time average, which is legitimate because the average is an integral over t at fixed $\boldsymbol{r}$.

Step 4: what survives. The magnetic terms do *not* vanish, because they are quadratic in the oscillating quantities and therefore have nonzero mean — this is the whole point. Collecting:

$$\rho\left(\frac{\partial \boldsymbol{v}_0}{\partial t} + (\boldsymbol{v}_0\cdot\nabla)\boldsymbol{v}_0\right) = -\nabla p_0 + \overline{(\boldsymbol{M}\cdot\nabla)\boldsymbol{H}} + \eta\Delta\boldsymbol{v}_0 + \tfrac{1}{2}\nabla\times\overline{(\boldsymbol{M}\times\boldsymbol{H})}. \tag{40.4.7}$$

Step 5: drop the subscript. Since the fast parts have served their purpose and been eliminated, the authors write $\boldsymbol{v}$ for $\boldsymbol{v}_0$ and p for p_0, giving

$$\rho\frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = -\nabla p + \overline{(\boldsymbol{M}\cdot\nabla)\boldsymbol{H}} + \eta\Delta\boldsymbol{v} + \tfrac{1}{2}\nabla\times\overline{(\boldsymbol{M}\times\boldsymbol{H})}, \tag{SM-26}$$

which is Eq. (SM-26). ■

40.5 Physical Interpretation

The oscillating torque shakes the fluid back and forth at 2ω, but the fluid is too sluggish to move appreciably in half a period; what it does respond to is the *rectified* part — the nonzero average that survives because the torque is quadratic in oscillating quantities. This is the same mechanism by which an alternating current produces a steady heating, or a sound wave produces a steady acoustic streaming: a nonlinear response to a zero-mean drive need not have zero mean.

Remark 40.5.1 (Why $\boldsymbol{M}$ and $\boldsymbol{H}$ are *not* split). The authors point out that no such slow/fast splitting is needed for $\boldsymbol{M}$ and $\boldsymbol{H}$, since these quantities oscillate rapidly and carry no slowly varying part at all. This is Assumption 40.3.2, and it is what makes the procedure clean: the magnetic quantities are purely fast, the hydrodynamic ones are predominantly slow, and the coupling between them survives only through the averaged products.

40.6 Connection to the Next Equation

Performing the average of (SM-25) and comparing the result with the viscous term gives the rotational viscosity, (SM-27).

Chapter 41

Equation (SM-27): The Rotational Viscosity

$$\Delta\eta = \tfrac{1}{4}\eta\varphi\xi^2 \cos^2\alpha\, \cos 2\alpha. \tag{SM-27}$$

41.1 Context

We now perform the average called for by (SM-26), substitute into the momentum equation, and read off the coefficient of $\nabla^2\boldsymbol{v}$. The procedure repeats Sec. 14.4 of Volume II, but with an averaged torque in place of a static one.

41.2 Step-by-Step Derivation

Step 1: average the torque. Apply the period average to (SM-25). The slowly varying factors Ω, $\cos^2\alpha$, $\cos 2\alpha$, $\sin 2\alpha$ are constants over one period and pass outside. There remain two elementary averages, established as Eqs. (27.4.5) and (27.4.6):

$$\overline{\cos^2\omega t} = \tfrac{1}{2}, \qquad \overline{\sin\omega t\cos\omega t} = 0. \tag{41.2.1}$$

Hence

$$\overline{\boldsymbol{M}\times\boldsymbol{H}} = -2\Omega\eta\varphi\xi^2\cos^2\alpha\Big(\overline{\cos^2\omega t}\,\cos 2\alpha + \overline{\sin\omega t\cos\omega t}\,\sin 2\alpha\Big)\hat{\boldsymbol{z}} \tag{41.2.2}$$

$$= -2\Omega\eta\varphi\xi^2\cos^2\alpha\Big(\tfrac{1}{2}\cos 2\alpha + 0\Big)\hat{\boldsymbol{z}} \tag{41.2.3}$$

$$= -\Omega\eta\varphi\xi^2\cos^2\alpha\,\cos 2\alpha\,\hat{\boldsymbol{z}}. \tag{41.2.4}$$

Since $\boldsymbol{\Omega} = \Omega\hat{\boldsymbol{z}}$, this may be written as a vector relation:

$$\boxed{\overline{\boldsymbol{M}\times\boldsymbol{H}} = -\eta\varphi\xi^2\cos^2\alpha\cos 2\alpha\,\boldsymbol{\Omega}} \tag{41.2.5}$$

This is the unnumbered result the authors record when they note that averaging (SM-25) leaves $\tfrac{1}{2}\cos 2\alpha$ from the first bracketed term and nothing from the second. ✓

Step 2: the disappearance of the second term, examined. It is worth pausing on why $\overline{\sin\omega t\cos\omega t} = 0$ kills the $\sin 2\alpha$ contribution. That term represents the part of the transverse magnetisation that is *in quadrature* with the field. A torque in quadrature with the driving

field does work over the first half of a cycle and takes it back over the second: it transports no net momentum to the fluid. Only the in-phase part, carried by $\cos 2\alpha$, survives averaging. This is the same distinction as between the reactive and resistive parts of an impedance.

Step 3: insert into the momentum equation. By (SM-26) the magnetic contribution to the force density is $\frac{1}{2}\nabla \times \overline{\boldsymbol{M} \times \boldsymbol{H}}$. With (41.2.5),

$$\tfrac{1}{2}\nabla \times \overline{\boldsymbol{M} \times \boldsymbol{H}} = -\tfrac{1}{2}\eta\varphi\xi^2 \cos^2\alpha \cos 2\alpha \, \nabla \times \boldsymbol{\Omega}. \tag{41.2.6}$$

Step 4: express the viscous term the same way. By the definition $\boldsymbol{\Omega} = \frac{1}{2}\nabla \times \boldsymbol{v}$ (3.4.1) and the identity (2.6.1) together with incompressibility (5.1.4),

$$\eta\Delta\boldsymbol{v} = \eta\nabla^2\boldsymbol{v} = -\eta\,\nabla \times \big(\nabla \times \boldsymbol{v}\big) = -2\eta\,\nabla \times \boldsymbol{\Omega}. \tag{41.2.7}$$

Step 5: add them. Summing (41.2.7) and (41.2.6):

$$\eta\Delta\boldsymbol{v} + \tfrac{1}{2}\nabla \times \overline{\boldsymbol{M} \times \boldsymbol{H}} = -2\eta\nabla \times \boldsymbol{\Omega} - \tfrac{1}{2}\eta\varphi\xi^2 \cos^2\alpha \cos 2\alpha \, \nabla \times \boldsymbol{\Omega} \tag{41.2.8}$$

$$= -2\Big(\eta + \tfrac{1}{4}\eta\varphi\xi^2 \cos^2\alpha \cos 2\alpha\Big)\nabla \times \boldsymbol{\Omega}, \tag{41.2.9}$$

where in the last line the factor 2 was extracted: $\frac{1}{2} = 2 \times \frac{1}{4}$. This is precisely the line printed in the paper,

$$\eta\Delta\boldsymbol{v} - \tfrac{1}{2}\eta\varphi\xi^2 \cos^2\alpha \cos 2\alpha \, \nabla \times \boldsymbol{\Omega} = -2\Big(\eta + \tfrac{1}{4}\eta\varphi\xi^2 \cos^2\alpha \cos 2\alpha\Big)\nabla \times \boldsymbol{\Omega}.$$

Step 6: read off the viscosity. Comparison of (41.2.9) with the field-free case, where the same combination is $-2\eta\nabla \times \boldsymbol{\Omega}$, shows that the fluid behaves exactly as a Newtonian liquid of viscosity $\eta + \Delta\eta$ with

$$\boxed{\Delta\eta = \tfrac{1}{4}\eta\varphi\xi^2 \cos^2\alpha \, \cos 2\alpha} \tag{SM-27}$$

■

41.3 Consistency Check Against the Master Formula

Volume II established the general relation $\Delta\eta = \frac{3}{2}\eta\varphi(\Omega - \Theta)/\Omega$, Eq. (14.4.9). Using (41.2.5) together with the torque balance $\overline{\boldsymbol{M} \times \boldsymbol{H}} = -6\eta\varphi(\boldsymbol{\Omega} - \boldsymbol{\Theta})$, Eq. (14.2.1):

$$\boldsymbol{\Omega} - \boldsymbol{\Theta} = \frac{\eta\varphi\xi^2 \cos^2\alpha \cos 2\alpha}{6\eta\varphi}\boldsymbol{\Omega} = \frac{\xi^2}{6}\cos^2\alpha \cos 2\alpha \, \boldsymbol{\Omega}, \tag{41.3.1}$$

so that

$$\Delta\eta = \frac{3}{2}\eta\varphi \cdot \frac{\xi^2}{6}\cos^2\alpha \cos 2\alpha = \frac{\eta\varphi\xi^2}{4}\cos^2\alpha \cos 2\alpha. \quad \checkmark \tag{41.3.2}$$

The two independent routes agree exactly. Equation (41.3.1) will be used again in Chapter 47 to obtain (SM-33).

41.4 Connection to the Next Equation

Expressing $\cos^2\alpha$ and $\cos 2\alpha$ through $\omega\tau_B$ turns (SM-27) into the central result of the paper, (SM-28).

Chapter 42

Equation (SM-28): The Central Result

$$\Delta\eta = \frac{1}{4}\,\eta\varphi\,\xi^2\,\frac{1-\omega^2\tau_B^2}{\left(1+\omega^2\tau_B^2\right)^2}. \tag{SM-28}$$

42.1 Context

This is the equation for which the paper was written. Everything before it is preparation and everything after it is generalisation. It says that the rotational viscosity of a ferrofluid in a weak alternating field is positive at low frequency, passes through zero at $\omega\tau_B = 1$, and is *negative* above it.

42.2 Step-by-Step Derivation

Step 1. From Eq. (33.4.1) with $x = \tan\alpha = \omega\tau_B$,

$$\cos^2\alpha = \frac{1}{1+\omega^2\tau_B^2}. \tag{42.2.1}$$

Step 2. From Eq. (37.5.8),

$$\cos 2\alpha = \frac{1-\tan^2\alpha}{1+\tan^2\alpha} = \frac{1-\omega^2\tau_B^2}{1+\omega^2\tau_B^2}. \tag{42.2.2}$$

Step 3. Multiply and substitute into (SM-27):

$$\Delta\eta = \frac{1}{4}\eta\varphi\xi^2 \cdot \frac{1}{1+\omega^2\tau_B^2} \cdot \frac{1-\omega^2\tau_B^2}{1+\omega^2\tau_B^2} \tag{42.2.3}$$

$$= \frac{1}{4}\eta\varphi\xi^2\,\frac{1-\omega^2\tau_B^2}{\left(1+\omega^2\tau_B^2\right)^2}. \tag{SM-28}$$

■

42.3 Analysis of the Function

Write $y \equiv \omega\tau_B$ and define the relative rotational viscosity plotted in the paper's Fig. 1,

$$R(y) \equiv \frac{4\Delta\eta}{\eta\varphi\xi^2} = \frac{1-y^2}{\big(1+y^2\big)^2}. \tag{42.3.1}$$

(a) Value at zero frequency. $R(0) = 1$, so

$$\Delta\eta(0) = \tfrac{1}{4}\eta\varphi\xi^2, \tag{42.3.2}$$

which is formally identical to the static formula (SM-3). The authors say (SM-28) "provides the correct value of $\Delta\eta$ at $\omega = 0$ [see (3)]". But there is a subtlety worth making explicit, because it is the deepest justification of the "acting field" convention of Sec. 33.2.

At $\omega = 0$ the field is not static — it is $H_0 \cos\omega t$ with ω merely very small. The instantaneous viscosity is then the static formula (SM-3) evaluated at the instantaneous field, and what a slow measurement returns is its average over the slow cycle:

$$\overline{\Delta\eta_{\text{stat}}} = \overline{\frac{\eta\varphi}{4}\left(\frac{mH_0\cos\omega t}{kT}\right)^2} \tag{42.3.3}$$

$$= \frac{\eta\varphi}{4}\frac{m^2H_0^2}{(kT)^2}\,\overline{\cos^2\omega t} \tag{42.3.4}$$

$$= \frac{\eta\varphi}{4}\cdot\frac{m^2H_0^2}{(kT)^2}\cdot\frac{1}{2} \tag{42.3.5}$$

$$= \frac{\eta\varphi}{4}\cdot\xi^2, \qquad \text{by (39.4.7)'s definition } \xi^2 = \frac{m^2H_0^2}{2(kT)^2} \tag{42.3.6}$$

exactly reproducing (42.3.2). *This is why ξ was defined over $H_0/\sqrt{2}$*: with that convention, and no other, the $\omega \to 0$ limit of the alternating-field theory agrees with the cycle-average of the static theory without stray factors of two. ✓

(b) The zero crossing. $R(y) = 0$ requires $1 - y^2 = 0$, i.e.

$$\boxed{\omega_0\tau_B = 1} \tag{42.3.7}$$

Below this frequency the ferrofluid is thickened by the field; above it, thinned below its zero-field value. The condition has a transparent meaning: it is the frequency at which the magnetisation lags the field by exactly $\alpha = \pi/4$, so that the transverse magnetisation, which lags by $2\alpha = \pi/2$, is exactly in quadrature with the field and delivers no average torque at all.

(c) The minimum. Differentiate (42.3.1) using the quotient rule, with $u = 1-y^2$ ($u' = -2y$) and $v = (1+y^2)^2$ ($v' = 2(1+y^2)\cdot 2y = 4y(1+y^2)$):

$$R'(y) = \frac{-2y\big(1+y^2\big)^2 - \big(1-y^2\big)\cdot 4y\big(1+y^2\big)}{\big(1+y^2\big)^4} \tag{42.3.8}$$

$$= \frac{\big(1+y^2\big)\Big[-2y\big(1+y^2\big) - 4y\big(1-y^2\big)\Big]}{\big(1+y^2\big)^4} \qquad \text{(factor out } (1+y^2)\text{)} \tag{42.3.9}$$

$$= \frac{-2y\big(1+y^2\big) - 4y\big(1-y^2\big)}{\big(1+y^2\big)^3} \tag{42.3.10}$$

$$= \frac{-2y - 2y^3 - 4y + 4y^3}{\big(1+y^2\big)^3} \tag{42.3.11}$$

$$= \frac{2y\big(y^2-3\big)}{\big(1+y^2\big)^3}. \tag{42.3.12}$$

This vanishes at $y = 0$ (the maximum, $R = 1$) and at $y^2 = 3$:

$$\boxed{\omega_m \tau_B = \sqrt{3} \approx 1.732} \tag{42.3.13}$$

That it is a minimum follows from the sign of R': for $0 < y < \sqrt{3}$ the factor $(y^2 - 3)$ is negative so R decreases; for $y > \sqrt{3}$ it is positive so R increases.

(d) The depth of the minimum. Evaluate (42.3.1) at $y = \sqrt{3}$:

$$R(\sqrt{3}) = \frac{1-3}{\big(1+3\big)^2} = \frac{-2}{16} = -\frac{1}{8}, \tag{42.3.14}$$

so that

$$\boxed{\Delta\eta\big(\omega_m\big) = -\frac{\Delta\eta(0)}{8}} \tag{42.3.15}$$

exactly as the authors state ("at this point $\Delta\eta(\sqrt{3}) = -\Delta\eta(0)/8$"). The negative excursion is thus *eight times smaller* than the positive low-frequency value — a quantitative fact that matters for experiment: the effect is real but modest, and Bacri *et al.* indeed measure a viscosity reduction of about 25% rather than a dramatic collapse.

(e) The large-frequency tail. For $y \gg 1$,

$$R(y) \approx \frac{-y^2}{y^4} = -\frac{1}{y^2}, \tag{42.3.16}$$

so $\Delta\eta$ returns to zero from below as ω^{-2}. This is (SM-29), the subject of the next chapter.

(f) Summary of the shape. R starts at 1, falls monotonically, crosses zero at $y = 1$, reaches $-1/8$ at $y = \sqrt{3}$, and climbs back towards zero as $-1/y^2$. This is exactly the curve of the paper's Fig. 1.

42.4 Physical Interpretation: What Is Actually Happening

The chain of causation, assembled from all of Volumes IV and V, runs as follows.

1. The shear tries to rotate the grains at Ω.
2. Rotating a grain shifts the frequency at which it sees each of the two circular components of the field — to $\omega - \Omega$ for the co-rotating one and $\omega + \Omega$ for the counter-rotating one (Sec. 35.4).

3. Because the Debye response is a nonlinear function of frequency, the two components no longer produce equal and opposite torques. The imbalance is a net transverse magnetisation $M_y \propto \Omega$, lagging the field by 2α (Sec. 37.5).

4. That transverse magnetisation, crossed with the field, gives a torque whose average is $\propto -\Omega \cos 2\alpha$ (Sec. 41.2).

5. At low frequency $\cos 2\alpha > 0$: the torque opposes the rotation, the grains turn slower than the fluid, the liquid must shear past them, and the viscosity rises.

6. At high frequency $\cos 2\alpha < 0$: the torque *drives* the rotation, the grains turn *faster* than the fluid, they hand angular momentum to it, and the viscosity falls.

Where does the energy come from? From the field. Bacri *et al.* summarise the route in their abstract: energy stored in the field is converted first into angular momentum of the grains, and then by the grains into hydrodynamic motion of the liquid. There is no violation of anything: the alternating field is maintained by a generator doing work on the solenoid, and a fraction of that work ends up as kinetic energy of the flow. Contrast the static case, where a permanent magnet — which does no work — could produce the field, and where $\Delta\eta$ is therefore required by thermodynamics to be positive (Sec. 15.6).

Why the effect must vanish at both ends. At $\omega \to 0$ the field is quasi-static and the thermodynamic argument applies, forcing $\Delta\eta > 0$. At $\omega \to \infty$ the moments cannot follow at all, $M \to 0$, and there is no torque of either sign. The negative region is squeezed between these two limits and is therefore necessarily bounded — hence the modest factor 1/8 of (42.3.15).

42.5 Connection to the Next Equation

The paper next records the explicit high-frequency asymptote, (SM-29).

Chapter 43

Equation (SM-29): The High-Frequency Asymptote

$$\Delta\eta = -\frac{\varphi}{\eta}\left(\frac{M_s H}{6\omega}\right)^2 \qquad \text{at} \quad \omega\tau_B \gg 1. \tag{SM-29}$$

43.1 Context

The authors record this limit because it shows something initially surprising: although the viscosity tends to zero at high frequency, *the way* it does so is independent of τ_B — the Brownian time cancels out. The high-frequency ferrofluid has forgotten how big its particles are.

43.2 Symbols

M_s is the spontaneous magnetisation of the grain material (G); H is the acting field $H_0/\sqrt{2}$, consistently with the convention of Sec. 33.2 — as the derivation below confirms.

43.3 Step-by-Step Derivation

Step 1: take the limit of (SM-28). For $\omega\tau_B \gg 1$ the numerator $1-\omega^2\tau_B^2 \to -\omega^2\tau_B^2$ and the denominator $(1+\omega^2\tau_B^2)^2 \to \omega^4\tau_B^4$, so

$$\Delta\eta \to \frac{\eta\varphi\xi^2}{4}\cdot\frac{-\omega^2\tau_B^2}{\omega^4\tau_B^4} = -\frac{\eta\varphi\xi^2}{4\,\omega^2\tau_B^2}. \tag{43.3.1}$$

Step 2: evaluate ξ^2/τ_B^2. Insert $\xi^2 = m^2H_0^2/2(kT)^2$ and $\tau_B = 3\eta V/kT$:

$$\frac{\xi^2}{\tau_B^2} = \frac{m^2H_0^2}{2(kT)^2}\cdot\frac{(kT)^2}{9\eta^2V^2} \tag{43.3.2}$$

$$= \frac{m^2H_0^2}{18\,\eta^2V^2} \qquad ((kT)^2 \text{ cancels}) \tag{43.3.3}$$

$$= \frac{\left(M_sV\right)^2H_0^2}{18\,\eta^2V^2} \qquad \text{using } m = M_sV \tag{43.3.4}$$

$$= \frac{M_s^2 H_0^2}{18\,\eta^2}. \qquad (V^2 \text{ cancels}) \tag{43.3.5}$$

The volume has disappeared — and with it the particle size, and with it τ_B.

Step 3: assemble. Substituting (43.3.5) into (43.3.1):

$$\Delta\eta = -\frac{\eta\varphi}{4\omega^2} \cdot \frac{M_s^2 H_0^2}{18\eta^2} \tag{43.3.6}$$

$$= -\frac{\varphi M_s^2 H_0^2}{72\,\eta\,\omega^2}. \tag{43.3.7}$$

Step 4: recognise the paper's form. Write the acting field $H = H_0/\sqrt{2}$, so $H_0^2 = 2H^2$:

$$\Delta\eta = -\frac{\varphi M_s^2 \cdot 2H^2}{72\,\eta\,\omega^2} = -\frac{\varphi M_s^2 H^2}{36\,\eta\,\omega^2} = -\frac{\varphi}{\eta} \cdot \frac{M_s^2 H^2}{36\,\omega^2} = -\frac{\varphi}{\eta}\left(\frac{M_s H}{6\omega}\right)^2, \tag{SM-29}$$

since $36 = 6^2$. ■

43.4 Physical Interpretation

Why τ_B drops out. At high frequency the magnetisation amplitude is $M \simeq \chi H_0/\omega\tau_B$, which falls with τ_B; but the torque per unit magnetisation grows with the lag, which rises with τ_B. The two dependences cancel. What remains is a competition between the magnetic energy density $M_s H$ and the frequency, moderated by the viscosity.

The sign and the scaling. $\Delta\eta < 0$ always in this regime, and $|\Delta\eta| \propto \omega^{-2}$. Doubling the frequency quarters the effect. Since $\Delta\eta$ must also vanish as $\omega \to 0$, there is necessarily an optimum, and Eq. (42.3.13) located it at $\omega\tau_B = \sqrt{3}$.

An experimental corollary. Because (43.3.7) contains η in the *denominator*, the negative viscosity at fixed high frequency is larger for a *less* viscous carrier. Together with the φ and M_s^2 dependences this is the design rule for a fluid in which the effect is large: high magnetisation, high concentration, low carrier viscosity — and, from (42.3.13), a particle size chosen so that τ_B places $\sqrt{3}/\tau_B$ in an accessible frequency range. Bacri *et al.* made exactly these choices.

43.5 Connection to the Next Equation

Having completed the first variant of the solution, the authors present the second: a direct calculation in the linearly polarised field, without any decomposition. That begins at (SM-30).

Chapter 44

Equation (SM-30): The Second Variant — Longitudinal Magnetisation

$$M_x = \chi H_0 \cos\alpha \, \cos(\omega t - \alpha), \qquad \tan\alpha = \omega\tau_B. \tag{SM-30}$$

44.1 Context

The authors remark that the same Eq. (SM-28) can be reached more quickly by working directly with the linearly polarised field (SM-14). The second variant is quicker but less transparent; the authors give both because each illuminates something the other hides. The first showed *why* the effect exists (the Doppler imbalance between two circular components); the second shows how little machinery is really needed.

We present it in full, both because completeness demands it and because it provides an independent check on the long calculation of Volume IV.

44.2 Assumptions

Same as before. In this chapter the fluid is at rest, $\mathbf{\Omega} = 0$; the vortex enters in Chapter 45.

44.3 Mathematical Background: forced linear relaxation

We shall need the steady periodic solution of

$$\tau_B \frac{\mathrm{d}y}{\mathrm{d}t} + y = A\cos\omega t. \tag{44.3.1}$$

Result 44.3.1. *The steady solution is*

$$y(t) = A\cos\alpha \, \cos(\omega t - \alpha), \qquad \tan\alpha = \omega\tau_B. \tag{44.3.2}$$

Derivation. Try $y = B\cos(\omega t - \alpha)$ with B and α to be determined. Then $\mathrm{d}y/\mathrm{d}t = -B\omega\sin(\omega t - \alpha)$ and the left-hand side of (44.3.1) is

$$B\Big[\cos(\omega t - \alpha) - \omega\tau_B \sin(\omega t - \alpha)\Big]. \tag{44.3.3}$$

To combine the two terms into a single cosine, use the identity

$$\cos u - \lambda \sin u = \sqrt{1+\lambda^2}\,\cos(u+\alpha) \qquad \text{with} \qquad \tan\alpha = \lambda, \tag{44.3.4}$$

which is proved by expanding the right-hand side: $\sqrt{1+\lambda^2}[\cos u\cos\alpha - \sin u\sin\alpha]$, and noting from Eq. (33.4.1) that $\sqrt{1+\lambda^2}\cos\alpha = 1$ and $\sqrt{1+\lambda^2}\sin\alpha = \lambda$. Applying (44.3.4) to (44.3.3) with $u = \omega t - \alpha$ and $\lambda = \omega\tau_B = \tan\alpha$ — the *same* α, which is the trick —

$$B\sqrt{1+\omega^2\tau_B^2}\,\cos\big(\omega t - \alpha + \alpha\big) = B\sqrt{1+\omega^2\tau_B^2}\,\cos\omega t. \tag{44.3.5}$$

Matching to the right-hand side of (44.3.1),

$$B\sqrt{1+\omega^2\tau_B^2} = A \qquad \Longrightarrow \qquad B = \frac{A}{\sqrt{1+\omega^2\tau_B^2}} = A\cos\alpha, \tag{44.3.6}$$

using Eq. (33.4.1) once more. This is (44.3.2). ■

44.4 Step-by-Step Derivation of (SM-30)

Step 1: the equation for M_x in a quiescent fluid. With $\boldsymbol{\Omega} = 0$ and (as the authors note) $\boldsymbol{\Theta} = 0$ — because in the linearly polarised field the mean particle spin vanishes by symmetry, Sec. 27.5 — Eq. (SM-7) reduces to

$$\frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t} = -\frac{1}{\tau_B}\big(\boldsymbol{M} - \boldsymbol{M}_0\big). \tag{44.4.1}$$

With $\boldsymbol{M}_0 = \chi\boldsymbol{H} = \chi H_0\cos\omega t\,\hat{\boldsymbol{x}}$ from (SM-15) and (SM-14), only the x component is driven:

$$\tau_B\frac{\mathrm{d}M_x}{\mathrm{d}t} + M_x = \chi H_0\cos\omega t. \tag{44.4.2}$$

Step 2: solve. This is (44.3.1) with $y = M_x$ and $A = \chi H_0$. By (44.3.2),

$$M_x = \chi H_0\cos\alpha\,\cos(\omega t - \alpha), \qquad \tan\alpha = \omega\tau_B, \tag{SM-30}$$

which is (SM-30). ■

44.5 Consistency with the First Variant

The first variant gave, for each circular component, $M_\pm = \chi H_0\cos\alpha$ with the same $\tan\alpha = \omega\tau_B$. Superposing the x components,

$$M_x = \tfrac{1}{2}\Big[M_+\cos(\omega t - \alpha) + M_-\cos(\omega t - \alpha)\Big] = \chi H_0\cos\alpha\cos(\omega t - \alpha), \tag{44.5.1}$$

identical to (SM-30). ✓ The two routes agree, as they must.

44.6 Physical Interpretation

Equation (SM-30) is the Debye response once again: amplitude reduced by $\cos\alpha$, phase retarded by α. The authors immediately draw the consequence: both the magnetic torque and the mean particle spin vanish. Indeed with $\boldsymbol{M} = M_x\hat{\boldsymbol{x}}$ and $\boldsymbol{H} = H_x\hat{\boldsymbol{x}}$ parallel, the cross product is identically zero. *In a quiescent fluid, a linearly polarised field does nothing at all mechanically*, however strong and however fast — the point already made qualitatively in Sec. 27.5 and now proved.

44.7 Connection to the Next Equation

Switching on the vorticity generates a y component, whose amplitude and phase obey (SM-31).

Chapter 45

Equation (SM-31): The Coupled Equations for the Transverse Component

$$\begin{aligned} a\big(\sin\theta - \omega\tau_B\cos\theta\big) &= \Omega\tau_B\chi H_0 \sin\alpha\cos\alpha, \\ a\big(\cos\theta + \omega\tau_B\sin\theta\big) &= \Omega\tau_B\chi H_0 \cos^2\alpha. \end{aligned} \tag{SM-31}$$

45.1 Context

The authors recall that a vortical flow destroys the symmetry of the problem: with $\boldsymbol{\Omega} = (0, 0, \Omega)$ a y component of the magnetisation appears, together with a z component of the particle spin. Substituting the trial form $M_y = a\cos(\omega t - \theta)$ into (SM-7) and (SM-8) then yields a coupled pair of equations for the amplitude and phase of that component.

45.2 Symbols

$a > 0$ is the amplitude of the transverse magnetisation (G) and θ its phase lag behind the field (rad). Both are unknowns to be determined; the answer, $\theta = 2\alpha$ and $a = \Omega\tau_B\chi H_0\cos^2\alpha$, is (SM-32).

45.3 Assumptions

1. Linear response in Ω: $a = O(\Omega)$, so any term containing a product of a with another small quantity is dropped.

2. The particle spin is $\boldsymbol{\Theta} \simeq \boldsymbol{\Omega}$ to the accuracy required. The magnetic correction to $\boldsymbol{\Theta}$ in (SM-11) is $O(\xi^2)$ and would contribute at order ξ^4 to the final viscosity — the same argument as in Sec. 35.3.

45.4 Step-by-Step Derivation

Step 1: write out the y component of the magnetisation equation. Equation (SM-7) with $\boldsymbol{\Theta} = \boldsymbol{\Omega} = \Omega\hat{\boldsymbol{z}}$ reads

$$\frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t} = \boldsymbol{\Omega} \times \boldsymbol{M} - \frac{1}{\tau_B}\big(\boldsymbol{M} - \boldsymbol{M}_0\big). \tag{45.4.1}$$

Compute the convective term for $\boldsymbol{M} = (M_x, M_y, 0)$:

$$\boldsymbol{\Omega} \times \boldsymbol{M} = \Omega\hat{\boldsymbol{z}} \times \big(M_x, M_y, 0\big) = \Omega\big(-M_y,\ M_x,\ 0\big), \tag{45.4.2}$$

using $\hat{\boldsymbol{z}} \times \hat{\boldsymbol{x}} = \hat{\boldsymbol{y}}$ and $\hat{\boldsymbol{z}} \times \hat{\boldsymbol{y}} = -\hat{\boldsymbol{x}}$. Since $\boldsymbol{M}_0$ has no y component, the y equation is

$$\frac{\mathrm{d}M_y}{\mathrm{d}t} = \Omega M_x - \frac{M_y}{\tau_B}. \tag{45.4.3}$$

(The x equation is $\mathrm{d}M_x/\mathrm{d}t = -\Omega M_y - (M_x - \chi H_0 \cos\omega t)/\tau_B$; its extra term ΩM_y is $O(\Omega^2)$ and is dropped, leaving (SM-30) intact.)

Step 2: substitute the ansatz. Put $M_y = a\cos(\omega t - \theta)$ and M_x from (SM-30) into (45.4.3), and multiply throughout by τ_B:

$$-a\omega\tau_B \sin(\omega t - \theta) = \Omega\tau_B\chi H_0 \cos\alpha \cos(\omega t - \alpha) - a\cos(\omega t - \theta), \tag{45.4.4}$$

which we rearrange as

$$a\cos(\omega t - \theta) - a\omega\tau_B \sin(\omega t - \theta) = \Omega\tau_B\chi H_0 \cos\alpha\ \cos(\omega t - \alpha). \tag{45.4.5}$$

Step 3: expand both sides in $\cos\omega t$ and $\sin\omega t$. Using the addition formulas $\cos(\omega t - \theta) = \cos\omega t\cos\theta + \sin\omega t\sin\theta$ and $\sin(\omega t - \theta) = \sin\omega t\cos\theta - \cos\omega t\sin\theta$, the left-hand side becomes

$$\begin{aligned} &a\big[\cos\omega t\cos\theta + \sin\omega t\sin\theta\big] - a\omega\tau_B\big[\sin\omega t\cos\theta - \cos\omega t\sin\theta\big] \\ &\quad = \cos\omega t \cdot a\big[\cos\theta + \omega\tau_B\sin\theta\big] + \sin\omega t \cdot a\big[\sin\theta - \omega\tau_B\cos\theta\big]. \end{aligned} \tag{45.4.6}$$

The right-hand side, with $\cos(\omega t - \alpha) = \cos\omega t\cos\alpha + \sin\omega t\sin\alpha$, becomes

$$\cos\omega t \cdot \Omega\tau_B\chi H_0\cos^2\alpha + \sin\omega t \cdot \Omega\tau_B\chi H_0\cos\alpha\sin\alpha. \tag{45.4.7}$$

Step 4: match coefficients. The functions $\cos\omega t$ and $\sin\omega t$ are linearly independent — no nontrivial linear combination of them vanishes for all t, as one sees by evaluating at $t = 0$ and $\omega t = \pi/2$. Hence their coefficients in (45.4.6) and (45.4.7) must match separately:

$$\cos\omega t: \qquad a\big(\cos\theta + \omega\tau_B\sin\theta\big) = \Omega\tau_B\chi H_0\cos^2\alpha, \tag{45.4.8}$$

$$\sin\omega t: \qquad a\big(\sin\theta - \omega\tau_B\cos\theta\big) = \Omega\tau_B\chi H_0\sin\alpha\cos\alpha. \tag{45.4.9}$$

These are exactly the two lines of (SM-31). ■

45.5 Physical Interpretation

The structure is that of a driven, damped oscillator resolved into in-phase and quadrature parts. The *driving* of the transverse component, on the right, is the longitudinal magnetisation being swept sideways by the vortex — the term ΩM_x in (45.4.3). The *response*, on the left, is the transverse magnetisation fighting its own relaxation. Note that the drive is proportional to Ω and to M_x: the vortex alone would do nothing without a magnetisation to sweep, and a magnetisation alone would do nothing without a vortex.

45.6 Connection to the Next Equation

Solving the pair for a and θ gives (SM-32).

Chapter 46

Equation (SM-32): Solving the Pair

$$\theta = 2\alpha, \qquad a = \Omega\tau_B \chi H_0 \cos^2\alpha. \tag{SM-32}$$

46.1 Step-by-Step Derivation

Step 1: divide one equation by the other. Dividing (45.4.9) by (45.4.8) eliminates both a and the common factor $\Omega\tau_B\chi H_0$:

$$\frac{\sin\theta - \omega\tau_B\cos\theta}{\cos\theta + \omega\tau_B\sin\theta} = \frac{\sin\alpha\cos\alpha}{\cos^2\alpha} = \frac{\sin\alpha}{\cos\alpha} = \tan\alpha = \omega\tau_B, \tag{46.1.1}$$

the last step by (SM-24). The division is legitimate provided $a \neq 0$ and the denominator does not vanish; both hold for $0 \le \alpha < \pi/2$.

Step 2: cross-multiply. Write $\lambda \equiv \omega\tau_B$ for brevity. From (46.1.1),

$$\sin\theta - \lambda\cos\theta = \lambda\big(\cos\theta + \lambda\sin\theta\big) \tag{46.1.2}$$

$$\sin\theta - \lambda\cos\theta = \lambda\cos\theta + \lambda^2\sin\theta \tag{46.1.3}$$

$$\sin\theta - \lambda^2\sin\theta = \lambda\cos\theta + \lambda\cos\theta \tag{46.1.4}$$

$$\sin\theta\big(1 - \lambda^2\big) = 2\lambda\cos\theta. \tag{46.1.5}$$

Step 3: identify the double angle. Divide both sides by $\cos\theta(1-\lambda^2)$:

$$\tan\theta = \frac{2\lambda}{1-\lambda^2} = \frac{2\tan\alpha}{1-\tan^2\alpha} = \tan 2\alpha, \tag{46.1.6}$$

where the last equality is the tangent double-angle formula, itself derived from (46.1.8) below. Hence

$$\boxed{\theta = 2\alpha} \tag{46.1.7}$$

(the solution in the principal range; θ and 2α can differ by a multiple of π, which is fixed by requiring $a > 0$ in Step 5).

Step 4: a needed identity. For completeness,

$$\tan 2\alpha = \frac{\sin 2\alpha}{\cos 2\alpha} = \frac{2\sin\alpha\cos\alpha}{\cos^2\alpha - \sin^2\alpha} = \frac{2\tan\alpha}{1-\tan^2\alpha}, \tag{46.1.8}$$

obtained by dividing numerator and denominator by $\cos^2\alpha$.

Step 5: find the amplitude. Substitute $\theta = 2\alpha$ into (45.4.8):

$$a\big(\cos 2\alpha + \tan\alpha\,\sin 2\alpha\big) = \Omega\tau_B\chi H_0\cos^2\alpha. \tag{46.1.9}$$

The bracket collapses beautifully:

$$\begin{aligned}\cos 2\alpha + \tan\alpha\sin 2\alpha &= \cos 2\alpha + \frac{\sin\alpha}{\cos\alpha}\cdot 2\sin\alpha\cos\alpha && (46.1.10)\\ &= \cos 2\alpha + 2\sin^2\alpha && (46.1.11)\\ &= \big(\cos^2\alpha - \sin^2\alpha\big) + 2\sin^2\alpha && (46.1.12)\\ &= \cos^2\alpha + \sin^2\alpha && (46.1.13)\\ &= 1. && (46.1.14)\end{aligned}$$

Therefore (46.1.9) reduces to

$$\boxed{a = \Omega\tau_B\chi H_0\cos^2\alpha} \tag{46.1.15}$$

which completes (SM-32). ■

46.2 The Two Variants Agree

Assembling,

$$M_y = a\cos(\omega t - \theta) = \Omega\tau_B\chi H_0\cos^2\alpha\,\cos(\omega t - 2\alpha), \tag{46.2.1}$$

which is identical to (SM-23), obtained in Volume IV by an entirely different route. The authors note the coincidence with their Eq. (SM-23) and conclude that the two solution procedures give the same $\Delta\eta(\xi, \omega\tau_B)$.

Remark 46.2.1 (Why the check matters). The first variant involved a decomposition, two separate solutions, a Doppler shift, a Taylor expansion in $\Omega\tau_B$ and a reassembly — many opportunities for a lost factor. The second involved one linear ODE and a 2×2 algebraic system. That they agree exactly, term for term and phase for phase, is strong evidence that neither contains an error. Independent derivation is the only real proof-reading available in physics.

46.3 Connection to the Next Equation

The paper concludes Sec. II by computing the mean particle angular velocity, which displays the mechanism in its most direct form. That is (SM-33).

Chapter 47

Equation (SM-33): The Mean Particle Spin

$$\Theta = \Omega\left(1 - \frac{\xi^2}{6}\,\frac{1-\omega^2\tau_B^2}{\left(1+\omega^2\tau_B^2\right)^2}\right). \tag{SM-33}$$

47.1 Context

The authors close their Sec. II by averaging Eq. (SM-11) over the fast time scale and combining it with (SM-26) to obtain (SM-33), which they describe as a striking illustration of how the field controls the viscosity. It is indeed the most vivid statement of the result: it says directly whether the grains turn faster or slower than the fluid.

Remark 47.1.1 (A typographical note). The equation as printed in the paper reads $\Theta = \Omega\big[1 - \frac{\xi^2}{6}(1-\xi^2\tau_B^2)/(1+\omega^2\tau_B^2)^2\big]$. The $\xi^2\tau_B^2$ in the numerator is a misprint for $\omega^2\tau_B^2$: dimensionally ξ is a pure number so $\xi^2\tau_B^2$ would carry units of s^2, and the derivation below — as well as consistency with (SM-28) and (SM-34) — gives $\omega^2\tau_B^2$. We use the corrected form throughout.

47.2 Step-by-Step Derivation

Step 1: average (SM-11). Taking the period average of $\boldsymbol{\Theta} = \boldsymbol{\Omega} + (\tau_s/I)(\boldsymbol{M}\times\boldsymbol{H})$, and noting that $\boldsymbol{\Omega}$ is a slow quantity so $\overline{\boldsymbol{\Omega}} = \boldsymbol{\Omega}$:

$$\overline{\boldsymbol{\Theta}} = \boldsymbol{\Omega} + \frac{\tau_s}{I}\,\overline{\boldsymbol{M}\times\boldsymbol{H}} = \boldsymbol{\Omega} + \frac{1}{6\eta\varphi}\overline{\boldsymbol{M}\times\boldsymbol{H}}, \tag{47.2.1}$$

using $\tau_s/I = 1/6\eta\varphi$ from Eq. (7.5.12). From here on we write $\boldsymbol{\Theta}$ for $\overline{\boldsymbol{\Theta}}$, as the paper does.

Step 2: insert the averaged torque. From (41.2.5),

$$\boldsymbol{\Theta} = \boldsymbol{\Omega} + \frac{1}{6\eta\varphi}\Big(-\eta\varphi\xi^2\cos^2\alpha\cos 2\alpha\,\boldsymbol{\Omega}\Big) \tag{47.2.2}$$

$$= \boldsymbol{\Omega}\left(1 - \frac{\xi^2}{6}\cos^2\alpha\cos 2\alpha\right). \tag{47.2.3}$$

Step 3: express through $\omega\tau_B$. By (42.2.1) and (42.2.2),

$$\cos^2\alpha\cos 2\alpha = \frac{1-\omega^2\tau_B^2}{\left(1+\omega^2\tau_B^2\right)^2}, \tag{47.2.4}$$

so that

$$\Theta = \Omega\left(1 - \frac{\xi^2}{6}\,\frac{1-\omega^2\tau_B^2}{\left(1+\omega^2\tau_B^2\right)^2}\right), \tag{SM-33}$$

which is (SM-33) in corrected form. ■

47.3 Physical Interpretation: the Whole Effect in One Line

The authors' commentary on this formula is the clearest statement of the physics in either paper, and it deserves to be reproduced in our own words and then checked against the algebra.

Permanent or slowly varying field, $\omega\tau_B < 1$. Then $1-\omega^2\tau_B^2 > 0$, the bracket is less than unity, and

$$\Theta < \Omega. \tag{47.3.1}$$

The particles rotate *slower* than the surrounding fluid. The liquid is forced to flow past them, extra kinetic energy is dissipated, and the additional viscosity is positive.

Fast alternating field, $\omega\tau_B > 1$. Then $1-\omega^2\tau_B^2 < 0$, the bracket exceeds unity, and

$$\Theta > \Omega. \tag{47.3.2}$$

The particles rotate *faster* than the fluid. They no longer retard the flow; they accelerate it. And $\Delta\eta < 0$. Shliomis & Morozov emphasise the reversal: instead of retarding the flow the grains now accelerate it, so that $\Delta\eta$ turns negative — the natural consequence, they observe, of part of the alternating field's energy being converted into kinetic energy of the liquid.

The two extreme cases as checks.

- *No field.* $\xi = 0$ gives $\Theta = \Omega$ exactly: the particles turn with the fluid and $\Delta\eta = 0$, as it must be.
- *Strong stationary field.* The formula above is a weak-field result, but its trend is right: as ξ grows at $\omega = 0$, Θ falls. The exact statement is the one derived in Volume II: in a saturating stationary field the moments are locked, $\Theta \to 0$, and $\Delta\eta \to \tfrac{3}{2}\eta\varphi$ — Eq. (SM-4).

47.4 Connection to the Next Equation

Comparing (SM-33) with the master formula closes the logical circle. That is (SM-34).

Chapter 48

Equation (SM-34): Closing the Circle

$$\Delta\eta = \frac{3}{2}\,\eta\varphi\,\frac{\Omega - \Theta}{\Omega}. \tag{SM-34}$$

48.1 Context

The authors write: "With the help of (33), expression (28) may be transformed into (34)." In this book the relation was derived long ago, in Sec. 14.4, as the general bridge between particle spin and viscosity; here it reappears as a *consequence* of the explicit solution, which is a nontrivial check that no factor has been lost anywhere in Volumes IV and V.

48.2 Step-by-Step Derivation

Step 1: form the slip from (SM-33). Subtracting (47.2.3) from $\boldsymbol{\Omega}$:

$$\boldsymbol{\Omega} - \boldsymbol{\Theta} = \boldsymbol{\Omega} - \boldsymbol{\Omega}\left(1 - \frac{\xi^2}{6}\cos^2\alpha\cos 2\alpha\right) = \frac{\xi^2}{6}\cos^2\alpha\cos 2\alpha\,\boldsymbol{\Omega}. \tag{48.2.1}$$

Step 2: divide by Ω and multiply by $\frac{3}{2}\eta\varphi$.

$$\frac{3}{2}\eta\varphi\,\frac{\Omega - \Theta}{\Omega} = \frac{3}{2}\eta\varphi\cdot\frac{\xi^2}{6}\cos^2\alpha\cos 2\alpha \tag{48.2.2}$$

$$= \frac{3\eta\varphi\xi^2}{12}\cos^2\alpha\cos 2\alpha \tag{48.2.3}$$

$$= \frac{1}{4}\eta\varphi\xi^2\cos^2\alpha\cos 2\alpha. \tag{48.2.4}$$

Step 3: recognise the right-hand side. This is precisely (SM-27), which by Chapter 42 equals (SM-28). Hence

$$\Delta\eta = \frac{3}{2}\eta\varphi\frac{\Omega - \Theta}{\Omega}, \tag{SM-34}$$

■

48.3 Why This Is More Than an Identity

Equation (SM-34) was derived in Volume II from angular-momentum bookkeeping alone, with *no* reference to the magnetisation dynamics, the field waveform, or the weak-field approximation. Its rederivation here from the explicit weak-field solution therefore checks the entire chain

$$\text{(SM-16)} \to \text{(SM-23)} \to \text{(SM-25)} \to \text{(SM-27)} \to \text{(SM-28)} \to \text{(SM-33)} \to \text{(SM-34)}.$$

A single misplaced factor of 2, or a $\cos^2\alpha$ where a $\cos 2\alpha$ belonged, would break the identity.

Its deeper significance is stated by the authors themselves in Sec. III: because (SM-34) rests only on the torque balance, a formula obtained under the weak-field assumption in fact holds for any field strength and any frequency. That is the bridge into the finite-amplitude theory, which will occupy Volume VI: the task there is not to redo the viscosity calculation but merely to compute $(\Omega - \Theta)/\Omega$ — equivalently the averaged torque — without the weak-field assumption.

Chapter 49

Status and Road Map

49.1 What has been established in Volume V

Equation	Statement	Where derived
(SM-25)	$\boldsymbol{M} \times \boldsymbol{H} = -2\Omega\eta\varphi\xi^2 \cos^2\alpha(\cos^2\omega t \cos 2\alpha + \sin\omega t \cos\omega t \sin 2\alpha)$	Ch. 39
(SM-26)	averaged momentum equation; two-scale expansion	Ch. 40
(41.2.3)	$\overline{\boldsymbol{M} \times \boldsymbol{H}} = -\eta\varphi\xi^2 \cos^2\alpha \cos 2\alpha\, \boldsymbol{\Omega}$	Sec. 41.2
(SM-27)	$\Delta\eta = \frac{1}{4}\eta\varphi\xi^2 \cos^2\alpha \cos 2\alpha$	Ch. 41
(SM-28)	$\Delta\eta = \frac{1}{4}\eta\varphi\xi^2 \dfrac{1-\omega^2\tau_B^2}{(1+\omega^2\tau_B^2)^2}$; zero at $\omega\tau_B = 1$, minimum $-\Delta\eta(0)/8$ at $\omega\tau_B = \sqrt{3}$	Ch. 42
(SM-29)	$\Delta\eta = -(\varphi/\eta)(M_s H/6\omega)^2$; τ_B cancels	Ch. 43
(SM-30)	$M_x = \chi H_0 \cos\alpha \cos(\omega t - \alpha)$, second variant	Ch. 44
(SM-31)	coupled equations for a and θ	Ch. 45
(SM-32)	$\theta = 2\alpha$, $a = \Omega\tau_B\chi H_0 \cos^2\alpha$; agrees with (SM-23)	Ch. 46
(SM-33)	$\Theta = \Omega[1 - \frac{\xi^2}{6}(1-\omega^2\tau_B^2)/(1+\omega^2\tau_B^2)^2]$	Ch. 47
(SM-34)	$\Delta\eta = \frac{3}{2}\eta\varphi(\Omega - \Theta)/\Omega$, recovered	Ch. 48

49.2 Cumulative progress

Shliomis & Morozov: Eqs. (SM-1)–(SM-34) complete — the whole of Secs. I and II of the paper. Bacri *et al.*: Eqs. (B-1)–(B-4) and (B-10) complete.

49.3 What remains

- **Volume VI** — Eqs. (SM-35)–(SM-43), Sec. III of the paper: the nonlinear magnetisation equation at arbitrary amplitude; the decomposition $\boldsymbol{M} = (nmF,\ nm\Omega\tau_B G,\ 0)$; the coupled equations (SM-36) for F and G; the rectangular waveform $\gamma(t)$; the averaged torque (SM-37); the general formula (SM-38)–(SM-39) and its reduction to (SM-34); the analytic weak-field result (SM-40) for square-wave modulation; the half-period identity (SM-41); the formal solution (SM-42)–(SM-43); and the numerical results of Figs. 3–5,

including the values $\omega_0\tau_B = 0.98$, $\omega_m\tau_B = 1.707$, $\Delta\eta_m = -0.105\,\Delta\eta(0)$ for the rectangular case.

- **Volume VII** — Eqs. (B-5)–(B-9): the effective-field method of Martsenyuk, Raikher and Shliomis derived from the Smoluchowski equation (11.6.1); the equation (B-6) for $\boldsymbol{\xi}_e$; the zero-approximation equation (B-7); the function F of (B-8); the averaged torque and the function g of (B-9); and the comparison with the measured isolines of Figs. 2 and 5, including the critical values $\xi_c = 126$, $\omega_c\tau_B = 20$ for $\eta_{\text{total}} = 0$.

End of Volume V. Volume VI begins with Eq. (SM-35).

THE NEGATIVE-VISCOSITY EFFECT IN FERROFLUIDS

A Complete First-Principles Reconstruction of Every Equation

VOLUME VI

Arbitrary Field Amplitude:
the Langevin Nonlinearity, the Square-Wave Solution,
and the Viscosity Map — Equations (SM-35)–(SM-43)

Where Volume V ended. Volume V completed Sec. II of Shliomis & Morozov. It computed the instantaneous torque (SM-25), separated the fast and slow time scales (SM-26), extracted the rotational viscosity (SM-27), and obtained the central weak-field result

$$\Delta\eta = \frac{1}{4}\eta\varphi\xi^2\,\frac{1-\omega^2\tau_B^2}{\left(1+\omega^2\tau_B^2\right)^2}, \tag{SM-28}$$

with its zero at $\omega\tau_B = 1$ and its minimum $-\Delta\eta(0)/8$ at $\omega\tau_B = \sqrt{3}$. It then re-derived the same result by the authors' second route and closed the circle by recovering the master formula (SM-34).

Volume VI takes up Sec. III. The restriction $\xi \ll 1$ is abandoned: the Langevin function must now be kept in full, the magnetisation equation becomes genuinely nonlinear, and superposition is no longer available. What survives is (SM-34), because it rests on torque balance alone. The task therefore reduces to computing the averaged torque without approximation — which the authors accomplish by replacing the harmonic field with a square wave, for which the equations can be integrated in closed form. Chapter numbers resume at 50, page numbers at 175.

Part XI

Arbitrary Field Amplitude

Chapter 50

Equation (SM-35): The Nonlinear Magnetisation Equation

$$\frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t} = \boldsymbol{\Omega} \times \boldsymbol{M} - \frac{1}{\tau_B}\left(\boldsymbol{M} - M_0 \frac{\boldsymbol{H}}{H}\right) - \frac{\tau_s}{I}\,\boldsymbol{M} \times \big(\boldsymbol{M} \times \boldsymbol{H}\big). \tag{SM-35}$$

50.1 Context

Section III of the paper opens by giving up the linear law (SM-15) and returning to the full equation (SM-13), now written with the Langevin magnetisation $M_0 = nmL[\xi(t)]$ in place of χH. Equation (SM-35) is therefore not a new equation — it is (SM-13) with the equilibrium term written out for arbitrary field strength. Its content is entirely in what it *no longer* assumes.

Three consequences follow immediately, and they organise the whole of this volume:

1. The equation is nonlinear in $\boldsymbol{H}$ through $L(\xi)$, so the circular-polarisation decomposition of Volume IV — which relied on superposition — is unavailable.

2. The equation is nonlinear in $\boldsymbol{M}$ through the last term, so even at fixed field the two magnetisation components couple.

3. A general analytic solution is therefore out of reach, and the authors proceed by a change of variables, a judicious choice of waveform, and finally numerics.

50.2 Symbols

Symbol	Meaning	Type	Units
$M_0 = nmL[\xi(t)]$	magnitude of the instantaneous-equilibrium magnetisation	scalar	G
$\boldsymbol{H}/H$	unit vector along the instantaneous field	vector	—
ξ	**in Sec. III**: the time-independent dimensionless field strength, $\xi = mH_0/kT$	scalar	—
$\gamma(t)$	periodic waveform of unit amplitude, $\xi(t) = \xi\gamma(t)$	function	—
$F(t)$	dimensionless longitudinal magnetisation, $M_x = nmF$	function	—
$G(t)$	dimensionless transverse magnetisation, $M_y = nm\Omega\tau_B G$	function	—
$\varepsilon = \pi/2\omega\tau_B$	dimensionless half-period, used from Ch. 54	scalar	—

Remark 50.2.1 (The third meaning of ξ). Volume IV warned that Sec. II of the paper uses the acting-field parameter $\xi = mH_0/kT\sqrt{2}$. Section III reverts to the amplitude-based $\xi = mH_0/kT$, which is also the ξ_0 of Bacri *et al.* The check is Sec. 54.7: at zero frequency (SM-40) must reduce to (SM-3) with the *static* Langevin parameter, and it does so only with the Sec. III convention. From this volume onwards, $\xi \equiv mH_0/kT$ unless explicitly stated otherwise.

50.3 The Waveform Factorisation

The authors write the time-dependent Langevin argument as

$$\xi(t) = \xi\,\gamma(t), \tag{50.3.1}$$

splitting it into a constant *strength* ξ and a dimensionless *waveform* $\gamma(t)$ of unit amplitude. For the harmonic field (SM-14), $\gamma(t) = \cos\omega t$; for the rectangular field used from Chapter 54 onwards, $\gamma(t) = \pm 1$ alternately.

The factorisation is useful because the two dependences play different roles: ξ controls how nonlinear the Langevin function is, while γ controls the time structure. Separating them makes it possible to change the waveform without touching the nonlinearity.

A property of L that makes the factorisation clean. The Langevin function is odd,

$$L(-x) = \coth(-x) - \frac{1}{-x} = -\coth x + \frac{1}{x} = -L(x), \tag{50.3.2}$$

using $\coth(-x) = -\coth x$ (both sinh and cosh have definite parity: sinh odd, cosh even). Consequently

$$M_0\frac{\boldsymbol{H}}{H} = nmL\big(\xi|\gamma|\big)\,\mathrm{sign}\big(\gamma\big)\hat{\boldsymbol{x}} = nmL\big(\xi\gamma\big)\hat{\boldsymbol{x}}, \tag{50.3.3}$$

so the awkward pair (magnitude, unit vector) collapses into the single signed expression $nmL(\xi\gamma)$. This is why the paper can write $M_0 = nmL[\xi(t)]$ without further comment even when the field reverses.

50.4 The Two-Component Decomposition

With the field along x and the vorticity along z, the magnetisation has only x and y components (nothing generates a z component: the torques lie in the xy plane's normal direction and the equations preserve $M_z = 0$). The authors introduce

$$\boxed{M_x = nm\,F(t), \qquad M_y = nm\,\Omega\tau_B\,G(t)} \tag{50.4.1}$$

Two things are built into this definition and both deserve comment.

(i) Why F and G are dimensionless. Both are magnetisations divided by the saturation value $nm = \varphi M_s$, so $|F| \le 1$ automatically. G carries an extra division by $\Omega\tau_B$.

(ii) Why the factor $\Omega\tau_B$ is extracted from M_y. Because M_y vanishes identically in a quiescent fluid — established in Sec. 44.6 for the harmonic case, and true generally by the symmetry argument of Sec. 27.5. The transverse magnetisation is therefore proportional to Ω for slow flows, and extracting that factor leaves G finite and Ω-independent as $\Omega \to 0$. The authors state the same thing as: the ratio M_y/M_x is of order $\Omega\tau_B$, since in a quiescent fluid only the x component survives.

Assumption 50.4.1 ($\Omega\tau_B$ is now the only small parameter)**.** Everything is computed to first order in $\Omega\tau_B$ and to *all* orders in ξ. This is the exact reverse of Volume V, where ξ was small and the frequency arbitrary.

50.5 Connection to the Next Equation

Substituting (50.4.1) into (SM-35) and separating components converts one nonlinear vector equation into two coupled scalar ones. Those are (SM-36).

Chapter 51

Equation (SM-36): The Coupled Equations for F and G

$$\begin{aligned}\tau_B \frac{\mathrm{d}F}{\mathrm{d}t} &= L\big[\xi\gamma(t)\big] - F(t),\\ \tau_B \frac{\mathrm{d}G}{\mathrm{d}t} &= F(t) - \big(1 + \tfrac{1}{2}\,\xi\gamma(t)F(t)\big)\,G(t).\end{aligned} \tag{SM-36}$$

51.1 Context

This pair is the exact content of (SM-35) for the geometry at hand, to first order in $\Omega\tau_B$. Everything remaining in Sec. III of the paper is the solution of these two equations for various waveforms γ.

51.2 Step-by-Step Derivation

Step 1: the convective term. With $\boldsymbol{\Omega} = \Omega\hat{\boldsymbol{z}}$ and $\boldsymbol{M} = (M_x, M_y, 0)$,

$$\boldsymbol{\Omega} \times \boldsymbol{M} = \Omega\hat{\boldsymbol{z}} \times \big(M_x, M_y, 0\big) = \Omega\big(-M_y,\ M_x,\ 0\big), \tag{51.2.1}$$

using $\hat{\boldsymbol{z}} \times \hat{\boldsymbol{x}} = \hat{\boldsymbol{y}}$ and $\hat{\boldsymbol{z}} \times \hat{\boldsymbol{y}} = -\hat{\boldsymbol{x}}$.

Step 2: the relaxation term. By (50.3.3),

$$-\frac{1}{\tau_B}\left(\boldsymbol{M} - M_0\frac{\boldsymbol{H}}{H}\right) = -\frac{1}{\tau_B}\Big(M_x - nmL(\xi\gamma),\ M_y,\ 0\Big). \tag{51.2.2}$$

Step 3: the nonlinear term. Expand the double cross product with the identity (1.7.1):

$$\boldsymbol{M} \times (\boldsymbol{M} \times \boldsymbol{H}) = \boldsymbol{M}\big(\boldsymbol{M}\cdot\boldsymbol{H}\big) - \boldsymbol{H}\big(\boldsymbol{M}\cdot\boldsymbol{M}\big) \tag{51.2.3}$$

$$= \big(M_x, M_y, 0\big)\,M_x H - \big(H, 0, 0\big)\big(M_x^2 + M_y^2\big), \tag{51.2.4}$$

since $\boldsymbol{H} = (H, 0, 0)$ gives $\boldsymbol{M}\cdot\boldsymbol{H} = M_x H$. Componentwise:

$$x: \qquad M_x^2 H - H\big(M_x^2 + M_y^2\big) = -H M_y^2, \tag{51.2.5}$$

$$y: \qquad M_y M_x H - 0 = M_x M_y H. \tag{51.2.6}$$

The x component is proportional to $M_y^2 = O\big((\Omega\tau_B)^2\big)$ and is therefore *second order*: it is dropped by Assumption 50.4.1. The y component is first order and must be kept.

Step 4: the x equation. Collecting the x components of (51.2.1), (51.2.2) and (51.2.5):

$$\frac{\mathrm{d}M_x}{\mathrm{d}t} = \underbrace{-\Omega M_y}_{O(\Omega^2\tau_B)} - \frac{M_x - nmL(\xi\gamma)}{\tau_B} + \underbrace{\frac{\tau_s}{I} H M_y^2}_{O(\Omega^2)}. \tag{51.2.7}$$

Both indicated terms are second order in $\Omega\tau_B$ and are dropped. Multiplying by τ_B and substituting $M_x = nmF$ from (50.4.1), the common factor nm cancels:

$$\boxed{\tau_B \frac{\mathrm{d}F}{\mathrm{d}t} = L\big[\xi\gamma(t)\big] - F(t)} \tag{51.2.8}$$

which is the first line of (SM-36). ✓ Note that it does not involve G at all: the longitudinal magnetisation is unaffected, at this order, by the flow. It can therefore be solved first, independently.

Step 5: the y equation. Collecting the y components:

$$\frac{\mathrm{d}M_y}{\mathrm{d}t} = \Omega M_x - \frac{M_y}{\tau_B} - \frac{\tau_s}{I} M_x M_y H. \tag{51.2.9}$$

Substitute $M_x = nmF$, $M_y = nm\Omega\tau_B G$ and $\tau_s/I = 1/6\eta\varphi$ [Eq. (7.5.12)]:

$$nm\Omega\tau_B \frac{\mathrm{d}G}{\mathrm{d}t} = \Omega\, nmF - \frac{nm\Omega\tau_B G}{\tau_B} - \frac{\big(nmF\big)\big(nm\Omega\tau_B G\big)H}{6\eta\varphi}. \tag{51.2.10}$$

Divide throughout by $nm\Omega$ (both nonzero):

$$\tau_B \frac{\mathrm{d}G}{\mathrm{d}t} = F - G - \frac{nm\,\tau_B\, H}{6\eta\varphi}\, FG. \tag{51.2.11}$$

Step 6: evaluate the nonlinear coefficient. This is the same kind of cancellation met in Sec. 33.4. Insert $\tau_B = 3\eta V/kT$ and $\varphi = nV$:

$$\frac{nm\tau_B H}{6\eta\varphi} = \frac{n\,m}{6\eta\,(nV)} \cdot \frac{3\eta V}{kT} \cdot H \tag{51.2.12}$$

$$= \frac{3\,n\,m\,\eta\,V\,H}{6\,\eta\,n\,V\,kT} \qquad \text{(collecting factors)} \tag{51.2.13}$$

$$= \frac{mH}{2kT} \qquad (n,\ V,\ \eta \text{ cancel};\ 3/6 = 1/2) \tag{51.2.14}$$

$$= \frac{\xi\gamma(t)}{2}, \qquad \text{since } \frac{mH(t)}{kT} = \xi\gamma(t) \tag{51.2.15}$$

which is dimensionless, as it must be, and depends on time only through the waveform.

Step 7: assemble. Substituting (51.2.15) into (51.2.11):

$$\tau_B \frac{\mathrm{d}G}{\mathrm{d}t} = F - G - \tfrac{1}{2}\xi\gamma FG \tag{51.2.16}$$

$$= F - \Big(1 + \tfrac{1}{2}\xi\gamma(t)F(t)\Big)G(t), \tag{51.2.17}$$

having factored G out of the last two terms. This is the second line of (SM-36). ■

51.3 Physical Interpretation

The F equation is pure Debye relaxation with a nonlinear drive. Its left side is the relaxation of the longitudinal magnetisation towards its instantaneous equilibrium $L[\xi\gamma(t)]$. Only the *drive* is nonlinear: the response is a simple exponential filter with time constant τ_B. At small ξ, $L \to \xi\gamma/3$ and the equation reduces to the linear relaxation equation (44.4.2) of Volume V.

The G equation contains the feedback. Compare the two equations: the F equation has the constant relaxation rate $1/\tau_B$, the G equation has the *enhanced* rate $(1 + \frac{1}{2}\xi\gamma F)/\tau_B$. The extra piece is the magnetic torque acting on the particles: a strongly magnetised grain in a strong field is mechanically pulled back into alignment, which destroys the transverse magnetisation faster than Brownian motion alone would. It is the same feedback that produced the denominator $1 + \zeta$ in Eq. (15.5.17), and indeed $\frac{1}{2}\xi\gamma F$ is the instantaneous version of $\zeta = \frac{1}{2}\xi L(\xi)$: in a stationary field $F \to L(\xi)$ and $\gamma = 1$, whereupon $\frac{1}{2}\xi\gamma F \to \frac{1}{2}\xi L(\xi) = \zeta$ exactly. ✓ This is a strong internal consistency check linking Volume II to Volume VI.

The drive of G is F. The transverse magnetisation is generated by the vortex sweeping the longitudinal one sideways. No F, no G; no Ω, no G either (that factor was extracted in (50.4.1)).

51.4 Connection to the Next Equation

The quantity we actually need is not G itself but the averaged torque, which involves $\overline{G\gamma}$. That is (SM-37).

Chapter 52

Equation (SM-37): The Averaged Torque at Arbitrary Amplitude

$$\overline{\big(\boldsymbol{M}\times\boldsymbol{H}\big)_z} = -\overline{M_y H_x} = -3\eta\varphi\,\Omega\,\xi\,\overline{G(t)\gamma(t)}. \tag{SM-37}$$

52.1 Context

Volume V established that the flow responds only to the period average of the torque (Chapter 40). That conclusion used nothing about the field amplitude, so it carries over unchanged. What changes is the value of the average, which must now be expressed through the unknown function G.

52.2 Step-by-Step Derivation

Step 1: the torque has only a z component. Exactly as in Sec. 39.4, Step 1: with $\boldsymbol{H} = (H, 0, 0)$ and $\boldsymbol{M} = (M_x, M_y, 0)$,

$$\big(\boldsymbol{M}\times\boldsymbol{H}\big)_z = M_x H_y - M_y H_x = -M_y H_x. \tag{52.2.1}$$

Step 2: substitute the definitions. With $M_y = nm\Omega\tau_B G(t)$ from (50.4.1) and $H_x = H_0\gamma(t)$,

$$\big(\boldsymbol{M}\times\boldsymbol{H}\big)_z = -nm\,\Omega\,\tau_B\,H_0\,G(t)\gamma(t). \tag{52.2.2}$$

Step 3: average over the period. The factors nm, Ω, τ_B and H_0 are constants on the fast time scale, so

$$\overline{\big(\boldsymbol{M}\times\boldsymbol{H}\big)_z} = -nm\,\Omega\,\tau_B\,H_0\,\overline{G\gamma}. \tag{52.2.3}$$

Step 4: evaluate the prefactor. Insert $\tau_B = 3\eta V/kT$ and $\varphi = nV$:

$$nm\tau_B H_0 = n\,m\,\frac{3\eta V}{kT}\,H_0 \tag{52.2.4}$$

$$= 3\eta\,(nV)\,\frac{mH_0}{kT} \qquad \text{(regrouping)} \tag{52.2.5}$$

$$= 3\eta\varphi\,\xi, \qquad \text{since } \xi = \frac{mH_0}{kT} \text{ in Sec. III} \tag{52.2.6}$$

Step 5: assemble. Substituting (52.2.6) into (52.2.3):

$$\overline{\left(\boldsymbol{M}\times\boldsymbol{H}\right)_z} = -3\eta\varphi\Omega\xi\,\overline{G\gamma}, \tag{SM-37}$$

■

52.3 Physical Interpretation

The whole of the finite-amplitude problem has now been compressed into the single number $\overline{G\gamma}$: the correlation between the transverse magnetisation and the driving waveform. If G is in phase with γ the correlation is positive, the torque opposes the vorticity, and the viscosity rises. If G is predominantly out of phase — which is what a large lag does — the correlation becomes negative and so does the viscosity. Everything hinges on a phase relationship, exactly as in the weak-field theory, where the same role was played by $\cos 2\alpha$.

52.4 Connection to the Next Equation

Converting the averaged torque into a viscosity gives (SM-38), and the average itself is defined in (SM-39).

Chapter 53

Equations (SM-38) and (SM-39): The General Viscosity Formula

$$\Delta\eta = \tfrac{3}{4}\,\eta\varphi\,\xi\,\overline{G(t)\gamma(t)}, \tag{SM-38}$$

$$\overline{G(t)\gamma(t)} = \frac{\omega}{2\pi}\int_0^{2\pi/\omega} G(t)\gamma(t)\,\mathrm{d}t. \tag{SM-39}$$

53.1 Context

Equation (SM-39) is simply the definition of the period average, restated for this particular product. Equation (SM-38) converts the torque into a viscosity by the route of Chapter 41.

Remark 53.1.1 (On the numerical coefficient). The scanned text of the paper renders the coefficient of (SM-38) indistinctly. Its value is fixed unambiguously by internal consistency: (SM-38) must, when combined with the relation $(\xi/2)\overline{G\gamma} = (\Omega - \Theta)/\Omega$ derived in Sec. 53.4 below, reproduce the master formula (SM-34). That requirement gives $\frac{3}{4}$, and the derivation of Sec. 53.3 obtains $\frac{3}{4}$ independently from the momentum equation. We therefore use $\frac{3}{4}$ throughout.

53.2 Symbols

Nothing new. Note that $\Delta\eta$ here depends on ξ both explicitly and through G, which is why the dependence on field strength turns out to be non-monotonic (Sec. 57.3).

53.3 Step-by-Step Derivation of (SM-38)

Step 1: the magnetic force density. By the averaged momentum equation (SM-26), the magnetic contribution is $\frac{1}{2}\nabla\times\overline{\boldsymbol{M}\times\boldsymbol{H}}$. Writing the averaged torque of (SM-37) as a vector — it points along $\hat{\boldsymbol{z}}$, and $\boldsymbol{\Omega} = \Omega\hat{\boldsymbol{z}}$, so

$$\overline{\boldsymbol{M}\times\boldsymbol{H}} = -3\eta\varphi\,\xi\,\overline{G\gamma}\;\boldsymbol{\Omega}, \tag{53.3.1}$$

we obtain

$$\tfrac{1}{2}\nabla\times\overline{\boldsymbol{M}\times\boldsymbol{H}} = -\tfrac{3}{2}\eta\varphi\xi\,\overline{G\gamma}\;\nabla\times\boldsymbol{\Omega}. \tag{53.3.2}$$

Step 2: add the ordinary viscous term. By Eq. (41.2.6), $\eta\Delta\boldsymbol{v} = -2\eta\nabla\times\boldsymbol{\Omega}$. Hence

$$\eta\Delta\boldsymbol{v} + \tfrac{1}{2}\nabla\times\overline{\boldsymbol{M}\times\boldsymbol{H}} = -2\eta\nabla\times\boldsymbol{\Omega} - \tfrac{3}{2}\eta\varphi\xi\overline{G\gamma}\,\nabla\times\boldsymbol{\Omega} \tag{53.3.3}$$

$$= -2\Big(\eta + \tfrac{3}{4}\eta\varphi\xi\overline{G\gamma}\Big)\nabla\times\boldsymbol{\Omega}. \tag{53.3.4}$$

Step 3: read off the viscosity.

$$\Delta\eta = \tfrac{3}{4}\eta\varphi\xi\,\overline{G(t)\gamma(t)}, \tag{SM-38}$$

■

53.4 Consistency: Recovering (SM-34)

The authors verify their formula by comparing it with the torque balance, and we follow them.

Step 1. From Eq. (14.2.1), the balance of magnetic and viscous torques gives $\overline{\boldsymbol{M}\times\boldsymbol{H}} = -6\eta\varphi(\boldsymbol{\Omega}-\boldsymbol{\Theta})$.

Step 2. Compare with (53.3.1):

$$-6\eta\varphi\big(\boldsymbol{\Omega}-\boldsymbol{\Theta}\big) = -3\eta\varphi\xi\overline{G\gamma}\,\boldsymbol{\Omega}. \tag{53.4.1}$$

Dividing by $-6\eta\varphi\,\Omega$:

$$\boxed{\frac{\Omega-\Theta}{\Omega} = \frac{\xi}{2}\,\overline{G(t)\gamma(t)}} \tag{53.4.2}$$

which is exactly the relation the authors state between their Eqs. (SM-39) and (SM-40).

Step 3. Substitute (53.4.2) into (SM-38):

$$\Delta\eta = \tfrac{3}{4}\eta\varphi\xi\overline{G\gamma} = \tfrac{3}{4}\eta\varphi\cdot 2\cdot\frac{\Omega-\Theta}{\Omega} = \tfrac{3}{2}\eta\varphi\,\frac{\Omega-\Theta}{\Omega}, \tag{SM-34}$$

recovering the master formula. ✓

What this establishes. Equation (SM-34) was derived in Volume II from angular-momentum bookkeeping, and in Volume V from the explicit weak-field solution. Here it is recovered from the *finite-amplitude* theory. The authors draw the correct conclusion: a formula obtained under the weak-field assumption in fact holds for any field strength and frequency, because it never used the weak-field assumption in the first place — only the torque balance.

53.5 Connection to the Next Equation

To get numbers we must solve (SM-36). The authors do so first analytically, in the weak-field limit and for a rectangular waveform. That is (SM-40).

Chapter 54

Equation (SM-40): The Square-Wave Solution

$$\Delta\eta = \frac{1}{4}\,\eta\varphi\,\xi^2\left(2 - \tanh^2\varepsilon - 2\,\frac{\tanh\varepsilon}{\varepsilon}\right), \qquad \varepsilon = \frac{\pi}{2\omega\tau_B}. \tag{SM-40}$$

54.1 Context

The authors note that Eqs. (SM-36) can be solved relatively easily if the field varies not harmonically but in a stepwise way — that is, if $\gamma(t)$ is a rectangular periodic function with period $T_0 = 2\pi/\omega$, as drawn in their Fig. 2. In weak fields the nonlinear term in the second equation may then be neglected and the problem solved in closed form.

The result is important for two reasons. First, it is the only closed-form finite-frequency result in Sec. III. Second, comparing it with the harmonic result (SM-28) shows how little the answer depends on the waveform — which is what licenses the use of a square wave for the numerics that follow.

Remark 54.1.1 (A misprint in the published equation). The published (SM-40) has $\tanh\varepsilon$ where the derivation below gives $\tanh^2\varepsilon$. That the square belongs there is confirmed by the paper's own numbers: with $\tanh^2\varepsilon$ the sign change occurs at $\omega\tau_B = 0.978$ and the minimum at $\omega\tau_B = 1.707$ with depth $-0.105\,\Delta\eta(0)$, which are precisely the three values the authors quote in the following paragraph. We verify this arithmetic explicitly in Sec. 54.7.

54.2 Symbols

Symbol	Meaning	Type	Units
$\gamma(t)$	rectangular waveform: $+1$ for $0 < t < \pi/\omega$, -1 for $\pi/\omega < t < 2\pi/\omega$	function	—
$T_h = \pi/\omega$	half-period of the field	scalar	s
$\varepsilon = T_h/2\tau_B = \pi/2\omega\tau_B$	half-period measured in units of $2\tau_B$	dimensionless	—
$u = e^{-2\varepsilon}$	abbreviation used in the algebra	dimensionless	—

Note the inverse relation between ε and frequency: $\varepsilon \to \infty$ is the static limit, $\varepsilon \to 0$ the

high-frequency limit.

54.3 Assumptions

1. Weak field, $\xi \ll 1$, so that $L(\xi\gamma) \simeq \xi\gamma/3$ [Eq. (10.3.8)] and the nonlinear term $\frac{1}{2}\xi\gamma FG$ in the second of (SM-36) — which carries an explicit factor ξ and is therefore of higher order — may be dropped.
2. Rectangular waveform.
3. Steady periodic state: all transients have died.
4. $\Omega\tau_B \ll 1$, as always.

54.4 Mathematical Background: antiperiodicity

Result 54.4.1 (Half-period antisymmetry)**.** *In the steady state, for a waveform satisfying* $\gamma(t+T_h) = -\gamma(t)$*,*

$$F\big(t+T_h\big) = -F(t), \qquad G\big(t+T_h\big) = -G(t). \tag{54.4.1}$$

Derivation. Consider the transformation $t \to t+T_h$, $F \to -F$, $G \to -G$. Under it, the first of (SM-36) becomes

$$\tau_B \frac{\mathrm{d}(-F)}{\mathrm{d}t} = L\big[\xi\gamma(t+T_h)\big] - (-F) = L\big[-\xi\gamma(t)\big] + F = -L\big[\xi\gamma(t)\big] + F, \tag{54.4.2}$$

using the oddness (50.3.2); multiplying by -1 returns the original equation. The second of (SM-36) is likewise invariant: the term $\frac{1}{2}\xi\gamma F$ is even under the transformation (both γ and F change sign), so $(1+\frac{1}{2}\xi\gamma F)G \to -(1+\frac{1}{2}\xi\gamma F)G$, matching the sign change of F on the right and of $\mathrm{d}G/\mathrm{d}t$ on the left. The transformed functions therefore solve the same equations; and since the steady periodic solution is unique (the equations are linear-in-G relaxation equations with a unique attracting periodic orbit), the solution must map to itself, giving (54.4.1). ■

This result does the work of halving the problem: it suffices to solve on $0 < t < T_h$, where $\gamma = +1$, and impose (54.4.1) as the boundary condition instead of an initial condition.

54.5 Step-by-Step Derivation

Throughout, $0 < t < T_h$ and $\gamma = +1$, and we write $u \equiv e^{-2\varepsilon} = e^{-T_h/\tau_B}$ — the two agree because $T_h/\tau_B = 2\varepsilon$ by the definition of ε.

Step 1: solve for F. With $L(\xi\gamma) \simeq \xi/3$ the first of (SM-36) is

$$\tau_B \frac{\mathrm{d}F}{\mathrm{d}t} + F = \frac{\xi}{3}, \tag{54.5.1}$$

a linear equation with constant coefficients. Its general solution is the particular solution $\xi/3$ plus the decaying homogeneous solution:

$$F(t) = \frac{\xi}{3} + \left(F_0 - \frac{\xi}{3}\right) e^{-t/\tau_B}, \qquad F_0 \equiv F(0). \tag{54.5.2}$$

Step 2: impose antiperiodicity on F. Setting $t = T_h$ in (54.5.2) and using $F(T_h) = -F_0$:

$$-F_0 = \frac{\xi}{3} + \left(F_0 - \frac{\xi}{3}\right) u \tag{54.5.3}$$

$$-F_0 - F_0 u = \frac{\xi}{3} - \frac{\xi}{3} u \tag{54.5.4}$$

$$-F_0\big(1+u\big) = \frac{\xi}{3}\big(1-u\big) \tag{54.5.5}$$

$$F_0 = -\frac{\xi}{3}\,\frac{1-u}{1+u} = -\frac{\xi}{3}\tanh\varepsilon, \tag{54.5.6}$$

the last step using

$$\tanh\varepsilon = \frac{e^{\varepsilon} - e^{-\varepsilon}}{e^{\varepsilon} + e^{-\varepsilon}} = \frac{1 - e^{-2\varepsilon}}{1 + e^{-2\varepsilon}} = \frac{1-u}{1+u}, \tag{54.5.7}$$

obtained by multiplying numerator and denominator by $e^{-\varepsilon}$.

Step 3: the explicit F. Substituting (54.5.6) into (54.5.2):

$$F(t) = \frac{\xi}{3} + \left(-\frac{\xi}{3}\tanh\varepsilon - \frac{\xi}{3}\right) e^{-t/\tau_B} \tag{54.5.8}$$

$$= \frac{\xi}{3}\Big[1 - A\,e^{-t/\tau_B}\Big], \qquad A \equiv 1 + \tanh\varepsilon = \frac{2}{1+u}, \tag{54.5.9}$$

where the value of A follows from (54.5.7): $1 + (1-u)/(1+u) = \big[(1+u)+(1-u)\big]/(1+u) = 2/(1+u)$.

Step 4: the equation for G. Dropping the nonlinear term (Assumption 1), the second of (SM-36) is

$$\tau_B \frac{\mathrm{d}G}{\mathrm{d}t} + G = F(t) = \frac{\xi}{3}\Big[1 - Ae^{-t/\tau_B}\Big]. \tag{54.5.10}$$

Step 5: solve it. The right-hand side has two pieces, and we find a particular solution for each.

(a) The constant $\xi/3$. A constant $G = \xi/3$ satisfies $\tau_B \cdot 0 + \xi/3 = \xi/3$. ✓

(b) The term $-(\xi A/3)e^{-t/\tau_B}$. This is *resonant*: e^{-t/τ_B} is itself a solution of the homogeneous equation, so the usual guess Be^{-t/τ_B} fails. The standard remedy is to multiply by t; try $G = Bte^{-t/\tau_B}$. Then

$$\tau_B \frac{\mathrm{d}}{\mathrm{d}t}\Big(Bte^{-t/\tau_B}\Big) + Bte^{-t/\tau_B} = \tau_B B\left(e^{-t/\tau_B} - \frac{t}{\tau_B}e^{-t/\tau_B}\right) + Bte^{-t/\tau_B} \tag{54.5.11}$$

$$= \tau_B B e^{-t/\tau_B} - Bte^{-t/\tau_B} + Bte^{-t/\tau_B} \tag{54.5.12}$$

$$= \tau_B B e^{-t/\tau_B}, \tag{54.5.13}$$

so matching to $-(\xi A/3)e^{-t/\tau_B}$ requires

$$B = -\frac{\xi A}{3\tau_B}. \tag{54.5.14}$$

(c) Assemble. Adding the homogeneous solution Ce^{-t/τ_B}:

$$G(t) = \frac{\xi}{3} - \frac{\xi A}{3\tau_B}\,t\,e^{-t/\tau_B} + C\,e^{-t/\tau_B}. \tag{54.5.15}$$

Step 6: impose antiperiodicity on G. At $t = T_h = 2\varepsilon\tau_B$,

$$G(T_h) = \frac{\xi}{3} - \frac{\xi A}{3\tau_B}\big(2\varepsilon\tau_B\big)u + Cu = \frac{\xi}{3} - \frac{2\xi A\varepsilon}{3}u + Cu, \tag{54.5.16}$$

and this must equal $-G(0) = -\big(\xi/3 + C\big)$. Hence

$$\frac{\xi}{3} - \frac{2\xi A\varepsilon}{3}u + Cu = -\frac{\xi}{3} - C \tag{54.5.17}$$

$$C\big(1+u\big) = -\frac{2\xi}{3} + \frac{2\xi A\varepsilon}{3}u \tag{54.5.18}$$

$$C = \frac{2\xi}{3}\cdot\frac{A\varepsilon u - 1}{1+u}. \tag{54.5.19}$$

Step 7: the average we need. By (SM-41), proved in Chapter 55, the average of $G\gamma$ over the full period equals the average of G over the half period on which $\gamma = +1$:

$$\overline{G\gamma} = \frac{1}{T_h}\int_0^{T_h} G(t)\,\mathrm{d}t. \tag{54.5.20}$$

Step 8: perform the integral. Three elementary integrals are needed:

$$\int_0^{T_h} \mathrm{d}t = T_h = 2\varepsilon\tau_B, \tag{54.5.21}$$

$$\int_0^{T_h} e^{-t/\tau_B}\mathrm{d}t = \tau_B\big(1-u\big), \tag{54.5.22}$$

$$\int_0^{T_h} t\,e^{-t/\tau_B}\mathrm{d}t = \tau_B^2\Big[1 - \big(1+2\varepsilon\big)u\Big]. \tag{54.5.23}$$

The last one is obtained by parts: with $\mathrm{d}v = e^{-t/\tau_B}\mathrm{d}t$, $v = -\tau_B e^{-t/\tau_B}$,

$$\int_0^{T} te^{-t/\tau_B}\mathrm{d}t = \Big[-\tau_B t e^{-t/\tau_B}\Big]_0^{T} + \tau_B\int_0^{T} e^{-t/\tau_B}\mathrm{d}t \tag{54.5.24}$$

$$= -\tau_B T e^{-T/\tau_B} + \tau_B^2\Big(1 - e^{-T/\tau_B}\Big), \tag{54.5.25}$$

and putting $T = T_h = 2\varepsilon\tau_B$, $e^{-T/\tau_B} = u$ gives $-2\varepsilon\tau_B^2 u + \tau_B^2(1-u) = \tau_B^2[1-(1+2\varepsilon)u]$. ✓

Therefore

$$\int_0^{T_h} G\,\mathrm{d}t = \frac{\xi}{3}\big(2\varepsilon\tau_B\big) - \frac{\xi A}{3\tau_B}\,\tau_B^2\Big[1-\big(1+2\varepsilon\big)u\Big] + C\tau_B\big(1-u\big). \tag{54.5.26}$$

Dividing by $T_h = 2\varepsilon\tau_B$:

$$\overline{G\gamma} = \frac{\xi}{3} - \frac{\xi A}{6\varepsilon}\Big[1-\big(1+2\varepsilon\big)u\Big] + \frac{C\big(1-u\big)}{2\varepsilon}. \tag{54.5.27}$$

Step 9: simplify the second term. With $A = 2/(1+u)$,

$$-\frac{\xi A}{6\varepsilon}\Big[1-\big(1+2\varepsilon\big)u\Big] = -\frac{\xi}{3\varepsilon}\cdot\frac{1-u-2\varepsilon u}{1+u} \tag{54.5.28}$$

$$= -\frac{\xi}{3\varepsilon}\cdot\frac{1-u}{1+u} + \frac{\xi}{3\varepsilon}\cdot\frac{2\varepsilon u}{1+u} \tag{54.5.29}$$

$$= -\frac{\xi\tanh\varepsilon}{3\varepsilon} + \frac{2\xi u}{3\big(1+u\big)}, \tag{54.5.30}$$

using (54.5.7) in the last line.

Step 10: simplify the third term. With (54.5.19),

$$\frac{C(1-u)}{2\varepsilon} = \frac{2\xi}{3}\cdot\frac{A\varepsilon u - 1}{1+u}\cdot\frac{1-u}{2\varepsilon} \tag{54.5.31}$$

$$= \frac{\xi}{3\varepsilon}(A\varepsilon u - 1)\frac{1-u}{1+u} \tag{54.5.32}$$

$$= \frac{\xi}{3\varepsilon}(A\varepsilon u - 1)\tanh\varepsilon \tag{54.5.33}$$

$$= \frac{\xi u A\tanh\varepsilon}{3} - \frac{\xi\tanh\varepsilon}{3\varepsilon}. \tag{54.5.34}$$

For the first piece, use $A\tanh\varepsilon = \dfrac{2}{1+u}\cdot\dfrac{1-u}{1+u} = \dfrac{2(1-u)}{(1+u)^2}$:

$$\frac{\xi u A\tanh\varepsilon}{3} = \frac{2\xi u(1-u)}{3(1+u)^2}. \tag{54.5.35}$$

Step 11: collect. Substituting (54.5.30), (54.5.34) and (54.5.35) into (54.5.27):

$$\overline{G\gamma} = \frac{\xi}{3} - \frac{2\xi\tanh\varepsilon}{3\varepsilon} + \underbrace{\frac{2\xi u}{3(1+u)} + \frac{2\xi u(1-u)}{3(1+u)^2}}_{\equiv S}. \tag{54.5.36}$$

Step 12: simplify S. Put the two terms over the common denominator $(1+u)^2$:

$$S = \frac{2\xi u}{3}\left[\frac{1}{1+u} + \frac{1-u}{(1+u)^2}\right] = \frac{2\xi u}{3}\cdot\frac{(1+u)+(1-u)}{(1+u)^2} = \frac{2\xi u}{3}\cdot\frac{2}{(1+u)^2} = \frac{4\xi u}{3(1+u)^2}. \tag{54.5.37}$$

Now note the identity

$$1-\tanh^2\varepsilon = 1 - \frac{(1-u)^2}{(1+u)^2} = \frac{(1+u)^2-(1-u)^2}{(1+u)^2} = \frac{4u}{(1+u)^2}, \tag{54.5.38}$$

the numerator having been expanded as $(1+2u+u^2)-(1-2u+u^2) = 4u$. Hence

$$S = \frac{\xi}{3}\Big(1-\tanh^2\varepsilon\Big). \tag{54.5.39}$$

Step 13: the average. Substituting into (54.5.36):

$$\overline{G\gamma} = \frac{\xi}{3} - \frac{2\xi\tanh\varepsilon}{3\varepsilon} + \frac{\xi}{3}\Big(1-\tanh^2\varepsilon\Big) \tag{54.5.40}$$

$$= \frac{\xi}{3}\left(2 - \tanh^2\varepsilon - \frac{2\tanh\varepsilon}{\varepsilon}\right). \tag{54.5.41}$$

Step 14: the viscosity. Insert (54.5.41) into (SM-38):

$$\Delta\eta = \frac{3}{4}\eta\varphi\xi\cdot\frac{\xi}{3}\left(2 - \tanh^2\varepsilon - \frac{2\tanh\varepsilon}{\varepsilon}\right) \tag{54.5.42}$$

$$= \frac{1}{4}\eta\varphi\xi^2\left(2 - \tanh^2\varepsilon - \frac{2\tanh\varepsilon}{\varepsilon}\right), \tag{SM-40}$$

since $3/(4\cdot 3) = 1/4$. ■

54.6 Consistency Checks and Verification Against the Paper's Numbers

Write

$$f(\varepsilon) \equiv 2 - \tanh^2\varepsilon - \frac{2\tanh\varepsilon}{\varepsilon}, \qquad \Delta\eta = \tfrac{1}{4}\eta\varphi\xi^2 f(\varepsilon). \tag{54.6.1}$$

(a) The static limit $\varepsilon \to \infty$ (i.e. $\omega \to 0$). Then $\tanh\varepsilon \to 1$ and $\tanh\varepsilon/\varepsilon \to 0$, so

$$f(\infty) = 2 - 1 - 0 = 1, \qquad \Delta\eta \to \tfrac{1}{4}\eta\varphi\xi^2, \tag{54.6.2}$$

which is exactly (SM-3), the stationary weak-field result. ✓ Note that here $\xi = mH_0/kT$ with no factor $\sqrt{2}$: a square wave at zero frequency is a *constant* field of magnitude H_0, not an oscillating one, so no root-mean-square correction arises. This is the promised confirmation that Sec. III uses the amplitude-based convention.

(b) The high-frequency limit $\varepsilon \to 0$. Use $\tanh\varepsilon = \varepsilon - \varepsilon^3/3 + O(\varepsilon^5)$, whence $\tanh^2\varepsilon = \varepsilon^2 + O(\varepsilon^4)$ and $\tanh\varepsilon/\varepsilon = 1 - \varepsilon^2/3 + O(\varepsilon^4)$. Then

$$f(\varepsilon) = 2 - \varepsilon^2 - 2\left(1 - \frac{\varepsilon^2}{3}\right) + O(\varepsilon^4) \tag{54.6.3}$$

$$= 2 - \varepsilon^2 - 2 + \frac{2\varepsilon^2}{3} \tag{54.6.4}$$

$$= -\frac{\varepsilon^2}{3} + O(\varepsilon^4). \tag{54.6.5}$$

So $\Delta\eta \to -\frac{1}{12}\eta\varphi\xi^2\varepsilon^2 < 0$, and since $\varepsilon \propto 1/\omega$ this is again the ω^{-2} decay of (SM-29). ✓

(c) The sign change. Solve $f(\varepsilon_0) = 0$ numerically:

ε	$\tanh\varepsilon$	$\tanh^2\varepsilon$	$2\tanh\varepsilon/\varepsilon$	$f(\varepsilon)$
1.55	0.9138	0.8350	1.1791	−0.0141
1.60	0.9217	0.8495	1.1521	−0.0016
1.61	0.9232	0.8523	1.1468	+0.0009
1.70	0.9354	0.8750	1.1005	+0.0246

The root is $\varepsilon_0 \approx 1.606$, corresponding to

$$\omega_0\tau_B = \frac{\pi}{2\varepsilon_0} = \frac{3.1416}{3.212} = 0.978 \approx 0.98. \quad \checkmark \tag{54.6.6}$$

This is precisely the value quoted in the paper for the rectangular case.

(d) The minimum. Evaluating f near $\varepsilon \approx 0.9$:

ε	$\tanh\varepsilon$	$\tanh^2\varepsilon$	$2\tanh\varepsilon/\varepsilon$	$f(\varepsilon)$
0.85	0.6911	0.4776	1.6260	−0.1036
0.90	0.7163	0.5131	1.5918	−0.1049
0.92	0.7259	0.5269	1.5779	−0.1048
0.95	0.7398	0.5473	1.5574	−0.1047

The minimum is $f_{\min} = -0.1049$ at $\varepsilon_m \approx 0.920$, i.e.

$$\omega_m \tau_B = \frac{\pi}{2\varepsilon_m} = \frac{3.1416}{1.840} = 1.707, \qquad \frac{\Delta\eta_m}{\Delta\eta(0)} = \frac{f_{\min}}{f(\infty)} = -0.105. \quad \checkmark \tag{54.6.7}$$

Both agree with the values printed in the paper to the quoted precision. *This is the verification promised in Sec. 54.1: the reading* $\tanh^2 \varepsilon$ *reproduces all three of the authors' numbers, whereas the literal printed* $\tanh \varepsilon$ *does not.*

54.7 Comparison of the Two Waveforms

Quantity	Harmonic (SM-28)	Rectangular (SM-40)
sign change, $\omega_0 \tau_B$	1	0.98
minimum, $\omega_m \tau_B$	$\sqrt{3} = 1.732$	1.707
depth, $\Delta\eta_m / \Delta\eta(0)$	$-1/8 = -0.125$	-0.105
static value $\Delta\eta(0)$	$\frac{1}{4}\eta\varphi\xi^2$	$\frac{1}{4}\eta\varphi\xi^2$

The two curves are not merely qualitatively but *quantitatively* similar, as the authors observe. That is the licence for what follows: since the answer barely depends on the waveform, one may as well use the waveform for which the equations integrate — the square wave.

Why the square wave gives a slightly shallower minimum. A square wave holds the field at full strength throughout the cycle, whereas a cosine spends much of each cycle near zero. One might expect the square wave to be *more* effective. What matters, however, is not the mean strength but the phase correlation, and the abrupt reversals of the square wave excite a broad spectrum of harmonics, only the fundamental of which is optimally phased. The net result is a 16% shallower minimum.

54.8 Connection to the Next Equation

The half-period reduction (54.5.20) used in Step 7 is stated by the authors as (SM-41), and we prove it next.

Chapter 55

Equation (SM-41): The Half-Period Reduction

$$\overline{G(t)\gamma(t)} = \frac{\omega}{\pi}\int_0^{\pi/\omega} G(t)\,\mathrm{d}t. \tag{SM-41}$$

55.1 Context

The authors note that Eqs. (SM-36) imply $G(t) = -G(t+\pi/\omega)$, so that instead of (SM-39) one may use (SM-41): it is sufficient to know the transverse magnetisation over half a period only. This halves the numerical work and, in the analytic solution of Chapter 54, made the problem tractable at all.

55.2 Step-by-Step Derivation

Step 1: antiperiodicity. Proved as (54.4.1) in Sec. 54.4: $G(t+T_h) = -G(t)$ with $T_h = \pi/\omega$. The waveform obeys the same relation, $\gamma(t+T_h) = -\gamma(t)$, by construction.

Step 2: the product is periodic with the *half* period.

$$G\big(t+T_h\big)\gamma\big(t+T_h\big) = \big(-G(t)\big)\big(-\gamma(t)\big) = G(t)\gamma(t). \tag{55.2.1}$$

A product of two antiperiodic functions is periodic on the half period.

Step 3: the full-period average equals the half-period average. Split the integral in (SM-39) at $t = T_h$ and substitute $t = t' + T_h$ in the second piece:

$$\int_0^{2T_h} G\gamma\,\mathrm{d}t = \int_0^{T_h} G\gamma\,\mathrm{d}t + \int_{T_h}^{2T_h} G\gamma\,\mathrm{d}t \tag{55.2.2}$$

$$= \int_0^{T_h} G\gamma\,\mathrm{d}t + \int_0^{T_h} G\big(t'+T_h\big)\gamma\big(t'+T_h\big)\,\mathrm{d}t' \tag{55.2.3}$$

$$= \int_0^{T_h} G\gamma\,\mathrm{d}t + \int_0^{T_h} G\gamma\,\mathrm{d}t' \qquad \text{by (55.2.1)} \tag{55.2.4}$$

$$= 2\int_0^{T_h} G\gamma\,\mathrm{d}t. \tag{55.2.5}$$

Dividing by the full period $2T_h = 2\pi/\omega$:

$$\overline{G\gamma} = \frac{\omega}{2\pi} \cdot 2 \int_0^{T_h} G\gamma \, \mathrm{d}t = \frac{\omega}{\pi} \int_0^{\pi/\omega} G\gamma \, \mathrm{d}t. \tag{55.2.6}$$

Step 4: drop γ. On the first half period $\gamma = +1$ by construction, so $G\gamma = G$ there, and

$$\overline{G\gamma} = \frac{\omega}{\pi} \int_0^{\pi/\omega} G(t) \, \mathrm{d}t, \tag{SM-41}$$

■

Remark 55.2.1 (The reduction is not merely convenient). For a harmonic waveform $\gamma = \cos \omega t$ the same antiperiodicity holds ($\cos \omega(t + \pi/\omega) = -\cos \omega t$), so (SM-41) applies there too — except that γ cannot then be dropped from the integrand, since it is not constant on the half period. It is the *rectangular* waveform that permits Step 4, and this is the main computational reason the authors adopt it.

55.3 Connection to the Next Equation

At finite amplitude the equations must be integrated with the full Langevin function. The formal solution is (SM-42), with the auxiliary functions (SM-43).

Chapter 56

Equations (SM-42) and (SM-43): The Formal Solution

$$G(t) = \frac{e^{g(t)}}{\tau_B}\left(-\frac{e^{g(\pi/\omega)}}{1+e^{g(\pi/\omega)}}\int_0^{\pi/\omega} F(t)e^{-g(t)}\,\mathrm{d}t + \int_0^t F(t')e^{-g(t')}\,\mathrm{d}t'\right), \tag{SM-42}$$

$$\begin{aligned} F(t) &= L(\xi)\left(1-\frac{2e^{-t/\tau_B}}{1+e^{-\pi/\omega\tau_B}}\right), \\ g(t) &= -\frac{t}{\tau_B}\left(1+\tfrac{1}{2}\xi L(\xi)\right) + \xi L(\xi)\,\frac{1-e^{-t/\tau_B}}{1+e^{-\pi/\omega\tau_B}}. \end{aligned} \tag{SM-43}$$

56.1 Context

These are the finite-amplitude counterparts of Chapter 54. The nonlinear term is now retained, and the Langevin function is used in full. The result is not a closed-form $\Delta\eta$ but a closed-form *quadrature*: G is expressed through integrals of known functions, which can then be evaluated numerically. This is what produces Figs. 3–5 of the paper.

56.2 Assumptions

1. Rectangular waveform, $\gamma = +1$ on $0 < t < \pi/\omega$.
2. Steady periodic state, with the antiperiodicity (54.4.1).
3. $\Omega\tau_B \ll 1$; *no* restriction on ξ.

56.3 Derivation of $F(t)$

With $\gamma = +1$ the first of (SM-36) reads

$$\tau_B\frac{\mathrm{d}F}{\mathrm{d}t} + F = L(\xi), \tag{56.3.1}$$

identical in structure to Step 1 of Sec. 54.5 but with $\xi/3$ replaced by the *exact* $L(\xi)$. Every step of Steps 1–3 there goes through unchanged with that substitution, giving

$$F(t) = L(\xi)\left[1-\frac{2}{1+u}e^{-t/\tau_B}\right], \qquad u = e^{-\pi/\omega\tau_B}, \tag{56.3.2}$$

which is the first line of (SM-43). ✓ (Note that $u = e^{-T_h/\tau_B}$ and $T_h = \pi/\omega$, so $u = e^{-\pi/\omega\tau_B}$ exactly as printed.)

Sanity check. At weak field $L(\xi) \to \xi/3$ and (56.3.2) reduces to (54.5.9). ✓

56.4 Derivation of $g(t)$

Step 1: put the G equation in standard linear form. With $\gamma = +1$, the second of (SM-36) is

$$\frac{\mathrm{d}G}{\mathrm{d}t} + \frac{1 + \tfrac{1}{2}\xi F(t)}{\tau_B}\, G = \frac{F(t)}{\tau_B}. \tag{56.4.1}$$

This is a first-order linear equation with a *time-dependent* coefficient, because F depends on t.

Step 2: the integrating factor. For $\mathrm{d}G/\mathrm{d}t + P(t)G = Q(t)$ the integrating factor is $\exp\!\int P\,\mathrm{d}t$. Define

$$g(t) \equiv -\frac{1}{\tau_B}\int_0^t \Big(1 + \tfrac{1}{2}\xi F(t')\Big)\mathrm{d}t', \tag{56.4.2}$$

so that the integrating factor is $e^{-g(t)}$ and $g(0) = 0$. (The minus sign in the definition is the authors' convention; it makes g negative and decreasing, so that e^{g} is a decaying factor.)

Step 3: evaluate the integral. Insert (56.3.2):

$$\int_0^t F\,\mathrm{d}t' = L(\xi)\int_0^t \left[1 - \frac{2}{1+u}e^{-t'/\tau_B}\right]\mathrm{d}t' \tag{56.4.3}$$

$$= L(\xi)\left[t - \frac{2}{1+u}\cdot\tau_B\Big(1 - e^{-t/\tau_B}\Big)\right]. \tag{56.4.4}$$

Therefore

$$g(t) = -\frac{1}{\tau_B}\left[t + \frac{\xi}{2}L(\xi)\,t - \frac{\xi L(\xi)\tau_B\big(1 - e^{-t/\tau_B}\big)}{1+u}\right] \tag{56.4.5}$$

$$= -\frac{t}{\tau_B}\left(1 + \tfrac{1}{2}\xi L(\xi)\right) + \xi L(\xi)\,\frac{1 - e^{-t/\tau_B}}{1+u}, \tag{56.4.6}$$

which is the second line of (SM-43). ■

Reading g. The first term is a steady decay at the enhanced rate $\big(1 + \tfrac{1}{2}\xi L\big)/\tau_B$ — the Brownian rate augmented by the magnetic restoring torque, exactly the feedback identified in Sec. 51.3. The second term is a transient correction that accounts for F not having reached its plateau immediately after each reversal of the field.

56.5 Derivation of (SM-42)

Step 1: integrate the linear equation. Multiply (56.4.1) by $e^{-g(t)}$. By construction $\mathrm{d}(-g)/\mathrm{d}t = \big(1 + \tfrac{1}{2}\xi F\big)/\tau_B$, so the left-hand side is an exact derivative:

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big(Ge^{-g}\Big) = e^{-g}\frac{\mathrm{d}G}{\mathrm{d}t} - g'e^{-g}G = e^{-g}\left[\frac{\mathrm{d}G}{\mathrm{d}t} + \frac{1 + \tfrac{1}{2}\xi F}{\tau_B}G\right] = \frac{F}{\tau_B}e^{-g}. \tag{56.5.1}$$

Step 2: integrate from 0 to t. With $g(0) = 0$,

$$G(t)e^{-g(t)} - G(0) = \frac{1}{\tau_B}\int_0^t F(t')e^{-g(t')}\,\mathrm{d}t', \tag{56.5.2}$$

that is

$$G(t) = e^{g(t)}\left[G(0) + \frac{1}{\tau_B}\int_0^t F(t')e^{-g(t')}\,\mathrm{d}t'\right]. \tag{56.5.3}$$

Step 3: fix $G(0)$ by antiperiodicity. Set $t = T_h = \pi/\omega$ in (56.5.3) and impose $G(T_h) = -G(0)$:

$$-G(0) = e^{g(T_h)}\left[G(0) + \frac{1}{\tau_B}\int_0^{T_h} Fe^{-g}\,\mathrm{d}t\right] \tag{56.5.4}$$

$$-G(0) - G(0)e^{g(T_h)} = \frac{e^{g(T_h)}}{\tau_B}\int_0^{T_h} Fe^{-g}\,\mathrm{d}t \tag{56.5.5}$$

$$G(0)\left[1 + e^{g(T_h)}\right] = -\frac{e^{g(T_h)}}{\tau_B}\int_0^{T_h} Fe^{-g}\,\mathrm{d}t \tag{56.5.6}$$

$$G(0) = -\frac{1}{\tau_B}\,\frac{e^{g(T_h)}}{1 + e^{g(T_h)}}\int_0^{T_h} Fe^{-g}\,\mathrm{d}t. \tag{56.5.7}$$

Step 4: assemble. Substituting (56.5.7) into (56.5.3) and factoring $1/\tau_B$:

$$G(t) = \frac{e^{g(t)}}{\tau_B}\left(-\frac{e^{g(\pi/\omega)}}{1 + e^{g(\pi/\omega)}}\int_0^{\pi/\omega} F(t)e^{-g(t)}\,\mathrm{d}t + \int_0^t F(t')e^{-g(t')}\,\mathrm{d}t'\right), \tag{SM-42}$$

which is (SM-42). ■

56.6 How the Computation Is Actually Done

The authors state the recipe: integrate (56.5.3) over the half period and substitute the result of the averaging (SM-41) into (SM-38). Explicitly, for prescribed ξ and $\omega\tau_B$:

1. Compute $L(\xi)$ from Eq. (10.2.16) and $u = e^{-\pi/\omega\tau_B}$.
2. Evaluate $F(t)$ from (56.3.2) and $g(t)$ from (56.4.6) on $0 \le t \le \pi/\omega$.
3. Compute the inner integral $\int_0^t Fe^{-g}\mathrm{d}t'$ by quadrature, and hence $G(t)$ from (SM-42).
4. Average: $\overline{G\gamma} = (\omega/\pi)\int_0^{\pi/\omega} G\,\mathrm{d}t$ by (SM-41).
5. Finally $\Delta\eta = \frac{3}{4}\eta\varphi\xi\,\overline{G\gamma}$ by (SM-38).

Three nested quadratures, no differential equations left to integrate. The results are the paper's Figs. 3–5.

56.7 Connection to the Next Chapter

What those figures show, and what they mean physically, is the subject of Chapter 57.

Chapter 57

The Numerical Results: Figures 3, 4 and 5

57.1 Context

Sec. III of the paper closes with three figures, which contain the finite-amplitude answer in graphical form. No new equations appear, but the content is essential and must be understood, since it is what Bacri *et al.* compare their measurements with.

57.2 Figure 3: $\Delta\eta$ versus frequency, at fixed amplitude

The quantity plotted is the relative viscosity

$$\frac{\Delta\eta\big(\xi,\omega\tau_B\big)}{\Delta\eta\big(\infty,0\big)}, \qquad \Delta\eta\big(\infty,0\big)=\tfrac{3}{2}\eta\varphi, \tag{57.2.1}$$

as a function of $\omega\tau_B$, for $\xi = 2, 5, 10, 15$. The normalisation is the saturation viscosity in a permanent field, Eq. (SM-4).

Three features, all of which we can now explain:

1. **Every curve starts positive and ends negative.** The zero-frequency value is the stationary result (SM-2); the high-frequency tail is negative, as Sec. 54.6(b) showed.

2. **The crossing frequency ω_0 decreases as ξ grows.** Stronger fields make the decrease of $\Delta\eta(\omega\tau_B)$ steeper. The reason is the feedback term of Sec. 51.3: at large ξ the effective relaxation rate of the transverse magnetisation is $(1+\frac{1}{2}\xi F)/\tau_B$, substantially faster than $1/\tau_B$. A faster effective relaxation means the phase lag reaches the critical value at a lower frequency.

3. **The minimum deepens with ξ.** At $\xi = 15$ the negative excursion approaches $-\Delta\eta(\infty,0)$, i.e. $\Delta\eta \approx -\frac{3}{2}\eta\varphi$ — comparable in magnitude to the entire saturation increase produced by a static field.

57.3 Figure 4: $\Delta\eta$ versus amplitude, at fixed frequency

Here $\Delta\eta/\Delta\eta(\infty,0)$ is plotted against ξ for $\omega\tau_B = 0, 0.1, 0.2, 0.4, 0.6, 0.8, 1.0, 1.2$. The striking feature, which the authors highlight, is that the dependence on ξ is *non-monotonic*: for $0 <$

$\omega\tau_B < 1$ every curve rises to a maximum and then falls into the negative region as the field is increased further.

Why non-monotonic? Two competing effects, both growing with ξ:

- the magnitude of the magnetisation, hence of the torque, grows — this pushes $|\Delta\eta|$ up;
- the effective relaxation rate $(1 + \frac{1}{2}\xi F)/\tau_B$ grows, which reduces the phase lag and eventually carries the system past the critical phase — this drives $\Delta\eta$ negative.

At small ξ the first wins; at large ξ the second does. The crossover is the maximum. Only the topmost curve, $\omega\tau_B = 0$, is monotonic — and that one is just the stationary formula (SM-2).

A prediction worth noting. In sufficiently strong fields the rotational viscosity is negative at *any* nonzero frequency. Only in the strictly stationary field is $\Delta\eta$ positive for all ξ, as thermodynamics requires (Sec. 15.6).

57.4 Figure 5: the map of viscosity

The most useful representation is the plane $(\xi, \omega\tau_B)$ with isolines $\Delta\eta = \text{const}$. The solid line $\Delta\eta = 0$ divides the plane into a positive region on the left (weak fields, low frequencies) and a negative region on the right. The numbers on the isolines are values of $\Delta\eta/\Delta\eta(\infty, 0)$, so that for instance the isoline labelled -1 corresponds to $\Delta\eta = -\frac{3}{2}\eta\varphi$.

This diagram is the direct theoretical counterpart of Fig. 4 of Bacri *et al.*, which plots the same isolines computed from the effective-field theory, and of their Fig. 5, which plots the measured ones. The comparison is the subject of Volume VII.

Chapter 58

Section IV of the Paper: Can the Total Viscosity Be Negative?

58.1 Context

The concluding section of Shliomis & Morozov asks the natural question: since $\Delta\eta < 0$ is possible, can it be large enough to make the *total* viscosity negative,

$$\eta_{\text{total}} = \eta + \Delta\eta < 0 \,? \tag{58.1.1}$$

The answer determines whether the effect is a curiosity or a route to a self-pumping fluid.

58.2 The condition, in the notation of Figure 5

Step 1. The condition $\eta_{\text{total}} = 0$ means $\Delta\eta = -\eta$.

Step 2. Divide by the normalisation used in Fig. 5, $\Delta\eta(\infty, 0) = \frac{3}{2}\eta\varphi$:

$$\frac{\Delta\eta\big(\xi, \omega\tau_B\big)}{\Delta\eta\big(\infty, 0\big)} = \frac{-\eta}{\frac{3}{2}\eta\varphi} = -\frac{2}{3\varphi}. \tag{58.2.1}$$

This is the unnumbered relation printed in the paper's Sec. IV.

Step 3: the number. At the realistic concentration $\varphi = 0.2$,

$$-\frac{2}{3 \times 0.2} = -\frac{2}{0.6} = -\frac{10}{3} \approx -3.33, \tag{58.2.2}$$

so $\eta_{\text{total}} = 0$ corresponds to the isoline $-10/3$ of Fig. 5. Since the figure's most negative plotted isoline is -1, the required point lies far outside the plotted region — in the domain of very high ξ and $\omega\tau_B$.

58.3 The Stability Argument

The authors approach the same question from a second direction, which is worth recording because it converts a statement about viscosity into a statement about *instability*.

The idea. A fluid with $\eta_{\text{total}} < 0$ cannot remain at rest: any infinitesimal disturbance grows rather than decays, because the "viscous" term now feeds energy into the disturbance instead of dissipating it. Therefore the neutral curve of the stability of mechanical equilibrium in the $(\xi, \omega\tau_B)$ plane is exactly the isoline $\eta_{\text{total}} = 0$.

The set-up. Reference 5 of the paper studies a ferrofluid in a plane channel between rigid walls at $x = 0$ and $x = l$, with $H_x = H_0 \cos \omega t$. For the one-dimensional perturbation

$$\boldsymbol{v} = \big[0,\ v(x,t),\ 0\big] \qquad \Longrightarrow \qquad \boldsymbol{\Omega} = \tfrac{1}{2}\nabla \times \boldsymbol{v} = \left(0,\ 0,\ \tfrac{1}{2}\frac{\partial v}{\partial x}\right), \tag{58.3.1}$$

where the vorticity was computed from the definition (3.4.1): for $\boldsymbol{v} = v(x,t)\hat{\boldsymbol{y}}$,

$$\nabla \times \boldsymbol{v} = \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z},\ \frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x},\ \frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}\right) = \left(0, 0, \frac{\partial v}{\partial x}\right). \checkmark \tag{58.3.2}$$

The mode structure. Seeking solutions of the form

$$v = w(t) \sin qx, \qquad M_y = f(t) \cos qx, \tag{58.3.3}$$

the no-slip boundary conditions $v(0,t) = v(l,t) = 0$ require $\sin ql = 0$, hence (with l taken as the unit of length)

$$q = \pi r, \qquad r = 1, 2, 3, \ldots \tag{58.3.4}$$

Derivation of the spectrum: $\sin(0) = 0$ automatically; $\sin(ql) = 0$ requires ql to be an integer multiple of π.

Which mode goes unstable first. The viscous damping rate of a mode is proportional to q^2 (from $\eta\nabla^2\boldsymbol{v}$, which for $\sin qx$ gives $-\eta q^2$), so the mode with the smallest q is the least damped and the first to become unstable. That is $r = 1$, i.e. $q = \pi$.

Why this mode matters physically. With $v = w(t) \sin \pi x$ on $0 < x < 1$, the velocity has the same sign across the whole channel, so the volume flux

$$\int_0^1 v \, \mathrm{d}x = w(t) \int_0^1 \sin \pi x \, \mathrm{d}x = w(t) \left[-\frac{\cos \pi x}{\pi}\right]_0^1 = w(t)\frac{2}{\pi} \tag{58.3.5}$$

is nonzero. The instability therefore produces a *net flow through the channel*: as the authors put it, if ξ and $\omega\tau_B$ lie in the unstable region the entire system works as a pump, driven by the alternating field with no moving parts and no applied pressure gradient.

58.4 The Critical Values, and Why They Are Out of Reach

For $q = \pi$ and $\varphi = 0.2$ the minimum on the neutral curve is found at

$$\xi_c = 126, \qquad \omega_c\tau_B \text{ correspondingly large.} \tag{58.4.1}$$

Bacri *et al.* quote the companion value $\omega_c\tau_B = 20$.

What $\xi = 126$ means physically. For the experimental grains, $\xi = 5.25 \times 10^{-3} H(\mathrm{Oe})$, so

$$H_c = \frac{126}{5.25 \times 10^{-3}} = 2.4 \times 10^4 \text{ Oe} = 24 \text{ kOe}. \tag{58.4.2}$$

That is a large but not impossible laboratory field. The real obstacle is different, and both papers identify it: such a value of ξ could be reached at a modest field only by using *larger* particles (since $\xi \propto d^3$), and larger particles bring two fatal problems — they aggregate, destroying the colloidal stability, and they make the fluid non-Newtonian. Moreover, at 10^4 Oe the assumption that the moment is rigidly frozen in the crystal begins to fail: the field can rotate the moment inside the grain, which decouples the magnetic from the mechanical degrees of freedom and switches the entire effect off. The authors state exactly this: the required ξ_c demands an amplitude at which the particles can no longer be treated as rigid dipoles.

The honest conclusion. Negative *rotational* viscosity is real, measurable, and was measured. Negative *total* viscosity — a spontaneously pumping ferrofluid — remains, on the basis of this theory, out of reach for practical ferrocolloids. What is within reach is the tunability that Bacri *et al.* emphasise: a viscosity adjustable between 220 and 50 cP by changing only the frequency of a fixed-amplitude field, which they suggest could form the basis of adaptive dampers.

Chapter 59

Status and Road Map

59.1 What has been established in Volume VI

Equation	Statement	Where derived
(SM-35)	nonlinear magnetisation equation at arbitrary amplitude; $\xi(t) = \xi\gamma(t)$	Ch. 50
(50.4.1)	$M_x = nmF$, $M_y = nm\Omega\tau_B G$	Sec. 50.4
(SM-36)	$\tau_B\dot{F} = L(\xi\gamma) - F$; $\tau_B\dot{G} = F - (1 + \frac{1}{2}\xi\gamma F)G$	Ch. 51
(SM-37)	$\overline{(\boldsymbol{M}\times\boldsymbol{H})_z} = -3\eta\varphi\Omega\xi\overline{G\gamma}$	Ch. 52
(SM-38)	$\Delta\eta = \frac{3}{4}\eta\varphi\xi\overline{G\gamma}$	Ch. 53
(SM-39)	definition of the period average	Sec. 53.1
(53.4.2)	$(\Omega - \Theta)/\Omega = \frac{\xi}{2}\overline{G\gamma}$, recovering (SM-34) at any amplitude	Sec. 53.4
(SM-40)	$\Delta\eta = \frac{1}{4}\eta\varphi\xi^2\big(2 - \tanh^2\varepsilon - 2\tanh\varepsilon/\varepsilon\big)$; misprint corrected and verified against $\omega_0\tau_B = 0.98$, $\omega_m\tau_B = 1.707$, -0.105	Ch. 54
(SM-41)	half-period reduction, from antiperiodicity	Ch. 55
(SM-42)	formal solution for G by integrating factor	Sec. 56.5
(SM-43)	$F(t)$ and $g(t)$ in closed form	Secs. 56.3–56.4
Figs. 3–5	frequency, amplitude and map representations; non-monotonic $\Delta\eta(\xi)$ explained	Ch. 57
Sec. IV	$\eta_{\rm total} = 0 \Leftrightarrow$ isoline $-2/3\varphi = -10/3$; $q = \pi$; the pump; $\xi_c = 126$	Ch. 58

59.2 Cumulative progress

Shliomis & Morozov (1994): complete. All 43 numbered equations, together with the unnumbered relations of Secs. II–IV, have now been derived from first principles.

Bacri, Perzynski, Shliomis & Burde (1995): Eqs. (B-1), (B-2), (B-3), (B-4) and (B-10) complete.

59.3 What remains: Volume VII

The final volume treats the experimental paper's own theoretical contribution — the effective-field method, which Bacri *et al.* adopt precisely because the phenomenological equation (SM-7) fails for finite ξ *and* finite $\omega\tau_B$ simultaneously (the limitation identified in Sec. 13.4):

- **(B-5)** $\boldsymbol{M} = \varphi M_s L(\xi_e)\boldsymbol{\xi}_e/\xi_e$: the effective-field ansatz, derived from the Fokker–Planck equation (11.6.1);
- **(B-6)** the evolution equation for $\boldsymbol{\xi}_e$, including the L_3 term;
- **(B-7)** the zero-approximation equation for ξ_e in the small parameter $\Omega\tau_B$;
- **(B-8)** the equation for the function F;
- **(B-9)** the mean magnetic torque and the function $g = \frac{1}{2}\xi_0\overline{\cos\omega t\, L(\xi_e)F(\xi_e)}$;
- the experimental analysis: the reduced viscosity $\eta_r(H, f)$, the calibrations $\xi_0 = 5.25 \times 10^{-3}H(\mathrm{Oe})$ and $\omega\tau_B = 10^{-2}f(\mathrm{Hz})$, the isoline comparison between Figs. 4 and 5, the observed 25% viscosity reduction, and the critical values $\xi_c = 126$, $\omega_c\tau_B = 20$.

End of Volume VI. Volume VII — the final volume — begins with Eq. (B-5).

THE NEGATIVE-VISCOSITY EFFECT IN FERROFLUIDS

A Complete First-Principles Reconstruction of Every Equation

VOLUME VII

(final volume)

The Effective-Field Method and the Experiment:
Equations (B-5)–(B-10), the Numerical Procedure,
and the Comparison with Measurement

Where Volume VI ended. Volume VI completed Shliomis & Morozov: the finite-amplitude equations (SM-35)–(SM-43), the square-wave solution, the viscosity map, and the conclusion that a negative *total* viscosity would require $\xi_c = 126$ — a field at which the rigid-dipole assumption itself fails. All 43 equations of that paper have now been derived from first principles.

Volume VII completes the project. It derives the effective-field method that Bacri *et al.* use in place of the phenomenological magnetisation equation, obtaining (B-5) through (B-9) from the Smoluchowski equation of Volume II; it shows how their numerical procedure works; and it goes through the experiment itself — the apparatus, the measured numbers, and the comparison between the theoretical and experimental isolines. Chapter numbers resume at 60, page numbers at 208.

Part XII

The Experimental Paper and the Effective-Field Method

Chapter 60

Orientation: What the Experimental Paper Adds

60.1 The state of the account

Of the ten numbered equations of Bacri, Perzynski, Shliomis and Burde, five have already been derived in earlier volumes, because they are the same equations as Shliomis & Morozov's written in different notation:

Equation	Statement	Equivalent to	Derived in
(B-1)	$6\eta\varphi(\boldsymbol{\omega}_p - \boldsymbol{\Omega}) = \boldsymbol{M}\times\boldsymbol{H}$	(SM-11)	Sec. 14.2, Ch. 24
(B-2)	$\Delta\eta = \tfrac{3}{2}\eta\varphi(\Omega - \omega_p)/\Omega$	(SM-34)	Sec. 14.4, Ch. 48
(B-3)	momentum equation with $\tfrac{1}{2}\operatorname{curl}(\boldsymbol{M}\times\boldsymbol{H})$	(SM-12)	Sec. 14.3, Ch. 25
(B-4)	$\mathrm{d}\boldsymbol{M}/\mathrm{d}t = \boldsymbol{\Omega}\times\boldsymbol{M} - \frac{1}{\tau_B}(\boldsymbol{M}-\boldsymbol{M}_0) - \frac{1}{6\eta\varphi}\boldsymbol{M}\times(\boldsymbol{M}\times\boldsymbol{H})$	(SM-13)	Sec. 13.3, Ch. 26
(B-10)	the grouping of the magnetic term with the viscous one	(SM-27) step	Sec. 14.4

Note the notational correspondences, fixed once and for all: their $\boldsymbol{\omega}_p$ is our $\boldsymbol{\Theta}$; their $\varphi M_s L(\xi)$ is our $nmL(\xi)$; their ξ_0 is the amplitude-based Langevin parameter, which is the ξ of Sec. III of Shliomis & Morozov (Sec. 50.2).

What remains, and what this volume derives, is Eqs. (B-5)–(B-9): the *effective-field method*, which replaces the phenomenological magnetisation equation by a better-controlled closure of the Fokker–Planck hierarchy.

60.2 Why a better closure is needed

Section 13.4 of Volume II identified precisely where the phenomenological equation (SM-7) becomes an approximation: the exact first-moment equation contains the second moment $\langle e_i e_j\rangle$, and (SM-7) follows only after replacing it by its isotropic value $\tfrac{1}{3}\delta_{ij}$ — legitimate for $\xi \ll 1$ and no other regime.

Bacri *et al.* state the consequence they encountered. Their experimental data agree with the Shliomis–Morozov theory on the whole, but with substantial discrepancies in detail. Specifically, the theoretical $\Delta\eta(\xi)$ of Fig. 4 of Shliomis & Morozov has a maximum at $\omega\tau_B < 1$ and no minimum at $\omega\tau_B > 1$, whereas the measurements show the inverse situation. Their diagnosis

is exactly the one made in Sec. 13.4: the phenomenological equation works well for a stationary field of any intensity (whence the good agreement between McTague's measurements and Shliomis's 1972 theory), and for an alternating field of small amplitude *and* small frequency — but for finite ξ and finite $\omega\tau_B$ together, a more rigorous magnetisation equation is required.

That equation was derived by Martsenyuk, Raikher and Shliomis from the Fokker–Planck equation for rotary diffusion, with the assistance of the effective-field method of Raikher and Shliomis. We now derive it.

60.3 Notation for this volume

Symbol	Meaning	Type	Units
$\boldsymbol{\xi}$	the *true* dimensionless field, $\boldsymbol{\xi} = m\boldsymbol{H}/kT$	vector	—
$\boldsymbol{\xi}_e$	the *effective* field: the fictitious static field whose equilibrium distribution equals the actual one	vector	—
$\xi_e = \lvert\boldsymbol{\xi}_e\rvert$	its magnitude	scalar	—
$\boldsymbol{n} = \boldsymbol{\xi}_e/\xi_e$	unit vector along the effective field	vector	—
$L_e = L(\xi_e)$	Langevin function of the effective field	scalar	—
L_3	the coefficient appearing in (B-6); shown in Sec. 62.6 to equal $L(\xi_e)$	scalar	—
$\xi_0 = mH_0/kT$	amplitude-based Langevin parameter	scalar	—
$F(\xi_e)$	the function of (B-8) governing the transverse response	function	—
g	normalised rotational viscosity, $g = \Delta\eta/\Delta\eta(\infty, 0)$	scalar	—
$\eta_r(H, f)$	measured reduced viscosity $[\eta(H, f) - \eta(0, 0)]/\eta(0, 0)$	scalar	—

Chapter 61

Equation (B-5): The Effective-Field Ansatz

$$\boldsymbol{M} = \varphi M_s \, L(\xi_e) \, \frac{\boldsymbol{\xi}_e}{\xi_e}. \tag{B-5}$$

61.1 Context

The paper introduces the method in one sentence: according to it, the nonequilibrium actual magnetisation is considered to be the equilibrium one, in the effective field $\boldsymbol{\xi}_e$. Equation (B-5) is that statement.

The idea deserves to be stated carefully, because it is a genuine physical hypothesis and not a mathematical identity.

Assumption 61.1.1 (Effective-field closure)**.** At every instant, the orientational distribution function has the *form* of an equilibrium Boltzmann distribution, but in a fictitious field $\boldsymbol{\xi}_e(t)$ that is not the applied field:

$$W(\boldsymbol{e}, t) = \frac{\xi_e}{4\pi \sinh \xi_e} \, \exp\big(\boldsymbol{\xi}_e \cdot \boldsymbol{e}\big). \tag{61.1.1}$$

The dynamics is then carried entirely by the three components of $\boldsymbol{\xi}_e(t)$.

61.2 Why this is a good closure

Recall the difficulty. The exact moment equation, derived as Eq. (13.2.1) and its consequences in Volume II, reads

$$\frac{\mathrm{d}\langle e_i \rangle}{\mathrm{d}t} = \big(\boldsymbol{\Omega} \times \langle \boldsymbol{e} \rangle\big)_i + \frac{1}{2\tau_B}\Big[\xi_i - \big\langle e_i e_j \big\rangle \xi_j\Big] - \frac{\langle e_i \rangle}{\tau_B}. \tag{61.2.1}$$

This equation is exact. No approximation whatever has been made in reaching it: it follows from the Smoluchowski equation (11.6.1) by multiplying by e_i and integrating, using Green's identity and the eigenvalue $\nabla_s^2 \boldsymbol{e} = -2\boldsymbol{e}$. Its only defect is that it is not closed: the first moment is driven by the second.

There are then three levels of treatment:

1. **Isotropic closure**, $\langle e_i e_j \rangle = \frac{1}{3}\delta_{ij}$: gives the phenomenological equation (SM-7). Exact only as $\xi \to 0$.

2. **Effective-field closure**, Assumption 61.1.1: expresses $\langle e_i e_j \rangle$ through $\langle \boldsymbol{e} \rangle$ by assuming the distribution retains its equilibrium *shape*. Exact in two limits — $\xi \to 0$, and any stationary field — and interpolating smoothly between them.

3. **Full Fokker–Planck**: solve the Smoluchowski equation (11.6.1) numerically for W itself. Exact, expensive, and not what either paper does.

The effective-field method is level 2. Its merit is that it costs three unknowns instead of one and is exact wherever the answer is known independently.

Remark 61.2.1 (When is the ansatz exact?). In a *stationary* field with no flow, the true distribution *is* Boltzmann in the true field, so $\boldsymbol{\xi}_e = \boldsymbol{\xi}$ and (61.1.1) is exact. In a weak field of any time dependence, the distribution is $W \simeq (1 + \boldsymbol{a} \cdot \boldsymbol{e})/4\pi$ to first order in the small anisotropy, and (61.1.1) reproduces exactly that form with $\boldsymbol{\xi}_e = \boldsymbol{a}$. The ansatz therefore has no error at either end of the range, and the question is only how much it errs in between — which experiment, in the end, answers.

61.3 Derivation of (B-5) from the Ansatz

Given (61.1.1), the magnetisation follows from $\boldsymbol{M} = nm\langle \boldsymbol{e} \rangle$, Eq. (4.5.1). The distribution (61.1.1) is identical in form to the equilibrium distribution (9.5.7) with $\xi \to \xi_e$ and the polar axis along $\boldsymbol{n} = \boldsymbol{\xi}_e/\xi_e$. Therefore the calculation of Chapter 10 applies verbatim, and Eq. (10.2.16) gives

$$\langle \boldsymbol{e} \rangle = L(\xi_e)\, \boldsymbol{n} = L(\xi_e) \frac{\boldsymbol{\xi}_e}{\xi_e}. \tag{61.3.1}$$

Multiplying by $nm = \varphi M_s$ [Sec. 10.4]:

$$\boldsymbol{M} = \varphi M_s L(\xi_e) \frac{\boldsymbol{\xi}_e}{\xi_e}, \tag{B-5}$$

■

61.4 The Second Moment

The whole point of the closure is that it determines $\langle e_i e_j \rangle$. We compute it now, since (B-6) cannot be derived without it.

Result 61.4.1. *For the distribution* (61.1.1),

$$\langle e_i e_j \rangle = \frac{L_e}{\xi_e}\, \delta_{ij} + \left(1 - \frac{3L_e}{\xi_e}\right) n_i n_j. \tag{61.4.1}$$

Step 1: the tensor structure. The distribution depends on $\boldsymbol{e}$ only through $\boldsymbol{e} \cdot \boldsymbol{n}$. The only tensors available to build $\langle e_i e_j \rangle$ are therefore δ_{ij} and $n_i n_j$, and it must be symmetric, so

$$\langle e_i e_j \rangle = A\, \delta_{ij} + B\, n_i n_j \tag{61.4.2}$$

for two scalars A, B depending on ξ_e alone.

Step 2: first condition — the trace. Contract $i = j$ in (61.4.2), using $\delta_{ii} = 3$, $n_i n_i = 1$ and $\langle e_i e_i \rangle = \langle 1 \rangle = 1$:

$$1 = 3A + B. \tag{61.4.3}$$

Step 3: second condition — the longitudinal component. Contract (61.4.2) with $n_i n_j$:

$$\big\langle (\boldsymbol{e}\cdot\boldsymbol{n})^2 \big\rangle = A + B. \tag{61.4.4}$$

So we need $\langle u^2\rangle$ with $u = \boldsymbol{e}\cdot\boldsymbol{n} = \cos\theta$.

Step 4: compute $\langle u^2\rangle$. Write the partition function of Eq. (9.5.8), $Z(\xi_e) = \int e^{\xi_e u}\mathrm{d}\Omega_e$, and recall from Chapter 10 that

$$\langle u\rangle = \frac{Z'}{Z} = L(\xi_e), \tag{61.4.5}$$

where the prime is $\mathrm{d}/\mathrm{d}\xi_e$, because differentiating $e^{\xi_e u}$ with respect to the parameter brings down a factor u. Differentiating again,

$$\frac{\mathrm{d}\langle u\rangle}{\mathrm{d}\xi_e} = \frac{\mathrm{d}}{\mathrm{d}\xi_e}\frac{Z'}{Z} = \frac{Z''}{Z} - \left(\frac{Z'}{Z}\right)^2 = \big\langle u^2\big\rangle - \langle u\rangle^2, \tag{61.4.6}$$

so

$$\big\langle u^2\big\rangle = L'(\xi_e) + L^2(\xi_e). \tag{61.4.7}$$

Now differentiate $L = \coth\xi - 1/\xi$:

$$L' = -\frac{1}{\sinh^2\xi} + \frac{1}{\xi^2}, \tag{61.4.8}$$

using $\mathrm{d}(\coth\xi)/\mathrm{d}\xi = -1/\sinh^2\xi$ (from the quotient rule applied to $\cosh/\sinh$ together with $\cosh^2 - \sinh^2 = 1$). And

$$L^2 = \left(\coth\xi - \frac{1}{\xi}\right)^2 = \coth^2\xi - \frac{2\coth\xi}{\xi} + \frac{1}{\xi^2} \tag{61.4.9}$$

$$= 1 + \frac{1}{\sinh^2\xi} - \frac{2\coth\xi}{\xi} + \frac{1}{\xi^2}, \tag{61.4.10}$$

where $\coth^2 = 1 + 1/\sinh^2$ (divide $\cosh^2 - \sinh^2 = 1$ by $\sinh^2$). Adding, the $1/\sinh^2$ terms cancel:

$$\big\langle u^2\big\rangle = \frac{1}{\xi^2} - \frac{1}{\sinh^2\xi} + 1 + \frac{1}{\sinh^2\xi} - \frac{2\coth\xi}{\xi} + \frac{1}{\xi^2} \tag{61.4.11}$$

$$= 1 + \frac{2}{\xi^2} - \frac{2\coth\xi}{\xi} \tag{61.4.12}$$

$$= 1 - \frac{2}{\xi}\left(\coth\xi - \frac{1}{\xi}\right) \tag{61.4.13}$$

$$= 1 - \frac{2L(\xi)}{\xi}. \tag{61.4.14}$$

Step 5: solve for A and B. From (61.4.4) and (61.4.14), $A + B = 1 - 2L_e/\xi_e$. Subtracting this from (61.4.3):

$$3A + B - \big(A + B\big) = 1 - \left(1 - \frac{2L_e}{\xi_e}\right) \quad\Longrightarrow\quad 2A = \frac{2L_e}{\xi_e} \quad\Longrightarrow\quad A = \frac{L_e}{\xi_e}. \tag{61.4.15}$$

Then from (61.4.3), $B = 1 - 3A = 1 - 3L_e/\xi_e$. Substituting into (61.4.2) gives (61.4.1). ■

Sanity checks. At $\xi_e \to 0$, $L_e \to \xi_e/3$, so $A \to 1/3$ and $B \to 1 - 1 = 0$: the isotropic result $\frac{1}{3}\delta_{ij}$ of Eq. (13.2.9). ✓ At $\xi_e \to \infty$, $L_e \to 1$, so $A \to 0$ and $B \to 1$: all moments aligned, $\langle e_i e_j \rangle = n_i n_j$. ✓

61.5 Connection to the Next Equation

Inserting (61.3.1) and (61.4.1) into the exact moment equation (61.2.1) closes the system and produces (B-6).

Chapter 62

Equation (B-6): The Effective-Field Evolution Equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(L_e\frac{\boldsymbol{\xi}_e}{\xi_e}\right) = \boldsymbol{\Omega}\times\left(L_e\frac{\boldsymbol{\xi}_e}{\xi_e}\right) - \frac{1}{\tau_B}\frac{L_e}{\xi_e}(\boldsymbol{\xi}_e-\boldsymbol{\xi}) - \frac{1}{2\tau_B\xi_e^2}\left(1-\frac{3L_3}{\xi_e}\right)\boldsymbol{\xi}_e\times(\boldsymbol{\xi}_e\times\boldsymbol{\xi}). \tag{B-6}$$

62.1 Context

This is the magnetisation equation of Martsenyuk, Raikher and Shliomis, written for the effective field rather than for $\boldsymbol{M}$. It replaces (SM-7)/(B-4), and it is the equation Bacri *et al.* actually integrate.

62.2 Assumptions

1. The Smoluchowski equation (11.6.1) — itself derived from probability conservation, inertialess torque balance and the Einstein relation.
2. The effective-field closure, Assumption 61.1.1.
3. Rigid dipoles, monodisperse, non-interacting: as always.

Note what is *not* assumed: nothing about the size of ξ, nothing about the frequency, nothing about $\Omega\tau_B$. Equation (B-6) is a closed nonlinear equation valid, within its closure, for arbitrary fields and flows.

62.3 Mathematical Background: the exact moment equation

We restate it because everything follows from it. From Volume II, Sec. 13.2, the three contributions to the first moment were the flow term (13.2.4), the diffusion term (13.2.6), and the magnetic term (13.2.8), before any closure. Assembling them with $D_r = 1/2\tau_B$ [Eq. (11.5.3)] and $m\boldsymbol{H}/kT = \boldsymbol{\xi}$:

$$\frac{\mathrm{d}\langle e_i\rangle}{\mathrm{d}t} = \big(\boldsymbol{\Omega}\times\langle\boldsymbol{e}\rangle\big)_i + \frac{1}{2\tau_B}\Big[\xi_i - \big\langle e_i e_j\big\rangle\xi_j\Big] - \frac{\langle e_i\rangle}{\tau_B}. \tag{62.3.1}$$

Check of the coefficient: the magnetic term in Volume II was $(m/\zeta_r)[H_i - \langle e_ie_j\rangle H_j]$. Using the Einstein relation $D_r = kT/\zeta_r$, $m/\zeta_r = mD_r/kT$, and $mH_i/kT = \xi_i$, this is $D_r[\xi_i - \langle e_ie_j\rangle\xi_j] = \frac{1}{2\tau_B}[\cdots]$. ✓

62.4 Step-by-Step Derivation

Step 1: insert the closure. Substitute (61.3.1) and (61.4.1) into (62.3.1). The second moment contracted with the field is

$$\langle e_i e_j\rangle \xi_j = \frac{L_e}{\xi_e}\delta_{ij}\xi_j + \left(1 - \frac{3L_e}{\xi_e}\right) n_i n_j \xi_j \tag{62.4.1}$$

$$= \frac{L_e}{\xi_e}\xi_i + \left(1 - \frac{3L_e}{\xi_e}\right) n_i(\boldsymbol{n}\cdot\boldsymbol{\xi}). \tag{62.4.2}$$

Hence, in vector form,

$$\frac{\mathrm{d}}{\mathrm{d}t}(L_e\boldsymbol{n}) = \boldsymbol{\Omega}\times(L_e\boldsymbol{n}) + \frac{1}{2\tau_B}\left[\boldsymbol{\xi} - \frac{L_e}{\xi_e}\boldsymbol{\xi} - \left(1 - \frac{3L_e}{\xi_e}\right)\boldsymbol{n}(\boldsymbol{n}\cdot\boldsymbol{\xi})\right] - \frac{L_e\boldsymbol{n}}{\tau_B}. \tag{62.4.3}$$

This is already a closed equation for $\boldsymbol{\xi}_e$. What remains is to rearrange it into the form printed by the authors.

Step 2: identify the target form. Expand the two magnetic terms of (B-6), using $\boldsymbol{\xi}_e = \xi_e\boldsymbol{n}$ and the triple product (1.7.1):

$$\boldsymbol{\xi}_e\times(\boldsymbol{\xi}_e\times\boldsymbol{\xi}) = \boldsymbol{\xi}_e(\boldsymbol{\xi}_e\cdot\boldsymbol{\xi}) - \boldsymbol{\xi}(\boldsymbol{\xi}_e\cdot\boldsymbol{\xi}_e) \tag{62.4.4}$$

$$= \xi_e^2\Big[\boldsymbol{n}(\boldsymbol{n}\cdot\boldsymbol{\xi}) - \boldsymbol{\xi}\Big]. \tag{62.4.5}$$

Therefore the last term of (B-6) is

$$-\frac{1}{2\tau_B\xi_e^2}\left(1 - \frac{3L_3}{\xi_e}\right)\xi_e^2\Big[\boldsymbol{n}(\boldsymbol{n}\cdot\boldsymbol{\xi}) - \boldsymbol{\xi}\Big] = -\frac{1}{2\tau_B}\left(1 - \frac{3L_3}{\xi_e}\right)\Big[\boldsymbol{n}(\boldsymbol{n}\cdot\boldsymbol{\xi}) - \boldsymbol{\xi}\Big], \tag{62.4.6}$$

the factors ξ_e^2 cancelling. And the middle term of (B-6) is

$$-\frac{1}{\tau_B}\frac{L_e}{\xi_e}(\xi_e\boldsymbol{n} - \boldsymbol{\xi}) = -\frac{L_e\boldsymbol{n}}{\tau_B} + \frac{L_e}{\tau_B\xi_e}\boldsymbol{\xi}. \tag{62.4.7}$$

So the right-hand side of (B-6), apart from the flow term, is

$$-\frac{L_e\boldsymbol{n}}{\tau_B} + \left[\frac{L_e}{\tau_B\xi_e} + \frac{1}{2\tau_B}\left(1 - \frac{3L_3}{\xi_e}\right)\right]\boldsymbol{\xi} - \frac{1}{2\tau_B}\left(1 - \frac{3L_3}{\xi_e}\right)\boldsymbol{n}(\boldsymbol{n}\cdot\boldsymbol{\xi}). \tag{62.4.8}$$

Step 3: compare with the derived equation. Rewriting (62.4.3) in the same order:

$$-\frac{L_e\boldsymbol{n}}{\tau_B} + \frac{1}{2\tau_B}\left[1 - \frac{L_e}{\xi_e}\right]\boldsymbol{\xi} - \frac{1}{2\tau_B}\left(1 - \frac{3L_e}{\xi_e}\right)\boldsymbol{n}(\boldsymbol{n}\cdot\boldsymbol{\xi}). \tag{62.4.9}$$

Step 4: match, term by term. The first terms are identical. Matching the coefficients of $\boldsymbol{n}(\boldsymbol{n}\cdot\boldsymbol{\xi})$ in (62.4.8) and (62.4.9) requires

$$L_3 = L_e = L(\xi_e). \tag{62.4.10}$$

Matching the coefficients of $\boldsymbol{\xi}$ requires

$$\frac{L_e}{\tau_B\xi_e} + \frac{1}{2\tau_B}\left(1 - \frac{3L_3}{\xi_e}\right) = \frac{1}{2\tau_B}\left(1 - \frac{L_e}{\xi_e}\right), \tag{62.4.11}$$

i.e., multiplying by $2\tau_B$,

$$\frac{2L_e}{\xi_e} + 1 - \frac{3L_3}{\xi_e} = 1 - \frac{L_e}{\xi_e} \quad\Longleftrightarrow\quad \frac{3L_e}{\xi_e} = \frac{3L_3}{\xi_e} \quad\Longleftrightarrow\quad L_3 = L_e, \tag{62.4.12}$$

the *same* condition. With (62.4.10) the two expressions agree identically, and (B-6) is established. ■

62.5 The Identification of L_3

The symbol L_3 in the printed equation is not defined in the letter, which cites the original derivation for it. The matching above determines it unambiguously: *within the effective-field closure,* $L_3 = L(\xi_e)$. Two independent confirmations follow:

1. Chapter 64 derives (B-8) from (B-6). The derivation reproduces the printed (B-8) exactly if and only if $L_3 = L_e$; any other value leaves an uncancelled residue. This is a genuinely independent check, since (B-8) is a different equation in the same paper.
2. The physical origin of the term is the second moment (61.4.1), whose anisotropic coefficient is $1 - 3L_e/\xi_e$ — exactly the combination appearing in (B-6) with L_3 in place of L_e. The subscript 3 evidently labels this coefficient as arising from the second-rank (three-component) part of the orientational distribution.

62.6 Physical Interpretation

The three terms on the right of (B-6) have distinct meanings.

- $\boldsymbol{\Omega} \times (L_e \boldsymbol{n})$: the magnetisation is carried around by the flow, as in every version of the theory.
- $-\frac{1}{\tau_B}\frac{L_e}{\xi_e}(\boldsymbol{\xi}_e - \boldsymbol{\xi})$: *longitudinal* relaxation. It drives the effective field towards the true field. If the true and effective fields were parallel this would be the only magnetic term, and (B-6) would reduce to a scalar relaxation equation — which is exactly what happens in Chapter 63.
- $-\frac{1}{2\tau_B \xi_e^2}(1 - 3L_e/\xi_e)\,\boldsymbol{\xi}_e \times (\boldsymbol{\xi}_e \times \boldsymbol{\xi})$: *transverse* relaxation, active only when $\boldsymbol{\xi}_e$ and $\boldsymbol{\xi}$ are not parallel. By (62.4.5) it is proportional to the component of $\boldsymbol{\xi}$ perpendicular to $\boldsymbol{\xi}_e$.

Anisotropic relaxation. The essential improvement over (SM-7) is that these two relaxation channels now have *different rates*. Reading off from (62.4.7) and (62.4.6), and dividing by L_e to convert to rates for $\boldsymbol{n}$:

$$\frac{1}{\tau_\parallel} \sim \frac{1}{\tau_B}\frac{L_e}{\xi_e} \cdot \frac{1}{\mathrm{d}L_e/\mathrm{d}\xi_e}, \qquad \frac{1}{\tau_\perp} \sim \frac{1}{2\tau_B}\left(1 - \frac{3L_e}{\xi_e}\right)\frac{\xi_e}{L_e}. \tag{62.6.1}$$

At $\xi_e \to 0$ both reduce to $1/\tau_B$ and the theory collapses back onto (SM-7), as it must. At large ξ_e they differ by a factor of order ξ_e: a strongly magnetised ferrofluid relaxes its magnetisation *direction* at a completely different rate from its *magnitude*. The phenomenological equation, which uses one time constant for both, cannot capture this — and it is precisely in the regime of large ξ and large $\omega\tau_B$, where both channels are active, that Bacri *et al.* found it failing.

62.7 Connection to the Next Equation

To use (B-6) we expand in the small parameter $\Omega\tau_B$. The zeroth order gives (B-7).

Chapter 63

Equation (B-7): The Zero Approximation

$$\tau_B \frac{\mathrm{d}\xi_e}{\mathrm{d}t} = -\left(\frac{\mathrm{d}\ln L_e}{\mathrm{d}\xi_e}\right)^{-1}\left(1 - \frac{\xi_0}{\xi_e}\cos\omega t\right). \tag{B-7}$$

63.1 Context

The paper explains the ordering: the problem contains a small dimensionless parameter $\Omega\tau_B$. A transient birefringence experiment gives $\tau_B \simeq 1.6$ ms for this fluid, so $\Omega\tau_B \ll 1$ is satisfied for any vorticity below 60 s^{-1}, which is always the case in their apparatus. For small $\Omega\tau_B$ the solution of (B-3) and (B-6) can be found by perturbation theory, and (B-7) is the zeroth-order equation.

Verifying the stated bound. $\Omega\tau_B \ll 1$ with $\tau_B = 1.6 \times 10^{-3}$ s means $\Omega \ll 625$ s^{-1}; the paper's statement that the inequality holds for Ω smaller than 60 s^{-1} corresponds to $\Omega\tau_B \lesssim 0.1$, a comfortable margin. ✓

63.2 Assumptions

1. $\Omega = 0$ at this order.
2. The applied field is $\boldsymbol{H} = (H_0\cos\omega t, 0, 0)$, so $\boldsymbol{\xi} = \xi_0\cos\omega t\,\hat{\boldsymbol{x}}$ with $\xi_0 = mH_0/kT$.
3. With no flow, the effective field has no reason to leave the field axis: $\boldsymbol{\xi}_e = \xi_e(t)\hat{\boldsymbol{x}}$, with ξ_e allowed to be of either sign.

63.3 Step-by-Step Derivation

Step 1: the transverse term vanishes. With $\boldsymbol{\xi}_e \parallel \boldsymbol{\xi}$, the cross product $\boldsymbol{\xi}_e \times \boldsymbol{\xi} = \boldsymbol{0}$ by Eq. (1.5.6), so the last term of (B-6) is identically zero. So is the flow term, since $\boldsymbol{\Omega} = \boldsymbol{0}$.

Step 2: what remains. With $\boldsymbol{n} = \hat{\boldsymbol{x}}$ constant,

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(L_e\hat{\boldsymbol{x}}\big) = -\frac{1}{\tau_B}\frac{L_e}{\xi_e}\big(\xi_e - \xi_0\cos\omega t\big)\hat{\boldsymbol{x}}, \tag{63.3.1}$$

i.e., as a scalar equation,

$$\frac{\mathrm{d}L_e}{\mathrm{d}t} = -\frac{L_e}{\tau_B \xi_e}\big(\xi_e - \xi_0 \cos\omega t\big). \tag{63.3.2}$$

Step 3: change the dependent variable. We want an equation for ξ_e, not for L_e. By the chain rule,

$$\frac{\mathrm{d}L_e}{\mathrm{d}t} = \frac{\mathrm{d}L_e}{\mathrm{d}\xi_e}\frac{\mathrm{d}\xi_e}{\mathrm{d}t}. \tag{63.3.3}$$

Substituting into (63.3.2) and solving for $\mathrm{d}\xi_e/\mathrm{d}t$:

$$\frac{\mathrm{d}\xi_e}{\mathrm{d}t} = -\frac{1}{\mathrm{d}L_e/\mathrm{d}\xi_e} \cdot \frac{L_e}{\tau_B \xi_e}\big(\xi_e - \xi_0 \cos\omega t\big). \tag{63.3.4}$$

Step 4: recognise the logarithmic derivative. Since

$$\frac{\mathrm{d}\ln L_e}{\mathrm{d}\xi_e} = \frac{1}{L_e}\frac{\mathrm{d}L_e}{\mathrm{d}\xi_e}, \qquad \text{so} \qquad \frac{L_e}{\mathrm{d}L_e/\mathrm{d}\xi_e} = \left(\frac{\mathrm{d}\ln L_e}{\mathrm{d}\xi_e}\right)^{-1}, \tag{63.3.5}$$

Eq. (63.3.4) becomes

$$\frac{\mathrm{d}\xi_e}{\mathrm{d}t} = -\left(\frac{\mathrm{d}\ln L_e}{\mathrm{d}\xi_e}\right)^{-1} \frac{1}{\tau_B \xi_e}\big(\xi_e - \xi_0 \cos\omega t\big). \tag{63.3.6}$$

Step 5: tidy. Multiply by τ_B and divide the bracket by ξ_e:

$$\tau_B \frac{\mathrm{d}\xi_e}{\mathrm{d}t} = -\left(\frac{\mathrm{d}\ln L_e}{\mathrm{d}\xi_e}\right)^{-1} \left(1 - \frac{\xi_0}{\xi_e}\cos\omega t\right), \tag{B-7}$$

which is (B-7). ■

63.4 Physical Interpretation

The fixed point. The right-hand side vanishes when $\xi_e = \xi_0 \cos\omega t$, i.e. when the effective field equals the true field. That is the quasi-static state, reached whenever the field varies slowly compared with τ_B. Everything interesting comes from the failure to reach it.

The relaxation rate is field-dependent. The prefactor $(\mathrm{d}\ln L_e/\mathrm{d}\xi_e)^{-1}$ is the new feature compared with (SM-7), where the corresponding factor was the constant 1. Its behaviour:

- Weak field: $L \simeq \xi/3$, so $\ln L \simeq \ln\xi - \ln 3$ and $\mathrm{d}\ln L/\mathrm{d}\xi = 1/\xi$, giving $(\mathrm{d}\ln L/\mathrm{d}\xi)^{-1} = \xi_e$. Then (B-7) reads $\tau_B \dot{\xi}_e = -(\xi_e - \xi_0\cos\omega t)$ — exactly the linear Debye relaxation of Volume V, Eq. (44.4.2). ✓
- Strong field: $L \to 1 - 1/\xi$, so $\mathrm{d}L/\mathrm{d}\xi \to 1/\xi^2$ and $\mathrm{d}\ln L/\mathrm{d}\xi \to 1/\xi^2$, giving $(\mathrm{d}\ln L/\mathrm{d}\xi)^{-1} \to \xi_e^2$. The restoring term is enormously enhanced: a strongly magnetised system is stiff, and its effective field tracks the applied one much more closely than the weak-field theory would suggest.

This stiffening is the single most important physical difference between the effective-field theory and the phenomenological one, and it is what shifts the computed curves into agreement with the measurements.

63.5 Connection to the Next Equation

At first order in $\Omega\tau_B$ the effective field acquires a transverse component, and its amplitude is governed by (B-8).

Chapter 64

Equation (B-8): The First-Order Response

$$\tau_B \frac{\mathrm{d}F}{\mathrm{d}t} = 1 - \frac{1}{2}\left(\frac{1}{L_e} - \frac{1}{\xi_e}\right)\xi_0 F\cos\omega t. \tag{B-8}$$

64.1 Context

The paper states: in the linear approximation in $\Omega\tau_B$, Eq. (B-6) yields

$$\boldsymbol{M}^{(1)} = M^{(0)} F(\xi_e)\,\tau_B\,\boldsymbol{\Omega}\times\hat{\boldsymbol{h}}, \qquad M^{(0)} = \varphi M_s L(\xi_e), \tag{64.1.1}$$

and the function F is determined by (B-8). This is the analogue of the function G of Volume VI: it carries the transverse response, and it is the object that will be correlated with the field to give the torque.

64.2 Symbols

$\hat{\boldsymbol{h}} = \hat{\boldsymbol{x}}$ is the unit vector along the applied field, so $\boldsymbol{\Omega}\times\hat{\boldsymbol{h}} = \Omega\hat{\boldsymbol{z}}\times\hat{\boldsymbol{x}} = \Omega\hat{\boldsymbol{y}}$: the first-order magnetisation is transverse, as it must be. F is dimensionless.

64.3 Set-up

Write the effective field to first order in $\Omega\tau_B$ as

$$\boldsymbol{\xi}_e = \xi_e\hat{\boldsymbol{x}} + \xi_e^{(1)}\hat{\boldsymbol{y}}, \qquad \boldsymbol{n} = \hat{\boldsymbol{x}} + \varepsilon\,\hat{\boldsymbol{y}}, \qquad \varepsilon \equiv \frac{\xi_e^{(1)}}{\xi_e} = O\big(\Omega\tau_B\big). \tag{64.3.1}$$

The magnitude $|\boldsymbol{\xi}_e| = \xi_e\sqrt{1+\varepsilon^2} = \xi_e + O(\varepsilon^2)$ is unchanged at first order, so L_e and every function of ξ_e retain their zeroth-order values.

Definition of F. By (B-5), the transverse magnetisation is

$$M_y = \varphi M_s L_e\,\varepsilon = M^{(0)}\varepsilon, \tag{64.3.2}$$

and comparison with (64.1.1), whose y component is $M^{(0)}F\tau_B\Omega$, gives

$$\boxed{\varepsilon = F\,\tau_B\,\Omega} \tag{64.3.3}$$

This *defines* F; the task is to find its equation of motion.

64.4 Step-by-Step Derivation

We take the $\hat{\boldsymbol{y}}$ component of (B-6), keeping terms of first order in ε.

Step 1: the left-hand side.

$$\left[\frac{\mathrm{d}}{\mathrm{d}t}\big(L_e\boldsymbol{n}\big)\right]_y = \frac{\mathrm{d}}{\mathrm{d}t}\big(L_e\varepsilon\big) = \tau_B\Omega\,\frac{\mathrm{d}}{\mathrm{d}t}\big(L_eF\big), \tag{64.4.1}$$

using (64.3.3) and treating Ω and τ_B as constants.

Step 2: the flow term.

$$\Big[\boldsymbol{\Omega}\times\big(L_e\boldsymbol{n}\big)\Big]_y = \Big[\Omega\hat{\boldsymbol{z}}\times L_e\big(\hat{\boldsymbol{x}}+\varepsilon\hat{\boldsymbol{y}}\big)\Big]_y \tag{64.4.2}$$

$$= \Big[\Omega L_e\big(\hat{\boldsymbol{y}}-\varepsilon\hat{\boldsymbol{x}}\big)\Big]_y \qquad \text{using } \hat{\boldsymbol{z}}\times\hat{\boldsymbol{x}}=\hat{\boldsymbol{y}},\ \hat{\boldsymbol{z}}\times\hat{\boldsymbol{y}}=-\hat{\boldsymbol{x}} \tag{64.4.3}$$

$$= \Omega L_e. \tag{64.4.4}$$

Note this term is *zeroth* order in ε: it is the source that creates the transverse component in the first place.

Step 3: the longitudinal relaxation term. With $\boldsymbol{\xi}_e = \xi_e\hat{\boldsymbol{x}} + \xi_e\varepsilon\hat{\boldsymbol{y}}$ and $\boldsymbol{\xi} = \xi_0\cos\omega t\,\hat{\boldsymbol{x}}$ (no y component),

$$\left[-\frac{1}{\tau_B}\frac{L_e}{\xi_e}\big(\boldsymbol{\xi}_e-\boldsymbol{\xi}\big)\right]_y = -\frac{1}{\tau_B}\frac{L_e}{\xi_e}\big(\xi_e\varepsilon\big) = -\frac{L_e\varepsilon}{\tau_B}. \tag{64.4.5}$$

Step 4: the transverse relaxation term. From (62.4.5),

$$\boldsymbol{\xi}_e\times\big(\boldsymbol{\xi}_e\times\boldsymbol{\xi}\big) = \xi_e^2\Big[\boldsymbol{n}\big(\boldsymbol{n}\cdot\boldsymbol{\xi}\big)-\boldsymbol{\xi}\Big]. \tag{64.4.6}$$

Compute the bracket:

$$\boldsymbol{n}\cdot\boldsymbol{\xi} = \big(\hat{\boldsymbol{x}}+\varepsilon\hat{\boldsymbol{y}}\big)\cdot\big(\xi_0\cos\omega t\,\hat{\boldsymbol{x}}\big) = \xi_0\cos\omega t, \tag{64.4.7}$$

$$\boldsymbol{n}\big(\boldsymbol{n}\cdot\boldsymbol{\xi}\big)-\boldsymbol{\xi} = \xi_0\cos\omega t\big(\hat{\boldsymbol{x}}+\varepsilon\hat{\boldsymbol{y}}\big)-\xi_0\cos\omega t\,\hat{\boldsymbol{x}} = \varepsilon\,\xi_0\cos\omega t\,\hat{\boldsymbol{y}}. \tag{64.4.8}$$

The $\hat{\boldsymbol{x}}$ components cancel exactly — as they must, since a vector triple product of this form is perpendicular to $\boldsymbol{\xi}_e$ — leaving a purely transverse result. Hence

$$\left[-\frac{1}{2\tau_B\xi_e^2}\left(1-\frac{3L_e}{\xi_e}\right)\boldsymbol{\xi}_e\times\big(\boldsymbol{\xi}_e\times\boldsymbol{\xi}\big)\right]_y = -\frac{1}{2\tau_B}\left(1-\frac{3L_e}{\xi_e}\right)\varepsilon\,\xi_0\cos\omega t, \tag{64.4.9}$$

where $L_3 = L_e$ from (62.4.10) has been used.

Step 5: assemble. Equating (64.4.1) to the sum of (64.4.4), (64.4.5) and (64.4.9), and substituting $\varepsilon = F\tau_B\Omega$:

$$\tau_B\Omega\frac{\mathrm{d}}{\mathrm{d}t}\big(L_eF\big) = \Omega L_e - \frac{L_e}{\tau_B}F\tau_B\Omega - \frac{1}{2\tau_B}\left(1-\frac{3L_e}{\xi_e}\right)F\tau_B\Omega\,\xi_0\cos\omega t. \tag{64.4.10}$$

Divide throughout by Ω (nonzero) and simplify each term:

$$\tau_B\frac{\mathrm{d}}{\mathrm{d}t}\big(L_eF\big) = L_e - L_eF - \frac{1}{2}\left(1-\frac{3L_e}{\xi_e}\right)\xi_0F\cos\omega t. \tag{64.4.11}$$

Step 6: expand the left-hand side. By the product rule,

$$\tau_B\frac{\mathrm{d}}{\mathrm{d}t}\big(L_eF\big) = \tau_BL_e\frac{\mathrm{d}F}{\mathrm{d}t} + \tau_BF\frac{\mathrm{d}L_e}{\mathrm{d}t}. \tag{64.4.12}$$

We must therefore know $\tau_B\,\mathrm{d}L_e/\mathrm{d}t$, which the zeroth-order equation supplies.

Step 7: evaluate $\tau_B\,\mathrm{d}L_e/\mathrm{d}t$ from (B-7). By the chain rule and (63.3.5),

$$\tau_B\frac{\mathrm{d}L_e}{\mathrm{d}t} = \frac{\mathrm{d}L_e}{\mathrm{d}\xi_e}\cdot\tau_B\frac{\mathrm{d}\xi_e}{\mathrm{d}t} \tag{64.4.13}$$

$$= \frac{\mathrm{d}L_e}{\mathrm{d}\xi_e}\cdot\left[-\left(\frac{\mathrm{d}\ln L_e}{\mathrm{d}\xi_e}\right)^{-1}\left(1-\frac{\xi_0}{\xi_e}\cos\omega t\right)\right] \tag{64.4.14}$$

$$= -\frac{\mathrm{d}L_e/\mathrm{d}\xi_e}{\mathrm{d}\ln L_e/\mathrm{d}\xi_e}\left(1-\frac{\xi_0}{\xi_e}\cos\omega t\right) \tag{64.4.15}$$

$$= -L_e\left(1-\frac{\xi_0}{\xi_e}\cos\omega t\right), \tag{64.4.16}$$

the last step because $\mathrm{d}\ln L_e/\mathrm{d}\xi_e = (1/L_e)\,\mathrm{d}L_e/\mathrm{d}\xi_e$, so the ratio is simply L_e. *This clean cancellation is the reason the authors wrote (B-7) in terms of the logarithmic derivative.*

Step 8: substitute and collect. Insert (64.4.12) with (64.4.16) into (64.4.11):

$$\tau_BL_e\frac{\mathrm{d}F}{\mathrm{d}t} - L_eF\left(1-\frac{\xi_0}{\xi_e}\cos\omega t\right) = L_e - L_eF - \frac{1}{2}\left(1-\frac{3L_e}{\xi_e}\right)\xi_0F\cos\omega t. \tag{64.4.17}$$

Move the second term of the left side across:

$$\tau_BL_e\frac{\mathrm{d}F}{\mathrm{d}t} = L_e - L_eF - \frac{1}{2}\left(1-\frac{3L_e}{\xi_e}\right)\xi_0F\cos\omega t + L_eF - \frac{L_eF\xi_0\cos\omega t}{\xi_e} \tag{64.4.18}$$

$$= L_e - \frac{1}{2}\left(1-\frac{3L_e}{\xi_e}\right)\xi_0F\cos\omega t - \frac{L_e\xi_0F\cos\omega t}{\xi_e}, \tag{64.4.19}$$

the terms $\mp L_eF$ having cancelled.

Step 9: divide by L_e.

$$\tau_B\frac{\mathrm{d}F}{\mathrm{d}t} = 1 - \xi_0F\cos\omega t\left[\frac{1}{2L_e}\left(1-\frac{3L_e}{\xi_e}\right)+\frac{1}{\xi_e}\right] \tag{64.4.20}$$

$$= 1 - \xi_0F\cos\omega t\left[\frac{1}{2L_e}-\frac{3}{2\xi_e}+\frac{1}{\xi_e}\right] \tag{64.4.21}$$

$$= 1 - \xi_0 F\cos\omega t\left[\frac{1}{2L_e} - \frac{1}{2\xi_e}\right] \tag{64.4.22}$$

$$= 1 - \frac{1}{2}\left(\frac{1}{L_e} - \frac{1}{\xi_e}\right)\xi_0 F\cos\omega t, \tag{B-8}$$

using $-3/2 + 1 = -1/2$ in the third line. This is (B-8) exactly. ■

Remark 64.4.1 (The check on L_3). Had we kept L_3 general in Step 4, the bracket in Step 9 would have been $\frac{1}{2L_e} - \frac{3L_3}{2L_e\xi_e} + \frac{1}{\xi_e}$, which equals the printed $\frac{1}{2L_e} - \frac{1}{2\xi_e}$ only if $L_3 = L_e$. Equation (B-8) as published therefore confirms the identification (62.4.10) independently of the matching argument of Sec. 62.5.

64.5 Consistency Check: the Weak-Field Limit

At $\xi_e \ll 1$, $L_e \to \xi_e/3$, so

$$\frac{1}{2}\left(\frac{1}{L_e} - \frac{1}{\xi_e}\right) \to \frac{1}{2}\left(\frac{3}{\xi_e} - \frac{1}{\xi_e}\right) = \frac{1}{\xi_e}, \tag{64.5.1}$$

and (B-8) becomes $\tau_B \dot{F} = 1 - (\xi_0/\xi_e)F\cos\omega t$. Meanwhile, in the same limit, the zeroth-order equation reads $\tau_B\dot{\xi}_e = -(\xi_e - \xi_0\cos\omega t)$ (Sec. 63.5). The transverse magnetisation is $M_y = M^{(0)}F\tau_B\Omega$ with $M^{(0)} = \varphi M_s\xi_e/3 = \chi H_e$ — i.e. exactly the structure of the weak-field theory of Volume V, in which M_y was likewise proportional to $\Omega\tau_B$ times a function of frequency. ✓

64.6 Connection to the Next Equation

With $\xi_e(t)$ from (B-7) and $F(t)$ from (B-8), the averaged torque and hence the viscosity follow. That is (B-9).

Chapter 65

Equation (B-9): The Mean Torque and the Function g

$$\overline{\boldsymbol{M}\times\boldsymbol{H}} = -6\eta\varphi\, g\,\boldsymbol{\Omega}, \qquad g = \frac{1}{2}\,\xi_0\,\overline{\cos\omega t\, L(\xi_e)F(\xi_e)}. \tag{B-9}$$

65.1 Context

The paper notes that the torque may be averaged over the period of field variation $2\pi/\omega$, because in their experiments this period is always much smaller than the characteristic hydrodynamic time $\rho d^2/\eta$. This is the same two-scale argument proved in Chapter 40; here we check the criterion with their numbers.

Verifying the scale separation. With $d = 0.1$ cm (the capillary diameter), $\rho \approx 2$ g cm^{-3} and $\eta = 0.77$ P,

$$\frac{\rho d^2}{\eta} = \frac{2\times10^{-2}}{0.77} = 2.6\times10^{-2}\ \text{s} = 26\ \text{ms}, \tag{65.1.1}$$

against field periods from 19 ms (at 52 Hz) down to 0.68 ms (at 1480 Hz). The separation is marginal at the lowest frequency and excellent at the highest, which — as in Sec. 40.3 — is the regime where the effect matters. ✓

65.2 Step-by-Step Derivation

Step 1: the torque has only a z component. As in Sec. 39.4, with $\boldsymbol{H} = H_0\cos\omega t\,\hat{\boldsymbol{x}}$ and $\boldsymbol{M} = (M_x, M_y, 0)$,

$$\big(\boldsymbol{M}\times\boldsymbol{H}\big)_z = -M_y H_x. \tag{65.2.1}$$

Step 2: substitute the first-order magnetisation. By (64.1.1) and (64.3.3), $M_y = M^{(0)}F\tau_B\Omega = \varphi M_s L(\xi_e)F\tau_B\Omega$, so

$$\big(\boldsymbol{M}\times\boldsymbol{H}\big)_z = -\varphi M_s L\big(\xi_e\big)F\big(\xi_e\big)\tau_B\Omega\cdot H_0\cos\omega t. \tag{65.2.2}$$

Step 3: average. The constants φM_s, τ_B, Ω, H_0 pass outside:

$$\overline{\big(\boldsymbol{M}\times\boldsymbol{H}\big)_z} = -\varphi M_s H_0\tau_B\Omega\,\overline{\cos\omega t\, L\big(\xi_e\big)F\big(\xi_e\big)}. \tag{65.2.3}$$

Step 4: evaluate the prefactor. Using $\varphi M_s = nm$ [Sec. 10.4], $\tau_B = 3\eta V/kT$ and $\varphi = nV$:

$$\varphi M_s H_0 \tau_B = n\,m\,H_0\,\frac{3\eta V}{kT} \tag{65.2.4}$$

$$= 3\eta\,(nV)\,\frac{mH_0}{kT} \tag{65.2.5}$$

$$= 3\eta\varphi\,\xi_0. \tag{65.2.6}$$

This is the same combination met at Eq. (52.4.4) of Volume VI. ✓

Step 5: assemble. Substituting (65.2.6) into (65.2.3), and writing the result as a vector along $\boldsymbol{\Omega} = \Omega\hat{\boldsymbol{z}}$:

$$\overline{\boldsymbol{M}\times\boldsymbol{H}} = -3\eta\varphi\xi_0\,\overline{\cos\omega t\,L_eF}\;\boldsymbol{\Omega} = -6\eta\varphi\left[\frac{\xi_0}{2}\overline{\cos\omega t\,L_eF}\right]\boldsymbol{\Omega}, \tag{65.2.7}$$

so that with the definition

$$g \equiv \frac{\xi_0}{2}\,\overline{\cos\omega t\,L\big(\xi_e\big)F\big(\xi_e\big)} \tag{65.2.8}$$

we obtain

$$\overline{\boldsymbol{M}\times\boldsymbol{H}} = -6\eta\varphi\,g\,\boldsymbol{\Omega}, \tag{B-9}$$

■

65.3 The Particle Spin and the Viscosity

The spin. Substitute (B-9) into the torque balance (B-1), $6\eta\varphi(\boldsymbol{\omega}_p - \boldsymbol{\Omega}) = \overline{\boldsymbol{M}\times\boldsymbol{H}}$:

$$6\eta\varphi\big(\boldsymbol{\omega}_p - \boldsymbol{\Omega}\big) = -6\eta\varphi g\boldsymbol{\Omega} \tag{65.3.1}$$

$$\boldsymbol{\omega}_p - \boldsymbol{\Omega} = -g\boldsymbol{\Omega} \tag{65.3.2}$$

$$\boxed{\boldsymbol{\omega}_p = \big(1-g\big)\boldsymbol{\Omega}} \tag{65.3.3}$$

exactly as the paper states.

The viscosity. From (65.3.3), $(\Omega - \omega_p)/\Omega = g$, and (B-2) gives

$$\boxed{\Delta\eta = \frac{3}{2}\,\eta\varphi\,g} \tag{65.3.4}$$

So g is nothing but the rotational viscosity in units of its saturation value:

$$g = \frac{\Delta\eta}{\frac{3}{2}\eta\varphi} = \frac{\Delta\eta\big(\xi_0,\omega\tau_B\big)}{\Delta\eta(\infty,0)} = \frac{\eta_r\big(\xi_0,\omega\tau_B\big)}{\eta_r(\infty,0)}, \tag{65.3.5}$$

which is why the paper plots g in Figs. 3 and 4 and compares it directly with the measured reduced viscosities of Fig. 5.

65.4 Physical Interpretation

Everything has again come down to a correlation. The number g measures how much of the transverse response L_eF is *in phase* with the driving field $\cos\omega t$. Positive correlation means the grains lag, the torque brakes them, and the viscosity rises; negative correlation means the grains are driven ahead of the flow and the viscosity falls. The structure is identical to $\overline{G\gamma}$ in Volume VI and to $\cos 2\alpha$ in Volume V. Three different formalisms, one mechanism.

65.5 Connection to the Next Equation

Substituting the averaged torque into the momentum equation gives (B-10) and closes the theoretical part of the paper.

Chapter 66

Equation (B-10): The Final Grouping

$$\eta\Delta\boldsymbol{v} - 3\eta\varphi g\,\operatorname{curl}\boldsymbol{\Omega} = -2\eta\left(1+\frac{3}{2}\varphi g\right)\operatorname{curl}\boldsymbol{\Omega}. \tag{B-10}$$

66.1 Step-by-Step Derivation

Step 1: the magnetic force density. By (B-3), the magnetic term of the averaged momentum equation is $\frac{1}{2}\operatorname{curl}\overline{\boldsymbol{M}\times\boldsymbol{H}}$. With (B-9),

$$\tfrac{1}{2}\operatorname{curl}\overline{\boldsymbol{M}\times\boldsymbol{H}} = \tfrac{1}{2}\operatorname{curl}\big(-6\eta\varphi g\boldsymbol{\Omega}\big) = -3\eta\varphi g\,\operatorname{curl}\boldsymbol{\Omega}, \tag{66.1.1}$$

which is the second term on the left of (B-10).

Step 2: the viscous term. By Eq. (41.2.6),

$$\eta\Delta\boldsymbol{v} = -2\eta\,\operatorname{curl}\boldsymbol{\Omega}. \tag{66.1.2}$$

Step 3: add.

$$\begin{aligned}\eta\Delta\boldsymbol{v} - 3\eta\varphi g\,\operatorname{curl}\boldsymbol{\Omega} &= -2\eta\,\operatorname{curl}\boldsymbol{\Omega} - 3\eta\varphi g\,\operatorname{curl}\boldsymbol{\Omega} &(66.1.3)\\ &= -2\eta\left(1+\frac{3\varphi g}{2}\right)\operatorname{curl}\boldsymbol{\Omega}, &(B\text{-}10)\end{aligned}$$

having extracted the factor -2η: $3\eta\varphi g = 2\eta\cdot\frac{3}{2}\varphi g$. ■

66.2 Reading the Result

The bracket $\big(1+\frac{3}{2}\varphi g\big)$ is the viscosity in units of η, so

$$\eta_{\text{total}} = \eta\left(1+\frac{3}{2}\varphi g\right), \qquad \Delta\eta = \frac{3}{2}\eta\varphi g, \tag{66.2.1}$$

in agreement with (65.3.4). Everything the ferrofluid does, in this theory, is contained in the single number $g(\xi_0,\omega\tau_B)$.

Chapter 67

The Numerical Procedure and the Theoretical Figures

67.1 The computation

The paper describes the procedure: for arbitrary values of the field amplitude and frequency the problem is solved numerically, by integrating the two differential equations (B-7) and (B-8) and then computing the integral over the period in (B-9). For prescribed ξ_0 and $\omega\tau_B$ the equations are solved by the fourth-order Runge–Kutta method with the initial conditions $\xi_e = \xi_0$ and $F \simeq 0$ at $t \simeq 0$.

Written out, the algorithm is:

1. Choose ξ_0 and $\omega\tau_B$. Set $t = 0$, $\xi_e = \xi_0$, $F = 0$.

2. Advance (B-7) and (B-8) together by fourth-order Runge–Kutta. They are coupled one way only — (B-7) does not contain F — so $\xi_e(t)$ could in principle be found first, but it is simplest to step both at once.

3. Integrate for several periods until the transients die and the solution is periodic. (The initial conditions are not the periodic ones; the choice $\xi_e = \xi_0$, $F = 0$ is merely convenient, and the attractor is unique.)

4. Over one final period, accumulate the average $\overline{\cos\omega t\, L(\xi_e) F}$ and form $g = \frac{1}{2}\xi_0 \times$ that average, by (65.2.8).

5. Repeat on a grid of $(\xi_0, \omega\tau_B)$.

Why ξ_e is well behaved. One might worry about (B-7) when $\cos\omega t$ passes through zero, since $\xi_0 \cos\omega t/\xi_e$ then vanishes and ξ_e is driven towards zero, where $(\mathrm{d}\ln L_e/\mathrm{d}\xi_e)^{-1} \to \xi_e \to 0$. In fact both the prefactor and the driving vanish together and the equation remains regular: near $\xi_e = 0$ it reduces to $\tau_B\dot{\xi}_e = -(\xi_e - \xi_0\cos\omega t)$, a perfectly ordinary linear equation which carries ξ_e smoothly through zero into negative values (the effective field simply reverses with the applied one).

67.2 Figure 3 of Bacri *et al.*: g versus field

The figure plots

$$g\big(\xi_0,\omega\tau_B\big)=\frac{\eta_r\big(\xi_0,\omega\tau_B\big)}{\eta_r(\infty,0)} \tag{67.2.1}$$

against ξ_0 (from 0 to 20), for $\omega\tau_B$ between 0.1 and 2. The qualitative content, now fully explicable:

- For $\omega\tau_B=0.1$ the curve rises monotonically towards ≈ 0.8: essentially the stationary behaviour of (SM-2), slightly reduced.
- For $\omega\tau_B=1$ the curve is nearly flat and small: this is the crossover, where the in-phase correlation in (65.2.8) nearly vanishes.
- For $\omega\tau_B=1.5$ and 2 the curves go *negative*, dip to a minimum near $\xi_0\approx 6$–8, and then rise back towards zero as ξ_0 grows further.

The non-monotonicity, explained. At large ξ_0 the effective field is stiff — Sec. 63.5 showed the restoring rate grows as ξ_e^2 — so ξ_e tracks the applied field ever more closely, the lag shrinks, and the correlation weakens. Hence $|g|$ falls again at very large fields. This is the same competition identified in Sec. 57.3 for the Shliomis–Morozov theory, but with the crucial difference that here the stiffening is computed from the Fokker–Planck-derived relaxation rather than assumed.

The discrepancy that motivated the whole exercise. Bacri *et al.* point out that with the phenomenological equation the computed $\Delta\eta(\xi)$ has a maximum at $\omega\tau_B<1$ and no minimum at $\omega\tau_B>1$, whereas the measurements show the opposite. Figure 3, computed from the effective-field theory, does show a minimum at $\omega\tau_B>1$, in agreement with the data. The improvement is entirely due to the anisotropic, field-dependent relaxation of Sec. 62.7.

67.3 Figure 4: the theoretical isolines

More convenient than a family of curves is a contour map: isolines $g=\text{const}$ in the plane $(\xi_0,\omega\tau_B)$, for g between $+0.8$ and -0.9. The isoline $g=0$ divides the plane into two regions,

$$g>0 \text{ for } \omega\tau_B\lesssim 1, \qquad g<0 \text{ for } \omega\tau_B\gtrsim 1, \tag{67.3.1}$$

so that the sign change occurs at $\omega\tau_B\sim 1$ almost independently of the field amplitude — a robust prediction, and the one that experiment can most cleanly test.

Chapter 68

The Experiment

68.1 The fluid

The magnetic fluid is prepared by Massart's method; the colloidal particles are cobalt ferrite with anisotropy constant $K = 2 \times 10^6$ erg cm^{-3}, grain magnetisation $M_s = 400$ G and typical size 10 nm. The large anisotropy freezes the moment into the crystal up to fields of order 10^4 Oe — the rigid-dipole condition, which we verified quantitatively in Sec. 17.6 by computing $\tau_N \approx 100$ s. To maximise the effect the suspension is concentrated: volume fraction $\varphi = 20\%$ in water.

The design logic, in one line. Every one of these choices follows from the theory. High M_s and high φ maximise $\Delta\eta = \frac{3}{2}\eta\varphi g$; high K guarantees rigid dipoles so that $g \neq 0$ at all; and the particle size places τ_B such that $\omega\tau_B \sim 1$ falls in the audio range, where a solenoid can actually be driven.

68.2 The apparatus and the measurement

A Poiseuille flow in a horizontal capillary of diameter 1 mm is used for the viscosity measurement. The pressure gradient is provided by the difference h in ferrofluid level between inlet and outlet, and the mass flow rate Q is weighed at the outlet. The tube sits inside a solenoid producing a field along the flow direction up to $H = 1$ kOe at frequencies $f = \omega/2\pi$ from 0 to 1 kHz.

Why this geometry. As shown in Sec. 3.5, a cylindrical Poiseuille flow has azimuthal vorticity, while the solenoid field is axial. Hence $\boldsymbol{H} \perp \boldsymbol{\Omega}$, i.e. $\beta = \pi/2$, which by (SM-2) maximises the effect. It also means the demagnetising field vanishes (Sec. 22.6), so the internal field equals the applied one.

Figure 1: the Newtonian check. Q is measured as a function of h for $H = 0$ and $H = 2000$ Oe. Both data sets fall on straight lines through the origin, which establishes that the fluid is Newtonian with and without field, and the slopes give the viscosities. This is a necessary check: the entire theory computes a *viscosity*, i.e. assumes a linear relation between stress and shear rate, and Sec. 15.5, Step 12 showed that the theory does predict Ω to cancel out of $\Delta\eta$. Figure 1 confirms it experimentally. ✓

68.3 The numbers

Quantity	Value	Source
solvent (water) viscosity	1 cP	—
$\eta(0,0)$, zero field	77 cP	Fig. 1 slope
$\eta(2000\text{ Oe},0)$, static	220 cP	Fig. 1 slope
$\Delta\eta$ at $f=0$, $H=2000$ Oe	$220-77=143$ cP	
η at $H\sim2000$ Oe, $f=50$ Hz	100 cP	
η at $H\sim2000$ Oe, $f=250$ Hz	77 cP	
η at $H\sim2000$ Oe, $f=700$ Hz	58 cP	
$\Delta\eta$ at $f\geq700$ Hz	$58-77=-19$ cP	
$\eta_r(\infty,0)$	$\simeq 2.5$	Fig. 2 asymptote

The central experimental result. At $f=250$ Hz the viscosity in a 2000 Oe field is 77 cP — *exactly the zero-field value*. The magnetic field, at that frequency, does nothing at all to the viscosity: $\Delta\eta=0$. Above that frequency it *reduces* the viscosity below its zero-field value, reaching 58 cP at 700 Hz. The reduction is

$$\frac{77-58}{77}=0.247\approx25\%, \tag{68.3.1}$$

the figure quoted in the paper. *This is the measurement of a negative rotational viscosity*, and it is the first such measurement reported.

Consistency with the theory's frequency scale. The sign change is observed near 250 Hz. With the calibration $\omega\tau_B=10^{-2}f(\text{Hz})$ from Sec. 17.5, that is $\omega\tau_B\approx2.5$. The theory of Sec. 67.3 predicts the change of sign near $\omega\tau_B\sim1$. The agreement is of the right order but not exact — unsurprising given the polydispersity of a real ferrofluid, which smears τ_B over a range, and the concentration $\varphi=0.2$, at which interparticle interactions are certainly not negligible.

68.4 Figure 2: the raw data

The reduced viscosity

$$\eta_r(H,f)=\frac{\Delta\eta}{\eta(0,0)}=\frac{\eta(H,f)-\eta(0,0)}{\eta(0,0)} \tag{68.4.1}$$

is plotted against H for $f=0,52,150,345,645,1480$ Hz. The behaviour:

- At $f=0$ the curve rises monotonically towards $\eta_r(\infty,0)\simeq2.5$ — this is the McTague effect, first measured in 1969 and explained by Shliomis in 1972, i.e. our Eq. (SM-2).
- At 52 Hz the shape is essentially unchanged.
- At 150 Hz the shape changes qualitatively: η_r first decreases, becomes negative, passes through a minimum, and only then rises towards its asymptote.
- At higher frequencies the same happens, with a deeper minimum.

The non-monotonic $\eta_r(H)$. This is precisely the behaviour predicted by the effective-field theory in Fig. 3 (Sec. 67.2) and *not* by the phenomenological one. It is the experimental signature of the field-dependent relaxation of Sec. 62.7.

68.5 Figure 5: the experimental isolines, and the comparison

For a quantitative comparison the data are replotted as isolines of g in the plane $(\xi_0, \omega\tau_B)$, using

$$\eta_r(\infty, 0) = 2.5, \qquad \xi_0 = \frac{M_s V H}{kT} = 5.25 \times 10^{-3} H \text{ (Oe)}, \qquad \omega\tau_B = 2\pi f \tau_B = 10^{-2} f \text{ (Hz)}. \tag{68.5.1}$$

Both calibrations were verified from first principles in earlier volumes: the first in Sec. 23.5, where $m = M_s V$ with $M_s = 400$ G and $d \simeq 10$ nm gives 5.1×10^{-3} and back-solving the paper's figure gives $d = 10.1$ nm; the second in Sec. 17.5, where $2\pi \times 1.6 \times 10^{-3} = 1.005 \times 10^{-2}$. ✓

The experimental isolines are drawn for the values

$$g = 0.216, \quad 0.1, \quad 0.044, \quad 0, \quad -0.036, \quad -0.052, \quad -0.068, \quad -0.092.$$

Comparing Fig. 5 with the theoretical Fig. 4, the authors report a strong similarity, and in particular that in both cases g changes sign at $\omega\tau_B \sim 1$.

What the comparison does and does not establish. It establishes that a single-particle theory — no interactions, monodisperse, rigid dipoles — captures the *topology* of the response in the $(\xi_0, \omega\tau_B)$ plane: where the sign changes, where the extrema lie, how the curves bend. It does not establish quantitative agreement in magnitude, and the authors do not claim it: the measured isolines reach only $g \approx -0.09$ whereas the theory permits g down to -0.9, and the measured $\eta_r(\infty, 0) = 2.5$ exceeds the dilute prediction $\frac{3}{2}\varphi = 0.3$ by a factor of eight (Sec. 15.6). At $\varphi = 0.2$ that is expected.

68.6 Conclusions of the Experimental Paper

The result. For $\omega\tau_B \gtrsim 1$ the viscosity of a ferrofluid in an alternating field can be *smaller* than in zero field. A decrease of 25% is observed.

The limit. An interesting limit would be to reach absolute zero viscosity, at which instabilities and flow recirculations would develop spontaneously without any external flow — the pump of Sec. 58.3. The marginal curve predicted by Shliomis & Morozov corresponds to $\xi_c = 126$ and $\omega_c\tau_B = 20$. Such values could be reached only with larger particles, for which both colloidal stability and Newtonian behaviour would be difficult to preserve. This is the same conclusion reached in Sec. 58.4, now stated from the experimental side: it is not the field that is out of reach but the particle.

The application. The result that survives is the tunability: the viscosity of this Newtonian magnetic fluid can be tuned between 220 and 50 cP by a field of 2000 Oe, monitoring *only* the frequency between 0 and 700 Hz. A factor of four in viscosity, controlled electronically, with no moving parts and no change of field amplitude. The authors suggest this could lead to a new generation of adaptive dampers — and that suggestion, thirty years on, remains the principal engineering legacy of these two papers.

Chapter 69

The Project Complete

69.1 What has been derived

Every equation in both papers has now been obtained from first principles, with no step omitted and no result assumed. The full index:

Equation	Content	Volume, chapter
(SM-1)	$\tau_s = \rho_s d^2/60\eta$	I, Ch. 7
(SM-2)	stationary rotational viscosity	II, Ch. 15
(SM-3), (SM-4)	weak- and strong-field limits	II, Ch. 16
(SM-5)	$\tau_B = 3\eta V/kT$	II, Ch. 17
(SM-6)	fluid motion with spin coupling	III, Ch. 19
(SM-7)	magnetisation relaxation	III, Ch. 20
(SM-8)	rotational motion of the particles	III, Ch. 21
(SM-9)	incompressibility and magnetostatics	III, Ch. 22
(SM-10)	Langevin magnetisation	III, Ch. 23
(SM-11)	inertialess spin	III, Ch. 24
(SM-12)	reduced momentum equation	III, Ch. 25
(SM-13)	reduced magnetisation equation	III, Ch. 26
(SM-14)	the alternating field	III, Ch. 27
(SM-15)	linear magnetisation law	III, Ch. 28
(SM-16)	circular decomposition	IV, Ch. 30
(SM-17)	rotating ansatz	IV, Ch. 31
(SM-18)	the three relations	IV, Ch. 32
(SM-19)	elimination of the phase	IV, Ch. 33
(SM-20)	the vortex breaks the symmetry	IV, Ch. 34
(SM-21)	rotational Doppler shift	IV, Ch. 35
(SM-22)	transverse magnetisation of each component	IV, Ch. 36
(SM-23), (SM-24)	M_y and $\tan\alpha = \omega\tau_B$	IV, Ch. 37
(SM-25)	instantaneous torque	V, Ch. 39
(SM-26)	two-scale averaging	V, Ch. 40
(SM-27)	$\Delta\eta = \frac{1}{4}\eta\varphi\xi^2\cos^2\alpha\cos 2\alpha$	V, Ch. 41

Equation	Content	Volume, chapter
(SM-28)	**the central result**	V, Ch. 42
(SM-29)	high-frequency asymptote	V, Ch. 43
(SM-30)–(SM-32)	the second variant	V, Chs. 44–46
(SM-33)	mean particle spin *(misprint corrected)*	V, Ch. 47
(SM-34)	the master formula	V, Ch. 48
(SM-35)	nonlinear magnetisation equation	VI, Ch. 50
(SM-36)	coupled F, G equations	VI, Ch. 51
(SM-37)	averaged torque at arbitrary amplitude	VI, Ch. 52
(SM-38), (SM-39)	general viscosity formula	VI, Ch. 53
(SM-40)	square-wave solution *(misprint corrected)*	VI, Ch. 54
(SM-41)	half-period reduction	VI, Ch. 55
(SM-42), (SM-43)	formal solution	VI, Ch. 56
(B-1)–(B-4)	torque balance, $\Delta\eta$, momentum, magnetisation	II–III
(B-5)	effective-field ansatz	VII, Ch. 61
(B-6)	effective-field evolution *(L_3 identified)*	VII, Ch. 62
(B-7)	zero approximation	VII, Ch. 63
(B-8)	first-order response	VII, Ch. 64
(B-9)	mean torque and g	VII, Ch. 65
(B-10)	final grouping	VII, Ch. 66

69.2 Three editorial findings

In the course of the derivations three points emerged where the printed papers and the mathematics disagree. All three were resolved in favour of the mathematics, and in each case the papers' own numbers confirm the correction.

1. **(SM-33)** is printed with $1 - \xi^2 \tau_B^2$ in the numerator. This is dimensionally impossible (ξ is a pure number) and inconsistent with (SM-28) and (SM-34); it should read $1 - \omega^2 \tau_B^2$. [Volume V, Sec. 47.1.]

2. **(SM-40)** is printed with $\tanh \varepsilon$ where the derivation gives $\tanh^2 \varepsilon$. The corrected form reproduces all three of the authors' own quoted values — $\omega_0 \tau_B = 0.98$, $\omega_m \tau_B = 1.707$, $\Delta\eta_m = -0.105 \Delta\eta(0)$ — and the printed form does not. [Volume VI, Secs. 54.1 and 54.6.]

3. **(B-6)** contains an undefined symbol L_3. Matching against the exact moment equation with the effective-field closure identifies it as $L_3 = L(\xi_e)$, and the independent derivation of (B-8) confirms it. [Volume VII, Secs. 62.5 and 64.5.]

69.3 The physics, in one page

If everything else were forgotten, this is what should remain.

A ferrofluid is a suspension of magnetic grains small enough to be kept apart by thermal motion. Each grain can *spin* independently of the liquid around it, and that single extra degree of freedom is the whole subject.

In a shear flow the liquid rotates locally at $\boldsymbol{\Omega} = \frac{1}{2}\nabla \times \boldsymbol{v}$, and without a field the grains simply turn with it: nothing happens. Apply a stationary field and the grains' moments are held; the grains cannot turn freely; the liquid must shear past them; the viscosity rises by $\Delta\eta = \frac{3}{2}\eta\varphi(\Omega - \Theta)/\Omega$, which saturates at $\frac{3}{2}\eta\varphi$ when the grains are locked. A permanent magnet does no work, so thermodynamics guarantees this is always an increase.

Apply instead an *alternating* field along a fixed axis. It picks out an axis but no sense of rotation, so in a fluid at rest half the grains swing one way and half the other and the mean spin is exactly zero. Introduce a shear, and the degeneracy breaks: a grain carried round by the flow sees the two circular components of the field at Doppler-shifted frequencies $\omega \mp \Omega$, responds differently to each, and acquires a net transverse magnetisation proportional to Ω and lagging the field by *twice* the single-component phase lag. Crossed with the field, that magnetisation gives an average torque proportional to $-\Omega \cos 2\alpha$ — and $\cos 2\alpha$ reverses sign when α passes $\pi/4$, that is when $\omega\tau_B = 1$.

Above that frequency the torque no longer brakes the grains: it drives them, they spin *faster* than the liquid, and they feed angular momentum into the flow. The rotational viscosity becomes negative. There is no paradox — the generator driving the solenoid is doing the work, and part of it arrives as kinetic energy of the fluid.

The effect is real, it was measured, and it is modest: a 25% reduction below the zero-field viscosity. Driving the *total* viscosity to zero, which would turn the fluid into a self-driven pump, would require grains so large that they would neither stay suspended nor keep their moments frozen. What remains, and what may yet be useful, is a liquid whose viscosity can be changed fourfold by turning a frequency dial.

End of the work.

Volumes I–VII, Chapters 1–69, pages 1–239.